

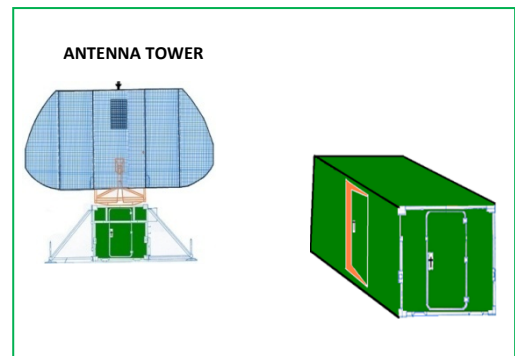
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AIR DEFENCE SYSTEM SCHOOL



MPDR 90ST



MPDR 90S / 45S



MPDR 45E / 60P /



MPDR 45L / 90SP

TYPE RATING COURSE MPDR (AVIONICS GROUND)

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CHAPTER No 1

GENERAL AND REVISIONARY

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CHAPTER No 01

GENERAL AND REVISIONARY

REGULATIONS CONCERNING THE PREVENTION OF ACCIDENTS DURING WORK ON RADAR EQUIPMENT

General

1. Radiation of electromagnetic RF energy by Radar equipment represents a potential source of danger for all personnel working in the vicinity of this equipment. The RF radiation from Radar antennas and equipment can cause severe damage through heating of body tissue. The eyes are particularly susceptible to radiation. Hence special care should be taken when working with Radar equipment. Radar personnel are subject to the special regulations concerning the medical care of soldiers, civil servants employees of the air force performing work which can injurious to the health.

2. Accident prevention is the best form of accident protection. Accident prevention regulations and guidelines always be carefully followed. Most accidents of Radar equipment are caused by careless handling. Warning signs must be attached to the operating during tests or repair work.

3. Persons working on or with a Radar set must be acquainted with the Radar set, the measuring equipment - and the safety prevention regulations. The operating personnel must be made familiar with these regulations before they commence their Radar training. Instruction must be repeated quarterly. Furthermore: personnel must be the instructed quarterly on fire-fighting with the aid of fire this extinguishers in and on electrical systems as well as on the associated danger of accidents. Moreover, personnel should be instructed at regular intervals, not exceeding a - year, in the handling of fire extinguishers. Practical training is necessary for this purpose.

(a) Persons not involved with the operation of a equipment should not be allowed near it when it is in operation.

(b) Before entering Radar positions, persons without expert knowledge must be informed of possible dangers and safety measures.

(c) First aid boxes must be close at hand for all work on or with Radar equipment. All persons operating or servicing Radar equipment must know how to give first-aid both to themselves and other

(d) For tactical reasons or in urgent cases of general, the troop commander may temporarily depart from specified safety regulations, though he carries the

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full responsibility for doing so. In any such case the basic principle is safety regulations may only be ignored and the fully risks temporarily taken if an even greater, already real danger can thereby be prevented.

4. General principles for accident prevention during work on all types of Radar equipment are that:

- (a) There should always be more than one person working on Radar equipment.
- (b) No superfluous tools should be carried in pockets during work. Rings, necklaces, identity disks, keys, coins, lighters and tobacco tins and any other metal objects must be removed from the body and from the clothing.
- (c) Before any work is carried out, all safety switches must be checked and the capacitors discharged by grounding them.
- (d) If a job can be performed with one hand only, do not use both hands.
- (e) Check and ground measuring equipment before use.
- (f) Ground Radar equipment and generators.
- (g) Petrol must not be used to clean electrical equipment.

5. It is forbidden to stand within the direct radiation cone of a Radar equipment when its transmitter is switched on. Specially identified emergency switches must be provided in the work area to allow the Radar system to be switched off in an emergency. The maximum permissible radiation power density for humans is 0.01 Watt per square centimeter of body surface. A certain distance from the reflector must therefore be maintained. This distance depends on the power output and type of equipment (concentration of the Radar beam).

6. The following safety measures must be taken when setting up, dismantling and operating Radar equipment and fire control systems:-

- (a) Turn all switches to "off" before connecting the power supply cable.
- (b) Only establish or break cable connections when the cable connectors have been isolated from the voltage.
- (c) Always first connect cable to the Radar equipment (load) and then to the power source (generator or ac line).
- (d) Always switch off the transmitter when placing the equipment in position.
- (e) Open all ventilation flaps on the Radar equipment and generator.
- (f) If fault occur, only perform those test which are expressly permitted in the equipment specification.

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- (g) Slide-in drawers, flaps or doors which lead to the inside of the individual modules should only be opened by a qualified Radar mechanic.
 - (h) Only the trained Radar mechanic may carry out work on the inside of equipment, in accordance with valid equipment specification.
 - (i) If explosive work is carried out with unscreened electrical detonators or inflammable liquids are refilled within a radius of 100 m, the Radar antenna must not be pointed in this direction with the transmitter switched "ON".
7. Work on Radar equipment with the transmitter switched on is only permitted in halls if the parts which make up the hall consist of non-reflecting material and if the beam is only directed at these nonreflecting parts. There should be no possibility of Radar beams being reflected by reflecting surfaces. An exception can only be made in cases where the equipment specifications prescribe tests which must be carried out with the transmitter switched on in front of the Radar antenna. Under no circumstances should the head come into close contact with the antenna aperture or the primary radiator (dipole; horn radiator).
8. Experience has shown that insufficient care is often taken during work on Radar equipment. Precautionary measures are of particular importance when working with electrical equipment. There are seldom any audible or visible signs of the dangers involved.
9. Negligence or thoughtlessness during work on Radar equipment can cause an electric shock which could result in severe injury or in death. All Radar equipment operates at dangerous high voltages.
10. The intensity of an electric shock depends on:
- (a) The magnitude of the voltage and the resistance of the human body.
 - (b) The path of the electrical current through the body.
 - (c) The time for which the current flows through the body.
11. Dry skin and objects offer greater resistance than damp ones. Contact with live parts is particularly dangerous if the equipment case is touched at the same time or if the body is connected to the chassis (ground) via wet or damp objects.
- (a) Dry shoes or dry clothing in contact with the chassis is less danger.
 - (b) Wet shoes or wet clothing in contact with the chassis is greater danger.
 - (c) Skin, particularly when damp, in contact with the chassis is highly dangerous.

12. The path of the electric current through the human body is determined by the points at which the body comes into contact with the voltage source. The current normally flows into the body via a hand and leaves again at the feet. It is very dangerous if the current flows from one arm to the other or flows diagonally through the body (left hand to right leg or right hand to left leg). In such cases the electric current passes through the chest and vital organs. This is immediately lethal in the case of high voltages. Even if the current only flows through the body for a very short time, severe burns are likely to occur. If the current flows through the body for a relatively long period, further muscle spasms (respiratory organs, heart, muscle) occur, this can result in death. For this reason tools and equipment parts should not be gripped tightly.

FIRST AID (Rescuing the victim)

13. An unconscious person who is still in contact with live parts or a person who cannot release the live parts due to muscle spasm must be freed from contact with the live conductor as quickly as possible. The manner of rescue depends on the magnitude of the voltage.

(a) In case of accidents with low voltages (up to 250 V) either switch off the power or pull the victim away from the conductor without switching off the power but after first isolating the helper. The helper isolates himself by standing on glass, dry clothing, wood, rubber, or the layers of dry paper and wrapping his hands in dry clothing (sometimes the conductor can be pushed away from the victim with a dry wooden pole).

(b) In the case of accidents with voltages of more than 250 V and of high-tension voltages (over 1000 V) keep distance from conductor since voltage can flash-over. Under no circumstances touch the victim, either directly or indirectly, before the power has been switched off.

(c) The muscle spasm ceases when the power is switched off. "If necessary, first make sure that the victim does not fall when the power is disconnected.

FIRE FIGHTING

14. If a fire breaks out in electrical systems, the system parts affected directly or indirectly by the fire must be switched out of circuit (their voltage disconnected).

15. If immediate disconnection is not possible, then keep distances of at least 2 m in case of voltages are more than 30 KVA and 1 m if voltages are less.

16. A suitable number of hand-held fire-extinguishers for the type and size of the electrical systems must be provided.

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17. Fire extinguishers and blankets must be provided at obvious and easily accessible points in the vicinity of electrical systems in appropriate mounts or protective containers, protected against sparks and arcing.

RADAR BASICS

18. The word Radar is an acronym derived from the phrase Radio Detection and Ranging and applies to electronic equipment designed for detecting and tracking objects (targets) at considerable distances.

19. Radar, the device which is used for detection and ranging of objects, independent of time and weather conditions, was one of the most important scientific discoveries and technological developments that emerged from WWII. Its development, like that of most great inventions was mothered by necessity. Behind the development of Radar, lay more than a century of radio development.

20 Radar is an addition to man's sensory equipment which affords genuinely new facilities. It enables a certain class of objects to be "seen" that is, detected and located at distances far beyond those at which they could be distinguished by the unaided eye. This seeing" is unimpaired by night, fog, cloud, smoke, and most other obstacles to ordinary vision. Radar further permits the measurement of the range of the objects it "sees" with a convenience and precision. Entirely unknown in the past it can also measure the instantaneous speed of such an object toward or away from the observing station in a simple and natural way.

21. Radar works by sending out radio waves from a transmitter powerful enough so that measurable amounts of radio energy will be reflected from the objects to be seen by the Radar to a radio receiver usually located, for convenience, at the same site as the transmitter. The properties of the received echoes are used to form a picture or to determine certain properties of the objects that cause the echoes. The Radar transmitter may send out c-w signals, or frequency-modulated c-w signals or signals modulated in other ways. Radar is an active device. It utilizes its own radio energy to detect and track the target. It does not depend on energy radiated by the target itself. The ability to detect a target at great distances and to locate its position with high accuracy are two of the chief attributes of Radar.

22. In pulse Radar, the transmitter is modulated in such a way that it sends out very intense, very brief pulses of radio energy at intervals that are spaced rather far apart in terms of the duration of each pulse. During the waiting time of the transmitter between pulses, the receiver is active. Echoes are received from the nearest objects soon after the transmission of the pulse, from objects farther away at a slightly later time, and so on. When sufficient time has elapsed to allow for the reception of echoes from the most distant objects of interest, the transmitter is keyed again to send another very short pulse, and the cycle repeats.

BASIC PRINCIPLE OF OPERATION

23. Radar measurement of range, or distance, is made possible because of the properties of radiated electromagnetic energy: To appreciate the capabilities and limitations of Radar and to be able to use it to full advantage, it is necessary to comprehend the characteristics and behavior of radio waves. Radio-frequency energy travels at the speed of light, approximately 162,000 nautical miles per second; therefore, the time required for a pulse to travel to the target and return to its source is a measure of the distance to the target. Since the radio-frequency energy makes a round trip, only half the time of travel determines the distance to the target. The speed of a pulse of radio-frequency energy is so fast that the pulse can circumnavigate the earth at the equator more than 7 times in 1 second.

24. Each pulse of energy transmitted during a few tenths of a microsecond or a few microseconds contains hundreds of complete oscillations. A **CYCLE** is one complete oscillation or one complete wave, i.e., that part of the wave motion passing zero in one direction until it next passes zero in the same direction (see figure 1). The **FREQUENCY** is the number of cycles completed per second. The unit now being used for frequency in cycles per second is the **HERTZ**. One hertz is one cycle per second; one kilohertz (kHz) is one thousand cycles per second; one megahertz (MHz) is one million cycles per second.

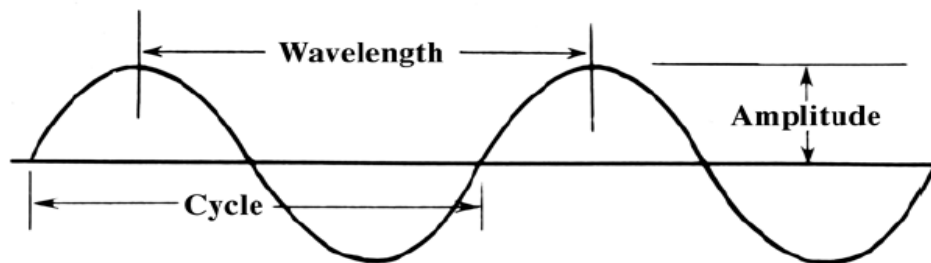


Fig 1-1 Wave

25. Wavelength is the distance along the direction of propagation between successive crests or troughs. When one cycle has been completed, the wave has traveled one wavelength. The amplitude is the maximum displacement of the wave from its mean or zero value. Since the speed of Radar waves is constant at 300,000 kilometers per second, there is a definite relationship between frequency and wavelength.

$$\text{frequency} = \frac{\text{speed of radar waves}}{\text{wavelength}}$$

26. The electromagnetic waves are reflected if they meet an electrically leading surface. If these reflected waves are received again at the place of their origin, then that means an obstacle is in the propagation direction. These principles can basically be implemented in a Radar system, and allow the determination of the distance, the direction and the height of the reflecting object.

27. The electronic principle on which Radar operates is very similar to the principle of sound-wave reflection. If you shout in the direction of a sound-reflecting object (like a rocky canyon or cave), you will hear an echo. If you know the speed of sound in air, you can then estimate the distance and general direction of the object. The time required for an echo to return can be roughly converted to distance if the speed of sound is known.

28. Radar uses electromagnetic energy pulses in much the same way, as shown in Figure 2. The radio-frequency (RF) energy is transmitted to and reflected from the reflecting object. A small portion of the reflected energy returns to the Radar set. This returned energy is called an ECHO, just as it is in sound terminology. Radar sets use the echo to determine the direction and distance of the reflecting object.

29. The word Radar is a contraction of **RA**dio **D**etecting **A**nd **R**anging. As implied by this contraction, Radars are used to detect the presence of an aim (as object of detection) and to determine its location. The contraction implies that the quantity measured is range. While this is correct, modern Radars are also used to measure range, speed and angle. The following figure shows the operating principle of primary Radar. The Radar antenna illuminates the target with a microwave signal, which is then reflected and picked up by a receiving device. The electrical signal picked up by the receiving antenna is called echo or return. The Radar signal is generated by a powerful transmitter and received by a highly sensitive receiver.

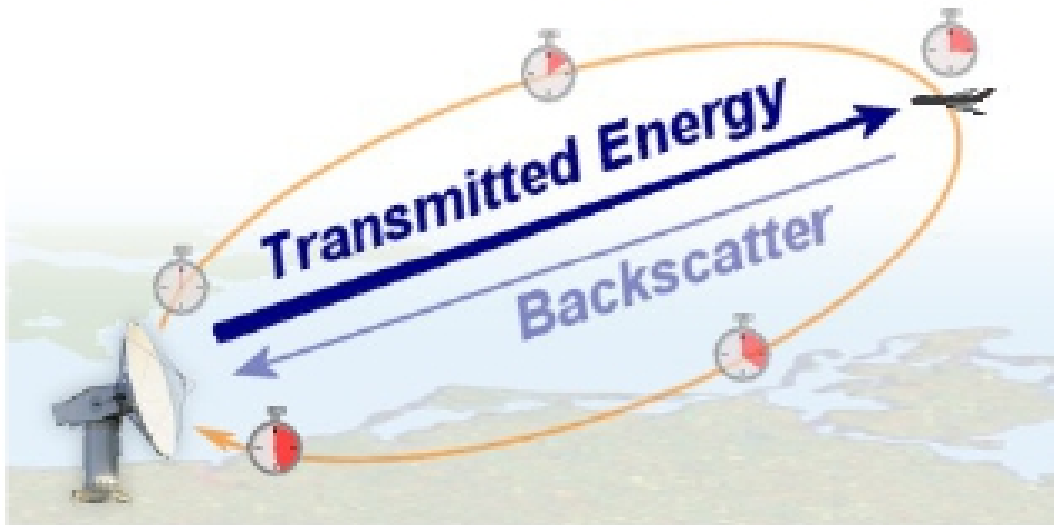


Fig 1-2 Radar principle

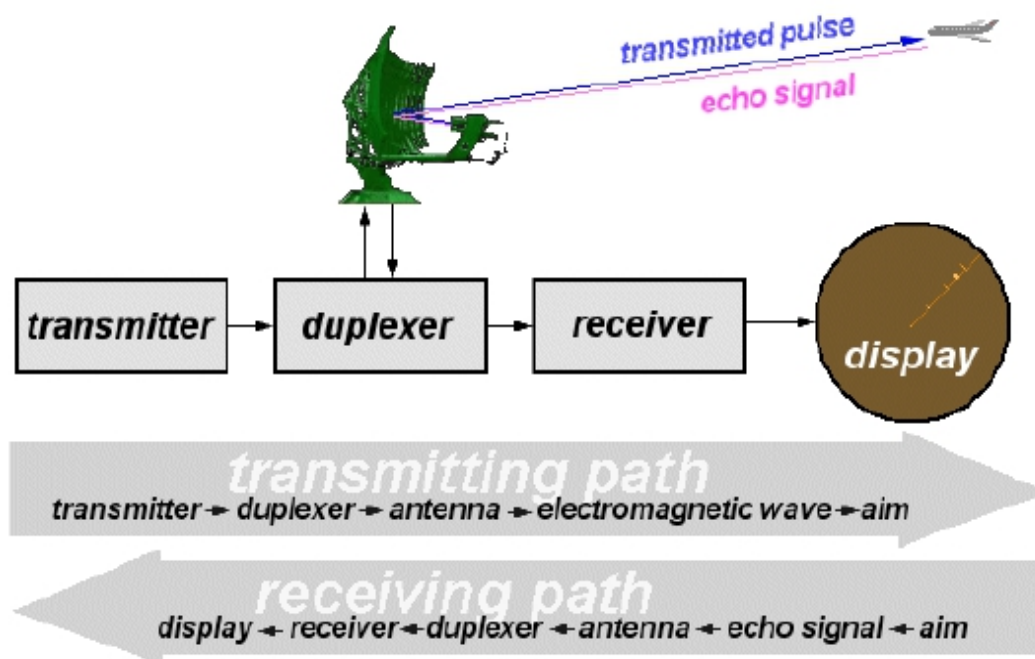


Fig 1-3 Block diagram of a primary Radar with the signal flow

Signal Routing

30. The Radar **transmitter** produces short duration high-power RF- pulses of energy.
31. The **duplexer** alternately switches the antenna between the transmitter and receiver so that only one antenna need be used. This switching is necessary because the high power pulses of the transmitter would destroy the receiver if energy were allowed to enter the receiver.
32. The **antenna** transfers the transmitter energy to signals in space with the required distribution and efficiency. This process is applied in an identical way on reception.
33. The transmitted pulses are radiated into space by the antenna as an electromagnetic wave.
34. This wave travels in a straight line with a constant velocity and will be reflected by an aim.
35. The **antenna** receives the back scattered echo signals.
36. During reception the **duplexer** lead the weakly echo signals to the receiver.
37. The hypersensitive **receiver** amplifies and demodulates the received RF signals.
38. The receiver provides video signals on the output.
39. The **indicator** should present to the observer a continuous, easily understandable, graphic picture of the relative position of Radar targets. The primary function of the indicator is to provide a visual display of the ranges and bearings of Radar targets from which echoes are received. In this basic Radar system, the type of display used is the Plan Position Indicator (PPI).
40. All targets produce a diffuse reflection i.e. it is reflected in a wide number of directions. The reflected signal is also called scattering. Backscatter is the term given to reflections in the opposite direction to the incident rays. Radar signals can be

displayed on the traditional plan position indicator (PPI) or other more advanced Radar display systems. A PPI has a rotating time base with the Radar at the center, which indicates the pointing direction of the antenna and hence the bearing of targets. It shows a map-like picture of the area covered by the Radar beam.

Signal Timing

41. Most functions of a Radar set are time-dependent. Time synchronization between the transmitter and receiver of a Radar set is required for *range measurement*. Radar systems radiate each pulse during transmit time (or Pulse Width t), wait for returning echoes during listening or rest time, and then radiate the next pulse, as shown in Figure 3. A so called synchronizer coordinates the timing for range determination and supplies the synchronizing signals for the Radar. It sent simultaneously signals to the transmitter, which sends a new pulse, and to the indicator, and other associated circuits.

42. The time between the beginning of one pulse and the start of the next pulse is called pulse-repetition time (PRT) and is equal to the reciprocal of PRF as follows:

$$PRT = \frac{1}{PRF}$$

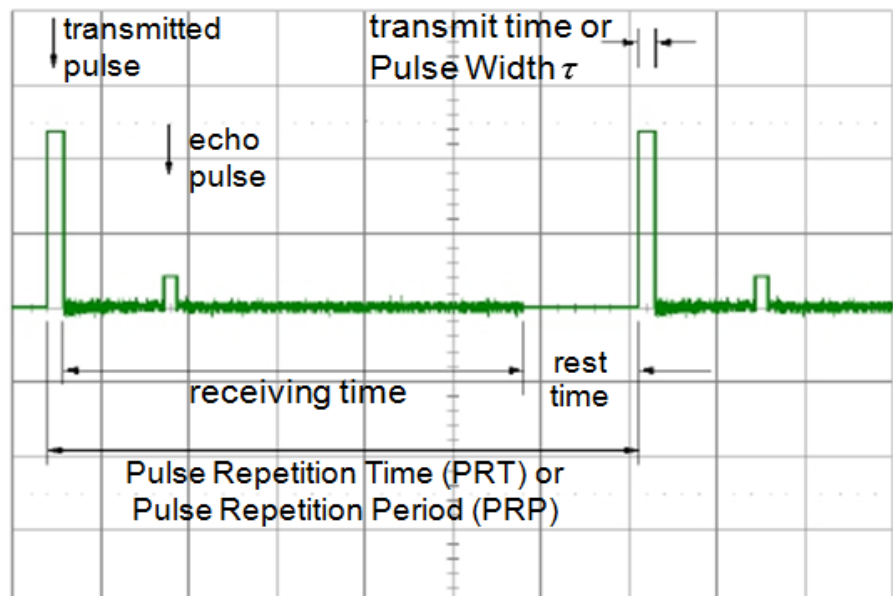


Fig 1-4 A typical Radar time line

43. The Pulse Repetition Frequency (PRF) of the Radar system is the number of pulses that are transmitted per second. The frequency of pulse transmission affects the maximum range that can be displayed.

Ranging

44. The distance of the aim is determined from the running time of the high-frequency transmitted signal and the propagation c_0 . The actual range of a target from the Radar is known as slant range. Slant range is the line of sight distance between the Radar and the object illuminated. While ground range is the horizontal distance between the Radar and its target and its calculation requires knowledge of the target's elevation. Since the waves travel to a target and back, the round trip time is divided by two in order to obtain the time the wave took to reach the target. Therefore the following formula arises for the slant range:

$$R = \frac{t_{\text{delay}} \cdot c_0}{2}$$

R is the slant range

t_{delay} is the time taken for the signal to travel to the target and return

$c()$ is the speed of light (approximately $3 \cdot 10^8$ m/s)

45. If the respective running time t_{delay} is known, then the distance R between a target and the Radar set can be calculated by using this equation.

Maximum Unambiguous Range

46. A problem with pulsed Radars and range measurement is how to unambiguously determine the range to the target if the target returns a strong echo. This problem arises because of the fact that pulsed Radars typically transmit a sequence of pulses. The Radar receiver measures the time between the leading edges of the last transmitting pulse and the echo pulse. It is possible that an echo will be received from a long range target after the transmission of a second transmitting pulse.

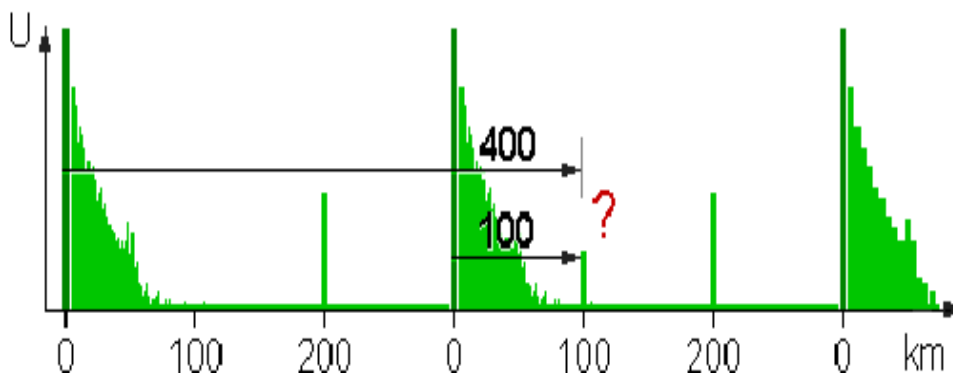


Fig 1-5 MTA echo in a distance of 400 km assumes a wrong range of 100 km

47. In this case, the Radar will determine the wrong time interval and therefore the wrong range. The measurement process assumes that the pulse is associated with the second transmitted pulse and declares a much reduced range for the target. This is called range ambiguity and occurs where there are strong targets at a range in excess of the pulse repetition time. The pulse repetition time defines a maximum unambiguous range. To increase the value of the unambiguous range, it is necessary to increase the PRT, this means: to reduce the PRF.

48. Echo signals arriving after the reception time are placed either into the transmit time where they remain unconsidered since the Radar equipment is not ready to receive during this time, or into the following reception time where they lead to measuring failures (ambiguous returns). The maximum unambiguous range for given Radar system can be determined by using the formula:

$$R_{unamb} = (PRT - \tau) \cdot c_0 / 2$$

49. The pulse repetition time (*PRT*) of the Radar is important when determining the maximum range because target return-times that exceed the PRT of the Radar system appear at incorrect locations (ranges) on the Radar screen. Returns that appear at these incorrect ranges are referred as ambiguous returns or second time around (second-sweep) echoes.

Radar Minimum Range

50. The minimum detectable range (or blind distance) is also a consideration. When the leading edge of the echo pulse falls inside the transmitting pulse, it is impossible to determine the "round trip time", which means that the distance cannot be measured. The minimum detectable range R_{min} depends on the transmitters pulse width τ , and the recovery time $t_{recovery}$ of the duplexer.

$$R_{min} = \frac{(\tau + t_{recovery}) \cdot c_0}{2}$$

51. The receiver does not listen during the transmitting pulse, because it needs to be disconnected from the transmitter during transmission to avoid damage. In that case, the echo pulse comes from a very close target. Targets at a range equivalent to the pulse width from the Radar are not detected.

Slant Range

52. Cause by the fact that the Radar unit measures a slope range, the Radar measures different ranges of two airplanes, which exactly one above the other flies (therefore having the same topographical distance to the Radar unit exactly). This false measurement could be corrected by software, or module in modern Radar sets with digital signal processing. These software modules then must also especially be adapted on the geographical coordinates of the Radar site, however. The calculation is very complicated and also requires some weather data to the correction.

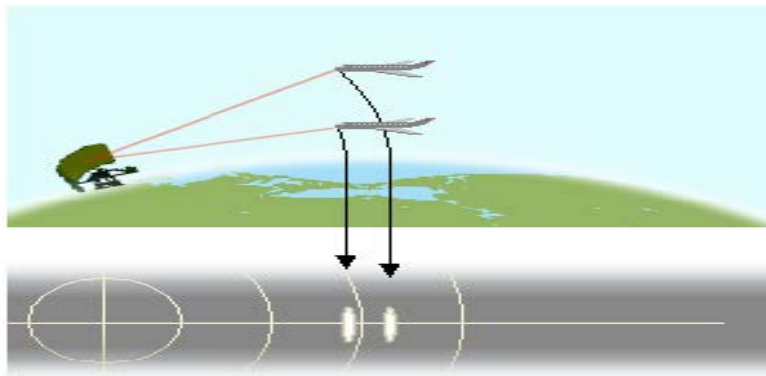


Fig 1-6 Different height causes a different slant range

DIRECTION DETERMINATION

Bearing

53. The direction to the target is determined by the directivity of the antenna. Directivity, sometimes known as the directive gain, is the ability of the antenna to concentrate the transmitted energy in a particular direction. An antenna with high directivity is also called a directive antenna. By measuring the direction in which the antenna is pointing when the echo is received, both the azimuth and elevation angles from the Radar to the object or target can be determined. The accuracy of angular measurement is determined by the directivity, which is a function of the size of the antenna.

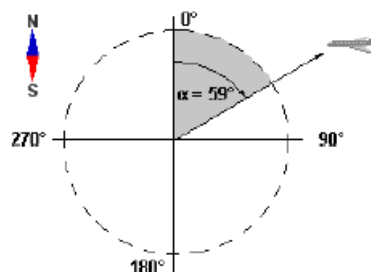


Fig 1-7 True Bearing

54. The **True Bearing** (referenced to true north) of a Radar target is the angle between true north and a line pointed directly at the target. This angle is measured in the horizontal plane and in a clockwise direction from true north. The bearing angle to the Radar target may also be measured in a clockwise direction from the centerline of your own Radar and is referred to as the relative bearing.

55. The rapid and accurate transmission of the bearing information between the turntable with the mounted antenna and the scopes can be carried out by servo systems and counting of azimuth change pulses. Servo systems are used in older Radar antennas and missile launchers and works with help of devices like synchro torque transmitters and synchro torque receivers. In other Radar units we find a system of Azimuth-Change-Pulses (ACP). In every rotation of the antenna a coder sends many pulses, these are then counted in the scopes. Some Radar sets work completely without or with a partial mechanical motion. These Radars employ electronic phase scanning in bearing and/or in elevation (phased-array-antenna).

Height

56. The height of a target over the earth's surface is called height or altitude. This is denominated by the letter H (like: Height). True altitude is the actual airplane distance above mean sea level.

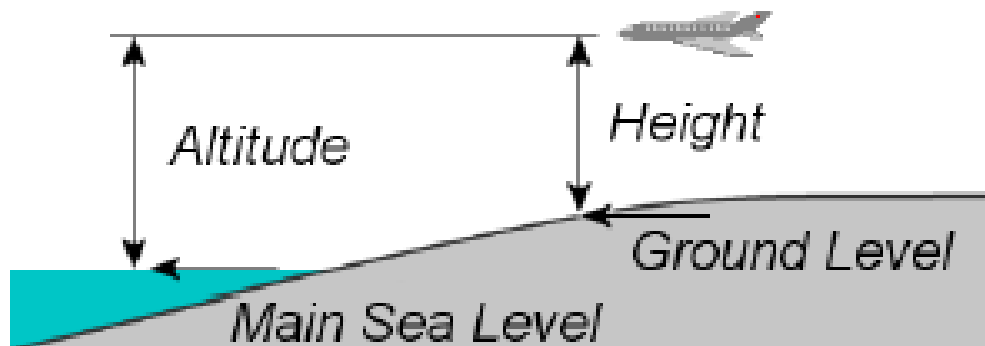


Fig 1-8 Definition of elevation angle

RADAR RESOLUTION

57. The target resolution of Radar is its ability to distinguish between targets that are very close in either range or bearing. Weapons-control Radar, which requires great precision, should be able to distinguish between targets that are only yards apart. Search Radar is usually less precise and only distinguishes between targets that are hundreds of yards or even miles apart. Radar resolution is usually divided into two categories; range resolution and angular (bearing) resolution.

Angular Resolution

58. Angular resolution is the minimum angular separation at which two equal targets at the same range can be separated. The angular resolution characteristics of a Radar are determined by the antenna beam width; two identical targets at the same distance are, therefore, resolved in angle if they are separated by more than the antenna beam width. An important remark has to be made immediately: the smaller the beam width, the higher the directivity of the Radar antenna, the better the bearing resolution.

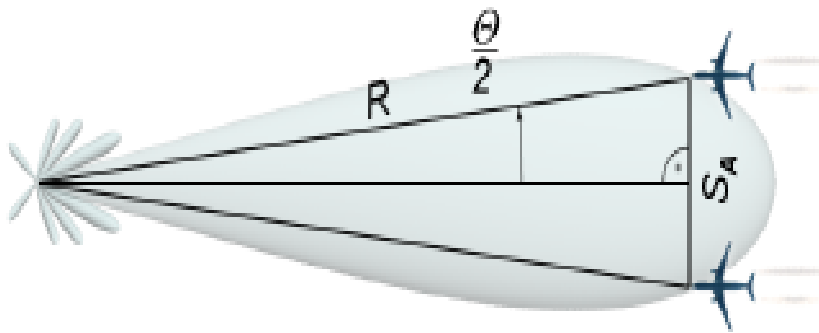
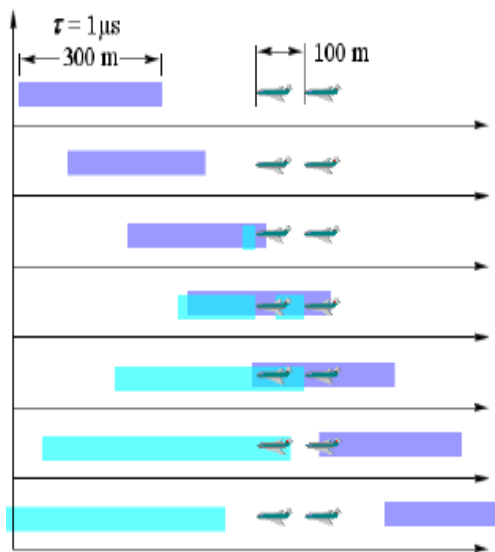
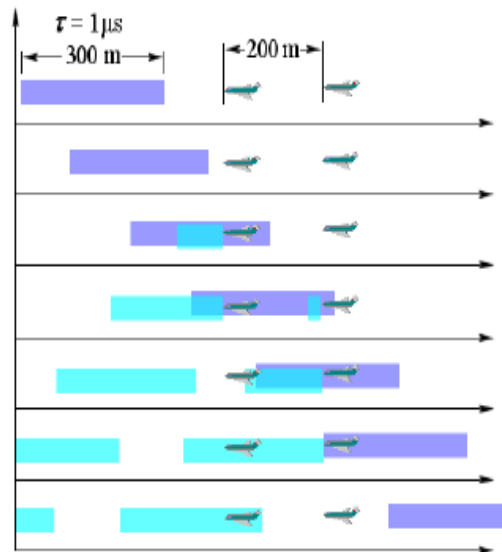


Fig 1-9 Angular resolution

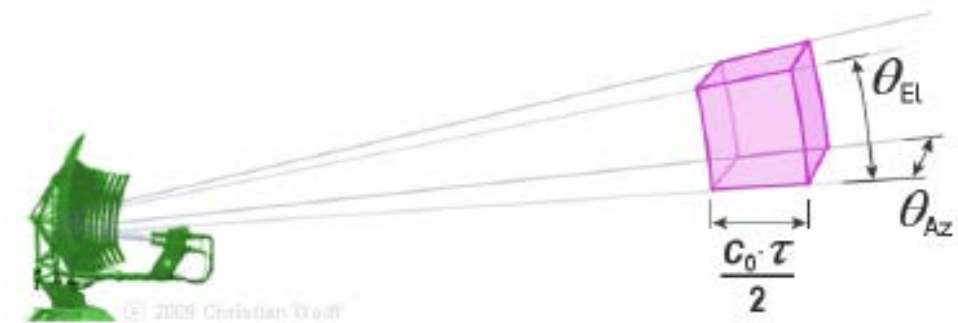
Range Resolution

59. Range resolution is the ability of a Radar system to distinguish between two or more targets on the same bearing but at different ranges. The degree of range resolution depends on the width of the transmitted pulse, the types and sizes of targets, and the efficiency of the receiver and indicator. Pulse width is the primary factor in range resolution. A well-designed Radar system, with all other factors at maximum efficiency, should be able to distinguish targets separated by *one-half the pulse width time*.

**Fig 1-10** Two objects with too small spacing**Fig 1-11** Two objects, when the spacing is large enough

Resolution Cell

60. The range and angular resolutions lead to the resolution cell. The meaning of this cell is very clear: it is impossible to distinguish two targets which are located inside the same resolution cell.

**Fig 1-12** The resolution cell

Theoretical Maximum Range Equation

61. The smallest signal that can be detected by the Radar is called Minimum Discernible Signal. Smaller powers than this P_{MDS} aren't usable since they are lost in the noise of the receiver and its environment. The theoretically maximum range of given Radar set can be calculated by:-

$$R_{\max} = \left[\frac{P_T G^2 \lambda^2 \sigma}{(4\pi)^3 S_{\min}} \right]^{1/4} \quad \text{or} \quad R_{\max} = \sqrt[4]{\frac{P_{tx} \cdot G^2 \cdot \lambda^2 \cdot \sigma_t}{(4\pi)^3 \cdot P_{MDS} \cdot L_s}}$$

RADAR PERFORMANCE FACTORS

Factors Affecting Maximum Range

62. As seen by the relation for $R_{\max}$. It is readily observed that $R_{\max}$ depends upon the transmitted power P_T , frequency of the transmitted signal ($\lambda = C/F$), the cross sectional area of the target σ and the minimum receiver signal ($S_{\min}$). Received echo power is inversely proportional to the fourth power of the range R . Thus in order to double the range, a 16-fold increase in peak power P_T is needed. Alternatively, the range may be increased by increasing the physical size of antenna. The range $R_{\max}$ varies directly as the square root of the capture area A of the antenna. The range may be increased by reducing the wavelength. Lastly the range $R_{\max}$ may be increased by reducing the minimum power $S_{\min}$. The decrease in $S_{\min}$ may be achieved by improving the noise figure of the Radar receiver. This method of improving the range by improving the receiving system is usually cheap and further results in reduction in size and weight than is obtained by increased transmitter power.

(a) Transmitters Power: The peak power of Radar is its useful power. Range capabilities of the Radar increase with peak power. Doubling the peak power increases the range capabilities by about 25 percent. **To double the maximum range you must increase the transmitted power 16-fold!** The inversion of this argument is also permissible: if the transmit power is reduced by 1/16 (e.g. failures into two of 32 transmitter modules), then the change on the maximum range of the Radar set is negligible in the practice:

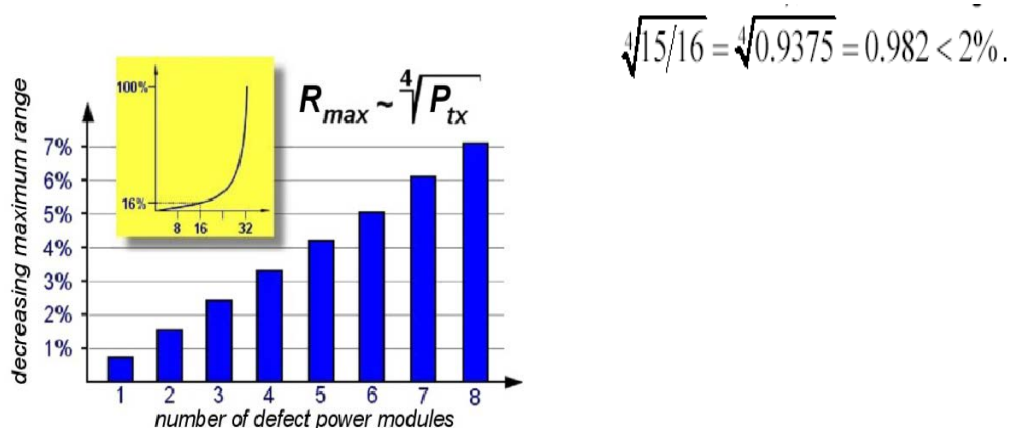
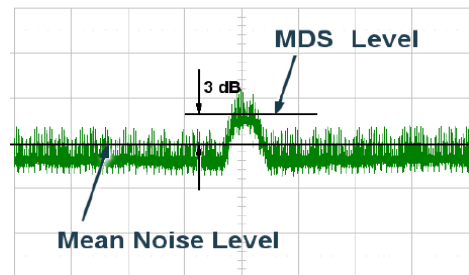


Fig 1-13 Decreased maximum range by defect or missing power modules in a solid state transmitter with 32 power modules

(b) MDS- Echo ($S_{\min}$): The minimum discernible signal is defined as the useful echo power at the reception antenna, which gives on the screen a discernible blip. The minimum discernible signal at the receiver input-jack leads to the maximum range of the Radar; all other nominal variables are considered as constant. A reduction of the minimal received power of the receiver gets an increase of the maximum range. For every receiver there is a certain receiving power as of which the receiver can work at all. This smallest workable received power is frequently often called MDS – Minimum Discernible Signal or Minimum Detectable Signal in Radar technology.

$$R_{\max} = \sqrt[4]{\frac{P_{tx} \cdot G^2 \cdot \lambda^2 \cdot \sigma_t}{(4\pi)^3 \cdot P_{MDS} \cdot L_s}}$$



(c) Noise: The value of the MDS echo depends on the Signal-to-Noise-Ratio, defined as the ratio of the signal energy to the noise energy. All Radars, as with all electronic equipment, must operate in the presence of noise. The main source of noise is termed *thermal noise* and is due to agitation of electrons caused by heat. The noise can arise from received atmospheric or cosmic noise and receiver noise - generated internally in the Radar receiver. The overall receiver sensitivity is directly related to the noise figure of the Radar receiver. It becomes clear, that a low noise figure receiver is accomplished by a good design in the very front-end components. An aspect to a very low noise figure receiver is achieved through minimizing the noise factor of the very first block. This component usual is characterized by a low noise figure with high gain. This is the reason for the often used denomination, **low noise preamplifier (LNA)**.

(d) Frequency: We know that, $R_{\max} \propto 1/\sqrt{\lambda}$ or $R_{\max} \propto \sqrt{f}$. This implies that increase in frequency increase the range. However this requirement is in conflict with the dependence of beam width of the antenna which is directly proportional to the wavelength λ . Moreover, higher the frequency of a Radar (radio) wave, the greater is the attenuation (loss in power), regardless of weather. Lower Radar frequencies (longer wavelengths) have, therefore, been generally superior for longer detection ranges.

(e) Pulse Repetition Rate / Frequency: The pulse repetition rate (PRR) determines the maximum measurable range of the Radar. Ample time must be allowed between pulses for an echo to return from any target located within the maximum workable range of the system. Otherwise, echoes returning from the more distant targets are blocked by succeeding transmitted pulses.

This necessary time interval determines the highest PRR that can be used. The PRR must be high enough, however, that sufficient pulses hit the target and enough echoes are returned to the Radar. The maximum measurable range can be determined approximately by dividing 81,000 by the PRR. As the pulse repetition frequency (PRF) is increased, the maximum value of two way travel time decreases and hence the maximum range decreases. Thus maximum allowable pulse repetition frequency is limited by the effective range of the Radar set. On the other hand, the PRF should not be below certain minimum value so that effective integration of weak echo pulses may take place, due to greater number of pulses received per second.

(f) Pulse Length: The longer the pulse length, the greater is the range capability of the Radar because of the greater amount of energy transmitted but consequently at cost of poor range resolution and increased minimum range unless achieved by pulse compression and use of two pulses (one for short ranges and other for long ranges) respectively as in modern Radar systems.

(g) Beam Width: The more concentrated the beam, the greater is the detection range of the Radar as energy will travel more distance.

(h) Antenna Rotation Rate: The more slowly the antenna rotates, the greater is the detection range of the Radar but it results in with poor picture resolution. Hits per scan be obtained by:

$$\frac{\text{Beam Width}}{\text{Degrees per Pulse}}$$

(j) Target Characteristics: Targets that are large can be seen on the scope at greater ranges, provided line-of-sight exists between the Radar antenna and the target. Conducting materials return relatively strong echoes while non-conducting materials return much weaker echoes.

False Alarm Rate

63. A false alarm is “an erroneous Radar target detection decision caused by noise or other interfering signals exceeding the detection threshold”. In general, it is an indication of the presence of a Radar target when there is no valid target. False alarms are generated when thermal noise exceeds a pre-set threshold level, by the presence of spurious signals (either internal to the Radar receiver or from sources external to the Radar), or by equipment malfunction. If the detection threshold is set too high, there will be very few false alarms, but the signal-to-noise ratio required will inhibit detection of valid targets. If the threshold is set too low, the large number of false alarms will mask detection of valid targets.

Probability of Detection:

64. The received and demodulated echo signal is processed by threshold logic. This threshold shall be balanced so that as of certain amplitude wanted signals being able to pass and noise will be removed. Since high noise exists in the mixed signal tops which lie in the range of small wanted signals the optimized threshold level shall be a compromise. Wanted signals shall on the one hand reach the indication as of minimal amplitude; on the other hand the false alarm rate may not increase.

$$P_D = \frac{\text{detected targets}}{\text{all possible blibs}} \cdot 100\%$$

Doppler Effect Associated with Moving Targets:

65. For a fixed target, the relationship between the phases of the transmitted pulse and the echo pulse considered at the site of Radar set remains the same for successive pulses. On the other hand, the relative phase of the echo signals received from a moving target continually changes relative to the phase of the transmitted pulse because of the continually changing distance of the target from the Radar set. This continuous change in the phase of the echoes received from a radially moving target causes the frequency to the echoes to differ from that of the outgoing pulses. The shift f_d in the frequency, referred to as the Doppler shift.

66. The MTI Radar uses the Doppler Effect for its operation. Many a times it is not possible to distinguish a moving target in the presence of static or permanent echoes of comparable appearance on the Radar screen. In a PPI display, there is a lot of clutter due to these stationary target echoes. Also, it is quite possible that a moving target has a range and bearing such that the echo from the moving target gets superimposed on the ground clutter; such a condition can exist in mountainous region or in close vicinity of modern cities cluttered with tall buildings. Basically, the moving target indicator (MTI) system compares a set of received echoes with those received during the previous sweep. Those echoes whose phase has remained constant are then cancelled out.

Blind Speeds:

67. If the target happens to have a velocity whose radial component results in a phase difference of exactly 2π rad between successive pulses. This is the same as having no phase shift at all. The target thus appears stationary, and echoes from it are canceled by the MTI action. A radial velocity corresponding to this situation is known as a blind speed, as are any integral multiples of it. It is readily seen that if a target moves a half-wavelength between successive pulses, the change in phase shift will be precisely 2π rad. Hence blind speed state that

$$V_b = PRF \frac{n\lambda}{2}$$

where V_b - blind speed
 λ - wavelength of transmitted signal
 and n - any integer (1, 2,...)

68. The target can travel with blind speed, by analyzing the transmitted frequency and pulse repetition frequency and adjust its radial velocity as per the analysis. However a variable pulse repetition frequency decided by a pseudo random generator used by the Radar transmitter will solve the problem of blind speed.

APPLICATIONS OF RADAR

General Applications

69. Ground-based Radar is applied chiefly to the detection, location and tracking of aircraft of space targets.

70. Ship-borne Radar is used as a navigation aid and safety device to locate buoys, shorelines and other ships. It is also used to observe aircraft.

71. Space-borne Radar is used for the remote sensing of terrain and sea, and for docking.

72. Airborne Radar is used to detect aircraft, ships and land vehicles. It is also used for mapping of terrain and avoidance of thunderstorms and terrain.

Major Applications

Air Traffic Control (ATC)

73. Radar used to provide air traffic controllers with position and other information on aircraft flying within their area of responsibility (airways and in the vicinity of airports). The high resolution Radar is used at large airports to monitor aircraft and ground vehicles on the runways, taxiways and ramps. The GCA (ground controlled approach) or PAR (precision approach Radar) provides on operator with high accuracy aircraft position information in both the vertical and horizontal. The operator uses this information to guide the aircraft to a landing in bad weather. The MLS (microwave landing system) and ATC Radar beacon systems are based on Radar technology.

Air Navigation

74. The weather avoidance Radar is used on aircraft to detect and display areas of heavy precipitation and turbulence. Low-flying military aircraft rely on terrain avoidance and terrain following Radars to avoid colliding with obstructions or high terrain. Military

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CHAPTER No 2

APPLICATION OF THE SYSTEM

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CHAPTER No 2

APPLICATION OF THE SYSTEM

BRIEF DESCRIPTION

1. The first and most important task of an effective Air Defence System is to detect and recognize hostile aircraft under all ground terrain and weather conditions as early as possible. These requirements can be met by building up an Air Surveillance system with different radar systems. One system is medium and long range radar for early warning and alert in the upper air space. The second system consists of short range radar equipment with low flying aircraft detection capability. It designates targets to weapons and fills gaps in the coverage of the long range radars. For this second purpose the SIEMEN's Low Level Air Control System (SILLACS) was designed.

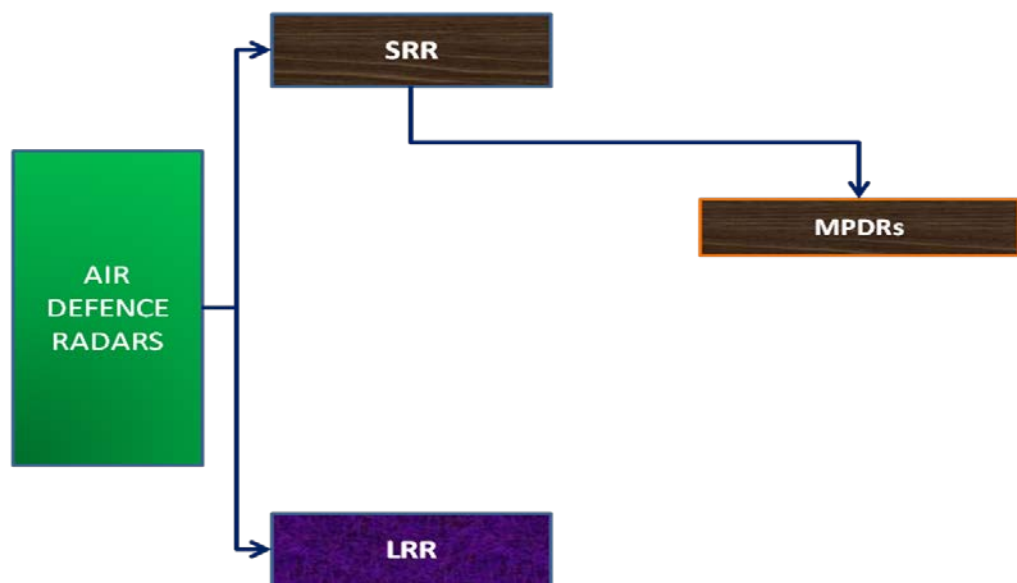


Fig 2-1 Air Defence Radars

2. The mobile system SILLACS consists of three main components. Up to eight MPDR radar sites for target detection, the Control and Reporting Center (CRC) for preparation of the target information of the radar sites and the Maintenance section in order to maintain the whole low level surveillance system during operation. The communication to the complete surveillance system was provided by the possibility of a computer to computer connection between the

control and reporting center of the SILLACS and a sector operating center, where all detection information of the complete surveillance system is present. Later on, CRC was replaced by MCC (Mission Control Center) with this up to twenty radar can be connected.

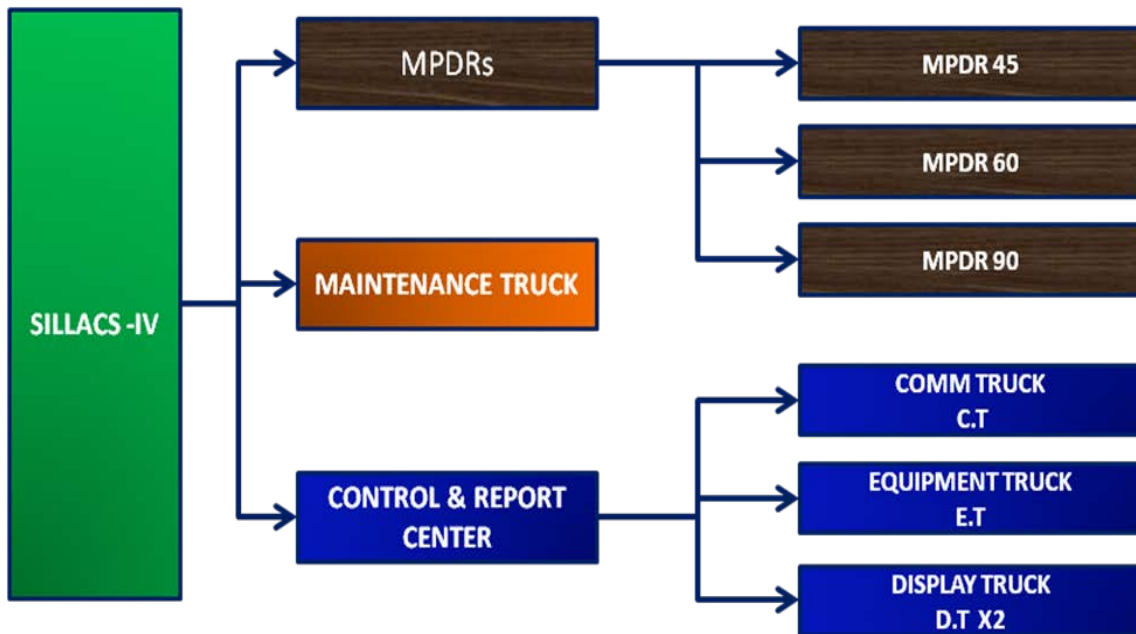


Fig 2-2 Components of SILLACS system

3. Radar System MPDRs (Mobile Pulse Doppler Radar) are employed for air surveillance within a certain range of the radar site (Fig. 2-3). Their main purpose is the detection of low flying aircraft and indication of the target information (range and azimuth) on the Plan Position Indicator (PPI) screen completely free of fixed target indication. Because of the suppression of slowly moving objects, such as clouds, trees in the wind, moving vehicles or waves, it is possible to quickly detect airborne targets and track them even in the immediate vicinity of the earth's surface, so that provided a suitable site is chosen. Flying below the surveillance volume is impossible.

4. The radar system incorporates Mark X-A IFF (Identification Friend or Foe) equipment to facilitate discrimination between hostile and friendly aircraft. Identification takes place by interrogation of the target and the receipt of coded responses. A special identification feature is provided for friendly targets.

5. The absence of any clutter on the screen as well as the digitizing of information facilitates the automatic transmission of target data to MCC, thus enhancing the effectiveness of air defense systems. On the other hand the radar operator has the capability to superimpose fixed targets for local orientation and to determine radar shadow areas. The system is also provided for automatic data exchange and for voice communication.



Fig 2-3 Radar system MPDR 45/E

6. The MPDRs radar system has been constructed with a view to ensuring easy transportation and minimum space requirement, and is particularly suitable for mobile service.

7. The system comprises two 5-ton trucks each with a 1.8-ton trailer. One truck (the radar equipment truck) contains the radar equipment; the other truck (the antenna assembly truck) carries the folding mast with the antenna, which can be operated at any mast height. The fully erected mast has a height of above 12 m. Each trailer (single axle generator set trailers) contains a mobile power generator

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set. With the aid of the generator set the system can operate entirely independently of public electric power supplies, but can also be run from such supplies if required. The wide climatic range of the MPDR 45/E radar system makes reliable operation possible at both high and low temperatures. A built-in air conditioner ensures a pleasant inside temperature. The equipment is also suitable for fixed service.

SYSTEM COMPONENTS RADAR SYSTEM MPDR 45/E

8. The Radar System MPDR 45/E consists of the radar set, the IFF interrogator set MSR 400/8, the radar antenna, the hinged mast, the gyroscope, the radar data processing equipment set and the communication equipment set.

9. Outside Surface of the trucks and all other parts mounted on the outside of the trucks have a camouflage coating. All outside parts which cannot be given this camouflage coating for electrical or mechanical reasons (e.g. sleeves on the antenna mast) are painted with a dark color.

RADAR EQUIPMENT TRUCK

10. A sound-proof partition wall divides the radar equipment truck into the equipment room and into the control room. The equipment room contains the radar set without indicator and the indicator power supply; the IFF interrogator unit without the IF controls unit, 6 (two spares) radio equipment sets without radio control unit and 1 radio relay equipment set. The equipment placed in cases with their covers removed. Most of the cases are strapped tightly to the unit mounting racks permanently installed in the van; therefore, they can be replaced easily. The equipment room contains the unit mounting rack of the installation kit which houses the radar set, the IFF interrogator unit and the radio relay equipment set. A desk plate which bears the equipment rack with the power distribution unit (5801235) on its rear end is installed at the right side of the truck of the equipment room. Beneath the desk plate is space for 6 radio equipment sets GRC 171/ M3SR (2 spares). The cables set containing the cables belonging to the cable set "outside" (5801245-617) and the antenna cable for radio relay and GRC 171 / M3SR radios.

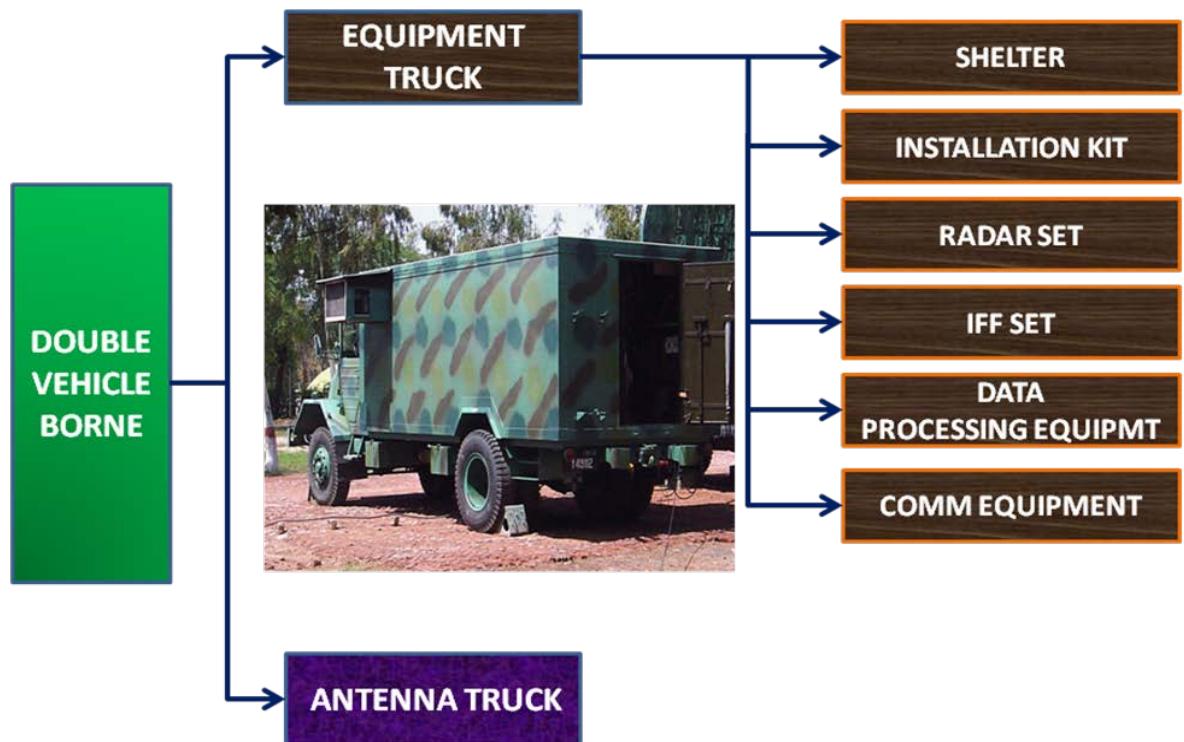


Fig 2-4 System Components in Radar Equipment Truck

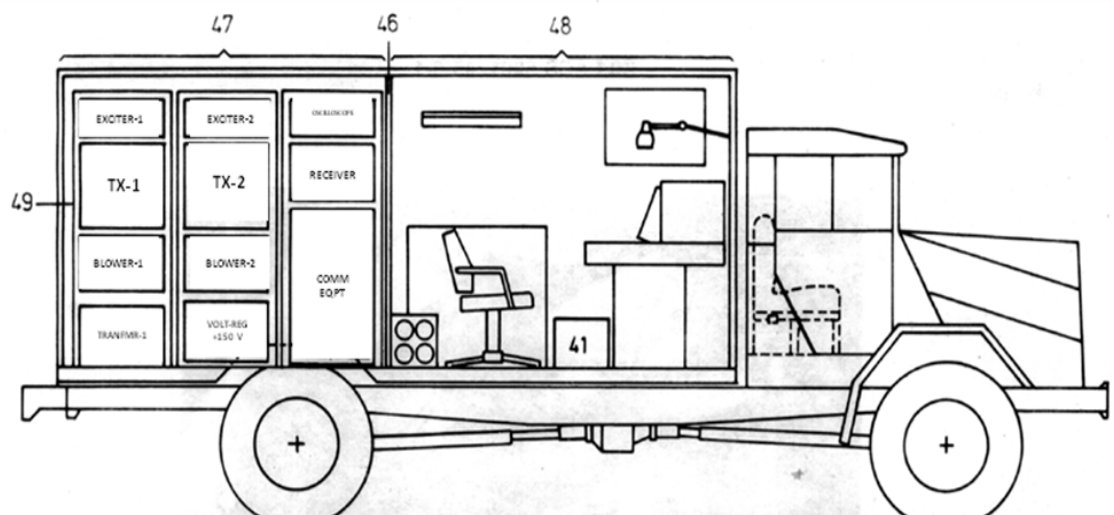


Fig 2-5 Location of parts in Equipment truck

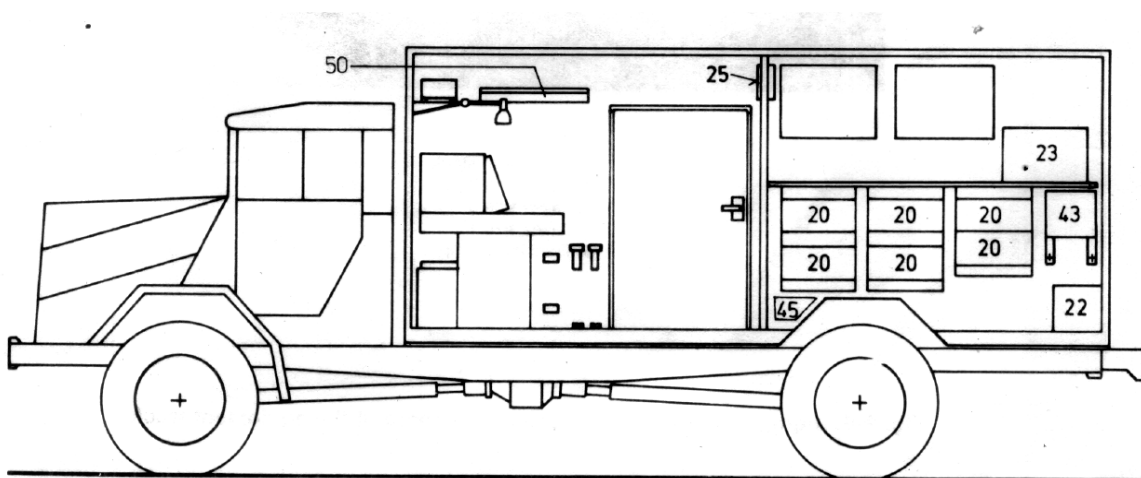


Fig 2-6 Location of parts in Equipment truck

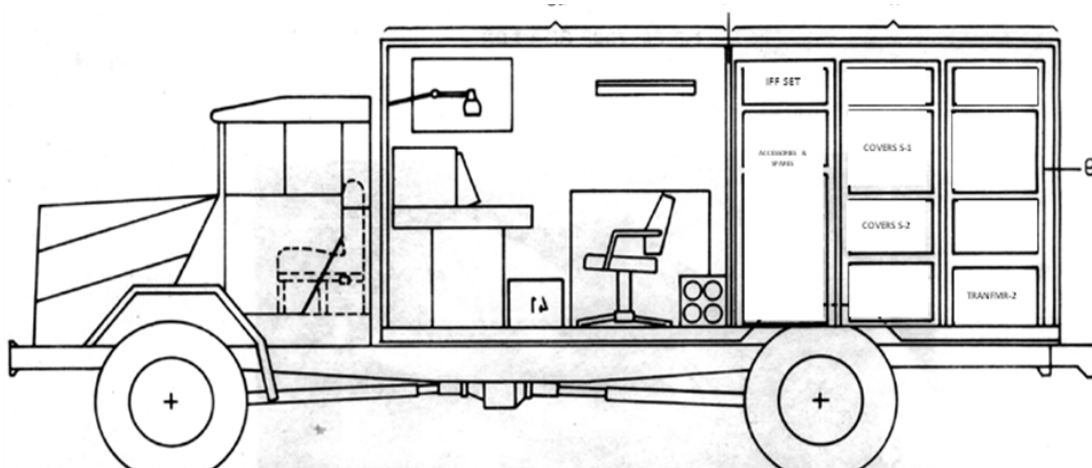


Fig 2-7 Location of parts in Equipment truck

11. The equipment room is air-conditioned with 2 air-conditioners (5801193-857) installed on the co-driver's side. They are pushed out on telescopic rails through an opening in the truck wall before operation. The RF terminal box is on the truck floor beneath the desk plate and on top is the power distribution box for the air-conditioners. For easy connection of the transmit/receive cable (ZM-1 cable) with a large radius of curvature to the radar equipment truck, the RF terminal box has been provided with a dual rotary joint whose external connection can rotate in two

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planes perpendicular to each other. A short flexible RF cable connects the internal connection of the dual rotary joint to the transmitter output at the blower case 1. Two switched rectifiers are mounted in the left rear corner of the left aisle and the IFF interrogator unit is mounted on the rear side of the unit mounting rack, the case "accessories and spares" and the cover of the equipment cases are arranged according to size 1, 2 and 3.

12. The transformer 2 is located at the rear end of the aisle. The switched rectifiers supply the power for the interior illumination either from the 220 V ac mains or from the 24 V dc mains. The equipment room and the control room are each illuminated by two transistor-chopper lamp, assemblies.

13. At the rear truck door are two ground augers in their retainer. The operator's console is fixed along the front wall of the control room. In front of it are two height-adjustable revolving chairs with four leg stands screwed to the floor when vehicle is in motion.

14. The console contains the indicator, the IFF control unit, the digital clock, the engineering channel module for telephone communication, and a central control and distribution panel for the 24 V dc power supply.

15. A second indicator can be installed with special equipment in the place of the two drawers. Above the console, charging rectifiers and two adjustable lamps are screwed to the rails with corresponding mountings.

16. On the left side beneath the console is the power supply "indicator" beneath it is the radar data processing unit. Space has been provided for the power supply of a second indicator under the right half of the desk plate. Beneath it, the power supply unit for the radar data processing set has been mounted.

17. A portable air-conditioner is available for the ventilation of the control room. Behind the opening at the right front of the control room is the AF connection box. The cables not belonging to the radar system for charging batteries and a FM cable for the transmission of data through land lines from the radar system to the MCC are connected to this box.

18. Two batteries, communication battery and the corresponding fuse cabinet to supply the dc leads inside the truck are mounted in the driver's cab under the driver's seat.

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19. All connectors and controls of the radar equipment are located on the front panels of the units. Most modules are plug-in type modules and can be removed towards the front. Extensive use is made of PCB's with IC's and silicon semiconductors. Apart from the cathode ray tube of the indicator and the test oscilloscope, electron tubes are used only in the transmitter output stages. The transmitter of the radar set and the receiver are air-cooled.

RADAR SET

20. The equipment belonging to the radar set is housed in the unit mounting racks inside the radar equipment truck and inside the operator's console. The radar set is fully operational in combination with the radar antenna and the "inside" and "outside" cable sets. The individual units of the radar set are designed as slide-in units which are housed in standardized portable cases. The rails in the cases housing the equipment are screwed to the cases with shock-absorbing devices. Three different case sizes are used. Owing to the dimensions of the cathode ray tube (length of the tube neck) the indicator is housed in a special case with the same frontal dimensions as the standardized case, but it is twice as deep. The electronic equipment cases are watertight when the covers are secured. They can be stacked. All cable connections are made at the front of the slide-in units by means of Plug connectors. The radar set (5801245-5) contains the following cases:

Case	Part No.	Dimensions in mm	Weight in kg
Transmitter	5801232	604X580x284	60
Exciter	5801231	604x190x284	22,7
Receiver	5801239-5	604x385X284	41
Indicator	5801240-5	604 X 385 x498	43,5
Blower unit 1	5801233-6	604 X 385 X 284	43
Blower unit 2	5801244-6	604 X 385 X 284	24
Test oscilloscope	5801238	604X 190X284	16.2
Regulator 150 V	5801234	604 X 385 X 284	44.3
Transformer 1	5801236-5	604 X 385 X 284	100
Power distributor unit	5801235	604X385x284	29
Power supply unit	5801241	604x190X284	21,7
Transformer 2	5801237	604 X 385 X 284	100
Total weight	approx, 800 Kg (Incl, cables)		

IFF INTERROGATOR SET

21. The IFF interrogator set MSR 400/8 (IFF = Identification Friend or Foe; MSR = Mobile Secondary Radar) provides discrimination between friendly and foe aircraft by transmitting coded interrogations in accordance With the mark X- A system. The moving target is identified as friendly when it responds to the interrogations it responds to the Interrogation with the agreed code. The IFF interrogator set MSR 400/8 is coupled With the Radar System MPDR 45/E in such a way that interrogation only takes place when the radar set has detected moving targets.



Fig 2-8 IFF Interrogator MSR 400/8

22. The controls of the MSR 400/8 are located on the IFF control unit (580117-32) which is a slide-in unit housed at the console of the radar set. The IFF Interrogator unit is housed in a watertight Light metal case with swing frame. The IFF interrogator (580117 -3) consist of following items.

Equipment	Part No.	Dimensions in mm (W ^x H ^x D)	weight in KG (with cover)
IFF interrogator MSR 400/8	580117-31	650X226X283	25
IFF Control Unit Compatible with and	580117-32 580117-63 & 580117-53	194X180X166	2

23. The IFF antenna and the SLS antenna are integrated into the radar set antenna and therefore supplied with the radar system MPDR 45E.

RADAR DATA PROCESSING EQUIPMENT SET

24. The radar data processing set (5801180-8) forms a connecting link between radar system and MCC where the situation information from several radar sites arrives.

25. The data processing equipment set (5801180-8) consists of following items.

Equipment	Part No.	Dimensions in mm (W^x H^x D)	Weight in KG (with cover)
Radar data processing unit	5801180-81	650X309X283	30
Power supply unit	5801189-8	650X418X283	55
Control panel	5801180-82	290X157X135	2.5
Cable set	5801180-83		
Rolling ball	5800684-811	257X160X137	2.3



Fig 2-9 Pictorial view Radar Data Processing Unit

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INSTALLATION KIT

26. The installation kit (5801245-611) includes all parts and equipment permanently mounted in the radar equipment truck as following.

Items	Qty	Part No
Wiring harness	01	5801245-6119
Partition WALL	01	5801245-
Revolving chairs	02	5801195-40
Transistor lamps	04	5800804-0.001
Adjustable lamps	02	6218
Air conditioner	03	5801193-857
Distribution box for air conditioners	01	5801245-6118
AF terminal box	01	5801245-6116
Communication terminal box	01	5801245-611021
RF terminal box	01	5801245-6115
RF dual rotary joint	01	5800686-1.038
Switched rectifiers	02	5800686-1.092
Battery charger	01	5800804-0.003
Power outlet and Switches	09	5800686-1.079
Grounding cables	02	5800895-235
Tool kit	01	5800686-1236
Built in floor	01	5801245-611079
Unit mounting rack	01	5801245-611008
UHF frames	02	5801245-6112
Desk plate	01	5801245-611010

OPERATOR'S CONSOLE

27. The Operator's Console (5801245-612) in the control room of the Radar Equipment Truck consists of a welded box-section frame with a console on top. The following items are located in the frame: power supply unit "indicator", space for the second power supply unit "indicator". The radar data processing unit and the power supply unit for the radar data processing equipment have separate rails for mounting; they ride on Teflon rails and can be pulled out from under the desk. The desk plate has openings for the rolling ball and the control panel for the radar data processing equipment. The desk plate has a second opening for the second control unit, this space is covered. The desk panel consists of a frame where the indicator,

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digital clock (580)193-18) and the control panels (control panel 24 V, 5801245-6124, IFF control panel, engineering channel module for the communication equipment set are plugged in and screwed tight. On the right side of the desk panel are 2 drawers for the headset, chest for storing the crystals and the manuals or, instead, space for a second indicator (5801245-616121).

ACCESSORIES AND SPARES

28. Accessories and spare parts are contained in the four drawers of the accessory case. They are as follows:

Items
1 Tube extractor 1
1 Module extractor
1 Lamp extractor
1 Surge diverter extractor
2 RF adaptors
2 AF adaptors
1 Adaptor, 96 contacts
16 Surge diverters
1 Blower hose
1 Crystal 1.2 MHz
Incandescent lamps
Fuses
Wrenches(Open end, hook, socket wrenches)

CABLE SET "INTERNAL"

29. The cable set "inside" (5801245-613) contains all the pluggable cable assemblies which are required for interconnection of the electronic equipment cases. Every cable is marked on both ends, with a code number and stock number consisting of letters or a combination of letters and numbers in addition to the part number. The connectors on the units have with a few exceptions - the same marking as the associated cables.

CABLE SET "EXTERNAL"

30. All interconnection cables between the radar equipment truck, antenna assembly truck and the generators (or power transmission lines) are part of the cable set "out-(5801245-617). They are rolled up on cable reels and stored on cable racks during transportation with the exception of the thick RF cable ZM1 for the radar set which is stored in the antenna truck and the main cables which are rolled on reels in the power supply trailers. Field trunk cables for field communication are not part of the Radar System MPDR 45/E, they must be supplied by the field communication detachment.

ANTENNA ASSEMBLY TRUCK

31. The antenna assembly truck (5801245-62) serves to transport the radar antenna and to support it when in operation. The truck chassis carries the hinged mast which supports the radar antenna. The hinged mast is elevated by four hydraulic rams and can be halted in any intermediate position. The radar antenna which takes down into sections is carried on fixed pedestal which incorporates the drive motor, the triple rotary joint and the angular transmission system.

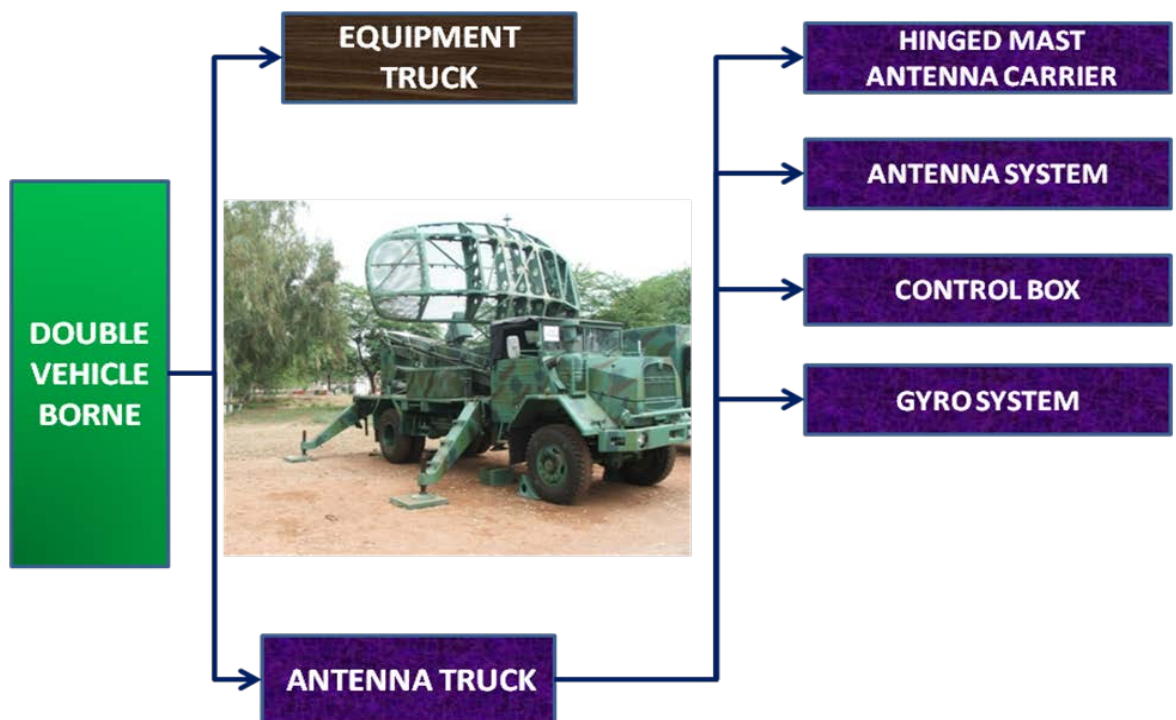


Fig 2-10 Antenna Assembly Truck

RADAR ANTENNA

32. The radar antenna (5800590-9) consists of the following components.

Equipment	Part No.	Dimensions in mm (W ^x H ^x D)	weight in KG (with cover)
L band reflector	5800590-1	5000X1900	155
Horn radiator with radar and IFF radiators	5800590-2	746X563X352	17
Boom	5800590-3	2269X750X373	17
Antenna pedestal	5800590-94	998X980X680	297
Sight telescope (beneath console)	5800590-5	210X14X15	2
SLS antenna	5800590-6	180X130	0.5
Set of RF cables	5800590-7		16

33. The radar antenna pedestal is attached by four bearing journals to the forked end of the upper mast arm. When the antenna assembly truck is moved, the side section of the three section L band reflector are detached and secured to the platform on the antenna truck sub frame. The centre section of the reflector with the SLS antenna and the boom with horn radiator are folded down and locked into position.



Fig 2-11 Antenna Assembly Truck

NORTH-SEEKING GYRO SYSTEM

34. The north-seeking gyro system (5801245-627) is mounted on the antenna assembly truck platform. It can be realigned horizontally by a few degrees in relation to the antenna carrier sub frame. The azimuth angle can be seen accurately on a scale and the deviation between North and the northerly alignment of the coordinate system used for data transmission taken into consideration. The gyroscope is easily removed and installed with the angular error introduced by removal and installation being $\approx 0.1^\circ$. The base plate is aligned on the antenna carrier sub frame in such a way that the 0 direction of the gyro- scope coincides with the 0 direction of the antenna Pedestal. The gyroscope is equipped with a synchro which, upon completion of the north-seeking operation, supplies the reference signal for alignment of the angle transmission system in the antenna pedestal.



Fig 2-12 North-Seeking Gyro System

CONTROL BOX "NORTH"

35. The control box "North" (5801245-628) is mounted beneath the walkway cover on the antenna assembly truck. It has a control switch which supplies power in a predetermined sequence to the leads on the antenna assembly truck (mast hydraulics drive motor, north seeking gyroscope and antenna drive motor). All other control and display elements of the control unit are intended for the north alignment of the angular transmission system in the antenna pedestal with the aid of a servo motor. The reference direction is determined by the north-seeking gyroscope.

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Fig 2-13 Control Box North

CABLE SET

36. The cable set (5800689-9) for the antenna assembly truck contains all the connecting cables between the antenna pedestal, the connector panel on the lower mast arm, the Control box "North", the north-seeking gyroscope and the mast hydraulics drive motor. The cables to the antenna pedestal are run through a cable duct rigidly attached to the hinged mast.

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SYSTEM COMPONENTS MPDR 45-L / 90SP

37. The Radar System MPDR 45-L / 90SP consists of the radar set, the IFF interrogator set, the radar antenna, the lattice mast, the gyroscope, the radar data processing equipment set and the communication equipment set. All the equipment of the Radar System MPDR 45-L / 90SP are accommodated on single truck.

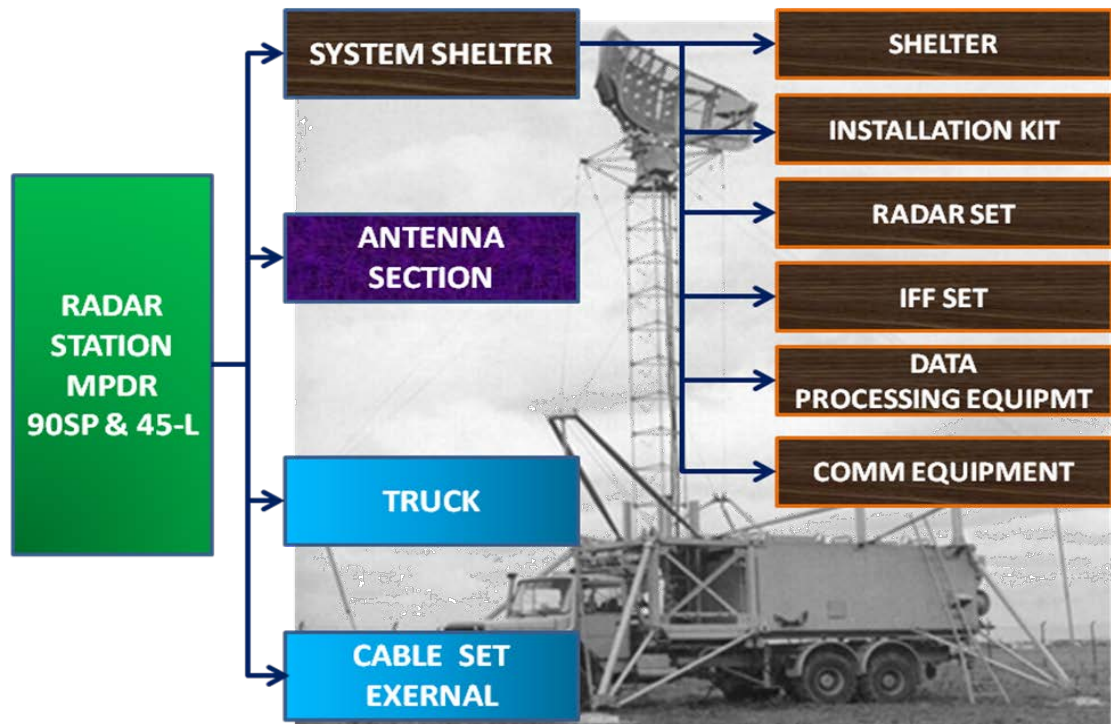


Fig 2-14 System Components MPDR 45-L / 90SP

38. "System shelter" is designation of the complete shelter with all instruments and internal equipment. "Shelter" is designation of the shelter only without instruments and internal equipment. The system shelter is divided into two compartments. All equipment except antenna system is installed by means of appropriate racks and mounts. The radar units are placed in the cases with their covers removed. Most of the cases are strapped tightly to the unit mounting racks, so that they can be replaced easily. The shelter consists of a welded constructional framework, which is lined both inside and outside with aluminum sheets (1.2 mm). The space between the double walls of sides and ceiling is filled with rigid expanded polyurethane for thermal isolation. The door is located at right side and emergency exit is at rear side of shelter. Space for three AC's and terminal boxes is provided in the walls.

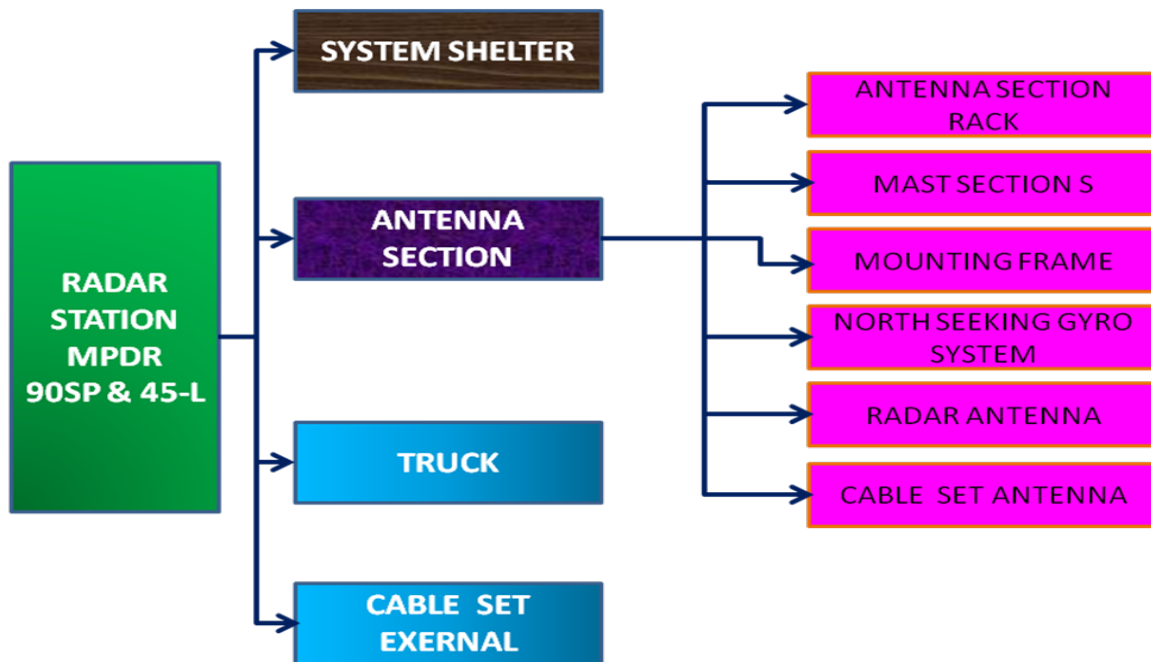


Fig 2-15 Components Antenna Section

39. The antenna section is best considered as an open container housing which contains the complete antenna system and an erectile and collapsible mast by hydraulic. The mast itself is a unique, lightweight and high-strength lattice structure which is stored compactly during transport within the above framework container and yet is rapidly deployed as a tower without assembly of additional parts. The erectable lattice mast carries the complete antenna system antenna pedestal, reflector and feed system. It is a semiautomatic telescopic radar mast which allows operation of the radar also in collapsed status. The mast consists of 17 sections with triangular cross-section and 80 cm height. They are formed by means of 25 mm diameter, light weight metal bars. Each section is individually erected and collapsed in the assembling and dismantling phase respectively. The bars of each section must be separately hoisted and then secured by means of internal guy-line cables. In this way one section after the other can be erected up to the total height of approx. 13.5 m. However, in the case of a plane geographical environment, the antenna can also be operated on top of antenna section rack. i.e. at a height of approximately 4 meter from the ground. During erection phase (and collapsing phase as well) the mast is subjected to a certain rotation and thus it is supported by means of a suitable turntable within the antenna section rack.

40. The antenna mast erection is facilitated by means of hydraulic equipment, consisting of a three internal hydraulic cylinders, which are simultaneously driven. They are fitted with a suitable gripping attachment such that the individual mast section can be erected automatically. Only locking and unlocking of the internal guy-lines and connecting and disconnecting of the gripping attachments are performed manually. To withstand the specified wind velocities the mast must be secured by means of guy ropes even in the assembling and dismantling phases. In any height position of the mast these 08 guy ropes (4 pairs of guy ropes) are tensed via internal guying struts by means of an automatic guy rope tensioning equipment.

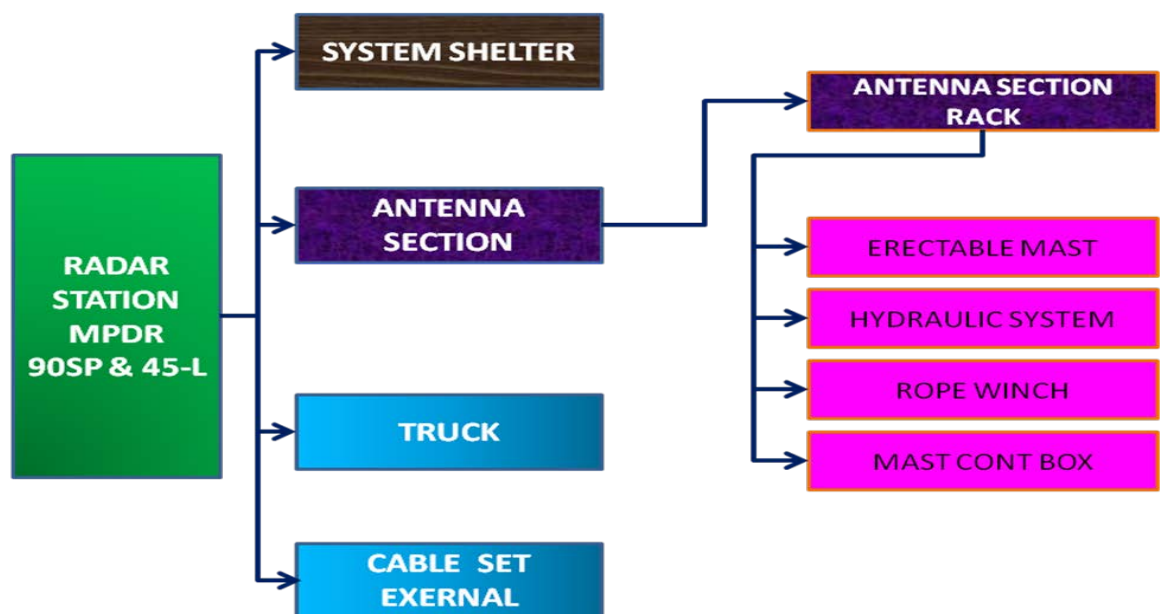


Fig 2-16 Components Antenna Section Rack

41. The lengths of guy ropes are already pre adjusted such that vertical assembling is performed in a minimum of time and with a maximum of efficiency. The guy ropes of the antenna mast are secured by the erectable ropes of the truck, such that no additional ground anchoring is required.

42. The mounting frame is located at the top of the mast and supports the antenna base. With the mast dismantled, the mounting frame is stored on the top of the antenna section rack. To prevent it shifting in transportation it is secured with four bolts.

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43. During transportation, if height of shelter is not a problem, then reflector center section is folded down towards the boom of the feed horn. Only the side sections are removed and stored separately. This configuration leads to minimum assembling / dismantling time at the radar site, while maintaining the required high mobility of the system. If minimum height of the transportable section is required, then the reflector center section is removed from the antenna base and stored as separate item. The antenna has an operational height of maximum 18 meters in fully erected position. RF connection between antenna base and shelter is provided by low loss RF coaxial cables.

SYSTEM MODE OF OPERATION**Overall System Mode of Operation**

44. Operation of Radar System MPDR 45/E is based on the Doppler method, in which the Doppler effect is utilized to discriminate between the echo signals from moving and fixed targets. The RF pulses radiated by the rotating surveillance antenna are reflected if they strike objects which are large compared with the RF wavelength. After a delay proportional to the range, the antenna intercepts these echo signals and feeds them to the receiver of the radar set.

45. If the target moves radially to the radar system, the frequency of the reflected pulse is displaced from the transmitted frequency (Doppler Effect). The frequency of the received echo signal then appears higher or lower than the transmitted frequency by an amount equal to the Doppler shift depending upon whether the reflection is received from a target which is approaching or receding from the radar system. This effect thus permits discrimination between moving and fixed targets. The magnitude of the Doppler shift is directly proportional to the radial velocity of the moving targets.

46. Radar System MPDR 45/E has a high sub-clutter visibility. This characteristic as well as a specially shaped antenna reflector and a cosec² vertical antenna radiation pattern make it possible to detect targets moving immediately above the surface of the earth, if the antenna is favorably sited.

47. The echo signals from moving targets appear on the screen of the indicator as bright luminous spots free from fixed echoes and echoes from slowly moving objects. The coverage area is scanned at each revolution of the antenna and the display on the screen simultaneously updated. Radar Set MPDR 45/E is linked with IFF interrogator Set MSR 400/8 (IFF, identification friend or foe). The antenna radiation patterns of both sets are substantially the same, since the IFF directional antenna and the radar antenna form a single mechanical unit (feed horn). The reflector of the surveillance antenna is used both by the same radar set and the IFF interrogator, so that the transmitter pulses from both equipments are radiated in the same space angle. The horizontal beam-width of the surveillance antenna is so narrow that the position of a detected target can be adequately determined in azimuth. Control and signal cables connect the surveillance antenna to the indicator and to the control units.

48. As the echo signals are time quantized in range channels, the range information is already available in digital form for processing and for transmission to MCC (Mission Control Center). The azimuth information is also available in digital form for remote transmission from a shaft encoder coupled with the antenna.

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49. The radar set includes a second transmitter chain (exciter, transmitter, blower unit) which is suitably tuned to another frequency, so that when reception is disturbed by another transmitter, the frequency can be quickly changed automatically or manually. In addition, this transmitter chain is available as a standby in the event faults in the active chain. In order to meet the high demands placed on the frequency stability of the radar set, crystal-controlled oscillators are employed. By exchanging the crystal in the oscillator in the exciter and retuning the final transmitter stages, any frequency in the L-band can be selected. Where necessary, the transmitter stages can be retuned using the indication on the built-in meter.

50. A sensitivity time control (STC) circuit prevents close targets from overdriving the receiver and second-time-around echoes from appearing to come from close targets. Electronically produced range rings, and a mechanical cursor with bearing scale around the indicator tube facilitate range and bearing determination and tracking of the moving target on the 30 cm diameter screen. In addition, the operator can superimpose the normal video in order to obtain information on the radar site and possible radar shadows resulting from this.

51. All adjustments and indications of the operating conditions (signaling) which are necessary for operation of the radar set occur on control panels arranged on the receiver and at the indicator.

52. Built-in monitoring circuits facilitate immediate detection of faults in the equipment and contribute to the operational reliability of the radar equipment. A generator set is available which supplies the power required by the Radar Set during operation. A second generator is available as a standby. The Radar Set is prepared for connection to 380 V AC, 3 phase power and further power supplies produce the operating voltages required for the different units of the Radar Set.

ANTENNA ASSEMBLY TRUCK

53. The antenna assembly truck is equipped with four pivoting outrigger supports which give the hinged mast antenna carrier a support base measuring 5 X 5 m². In the travel position the outrigger supports, are raised. They can be actuated hydraulically and are locked mechanically at the limits of their movement. The vehicle can be aligned horizontally with the aid of four auxiliary supports. These auxiliary supports are mounted vertically at the four corners of the sub frame carrying the hinged mast. The rear section of this sub frame is attached rigidly to the truck's chassis frame. The front section is secured flexibly to the chassis frame through plate springs. This prevents the truck chassis from losing its ability to flex

when driving across country. When the mast is in operation a frame interlock and a suspension locking mechanism connects the sub-frame to the chassis to form a single load-bearing unit. The sub-frame is provided with a non-slip sheet metal walk-way covering platform on which the various accessories and control units are located. The mast base is rigidly connected to the sub-frame, and holds the lower end of the hinged mast and the lower arm elevating rams.



Fig 2-17 Antenna assembly truck

54. The hinged mast itself consists primarily of the upper and lower arms and an intermediate section linking them. The two arms consist of two parallel box sections connected at the ends by pivots to form a parallelogram linkage. This parallel location ensures that when the mast is moved the antenna rotating axis always remains vertical to the sub-frame. The hinged mast is elevated by four hydraulic rams, two acting between the base and the lower arm and two between the intermediate section and the upper arm. The head of the upper arm is forked to accept the antenna pedestal with drive motor. A cable protecting arm is mounted at the side of the hinged mast to hold the RF cable. It is secured to the mast sections by pivot joints and connects the feed point at the mast base with the feed point in the antenna pedestal. All movements of the hinged mast and also vehicle leveling and supporting, suspension locking engagement and frame interlocking take place

hydraulically. An oil pump installed in the sub-frame generates the necessary oil pressure, and this is conveyed to the rams via control valves, pipes and ultra-high pressure hoses. The oil pump is driven by an electric motor which draws its power from radar set through control box north. The remaining electrical system for the mast assembly is fed from the 24 volt truck battery. Limit switches are provided on the outrigger supports and the upper mast arm to monitor their movements and control the flow of oil to the rams on the mast and those in the sub-frame by way of solenoid hydraulic valves in order to prevent damage caused by incorrect operation of the controls.



Fig 2-18 Outrigger support positions

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CHAPTER No 3

GENERAL SYSTEM DATA OF MPDRs RADAR

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CHAPTER No 3**GENERAL SYSTEM DATA OF RADAR SYSTEMS MPDRs****1. Overall characteristics**

Mode of operation	MTI Digital	All MPDRs
Frequency range	1215 ~ 1365 MHz (in L band)	All MPDRs
Detection range (RCS of a target of 1m ²)	1.5 ~ 45 Km	All MPDR 45
	1.5 ~ 60 Km	MPDR 60P
Detection range (RCS of a target of 2m ²)	1.8 ~ 92.5 Km	All MPDR 90
Detection altitude	14760 ft	All MPDRs except 90SP
	20000 ft	MPDR 90SP
Detection probability	90%	All MPDR 45
	80%	All MPDR 90 & 60P except 90P
	70%	MPDR 90P
Sub clutter visibility	≥55 dB	All MPDRs
Target resolution (in range)	0.27 NM (500 m)	All MPDR 45
	0.25 NM (460 m)	All MPDR 90
Target resolution (in azimuth)	4°	All MPDR with small reflector
	3.6°	MPDR 90S & 90ST
Location accuracy (in range)	±250 m	All MPDR 45
	± 230 m	All MPDR 90
Location accuracy (in azimuth) (in azimuth)	± 2°	All MPDR with small reflector

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2. Transmitter

Pulse duration	3.3 μ sec	All MPDRs
Average power	200 watts	All MPDR 45
	100 & 133 watts	MPDR 60P
	67 & 100 watts	All MPDR 90
Peak power	20 Kw	All MPDRs
Duty cycle	1% (1:100)	All MPDR 45
	0.50% & .33% (1:200 & 1:300)	MPDR 60P
	0.50% & 0.75% (1:200 & 1:133)	All MPDR 90
Hits per scan	About 120	All MPDR 45
	34 PRF 1.5 KHz 63 PRF 2 KHz	MPDR 60P
	34 PRF 1.5 KHz 21 PRF 1 KHz	All MPDR 90

3. Receiver

Input sensitivity (MDS)	-110 dBm	All MPDR 45
	-115 dBm	MPDR 60P
	-117 dBm	All MPDR 90
Time for frequency change	3 sec	All MPDR
Minimum radial velocity	150 Km	All MPDR 45
	100 Km	MPDR 60
	100 Km	All MPDR 90
ECCM features	Constant False Alarm Rate	All MPDR
	Amplitude versus azimuth	
	Pulse length discriminator	
	Inter pulse suppression	
	Frequency change	
	Constant false alarm rate	

4. Plan Position Indicator

Indicator tube	12" (30 cm) diameter cathode ray tube with magnetic deflection	All MPDR
Types of presentation	Polar coordinates with reference to north	All MPDR
Minimum detectable range	1 n.m	All MPDR 45
	1.5 n.m	All MPDR 90
Range markers	5 at interval of 5 n.m	All MPDR 45
	5 at interval of 10 n.m	All MPDR 90
	7 at interval of 5 n.m	MPDR 60
Heading line	Indicates front direction of antenna pedestal	All MPDR
Angle transmission	Resolver coupled to the antenna by a synchro system	All MPDR

5. IFF SET MSR 400/8

Transmitting frequency	1030 $\pm$ 0.2 MHz	All MSR 400/8
Receiving frequency	1090 $\pm$ 3 MHz	
Pulse power	400 watts	
MDS for correct decoding	-72 dBm $\pm$ 3 dBm	
Decoder	Passive two channel	
Code selection	2 codes pre-selectable	
Interrogation Modes	Mode 1,2 & 3/A	
SLS method	ISLS three pulse method RSLS two channel receiver	

6. Data Processing Equipment

Max. number of target data which can be processed and transmitted in automatic mode	112±8 per antenna rotation for 1200 bit/s	-
Number of plots in manual mode	1/4 n.m	
Operating modes	03	

7. Antenna

Antenna rotation	16±1 rpm (50 Hz power supply) 19.3±1 rpm (60 Hz power supply)	-
Angle transmission	Synchro fine 1:11 Synchro coarse 1:1 Shaft encoder 1024 pulses every 360°	
Isolation between radar and IFF antenna	About 40 dB	
Radiation pattern	Approx. Cosec ²	All MPDRs
Mechanical tilt angle of vertical beam	-1.5° ~ +8.5°	All MPDR with small reflector
	-5° ~ +5°	MPDR 90S & 90ST
Height of hinged mast to centre of antenna	12 m	MPDR 45E/60P/90P
Overall height of hinged mast	app. 13.1 m	
Reflector dimensions	WXH 5.0 X 1.9 m	All MPDR with small reflector
	WXH 7.2 X 3.28 m	MPDR 90S & 90ST
Total weight with cables of Antenna system	App. 523 kg	All MPDR with small reflector
	App.1000 kg	MPDR 90S & 90ST

8. Climatic conditions

Temperature limits between which operation is still permissible	- 25°C to 50°C outside 28°C inside shelter	All MPDR
Max. operating height	9900 ft ASL	
For non-rotating antenna without ice load	165 km/h	
For rotating antenna without ice load	140 km/h (As per T.O) 75 km/h (As per SOP)	
For rotating antenna without ice load	140 km/h (As per T.O) 75 km/h (As per SOP)	
During erection / lowering of the antenna mast	60 km/h	
Operating voltages	380 V 3Ø +5% / - 15%	
Operating frequency	47~63 Hz	

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CHAPTER No 4

DESCRIPTION OF EQUIPMENT “RADAR SET”

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♣ Plan Position Indicator Unit	
♣ Power Supply Generation And Distribution	

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♣	Power Distributor Unit	
♣	Voltage Regulator +150V	
♣	Test Oscilloscope	

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CHAPTER No 04

DESCRIPTION OF EQUIPMENT “RADAR SET”

SPECIAL TERMS USED IN MPDR RADAR SET

1. The mode of operation of Radar Set MPDR 45/E is based on the pulse Doppler principle. The surveyed air space is continuously scanned with the aid of short pulses radiated by a rotating radar antenna. If these pulses strike fixed or moving objects, some of the energy is reflected as a weak echo pulse and returns to the antenna after a delay proportional to the range. In order to prevent ambiguous indication of echo pulses, the spacing between the transmitter pulses must be greater than the longest delay due to 25 n.m range.
2. The range is determined both by the system parameters and by the pulse repetition frequency (PRF) employed. The indication of second-time-around echoes is suppressed by suitably limiting the range. The transmitter power and the antenna gain must therefore be designed to provide an echo power of the smallest target cross section of interest (standard target) that exceeds the noise power of the receiver.
3. Two definitions must be given here:
 - (a) The false alarm rate (FAR): This number states how often the indication threshold is exceeded by the receiver noise alone; it is in the order of 10^{-4} for each range element for MPDR 45/E.
 - (b) The detection probability (Pd): This number shows the probability of an echo signal exceeding the indication threshold of the receiver during time-on-target. It depends on "the signal-to-noise ratio in the receiver and on the threshold determined by the FAR. For MPDR 45/E the detection probability is 90% for a standard target at maximum range. Only the echoes from moving targets are to be displayed on the indicator screen, i.e. high clutter attenuation is required. In Radar System MPDR 45/E the Doppler shift is used to discriminate between echoes from radially moving targets and echoes from fixed targets (clutter). Digital MTI filters (moving target indication filter) are used for suppressing clutter and slow moving objects echoes.
4. The value of clutter attenuation of a radar set is defined by the level difference of a received fixed target echo and a received moving target echo leading to the same detection probability on the PPI screen. A suitable combination of the MTI

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filters and a signal limiter in the IF part of the receiver results in, an extremely high clutter attenuation of MPDR 45/E.

5. Moving target echoes arriving simultaneously with large clutter echoes may be suppressed due to the limited dynamic range of the receiver and therefore may not be indicated. This consideration shows that the clutter attenuation alone is not a suitable parameter for the assessment of a radar set. This is the reason for the introduction of the term sub-clutter visibility (SCV). It states the maximum permissible magnitude of a clutter echo arriving simultaneously with the echo of the standard target to remain visible on the screen; i. e. the standard moving target is still indicated when it is within the same range element together with the fixed target echo which is less than or equal to the amount of the sub-clutter visibility.

6. Due to the fact that the power of radar target echoes is inversely proportional to the fourth power of the range, the differences between the strength of echo signals can be very large. This can easily lead to overloading of the receiver by echoes from large short-range fixed targets and thus suppress the indication of a smaller moving target at the same range. So a time-dependent sensitivity control is employed in the receive path. The sensitivity time control (STC) reduces the gain in the receiving path for short ranges, the gain rises to its full value towards the longest range.

7. Modern large transport aircrafts have a large radar target cross section, this means that pulse Doppler radars can detect them at ranges which exceed the radar's uniquely defined detection range, and thus these aircraft can be displayed on the PPI. The echoes of these out of range objects are called MTA (Multiple Time Around) echoes. Without special measures the MTA echoes are not distinguishable from echoes which lie within the uniquely defined range and cannot be suppressed. A suitable possibility to recognize these MTA echoes is the change of the PRF. The variation of the PRF causes the MTA signals to be received in different range gates, while echoes resulting from objects within the uniquely defined detection range always appear in the same gates, depending only on their distance. To avoid indication of targets out of the defined range (Multiple Time Around echoes, MTA echoes) on the PPI, the PRF is periodically altered by a little amount. Changing the PRF after a fixed number of transmit pulses by some Hz is called *PRF Jittering* and Changing the PRF after a fixed time by higher amount of Hz is called *PRF staggering*.

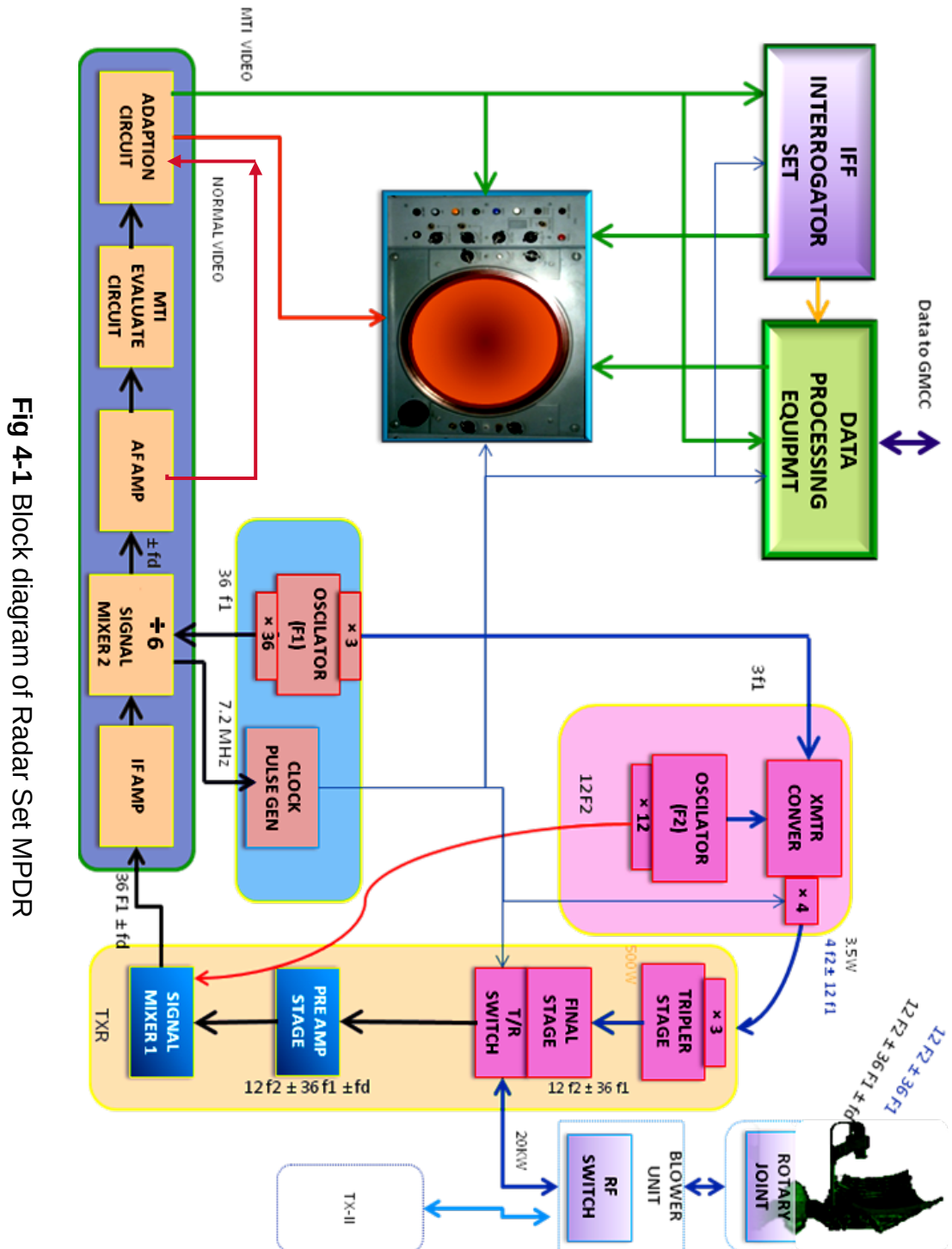


Fig 4-1 Block diagram of Radar Set MPDR

MODE OF OPERATION OF UNITS FORMING RADAR SET MPDR

8. Radar System MPDR 45/E consists of the radar antenna (5800590-9), radar set (5801245-5), IFF interrogator set (5801117-3), radar data processing equipment set (5801180-8) and the communication equipment set (5801245-618). The timing of all operational sequences in the radar system is controlled from a central clock pulse generator (system timing unit, here referred to as clock pulse generator) which supplies all timing signals required for the internal control of functional sequences. The clock pulse generator in the receiver (5801239-5) controls the keying of the transmitter path and also produces the start pulse for the indicator time base as well as for the range rings on the screen of the indicator (PPI, 5801240-5). It also supplies the synchronizing pulses for the IFF interrogator. The necessary control pulses are derived by dividing the frequency of crystal-controlled oscillator f_1 , the transmitter frequency f_t is produced from oscillator frequencies f_1 and f_2 by frequency conversion and subsequent multiplication. The frequency f_2 is mixed with the tripled frequency f_1 in the transmitter converter and the resulting frequency $f_2 \pm 3f_1$ is multiplied to the required transmitter frequency $f_t = 12 (f_2 - 3f_1)$. The transmitter stages raise the power to the necessary level. The transmitter pulse is triggered by keying pulses which successively key the frequency multiplier and transmitter stages. *The transmitter pulse repetition frequency (PRF) determines the maximum range of the radar set.*

9. The RF pulses from the transmitter are fed to the TR switch which isolates the receiver from the transmitter. The switching pulses from the clock pulse generator alternately apply the transmitter to the antenna (TX period) and then the antenna to the receiver (receive period) in synchronism with, the PRF (pulse repetition frequency). The TR switch thus permits the common use of the antenna for transmit and receive functions. A rotary joint couples the transmitter energy from the fixed antenna pedestal into the antenna which radiates it in a beam. The speed of rotation of the antenna is so chosen that, for a given antenna horizontal beam width, each target is struck by sufficient pulses during one revolution to ensure good sub-clutter visibility.

10. In the period between two successive transmitted pulses all objects are detected which are located in the path of the beam transmitted from the radar antenna. Part of the energy reflected by the objects returns to the antenna, the received frequency f_r , of the echo signals from moving targets is shifted from that of echoes from fixed targets by the Doppler frequency f_d ($f_r = f_t \pm f_d$). With the TR switch set to receive, the echo signals are fed to the input of the receiver. In order to ensure adequate separation of the image frequency, double super heterodyne operation is employed for the receiver. In the first signal mixer the echo signals are converted to the first IF by heterodyning with the 12th harmonic of the frequency of oscillator-F2 and subsequently amplified.

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11. After the second frequency conversion in the second signal mixer with the 36th harmonic of the frequency of oscillator F1, the echo signals appear at the video frequency. The transmitters including exciters are duplicated, the oscillator crystals having different frequencies, so that selection can be made between two transmitters operating at different frequencies. When the frequencies change, the receiver intermediate frequency remains unchanged as a result of the special method of frequency processing, and re-tuning of the receiver unit is unnecessary.

12. The signals from the first signal mixer are fed via two paths to the indicator: One path (moving target or MTI path) leads via the second signal mixer to the MTI signal processor modules where fixed targets are suppressed and moving targets are outputted. In order to get good sub-clutter visibility on the screen, the MTI filter must provide high attenuation for the fixed target echoes but allow the Doppler shifted moving target signals to pass with as little attenuation as possible. On the other path (fixed target or normal video path) part of the echo signal energy is fed to a permanent echo amplifier after the second frequency conversion, and fed to the indicator as required. By means of a switch on the indicator the permanent echoes can be superimposed on the display if required. In addition, the amplitude of the permanent echoes on the display can be adjusted and thus an optimum brightness of the permanent echoes superimposed on the moving targets can be obtained.

13. The facility for indicating and controlling the brightness of permanent echoes makes it possible for targets flying tangentially to the radar equipment to be detected in the permanent echo display. It is principally the video signals of moving targets which are displayed on the indicator screen (plan position indicator, PPI) for determination of range and track of the targets. Determination of target range on the PPI is effected with the aid of electronically produced range rings, the spacing of which is determined by the mile marker pulses from the clock pulse generator in the receiver. The accurate presentation of the azimuth of a moving target on the PPI is ensured by the synchronism between the antenna and the PPI trace.

14. Due to the afterglow of the PPI, a track is formed behind each target from which the direction of movement and velocity of the target can be estimated.

15. The direction is given by the position of the track and the velocity by its length. The position of the antenna is represented by the center of the screen. The video signals are displayed by intensity modulating the CRT beam, while simultaneously deflecting it from the center of the screen to the edge. In addition to the radial deflection, the CRT beam also rotates in synchronism with the antenna and so leads to a PPI display on the screen.

MARKING OF INTERCONNECTIONS

16. In addition to the part number and stock number the cables between units forming the radar set are marked at both ends. Connectors on the equipment and the corresponding cable connectors have the same designation, with a few exceptions.

17. The following letters are used to identify the connectors and cables interconnecting the units:

- A** Evaluator unit (in case of MPDR 90)
- B** IFF control unit
- C** Radar data processing unit
- D** Power supply unit for radar data or processing unit
- E** Receiver
- F** Control panel for radar data processing Unit
- G** Blower unit 2
- H** Transformer 1
- I** IFF interrogator
- L** Transmitter
- M** Antenna Mast
- N** Indicator power supply unit
- O** Test oscilloscope
- P** Indicator (PPI)
- Q** Antenna pedestal
- R** Voltage regulator
- S** Exciter (signal generator)
- T** Transformer 2
- U** Blower unit 1
- V** Power distributor
- W** Terminal block assembly
- X** Rolling ball
- Y** Radar data processing unit
- Z** RF terminal box

18. A line e.g. that between the receiver and the IFF interrogator carries the designation EI. If several lines are connected between these cases a number (1, 2....) is added to the code letter. If units are duplicated in the system (exciter S, transmitter L) the code letter is preceded by an ordinal number (transmitter left 1.L, transmitter right 2.L).

Basic Frequency and Clock Generator

Generations of basic transmission signal and LOs frequency signal

19. In order to achieve high sub-clutter visibility (SCV) of moving targets not only the transmitter frequency but also the local-oscillator frequencies for the two signal mixers must be very stable in mixer 1 and mixer 2. The local oscillator frequency and the transmitter frequency are separated by a fixed amount which determines the IF of the receiver. The transmitter and the local-oscillator frequencies are derived from the same oscillators F1 and F2. Oscillator F1 is accommodated in the receiver, oscillator F2 in the exciter (signal generator).

20. When the radar system is connected to the supply voltage the oscillator F1 starts operation and generates its basic frequency signal (1.2 MHz), then f_1 frequency multiplied by a factor of 36 is fed into mixer 2 where the master clock frequency of 7.2 MHz is generated through division by a factor of 6 and same is utilized as local oscillator frequency for the down-conversion of echo signal. The basic frequency for the transmission path is obtained via multipliers 3f1 in this oscillator F1. From the basic 7.2 MHz clock, the module clock pulse 1 and associated modules generates all control pulses for the radar system MPDR.

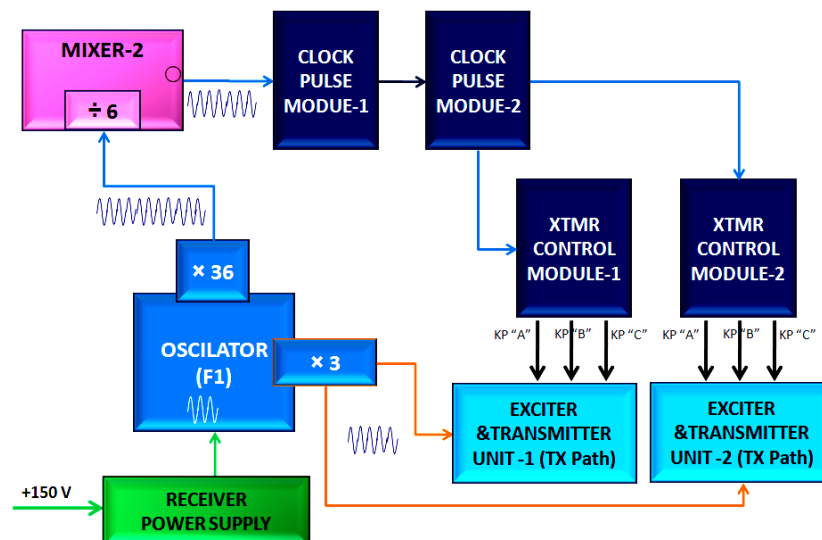


Fig 4-2 Block diagram of Basic Frequency and Clock Generation

Generations of the timing / clock signals

21. The clock pulse generator in the receiver is the control center of the radar set. All control pulses are derived here from master clock frequency of 7.2 MHz. To suppress echoes from targets out of the defined range (MTA echoes, MTA Multiple Time Around) and to avoid a reduction of SCV (Sub-clutter Visibility) the PRF is block staggered. This means 20 pulses are transmitted with no change in the PRF of 3 kHz, then the PRF is changed to 2.94 kHz for 20 pulses and after this a third PRF of 2.88 kHz is applied. This is repeated all the time the radar set is working.

22. Some of the most important functions of the module are supplying the trigger pulses for generating the transmit pulse, starting the deflection beam on the display, monitoring MTI signal processing, generating the staggered PRF and supplying synchronizing pulses to peripheral equipment such as the test oscilloscope, radar data processing unit and IFF interrogator

23. The availability of input basic clock of 7.2 MHz is monitored in clock pulse module, as long as this signal is present, a built-in LED V2 lights up. If the 7.2 MHz is missing then LED V2 goes out and fault signal generated. Apart from the basic clock of 7.2 MHz derived in the mixer by 6: 1 division of the 43.2 MHz oscillator output frequency of the oscillator (5801239-530), the signals STC-AUS and JITTER OFF are applied to clock pulse 1 as input signals. The input signal JITTER OFF is only used for test purposes. With JITTER OFF at "L" the staggered PRF is kept at 3 KHz.

24. Apart from the 3 above named signals used for controlling clock pulse 1, all other signals of clock pulse 1 are output signals which in turn control signal processing for the entire radar equipment. The required output signals are generated in principle as follows: counters that operate as dividers generate, from the 7.2 MHz basic clock, pulse diagrams that form addresses for four PROMS in the module clock pulse 1, according to the PROM address generated, the corresponding output signal is then produced. One of the PROMs is used as address PROM. This in turn again loads counters, which generate the addresses for the two PROMs, whose output signals are the staggered PRF (3 KHz, 2.941 KHz, and 2.885 KHz) and as long as radar set is operating in staggered PRF mode the built-in LED "VI" lights up with a keying ratio of about 1:3, which simulates medium brightness to the eye. As mentioned above, with JITTER-OFF at "L" the staggered PRF can be kept at 3 kHz for testing. In this mode "VI" no longer lights up. The chopper pulse which is passed to the power supplies of the receiver and the exciter is also generated in the clock pulse module 1.

25. The Clock pulse module 2 supplies indicating pulses to the video mixer in the PPI unit, enabling range markers to be produced on the PPI. The trigger for the range markers from clock pulse module 1 is mixed in clock pulse module 2 with the range markers to produce the start pulse for the deflection beam on the indicator.

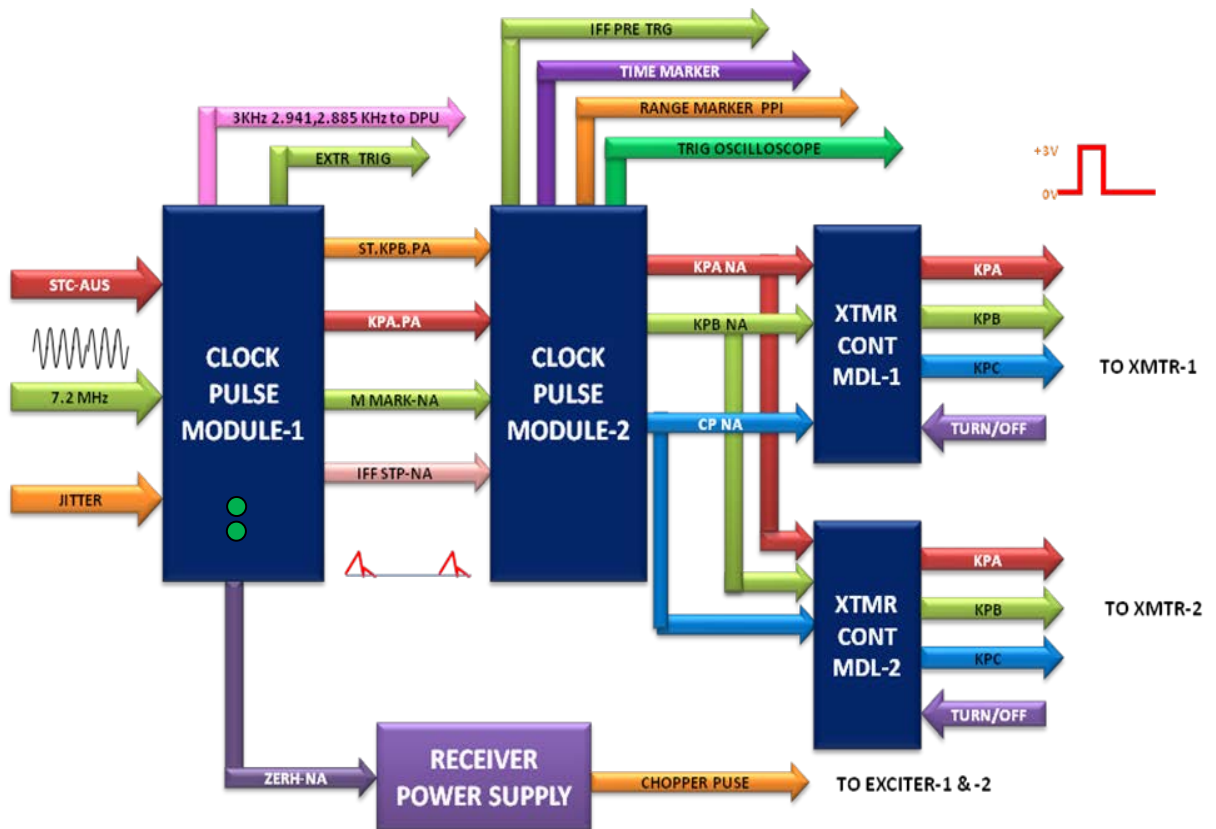
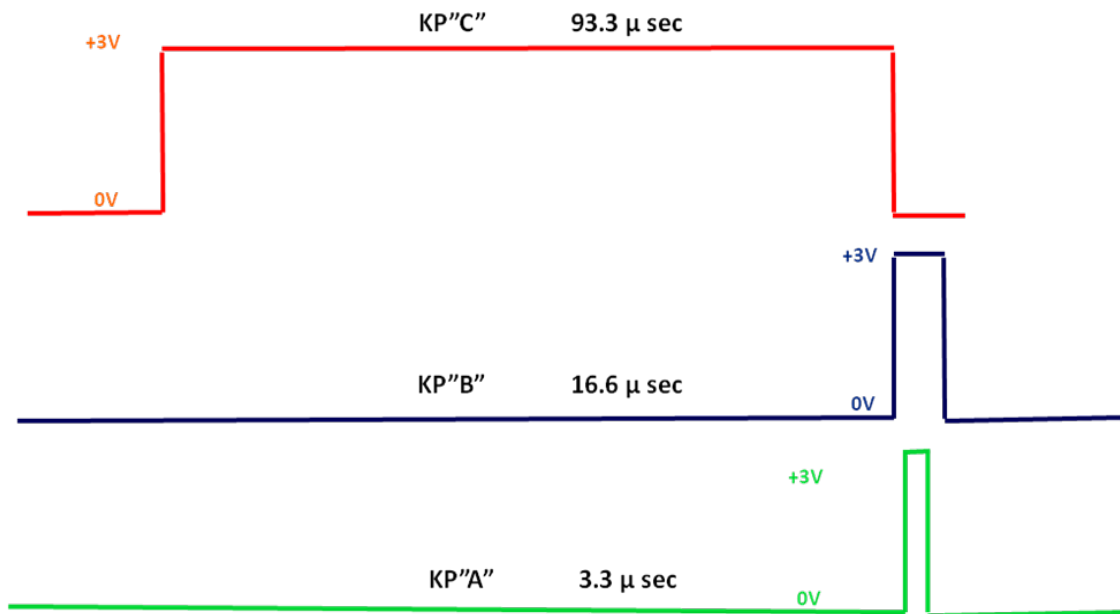


Fig 4-4 Block diagram of Clock Generation

26. The clock pulse module 2 supplies the radar zero pulse and the pre-trigger to the IFF interrogator. In the clock pulse module 2 the TTL levels are matched to the levels of XMTR control module, the PPI and the oscilloscope. In order to synchronize the transmitter with the signal processor and the display of the indicator, the transmitter is driven by the keying pulses A and B as well as by the control pulse C from the XMTR control. After appropriate amplification in the trigger and pulse amplifiers in the transmitter units, these pulses cause the high voltages to be applied to the transmitter stages. Presence of all these pulses can be verified at various stages in the receiver and transmitter units.

**Fig 4-5** Sequence of clock pulse generation

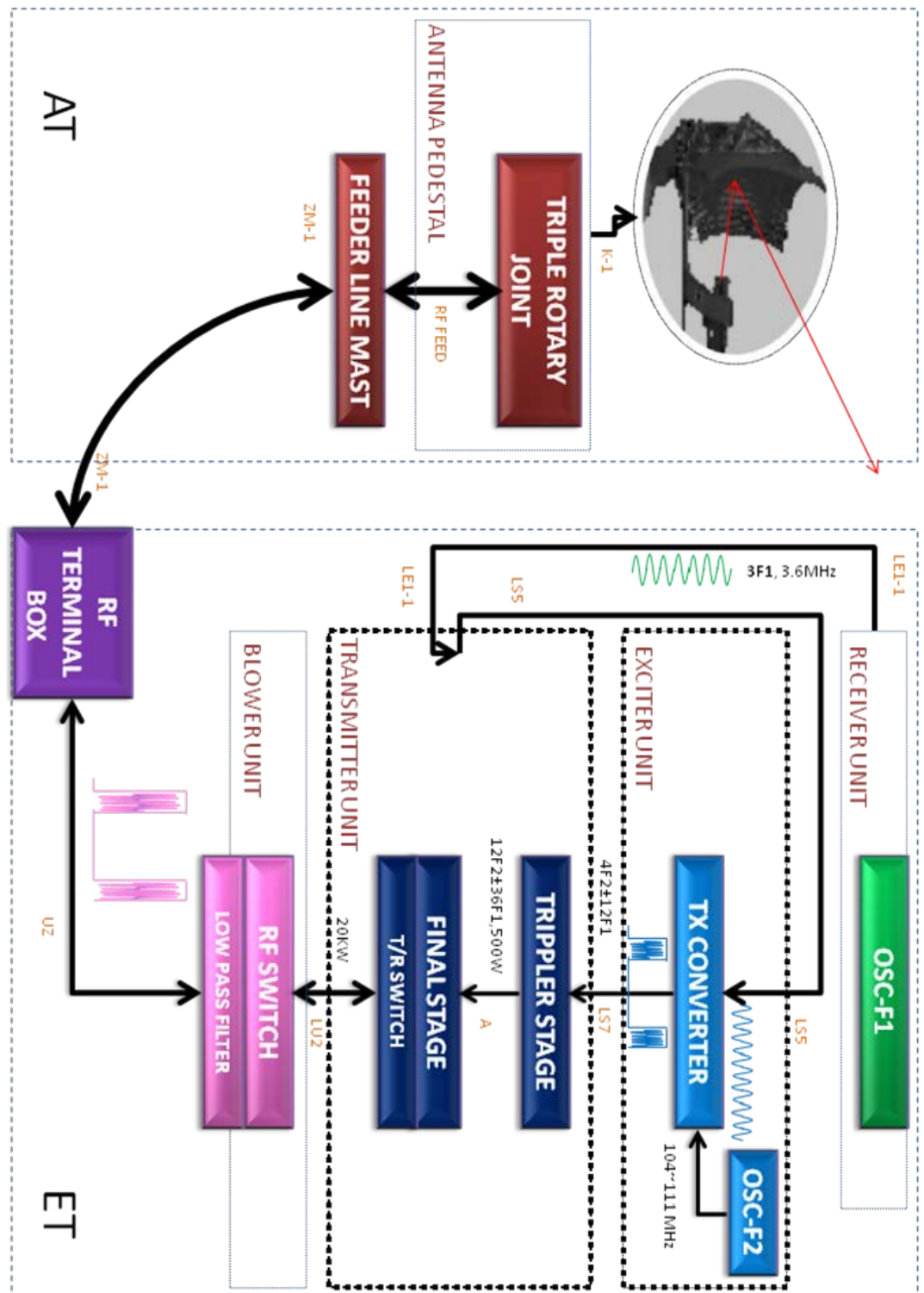


Fig 4-5 Complete transmission path of radar set MPDR45-E

TRANSMISSION PATH

Exciter (Signal Generator)

27. There are two signal generator units in MPDR, one act as active and other serves as standby and becomes operative in the event of frequency change or fault in the active channel. It up-convert the transmitter signal by mixing and subsequent multiplication of oscillator frequency $3F_1$ coming from receiver and F_2 generated within exciter unit, convert the CW signal into pulsed form upon arrival of keying pulse "B" and amplifies the pulsed form signal to drive the transmitter unit. It is located above the transmitter unit and consists of three sub assemblies.

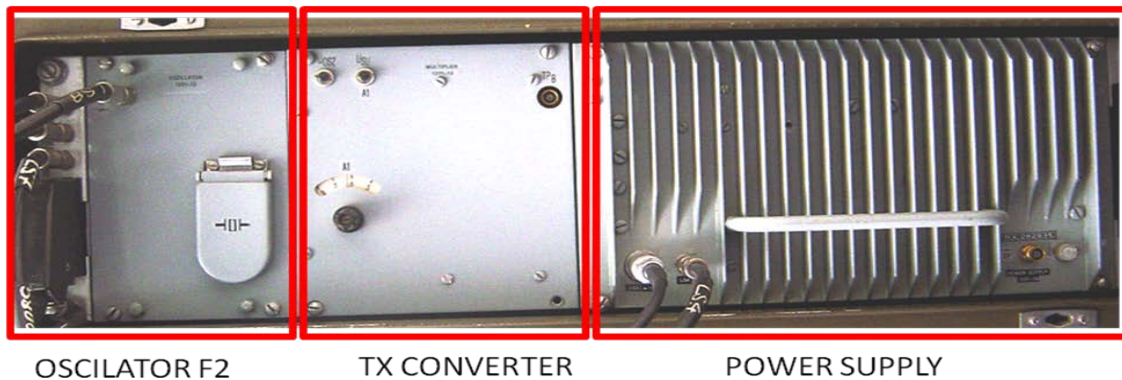


Fig 4-6 Pictorial view of Signal Generator Unit

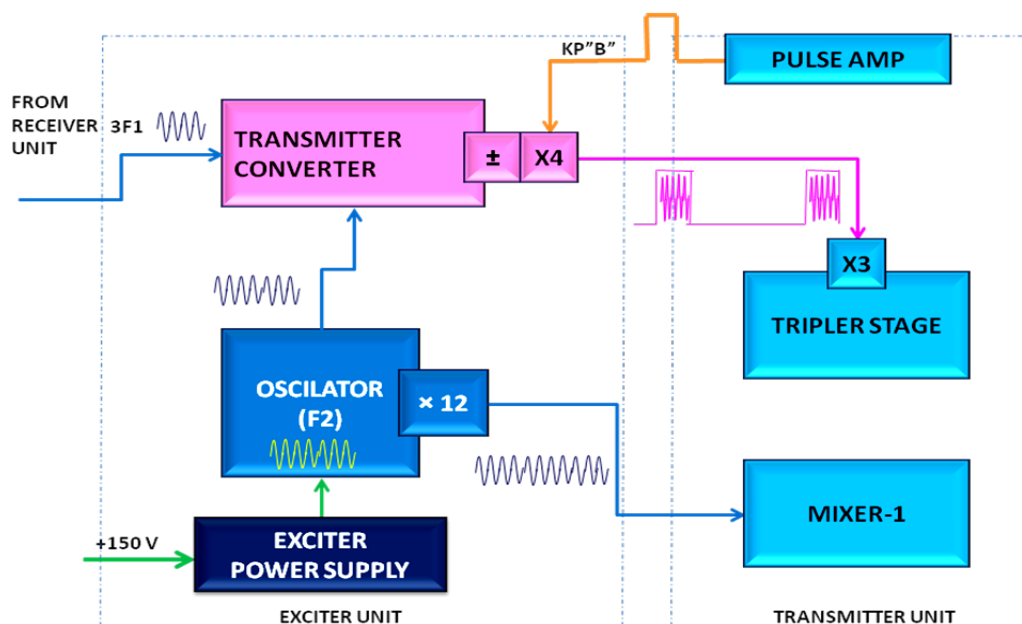


Fig 4-7 Block diagram of Signal Generator Unit

Oscillator F2 (Signal Generator)

28. It is a crystal controlled oscillator, the range of this oscillator is from 104-111 MHz, a particular frequency between this range can be obtained by changing the crystal in oscillator 105 (F2). There are total 70 crystals; utilizing those complete operating frequency range of radar set MPDR is obtained. For the checking of the oscillator F2 a sample of the oscillatory current is coupled out to the built-in meter of the TX unit. The associated test key 'U_{Osc}' is located on the front panel of the exciter. One output of this oscillator (104~111MHz >90 mw) is fed to transmitter converter for generation of transmission signal and in order to produce local oscillator frequency for receive path, the frequency f2 of oscillator 105 is multiplied in a 12 X multiplier within the oscillator module and then fed via an attenuator and coaxial cable LS8 to the signal mixer for down conversion of echo signal.

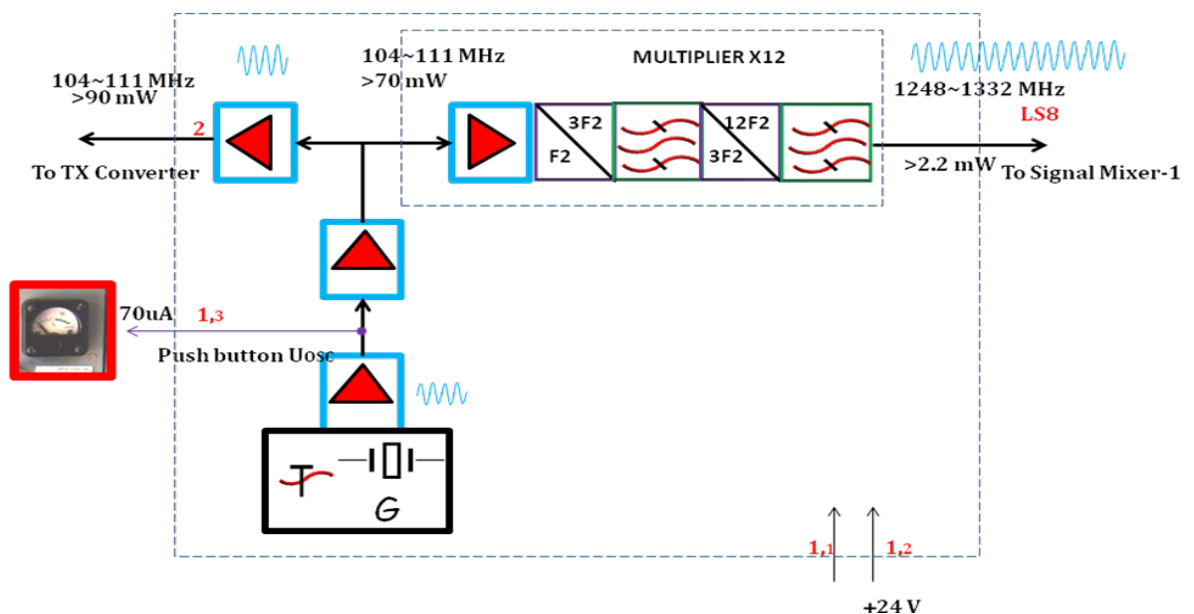


Fig 4-8 Functional diagram of Oscillator F2

Transmitter converter (Signal Generator)

29. Transmitter converter (fig 4-9) consists of modulator 105, side band filter and quadrupler.

30. The one input to the modulator is $3f_1$ from RX via transmitter unit and the other is from the oscillator F2 of the exciter. The two frequencies $3f_1$ from RX and F2 from oscillator are heterodyned in modulator 105 and the sum ($f_2 + 3f_1$) and the difference ($f_2 - 3f_1$) then appear at the output of modulator 105. The composite signal is fed to a four-stage sideband filter (5801231-1312) which can be externally

tuned to one of the sidebands. As both upper and lower side bands lie in the TX frequency range so by using both side bands the TX frequency range is covered with only half the number of crystals otherwise, necessary, since frequency of both the sum and difference frequencies fall in the tuning range of the transmitter stages.

31. By pressing associated test key 'U_{su}' located on the front panel of the exciter the output of the SB filter can be checked by feeding it to the built-in meter of the TX unit. The filter output voltage is first amplified, and then passed to the frequency quadrupler (5801231-1313). In order to produce the transmitter frequency, the output frequency from side band filter is multiplied by '4' times in quadrupler, which also increases the power level by a factor of eight. Pulse amplifier within the transmitter converter is provided for keying the quadrupler stages to convert the CW signal into pulsed form. This pulse amplifier amplifies the incoming keying pulse "B" from pulse amplifier-15 of transmitter unit, which have duration of 16.6 μ sec and fed it to quadrupler. Availability of KPB can be verified through test jack "TPB" located on the front panel of the exciter. The output of quadrupler $4(F2 \pm 3F1)$ with a duration of 16.6 μ sec (pulsed form) and output power is 3.5 W is fed to tripler unit of transmitter unit via cable LS-7. The quadrupler in the transmitter converter has the longest ON-time; it receives its ON-pulse from the pulse amplifier which is driven directly by keying pulse "B".

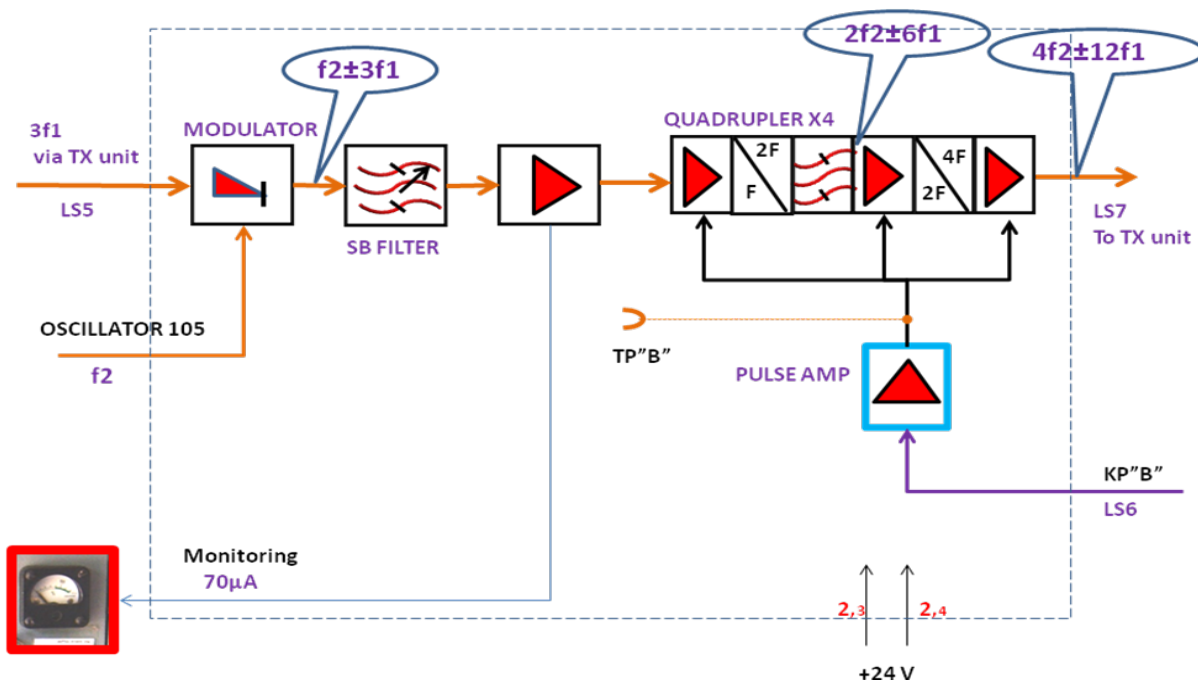


Fig 4-9 Functional diagram of TX converter

Power Supply (Signal Generator)

32. The power supply exciter (5801231-14) delivers all DC voltages required by the sub assemblies of exciter and the transmitter units. Input of exciter power supply +150V from voltage regulator, this comes via transmitter unit. The outputs of the exciter power supply are 6.3 VAC 12A for thyatron heater, 6.3 VAC 3A for final stage (pre amplifier), 6.3 VAC 7.5A for final stage heater (final amplifier), 6.0 VAC 1A for heater of tripler stage tubes, +24 V for all circuits of exciter and transmitter units.

33. The inverter is fed with 150 VDC from the +150 V. regulator. It is fed through cable LS3. Inverter is driven by the AC control voltages at a frequency of approximately 2.5 KHz supplied by the current limiter. The square wave voltages at the output of the inverter are fed to the primary windings of the output transformer in the output section. The exciter module is designed as a controlled inverter; it is also fed with 150 VDC and triggered by 3 KHz clock of the chopper control. It drives the current limiter and it also supplies the operating voltage for the current limiter. The current limiter controls the conduction of the inverters in inverter module according to the output load. In case of over current drawn, conduction of the inverter transistor ceases and this state is signaled by fault lamp "excessive current" located at the front of power supply. The time switch module has the task to provide delay in switching on the transmitter while the electron tubes are being warmed-up, as switching on the transmitter before warming up of tubes can cause damage to the tubes. The time switch module is fed with a square-wave voltage of 34 vpp from the primary winding of the current transformers. Time switch module contains 6 capacitors, those take 3 minutes approx to be charged; resultantly a delay of 3 minutes is achieved in switching on transmitter stages. The 24V, voltages are rectified and filtered before being fed to exciter and transmitter. The filament voltages required by the electron tubes of transmitter unit are fed directly to the output connector LS-2.

34. Presence of Chopper pulse of 3 KHz to trigger exciter module can be verified through a test jack located on exciter power supply. *It is imperative to mention that exciter module installed in power supply of exciter unit is similar to the exciter module installed in receiver power supply. If situation arises that the exciter module of receiver power supply becomes defective then one of exciter module of the exciter units may be utilized to recover receiver power supply.*

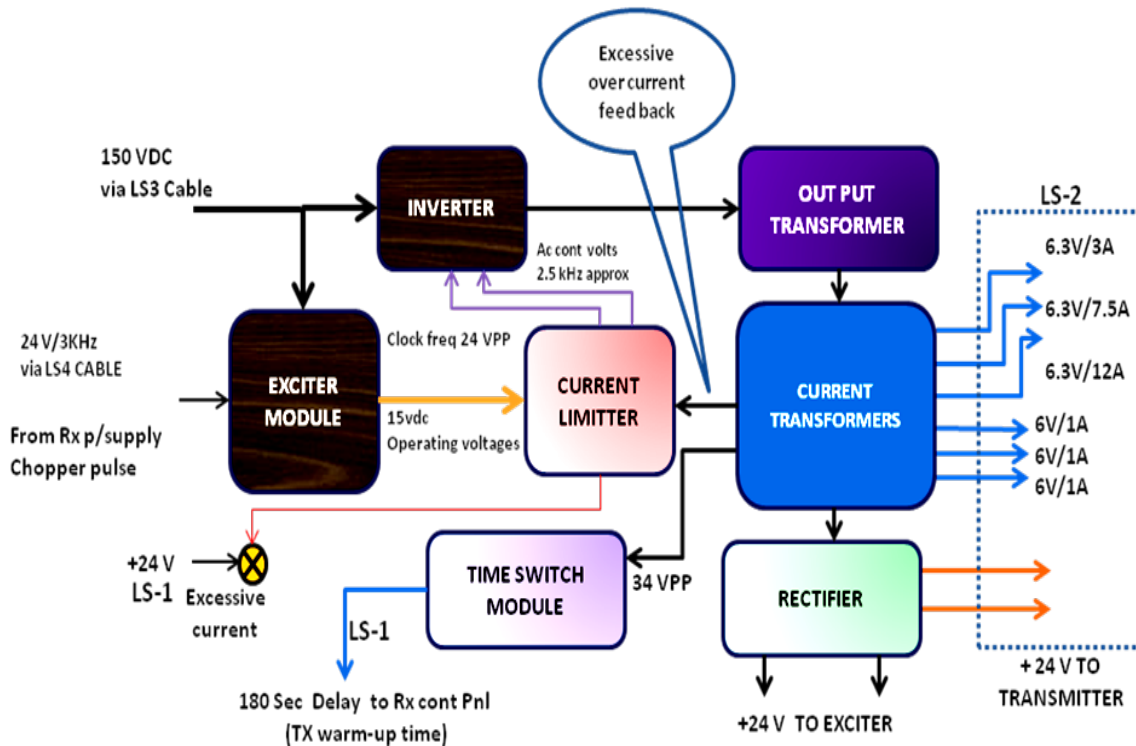


Fig 4-10 Block diagram of power supply exciter

1. Tripler Stage
2. Final Stage
3. H.V Power Supply
4. Pulse Amp-15
5. Pulse Amp-12
6. Trigger Amp-16
7. Monitor module
8. TX Monitor Assy
9. Elect Relay Module

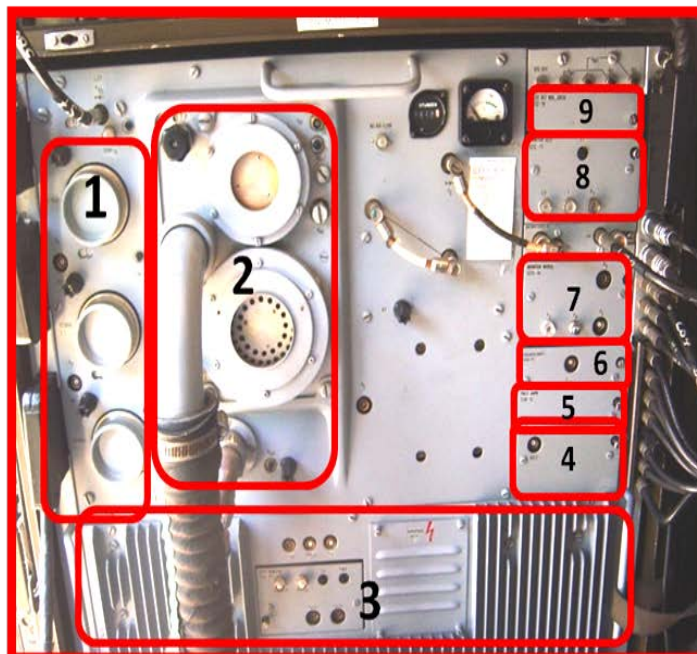


Fig 4-10 Pictorial view of Transmitter Unit

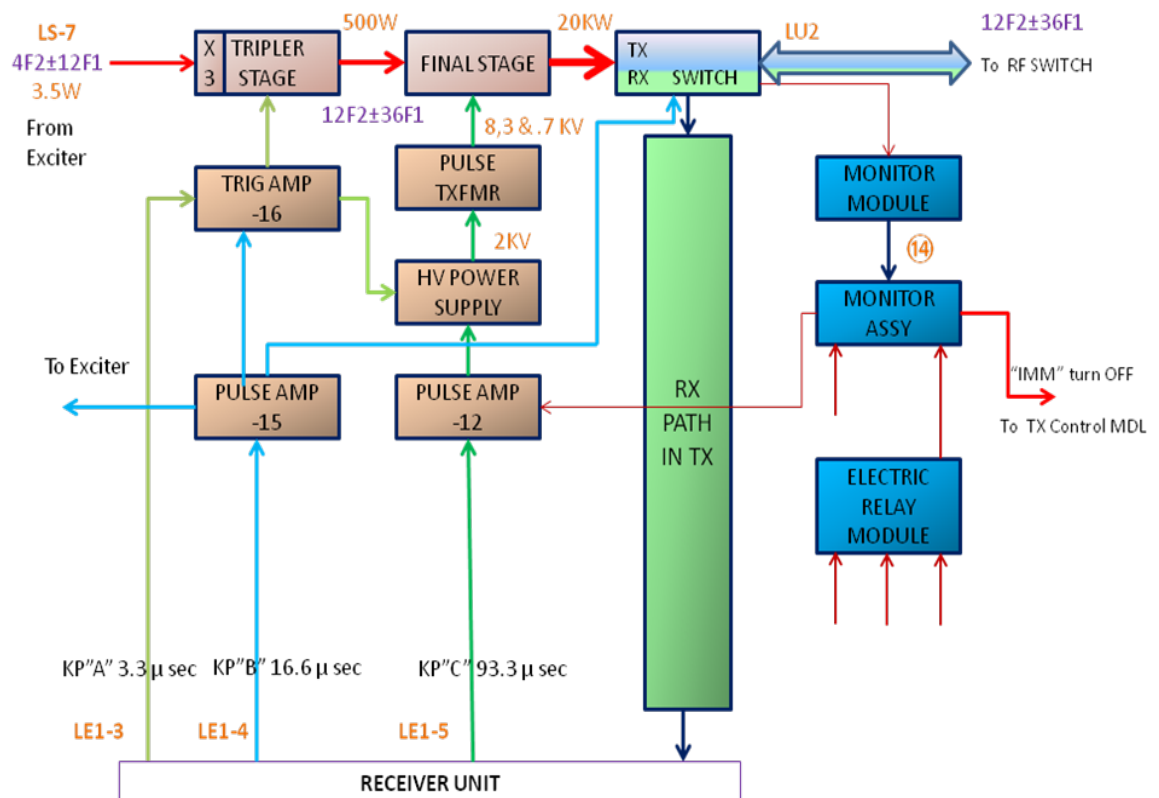


Fig 4-11 Block diagram of TX Unit

TRANSMIT PATH IN THE TRANSMITTER

Trippler Complete

35. The complete tripler consists of two units, "TRIPPLER" and "KEYING UNIT".

Trippler

36. The trippler converts the frequency (405 to 455 MHz) of the signal supplied by the exciter to the transmitter frequency of 1215 to 1365 MHz. In addition; power is amplified to the value required for driving the final stage. The trippler consists of one multiplier stage and two amplifier stages. The three stages are identically designed and capacitively tuned from the outside. The anode voltages fed from keying unit are keyed. Heater voltages of 6VAC/1A are supplied from exciter power supply. The signal present at cathodes of each stage can be fed to the meter installed in the TX unit via push buttons JK1 to JK3. If the tube covers are removed, the anode voltages are switched off by a protection circuit. Output of exciter which is 4F2±12F1 (405 to 455 MHz) with power of 3.5 W is fed to trippler stage, where the frequency is tripled

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to 1215 to 1365 MHz. The first stage in amplifier chain, amplifies the signal up to 50-60 Watts and 2nd amplifier stage amplifies further to 500 Watts (peak power). The ON time of the tripler is determined by the keying unit.

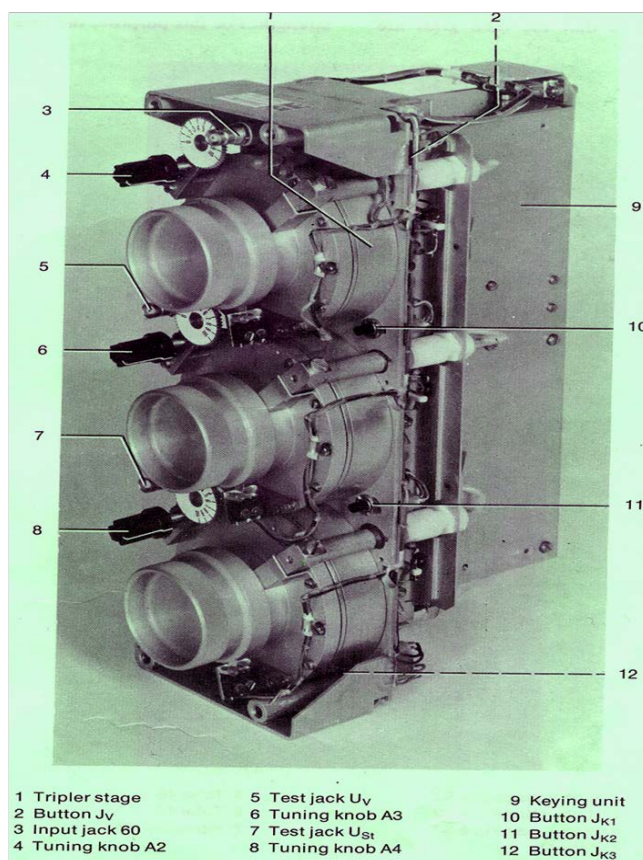


Fig 4-12 Pictorial view of Tripler stage complete

Keying Unit

37. The keying unit generates the anode pulse voltages of 1KV and 2KV required by the electron tubes in tripler. It is driven by the trigger amplifier with the keying pulse "A" (+3V, 3.3 μ sec). Since the transmit pulse duration for the final stage must lie within the ON-times of the preceding quadrupler, in the transmitter converter and tripler, the tripler is driven for about twice as long as the final stage. This ensures that the input of the final stages is provided before the stage is keyed by the application of H.T voltages. That's why keying pulse 'A' is extended first to approx. 4.5 μ sec prior feeding power transistor. This pulse can be monitored at test jack "U_{st}". A regulated +150 VDC form voltage regulator are fed to the collector of power transistor via HI-Pot supply. During the pulse, 1KV and 2 KV are produced at the

secondary of transformer. 2KV anode voltages can be checked at test jack U_v . If, the anode circuit is overloaded (for example short circuited), temperature of power transistor increases, thermal switch actuated, power transistor is cut off and indicator "excessive temp" located on front of tripler comes on. The current drawn by tripler can be checked with push button ' J_v '.

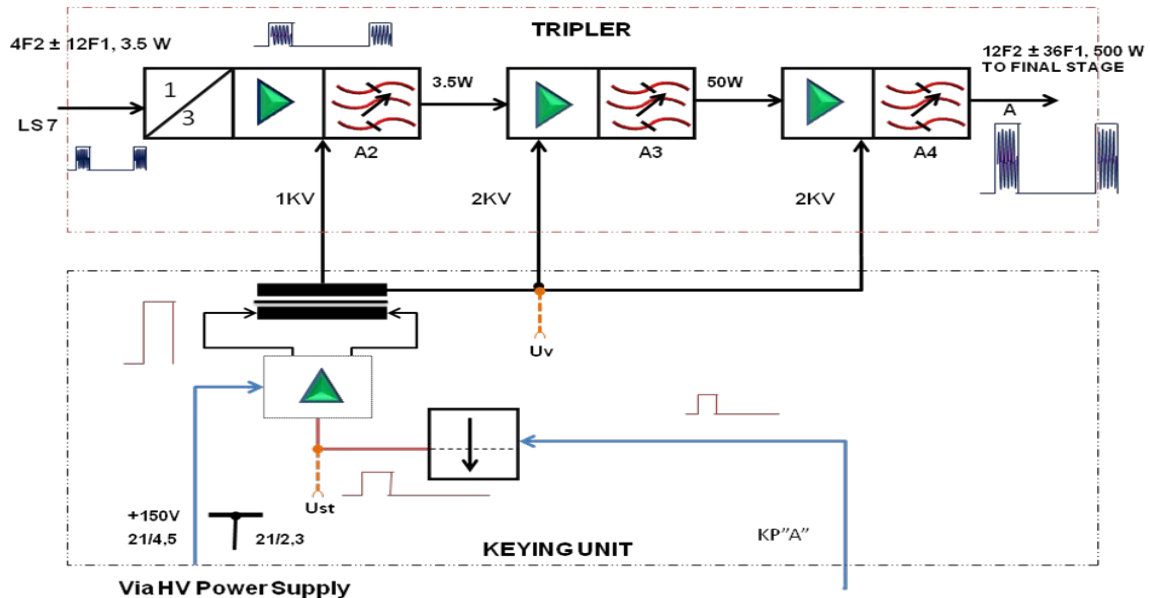


Fig 4-13 Functional diagram of Tripler stage complete

Final Stage Complete

38. Final stage complete consists of two sub units, "final stage and TR (transmit/receive) switch"

Final Stage

39. In the final stage, the RF pulse power of 500 W in frequency range of 1215 to 1365 MHz supplied by the tripler is amplified to RF pulse power of ≥ 20 KW. The final stage consists of pre- amplifier and final amplifier stage. The anodes of both stages as well heater and cathode connections of final amp are cooled with compressed air. The ON time of the Final stage is determined by the pulse forming network (PFN) in the high voltage power supply.

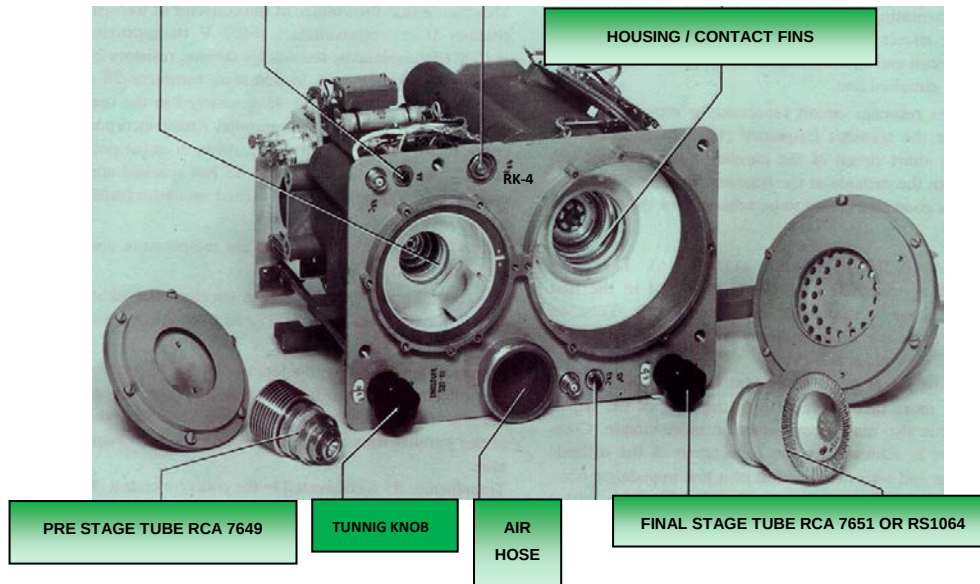


Fig 4-14 Pictorial view of Final stage

40. The anode voltages of pre amplifier are 3 KV, screen grid voltages are 700 V, and both voltages are keyed. The heater voltage 6.3 AC/3A is supplied by exciter power supply. This stage amplifies the RF pulse to 3.5 KW. This stage can be tuned capacitively from outside. The anode voltage of final amplifier is 8 KV, screen grid voltages are 700 V, and both voltages are keyed. The heater voltages (6.3 VAC/7.5 A) are supplied by exciter. The gain of final amplifier is 9 db and RF pulse power of 20 KW is available at the output. The cathode voltages can be displayed on oscilloscope from test jacks JK 4 and JK 5.

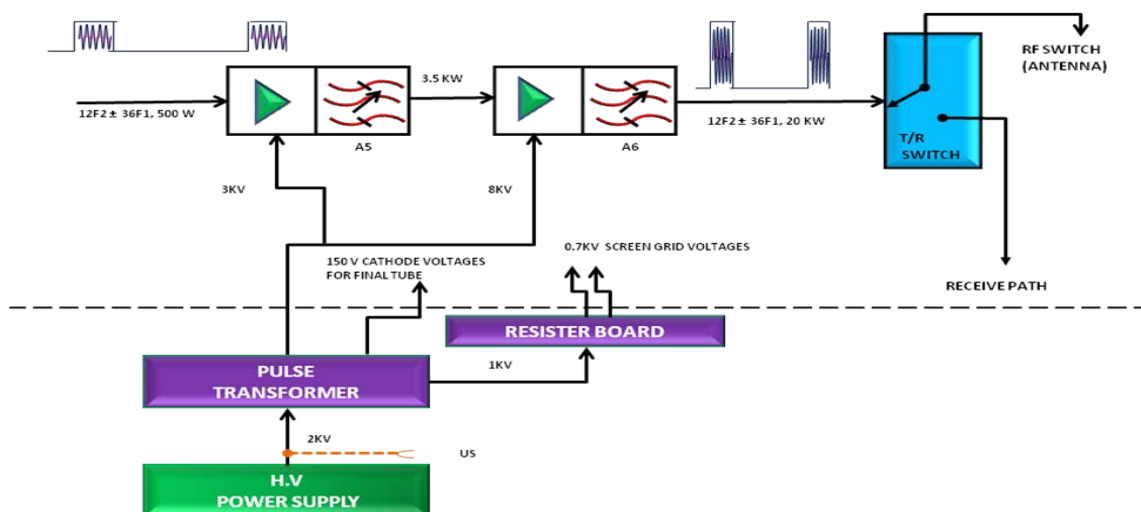


Fig 4-15 Functional diagram of Final stage complete

TR Switch

41. As common antenna is used by radar set for transmission and reception it is necessary to connect either the TX or RX to the antenna, as required. This is done with the aid of transmit/receive (T/R) switch. Switching is carried out by means of pin diodes (positive intrinsic Negative diodes). When the diodes conduct, the path between the RX and the antenna is at high impedance and the path between the TX and antenna is conducting. If a blocking voltage of +150 V is applied, the situation is reversed. The keying pulse for control of pin diodes comes from the pulse amplifier. The level of keying pulse is -6 V during transmission and +150 V during reception. Presence of keying pulse can be verified by pressing key "U_{SE}". Two directional couplers ("forward" and "reflected") are arranged at the transmitter output following the TR switch to facilitate tuning the transmitter output to maximum power and measurement of the power delivered by the transmitter as well as that reflected by the antenna or its feeder line. The directional couplers extract a sample of the forward and reflected power out of the antenna feeder line and feed it to the transmitter monitor.

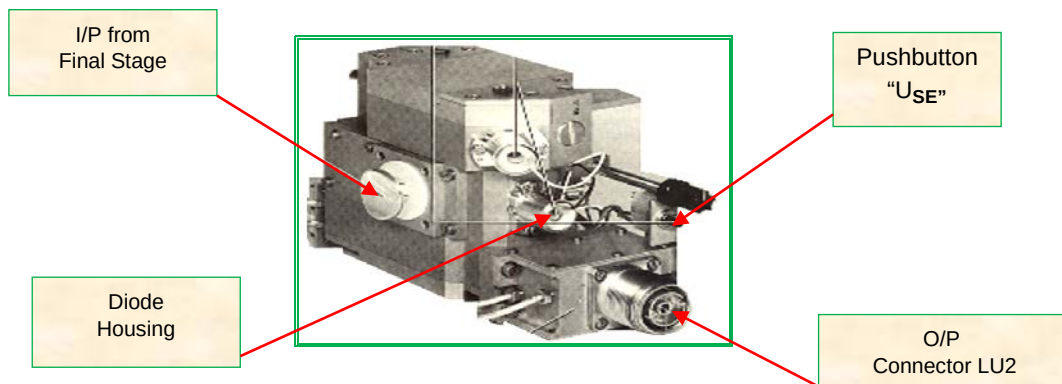


Fig 4-16 Pictorial view of T/R switch

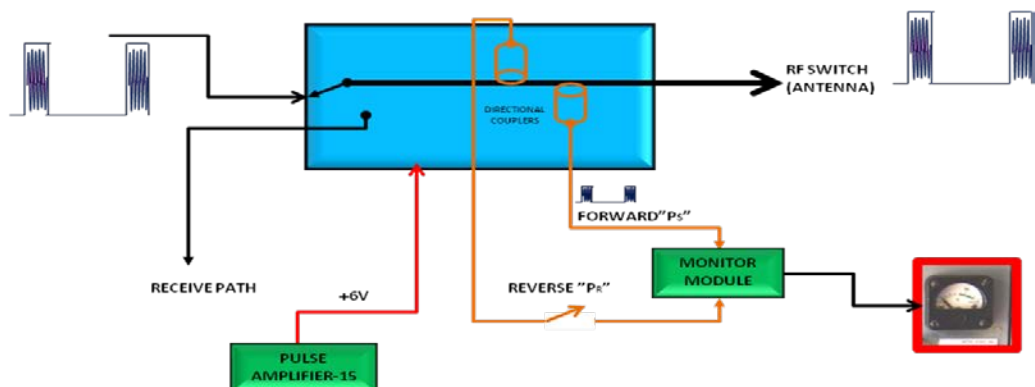


Fig 4-17 Functional diagram of T/R switch

High Voltage Power Supply

42. The high pulse power produced by the final stage necessitates anode voltages of several kilovolts, which are produced by H.V power supply in conjunction with "pulse transformer". It is mainly consists of a charging transformer, a charging inverter, a pulse forming network, a thyatron and a monitoring module. The control pulse ($93.3 \mu\text{sec}/ 3 \text{ KHz}$) is fed to amplifier, where it is amplified. The emitter voltages can be checked at test jack TRIG_{LU}; the amplified pulse drives parallel connected converter transistors. The collector circuit of these transistors is connected to +150 VDC via charging transformer. A pulsed voltage with PRF of converted control pulse is generated from +150 VDC. The collector voltage of converter transistor is monitored from test jack U_{LU}. The lamp TEMP/LU glows in case converter transistors are over heated. The PFN gets charged to 2 KV as soon as trigger KPA arrives, thyatron fires and PFN is discharged through primary of pulse transformer. Thyatron anode voltages can be tested at test jack U_{thy}. The lamp TEMP/ LFZK glows in case PFN is overheated.

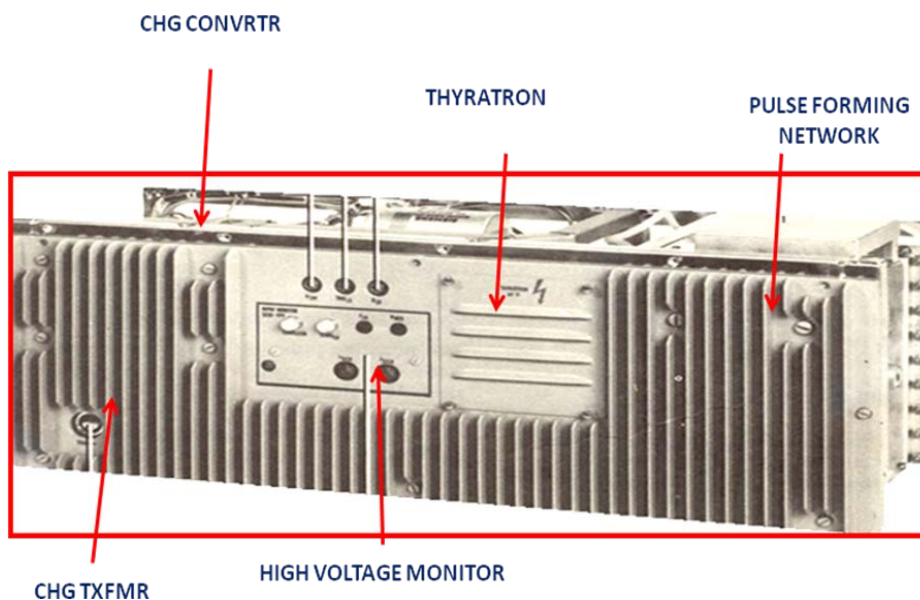


Fig 4-18 Pictorial view of High Voltage Power Supply

Pulse Transformer

43. The pulse transformer transforms the high voltage pulse coming from the high voltage power supply (5801232-17) to the necessary voltages for the tubes of the final amplifier. A pulse voltage of 2 KV is transformed up to the anode voltages of 3 KV and 8 KV. A further tap on secondary side for screen grid voltages of 0.7 KV is provided and moreover, 150 V for the cathode of final amplifier tube are also provided through this pulse transformer.

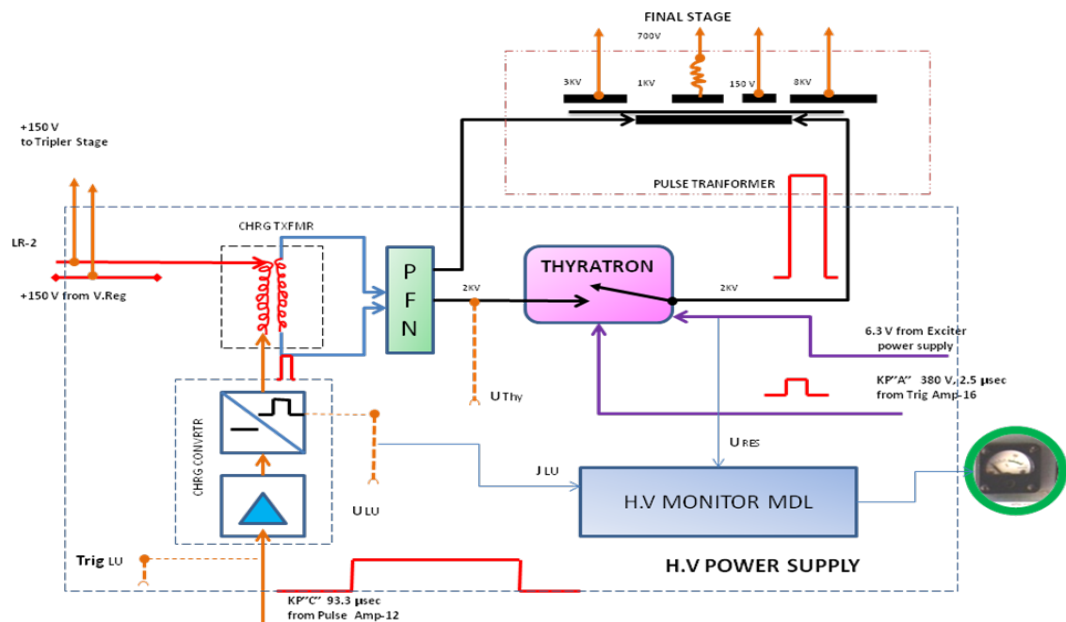


Fig 4-19 Functional diagram of High Voltage Power Supply

Control & Monitoring Modules

Pulse Amplifier -12

44. In the pulse amplifier the control pulse coming from the TX control module of RX are amplified and shaped so that they can be used for control of the charging converter transistors in the high voltage power supply of TX unit. The control pulse (+3V, 93 μsec) is amplified and is passed, provided a positive voltage supplied by monitor assembly is present via an AND gate to the pulse shaping stage

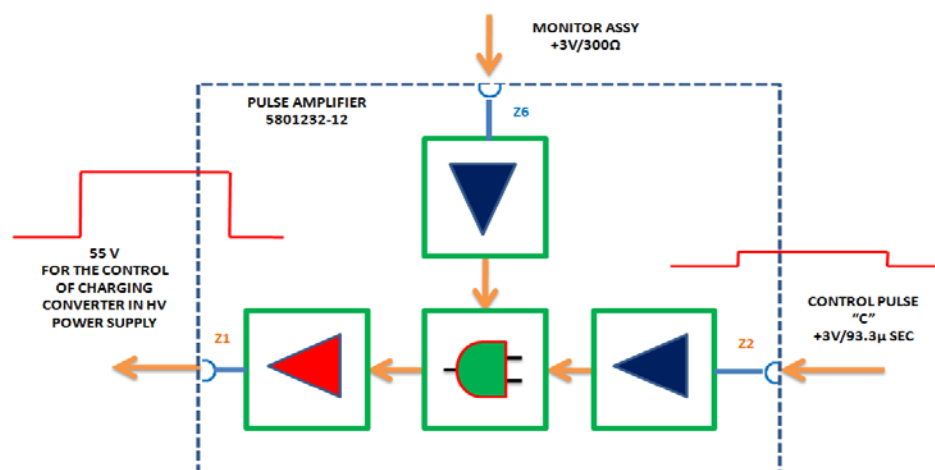


Fig 4-20 Functional diagram of Pulse Amplifier -12

Pulse Amplifier -15

45. The pulse amp consists of several amplifier stages whose output pulses are used to drive various modules in exciter and in TX unit. The keying pulse 'B' (+3V/16.6 μ sec) arrives from the TX control module of the RX at connection Z6. Same keying pulse 'B' is available at connection Z5 which, is fed to the pulse amplifier in the exciter. After amplification in two amplifier stages, a pulse of -12 V for triggering the STC stage is available at connection Z3. The pulse also passes via output connection Z1 to pin diodes of T/R switch through another amplifier stage. After another amplifier stage, a positive pulse is passed via output connection Z2 to trigger amplifier it drives a gate, which prevents keying of the final stage, if the pin diodes in the T/R switch receive no drive.

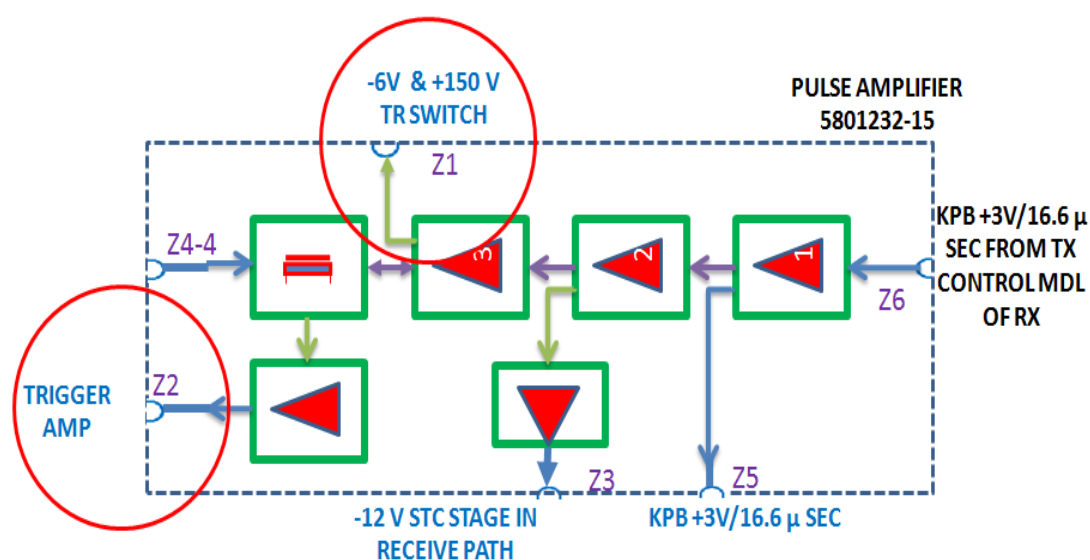


Fig 4-21 Functional diagram of Pulse Amplifier -15

Trigger Amplifier -16

46. It generates the control pulse for thyatron in the high voltage power supply and for the keying unit. The keying pulse 'A' comes from RX. A positive pulse is sent to keying unit at output connection Z5. Same pulse is differentiated and fed to mono stable multi vibrator. The output is fed to an AND gate, where keying pulse "B" also arrives. Output of gate will be available when keying pulses "A" and "B" both are present. A pulse of +380V/2.5 μ sec is fed to thyatron grid. The output pulse can be monitored on test jack U_{st} . When the TX is firing, terminal Z44 is connected to ground in the relay module. When the system is standby, +150V are fed and thyatron fires. Thus pulse forming network is discharge.

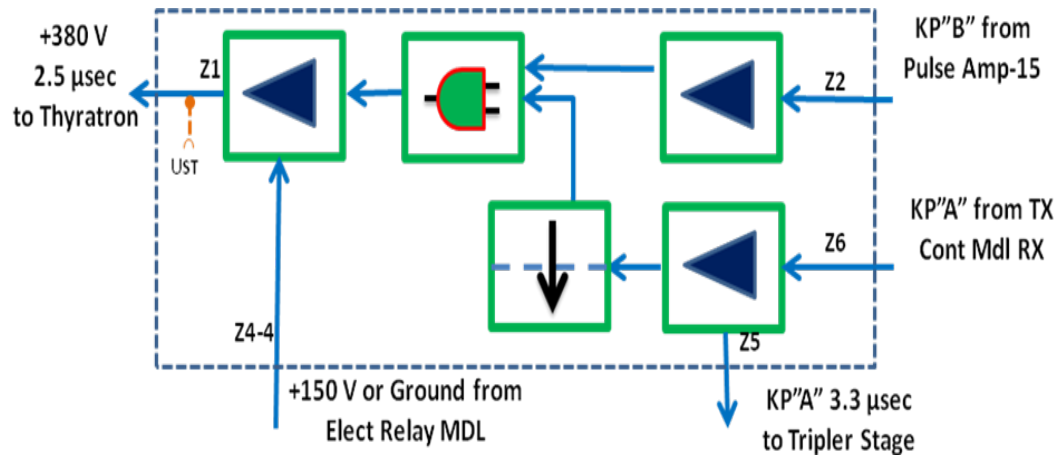


Fig 4-21 Functional diagram of Trigger Amplifier -16

Blower Units 1 and 2

47. For each of the transmitter chains the radar equipment includes a blower unit (5801233-6 and 5801244) incorporating a blower assembly (5801233-62). The RF switch module (5801233-63) is also accommodated in blower unit 1. The blowers extract the heat generated in the final stages of the transmitter. The cold air is fed via air hoses to the transmitter.

Blower unit 2



Blower unit 1



Fig 4-22 Pictorial view of Blower Units 1 and 2

48. By means of the coaxial switch incorporated in the RF switch module of blower unit 1, the antenna is transferred to the standby transmitter in the event of a frequency change or a fault. The same switch is used for test purposes to connect a noise generator via the TR switch to the receiver, which is then disconnected from the antenna. An electronic switch (5801233-636) connects the noise generator with the receiver input via the coaxial switch (RF switch). The noise power is keyed (keying time 470 ms) by a noise modulator (5801233-631) in order to prevent the automatic gain control (AGC) from responding during testing of the receive path. Before reaching the antenna, the transmit pulse must pass through the harmonic filter (5801232-4; Fig. 1-6) which is included in the feeder line from the blower unit to the antenna.

RF Terminal Box

49. RF Terminal Box serves as a connecting unit between equipment truck and antenna truck & power generator / mains. For easy connection of the transmit/receive cable (ZM-1 cable) with a large radius of curvature to the radar equipment truck, the RF terminal box has been provided with a dual rotary joint whose external connection can rotate in two planes perpendicular to each other. A short flexible RF cable connects the internal connection of the dual rotary joint to the transmitter output at the blower case 1.



Fig 4-23 Pictorial view of RF terminal box

Feed Cables with Rotary Joint

50. The feed cables pass the radar and IFF transmitter signals to the appropriate radiators (for radar, IFF and SLS) where they are transmitted. The signals received by the antenna pass the opposite way along the feed lines to the radar and IFF receivers. The feed cables are carefully matched so that the voltage standing wave ratio is small. For every radar cable the VSWR in the range 1215 to 1365 MHz is less than 1.05. For the IFF and SLS cables the VSWRs in the range 1030 to 1090 MHz are both less than 2. In order to avoid reflections and thus serious transmission losses, good contact between the plugs, coaxial cables and the appropriate rotary joint lines is essential.

51. The radar signals are coupled out of one of the three coaxial cables at the upper end of the triple rotary joint (so called because it has three coaxial lines) and led via a 2.5 m long coaxial cable K-1 to the horn radiator, where they are coupled in with the aid of a short probe. The horn radiator is excited horizontally (horizontal polarization). The IFF interrogation signals are fed from another of the triple rotary joint cables to the horn radiator (in the same way as the radar signals) via a coaxial cable K-2, in this case 2.3 m long and leading to the vertically positioned probes in the horn radiator. Here the IFF interrogation signals are transmitted and received in vertical polarization. Mutual interference is low due to the differing frequencies and the cross polarization of the IFF and radar signals. A third coaxial cable K-3, approx. 2.6 m long, connects the inner conductor of the rotary joint to the SLS antenna. The SLS antenna is a $\lambda^{1/4}$ dipole with base-plate, the base plate forming the outer conductor of a coaxial conductor.

52. The triple rotary joint ensures almost lossless transfer of transmit energy from the fixed antenna pedestal to the rotating antenna. The joint consists of a fixed and a revolving section, the fixed section being attached to the antenna pedestal and the revolving section supported on a ball race for positive location - being bolted to the antenna. The signal energy is passed from one section to the other of the rotary joint by means of three rigid coaxial cables located concentrically one inside the other. The outermost coaxial line can transmit the most energy on account of its larger cross-section, and is therefore employed for the radar signals. The middle coaxial cable transmits the IFF signals, the innermost coaxial line the SLS signals. In order to prevent the currents flowing along the outer conductors being interrupted in spite of the mechanical separation of the fixed and rotating cables a capacitive electrical connection is established for each between the fixed and rotating outer conductors of each cable by suitable $\lambda^{1/4}$ transformation without recourse to direct coupling. The feed to the three coaxial conductors at the lower end of the rotary joint is by way of three plug-in connectors.

Antenna Pedestal

53. The antenna pedestal (5800590-94) contains the parts needed to drive the antenna and in addition the elements which register and process information on the angular position of the antenna. The triple rotary joint projects from the antenna pedestal into the rotating antenna assembly. The drive motor revolves the antenna by way of the gearing. The antenna can in addition be turned by a hand-wheel on the drive motor for test or repair purposes.

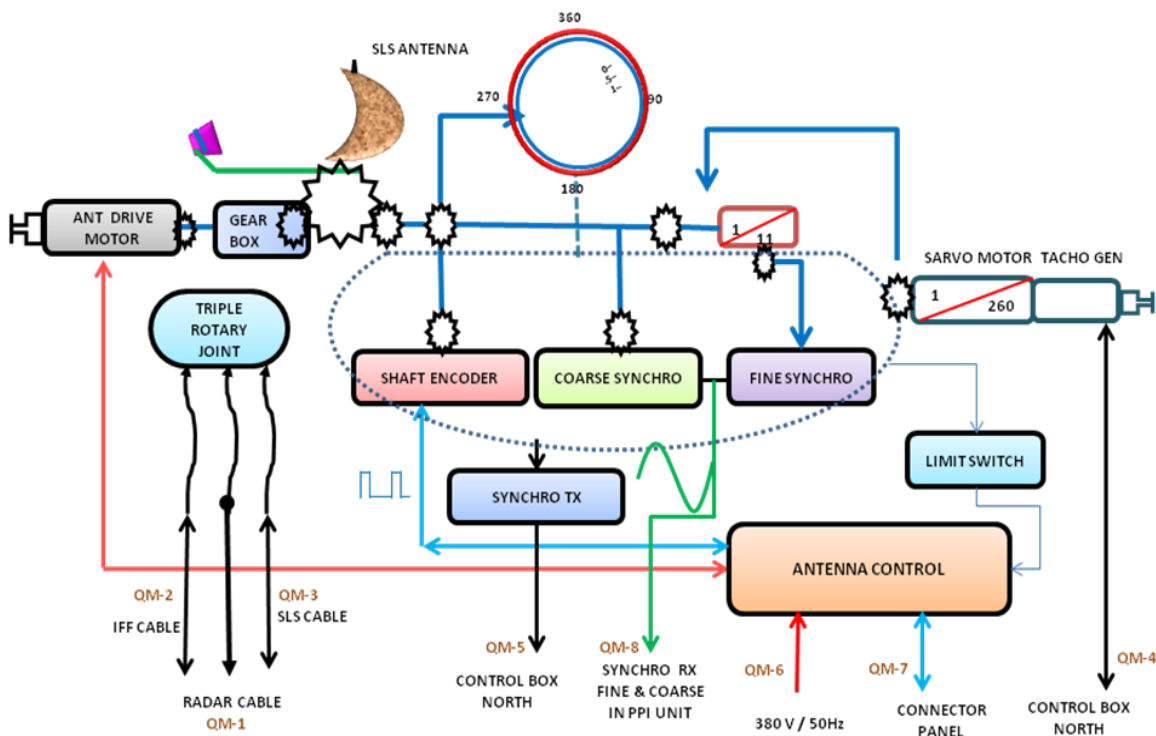


Fig 4-24 Functional Parts inside antenna pedestal

54. The synchro drum is the assembly incorporating the shaft encoder and the coarse and fine synchros. It is turned by the servo motor or by hand for alignment or north-seeking. The tachometer generator mounted on the servo motor shaft ensures, by way of the servo motor control in the control box 'North' that the servo motor always runs uniformly regardless of load. The shaft encoder supplies signals as a function of the antenna angle with 1024 equal steps suitable for the remote transmission of angle data. The shaft encoder is an incremental device which produces an electrical signal in the form of a square wave pulse train by photoelectric means. The number of pulses (1024 in equal steps, starting with 0 when passing through north) is a measure of the angle of rotation. After amplification and pulse shaping the signal is fed over cable to the angle counter in the DPU. The 'coarse' and 'fine' synchros

transmit angle and rotation of the antenna to the trace on the PPI screen, so that the antenna and the trace are at all times synchronized and at the same relative angle.

55. The sighting window with mirror and scale enables the synchro drum to be set to a precisely determined angle (e.g. to North) with the aid of the sight telescope during the north-seeking procedure, and also permits angles to be read off between triangulation points or other landmarks. A spirit level is provided for horizontal alignment of the antenna pedestal, and permits a reading accurate to within 0.1° .

56. A synchro system transmits the angular position of the revolving antenna synchronously to the trace on the PPI screen. The synchro system comprises the coarse synchro transmitter in the antenna pedestal (5800590-94) and the coarse synchro receiver in the PPI unit. In order to enhance the accuracy of the synchro system, it is paralleled by a second, 'fine' synchro system running at a speed eleven times higher than the coarse synchro system (gear ratio 1:11). A further electrical shaft is provided by the synchro transmitter in the antenna pedestal, which is linked electrically with the synchro receiver in the north seeking gyroscope (5800689- 71). If a difference in bearing occurs between the synchro transmitter and receiver positions, an error voltage signal is generated and indicated on a meter at the control box 'North' (5801245-628). With the aid of the servo motor the synchro drum can be rotated in the desired direction to bring the axis of the synchro transmitter connected to the synchro drum to the same angle as the axis of the synchro receiver in the gyroscope. For this purpose the pointer in the meter on control box 'North' is adjusted then at zero; there is no error voltage in the synchro system, that is to say the synchro drum is aligned to North. Since the synchro drum, to which many electrical cables lead, cannot be rotated by more than just over 360° , any attempt to reach a setting beyond 360° automatically trips a limit switch and causes the drum to run back to the other limit stop, from which point it can then be advanced again to the desired position.

57. A sighting window with scale can be used for manually aligning the synchro drum to north. The inner section with vernier is rigidly attached to the antenna, the outer section with the scale is coupled rigidly to the synchro drum housing.

58. The shaft encoder forms a zero pulse (North direction) and 10204 pulses each rotation dependent on the angular position of the antenna. Its output signals are used as remote angle data for the radar data processing equipment. A mounting device for a sight telescope (5800590-5) is provided on the side of the central reflector section; it can be tilted vertically, and is used to align the antenna to landmarks if north-seeking without the gyro has to be carried out and it can also be used to check the adjustment of the synchro system. The power supply cables for

antenna drive are monitored by an automatic circuit breaker in the radar system power distributor box (5801235). In the event of a severe overload the circuit breaker trips magnetically or thermally. If the antenna motor overheats, the supply to the motor is interrupted by the antenna control unit. All plug-in connections on the radar antenna are watertight. Two jacks at the antenna pedestal permit a field telephone to be connected for maintenance work.

ANTENNA SYSTEM

Construction

59. The radar antenna (5800590-9) is designed to radiate high-frequency L band pulses from the radar system and the IFF interrogating unit directionally and, on account of the antenna's rotating movement, in a fan-shaped pattern round the antenna location, and to receive them directionally. The radar antenna (5800590-9) is mounted on the upper arm of the folding hinged mast (5800698-1). A gearbox transfers rotary movement from the drive motor to a pinion which rotates the turntable (mounted on ball bearings) carrying the antenna at 16 revolutions per minute. The turntable carries the horizontal boom (5800590-3), at the outer end of which is mounted the horn waveguide (5800590-2) and at the other end of which the L band reflector (5800590-1) is installed. The horn waveguide contains probes which excite the horn horizontally for radar and vertically for IFF interrogation. The reflector is mounted opposite to the horn, and disperses the high-frequency energy radiated by the horn into the surrounding space. The SLS antenna is located at the top of the reflector (5800590-6). The pulses transmitted by the SLS antenna (so-called control pulses) prevent the interrogated aircraft replying to interrogation received by the aircraft transponder via the side lobes of the IFF antenna. As a result, only the aircraft in the main lobe of the antenna is interrogated. This prevents ambiguous evaluation by the interrogating ground station and over interrogation of the aircraft transponder. **The 5 m wide, 1.90 m high** reflector can be divided into three sections after releasing locking levers. It is designed, in such a way that torques occurring during rotation as a result of wind pressure are kept to a minimum.



Fig 4-25 Pictorial view of antenna system

60. The horn radiator is fed with the radar and IFF signals via separate cables. In contrast, the reflector is used by the radar and IFF interrogation systems both for transmitting and receiving. The horn radiator is aligned to direct its signal energy at the reflector, which reflects it and radiates it predominantly at a $+3.5^\circ$ angle of elevation. The reflector is of barrel outline. It is parabolically curved in the horizontal plane. In the vertical plane the contour deviates from the parabolic form in that the underside of the reflector is brought forwards to produce the desired cosec² antenna radiation pattern. The slope of the lower reflector sections determines the top cutoff angle of the radiation pattern (approx. $+40^\circ$), whereas the slope of the upper sections of the reflector primarily determines the direction of maximum radiation ($+3.5^\circ$ of elevation). If the positions of the reflector or of the horn radiator are altered, the antenna's direction of maximum radiation will be affected. The reflector is illuminated by the beam in such a way that the power toward the edge of the reflector in the horizontal plane falls to **approx. 1/30th** of the value at the center of the reflector. This ensures a high degree of side lobe attenuation for the horizontal radiation pattern. In the vertical plane the power at the upper and lower reflector edges falls to **about 1/10th** of the value at the center.

61. Radar and IFF antennas have a directional characteristic, whereas the SLS antenna is omnidirectional. The SLS antenna consists of a dipole mounted at the highest point of the reflector on its axis of symmetry. This transmits the control signal 2μsec after the first interrogation pulse transmitted by the IFF antenna.

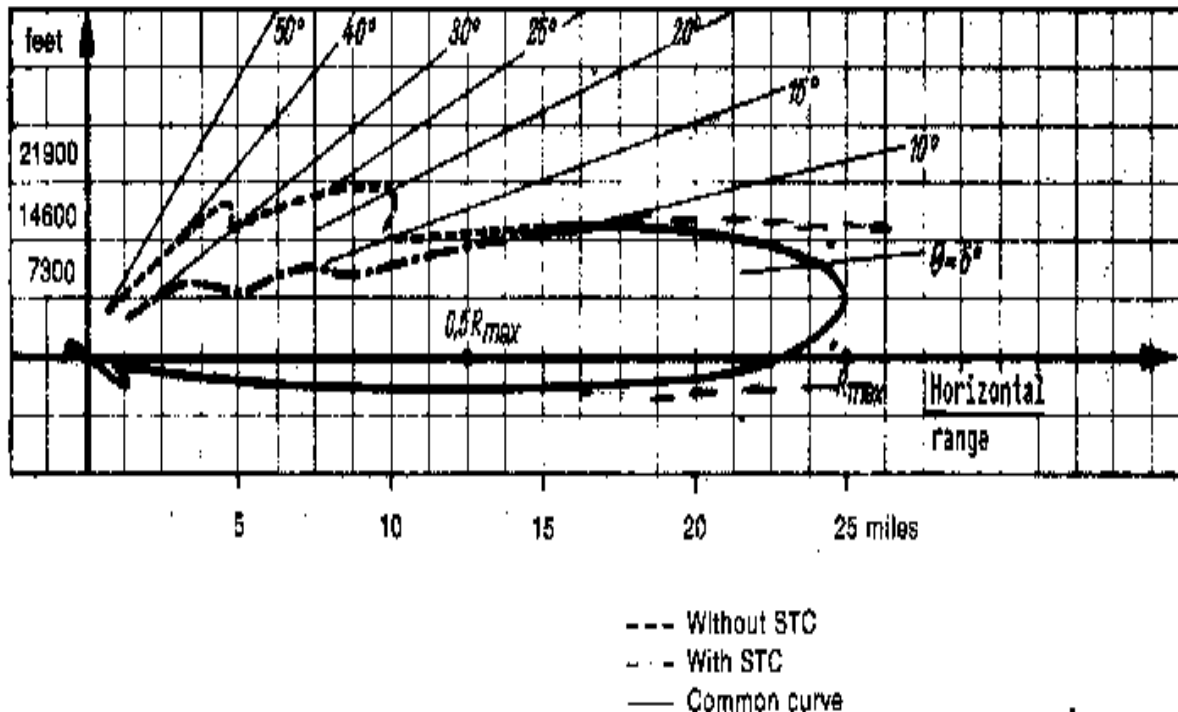


Fig 4-26 Cossec² vertical coverage diagram

62. The directional radiation pattern of the radar antenna is governed by the intended application of the radar system. As a target identification system for low-flying aircraft, the system covers an area of low altitude to just above the ground. In addition the echo signals reflected from an aircraft at constant altitude are intended to provide a largely constant input power at the receiver for the entire flying time within the coverage area. These requirements are fulfilled by a vertical radiation pattern in which the field strength is proportional to cosec² of the angle of elevation. A cosec² antenna radiation pattern has a response of almost uniform height with a sharp top cutoff. The direction of a target is determined with a strongly directional antenna by the position of the main lobe in a horizontal plane around the antenna site. The horizontal half power beamwidth (3 dB width) of the radar and IFF antennas in the direction of maximum radiation is narrow in order to obtain good target resolution among closely adjacent targets in the azimuth direction. The half power beamwidth for angles of elevation from 0° to 15° is less than 3.9° over the entire

radar frequency range, and less than 5.2° in the IFF range. Up to an angle of elevation of 40° the side lobes remain at least 26 dB down relative to the radiation maximum (0 dB) for radar and at least 24 dB down for IFF relative to the radiation maximum (0 dB). The antenna gain for radar is approx. 28 dB (at 3.5° elevation) and approx. 25 dB for the IFF antenna.

63. The vertical radiation pattern (Fig. 4-26) for the radar antenna, whose direction of maximum radiation is at 3.5° , is a close approximation to the cosec-squared pattern up to approx. 15° . Thereafter, the pattern is somewhat higher than the cosec-squared pattern due to the requirements of the sensitivity time control in the receiver. At approx. 40° the pattern falls slowly but continuously away. The portion of the radiation transmitted into the ground is relatively small on account of the steep lower edge of the pattern.

64. The entire vertical pattern can be tilted from -5° to $+5^\circ$, giving -1.5° to $+8.5^\circ$ range of adjustment (see elevation scale on reflector center section bearing). In the vertical plane the aperture angle has been chosen to scan airspace only such as would be occupied by low-flying aircraft.

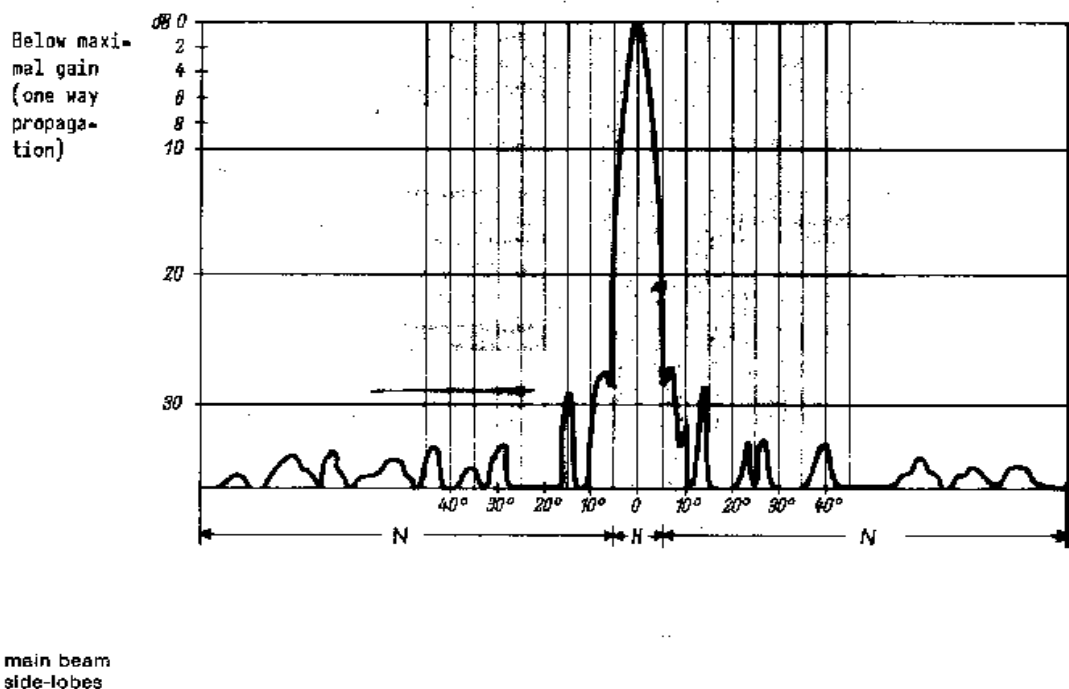


Fig 4-27 Horizontal radiation pattern

ANGLE TRANSMISSION

65. Rotation of the deflection beam round the central point of the PPI screen is synchronized with antenna rotation and thus permits the bearing of the target to be accurately established. The antenna angle is transmitted by synchro systems to a resolver in the PPI unit, which modulates the deflection currents of the PPI unit as a function of the antenna angular position. For accurate transmission of antenna angle information, a 'coarse' synchro system is coupled to a 'fine' synchro system. The 'coarse' synchro system achieves alignment of the synchro transmitter and receiver to within approx. 4° , at which point automatic changeover takes place to the 'fine' synchro system, which permits alignment to within a few tenths of a degree. The rotors of the coarse and fine synchros are coupled to the antenna axis by 1:1 (coarse) and 1: 11 (fine) gear. Detail operation of the synchro system is as follows: the synchro transmitter in the antenna pedestal and the synchro receiver in the PPI unit are similar in construction.

66. The rotor section of the synchro has one coil, the stator section three coils at 120° included angles. If an AC signal is supplied to the rotor coil of the synchro system, voltages are induced in the three stator coils at values which depend on the instantaneous position of the rotor. The stator coils of synchro transmitter and receiver are interconnected. Hence, a signal resulting from the three individual stator coil voltages - the so-called error voltage appears at the coarse and fine synchro receiver rotor coil terminals, with amplitude which depends on the relative positions of the synchro transmitter and receiver rotors. The error voltage is reduced to zero when the rotors assume a particular position relative to each other. The antenna is connected to the shaft encoder, the rotors of the coarse and fine synchro transmitters and the inner disc of the sighting window via gear trains. When the antenna revolves, all these components are also rotated. The stators of the coarse and fine synchro transmitters are combined with the shaft encoder housing to form a single assembly, the synchro drum. The outer rim of the sighting window scale and the servo motor are connected to the synchro drum (broken double. When aligning to north with optical aids, the synchro drum can also be rotated by hand. When north alignment is performed with the gyro, the drum is rotated by the servo motor. These rotary movements are transmitted to the rotor of the synchro transmitter in the synchro drum, which then transmits its angular position to the synchro receiver in the gyro. When the antenna has been aligned with North, the rotor positions in the synchro transmitter of the drum and the synchro receiver in the gyroscope must be identical. For alignment of the angle transmission system with North, the common housing (synchro drum) for all three angle transmitters (synchros and shaft encoder) is rotated in relation to the drive housing.

67. The horizontal bearing for antenna and sight telescope can be read from the 360° scale in the sighting window with the aid of a vernier accurate to 0.1° . The sight

telescope can be adjusted vertically without scale reference. The horizontal bearing of the sight telescope and the antenna's direction of maximum radiation coincide.

68. A solenoid switch in the antenna pedestal supplies a signal whenever the magnet mounted on the turntable is directly opposite the switch. This signal is superimposed on the PPI screen as a heading line (azimuth marker) and provides a check on the North bearing.

69. The shaft encoder supplies a North pulse U_{a0} 0.088° wide and two rectangular pulse trains 90° apart with 1024 pulses every 360° . The signals are generated with the aid of an encoding disc which permits light emission from a lamp to strike a photo-electric cell.

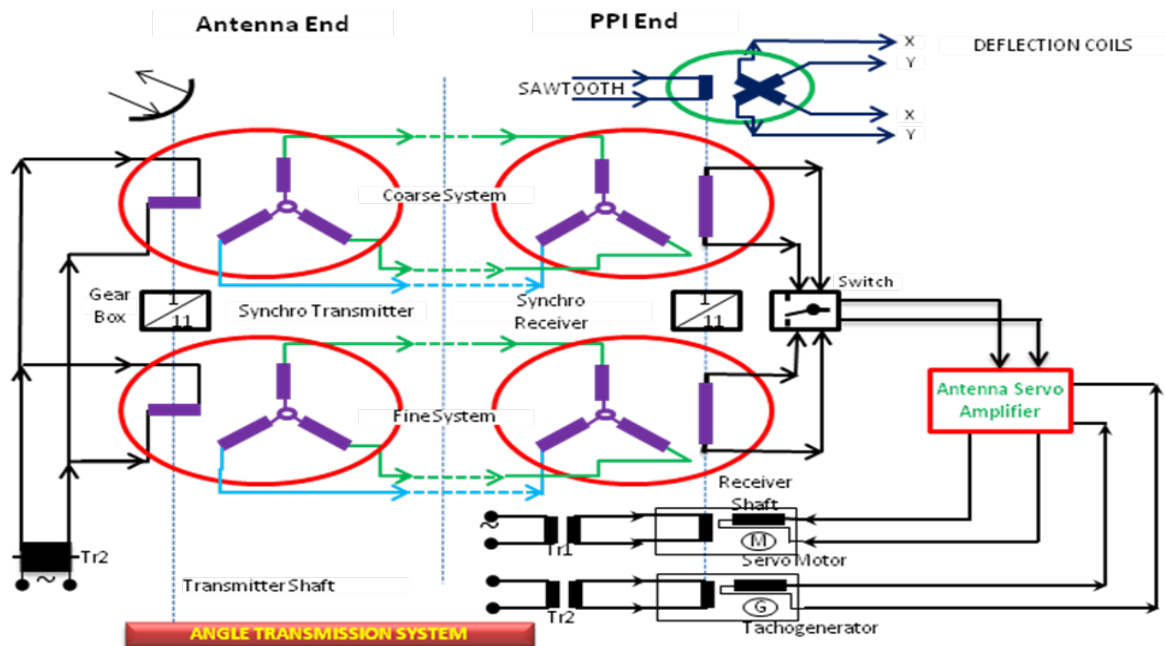


Fig 4-28 Angle transmission/ synchronization B/W ant and PPI

ANTENNA ASSEMBLY TRUCK

Main Assemblies

70. The chassis (5801245-621) of the antenna assembly truck (5801245-62) carries, apart from the antenna (5800590-9), the hinged mast (5801245-622), the control box 'North' (5801245-628), the north seeking gyroscope NKA 6-4 (5800689-7) and the wiring harness (5800689-9). The hinged mast antenna carrier comprises:

- (a) The sub-frame with four outrigger supports, four auxiliary supports (leveling rams), one suspension lock and one frame interlock on each side of the truck, and the mast base.
- (b) The hinged mast with four elevating rams and a cable duct.
- (c) The hydraulic system.
- (d) The electrical system.

Sub-Frame

71. The sub-frame of the hinged mast is the load-bearing structure for the mast and the outrigger supports. It is fabricated by electric arc welding from folded steel sheet to form closed box sections and thus constitutes a unit with exceptional horizontal arc flexural strength. The boxed construction of the sub-frame makes it possible to transmit forces from the mast to the outrigger supports with a minimum of deformation. When the mast is operated, the frame interlocks and suspension locks connect the sub-frame and truck chassis together to form a strong unit. The suspension locks are also engaged hydraulically. The four auxiliary supports at the corners of the mast subframe are used to level the truck on a sloping surface and to raise it slightly. These supports can compensate for a lateral slope of up to 15% (approx. 8.5°). They are controlled individually from the control unit on the operator's control panel. A spirit level mounted there indicates when the truck is horizontal.



Fig 4-28 Pictorial View Antenna Assembly Truck

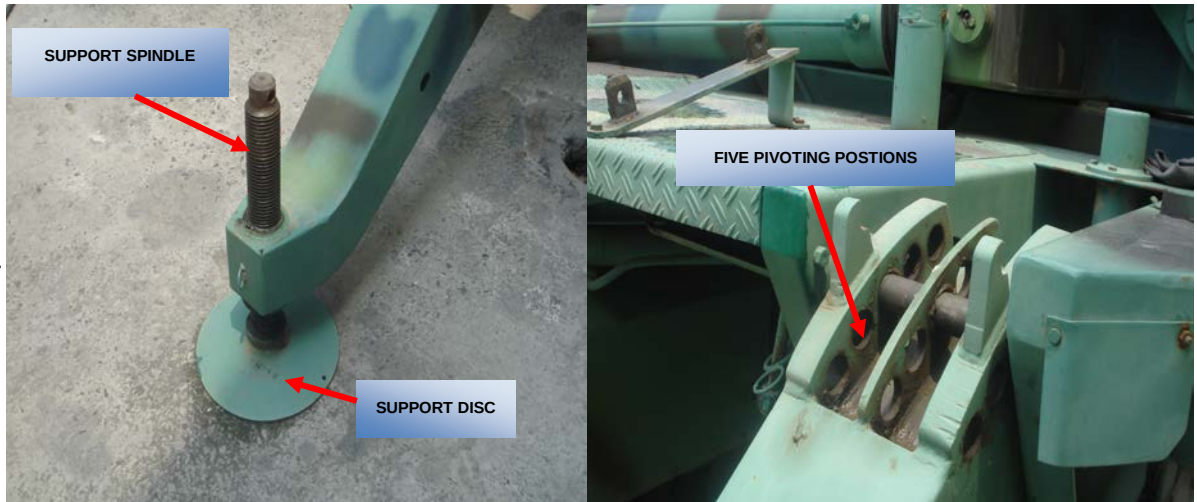


Fig 4-29 Pictorial View Outrigger Support

72. Once the horizontal attitude has been obtained with the aid of the auxiliary supports, it is held by four pivoting outrigger supports. When these are in use, the vehicle has a support base area of $5 \times 5 \text{ m}^2$. The outriggers pivot on journal bearings on the mast subframe and are also of box-section construction. At their outer ends they carry substantial support spindles provided with ball-jointed discs. The outrigger supports are raised by single-acting hydraulic rams operating via ropes and pulleys to the transport position, and locked there mechanically by locking pins. Unlike the auxiliary support rams, these hydraulic rams are operated by a common control valve. They are raised hydraulically and lowered by their own weight. The lowering movement is initiated by coil springs which are compressed as the outriggers are raised. These coil springs force the outrigger supports out as they are lowered. When lowered, the outrigger supports can be locked into five "pivot" positions (Fig. 4-29) to allow for ground surface irregularities. The support spindles can then be used for final alignment of the truck along its horizontal axis. The mast base is located centrally between the outrigger supports, that is to say in the middle of the mast subframe. The two halves of the lower mast arm and its elevating rams are attached to the mast base, which in turn is bolted to a recessed mounting on the sub-frame. In this way it is possible to detach the complete mast with base from the subframe for repairs. The lower rotary joint for the rigid RF cable is also mounted on the mast base.

73. The entire subframe to the left and right of the mast is covered with a non-slip sheet metal walkway (platform). Storage facilities for various accessory items are provided on or under this platform. The control unit for mast operation with control levers for the hydraulic valves is located on the left side of the truck, next to the

outrigger support. Behind it is the control box 'North' for operation of the radar antenna and of the gyroscope, in a box which forms part of the walkway. Next to this control box is the north seeking gyro- scope with its power supply. On the opposite side of the truck are support blocks and a large equipment case for an equipment kit. Below the platform in the vicinity of this case are three spare cans, two containers for fire extinguisher and miscellaneous small components. To the right and left of the folded mast on the rearmost third of the truck chassis are mountings for the antenna reflector side sections. Other accessory items such as pickaxe and spade are secured to the outrigger supports.

Hinged Mast

74. The hinged mast (5801245-622) consists primarily of three parts: the lower arm, the intermediate section and the upper arm. The two arms are made up two parallel mast sections pivoted together at the upper and lower ends to form parallelogram linkages. This positive parallel movement ensures that as the mast is raised and lowered the rotating axis of the antenna always remains perpendicular to the vehicle's support base.

75. A cable duct is attached to the left side of the mast to provide protection for the rigid RF coaxial line, which is sensitive to impact and mechanical damage. The cable clip mounted above the control box 'North' is used to secure the remaining RF cable ZM-1 leading to the radar equipment truck. A swiveling cable arm protects the cable against kinking. In addition, various cables (IFF, SLS, antenna angle information, power supply) lead from the antenna assembly truck control and connector panel to the connector panel on the antenna pedestal. These cables are attached to the outside of the cable duct in the form of a wiring harness.

76. A control cam with two limit switches is mounted on the lowest link of the upper arm. The upper limit switch operates electro-hydraulically to control the leading (during elevation) and lagging (during folding) of the upper arm in relation to the lower mast arm, and thus prevent damage to the antenna or to the truck cab. The lower limit switch interrupts the complete mast lowering action shortly before the antenna pedestal descends to the height of the truck cab, so that the horn radiator and reflector can be turned to the positions in which they are stowed. When the equipment is in the transport position, the center section of the antenna reflector rests on the upper arm when folded down, and is secured there by a clamp. The antenna boom with horn radiator is also folded down, and housed in the recess formed by the upper arm fork, where it is locked into position.

Hydraulic System

77. The hydraulic system drive assembly consists of an axial-piston pump and powered by a 5 kW electric motor running at 1450 rev/min. The pump and motor are located behind the wall of the cab on the left side of the truck. If none of the control levers have been moved, the oil delivered by the pump passes via a check valve initially in an unrestricted circuit to the return filter and thence back to the oil tank, which holds app. 45 liters.

78. The truck is raised slightly by the auxiliary support leveling rams and aligned horizontally if parked on an uneven surface. The four outriggers are coupled via the restricted cross-section of the return line, so that the outrigger with the least internal friction extends first.

79. A so-called output control circuit prevents the cab from damage when the mast is elevated or lowered (movement. sequence 5). This circuit ensures that the upper arm moves faster than the lower arm when elevating the mast, but lags behind it as the mast is lowered. This process is controlled by a relay actuated by solenoid valve 2 in response to the angle of elevation of the upper arm as detected by the upper limit switch at the control cam.

80. When the mast is folded, the closed limit switch and the associated switching relay keep the solenoid valve in its normal position. The full oil flow can then reach the two upper elevating rams. As soon as the upper arm has cleared the angular range defined by the control cam, the switching relay releases, the solenoid valve moves to its operated position and the oil flow is then directed to a flow regulator which diverts a constant oil flow of 3 liters/min to the upper rams. The larger residual oil flow is then used to extend the lower arm elevating rams.

81. During mast retraction, when the preset angular position for the upper arm is reached, the solenoid valve is reset to its normal position. It then directs the return oil flow from the upper rams via a restrictor, whereas the return oil flow from the lower rams continues unrestrictedly. This greatly retards the retracting speed of the upper rams, so that the upper mast lags behind the lower mast.

82. Sudden retraction of the mast rams or uncontrolled mast section movements in the event of fracture of or leakage from the oil pressure lines are prevented by inhibit valves (check valves with hydraulic release) on the elevating rams, which oppose the escape of oil from the rams.

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83. It is not possible to elevate the mast with only one ram. Pressure relief valves in the individual control valves protect the system against overloading.

84. If the oil pump drive should fail, the elevated mast can be folded with the aid of hand pump while pressing solenoid valves 1 and 2. Both the hand pump and the solenoid valves are located on the left side of the vehicle adjacent to the control, unit.

85. When using the hand pump, it is important to comply exactly with the correct movement sequence for retracting the mast (movements 4, 3, 2 and 1).

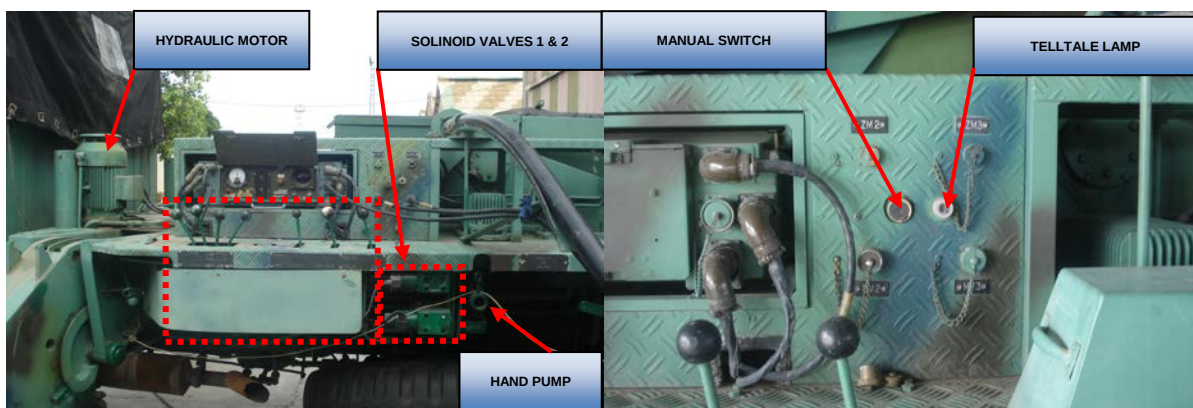


Fig 4-30 Pictorial View of Hydraulic System

Electrical System

86. The drive motor (5 kW three-phase, 380 V, 50 Hz, 1500 rev/min) for the hydraulic pump is supplied with power via the power distributor unit and transformer 1 in the equipment truck, using either the 380 V three-phase generator set on a single axle trailer, or the public power network. The motor master switch is on the control box 'North'. The electrical control equipment takes its power from the truck's 24V battery. All earth (ground) lines are led back separately inside of the cables from the mast assembly to the truck.

87. The battery power supply passes via an 8-Amp fuse and the oil pressure switch to the mast assembly, to provide control and operating voltages. The oil pressure switch supplies the power to the mast assembly only when oil pressure is being built up by the hydraulic pump assembly. A green telltale lamp (at the connector panel) in the control circuit indicates that the electrical system is

operational. If the outrigger supports have not been lowered, the reflector center section not elevated or the horn radiator not raised, solenoid valve 1 is kept in its normal position by control relay 1. This interrupts the oil pressure supply to the mast and at the same time the oil flow to the outrigger supports is shutoff. This control valve position is maintained by limit switches on the outrigger supports, the reflector center section and the horn radiator. Only when all these switches, which are wired in parallel, have opened and thus interrupted the control voltage, can the solenoid valve control relay revert to its normal position. The solenoid control valve is energized and thus permits oil pressure to reach the mast rams. At the same time the oil flow to the outrigger supports is shut off. There are two limit switches on each of the four outrigger supports. One of these switches detects the completion of the swing-down movement; the other is tripped when the locking pin is inserted. The erecting movements of the reflector center section and the horn radiator are similarly protected. The mast cannot be moved until the mast subframe has been correctly supported and interlocked.

88. A further limit switch on the upper arm is actuated by a control cam and protects the horn radiator, reflector and cab against damage when the mast is lowered by shutting off the oil pressure flow to the mast. The oil pressure can be restored by means of manual button on the control unit paneling. The interruption is intended to draw operating personnel's attention to the need to swivel the antenna reflector before lowering the mast further so that, the horn radiator enters the mast fork recess correctly. The above control system has the effect of causing the upper arm to lead the lower arm, but to lag behind it when the mast is lowered. The process is controlled hydraulically by solenoid valve 2 (the upper of the two solenoid valves) below the control box 'North', which in turn receives its command signals from the limit switch (32) which is actuated by the above-mentioned control cam and thus operates as a function of the upper arm elevation angle.

89. In the event of a power failure, the two solenoid valves 1 and 2 can be manually operated by push buttons at their location below the control box 'North'. A contact switch on the frame interlocking mechanism interrupts the truck's starter motor circuit so that the vehicle cannot be started when the interlocks are engaged. A telltale lamp in the cab lights if the locks have not been released when the starter is operated.

CONTROL BOX 'NORTH'

90. The control box 'North' (5801245-628, Fig. 4-31) controls the processes required to coordinate the operation of the radar antenna with the radar system.

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91. The following processes can be controlled by the program switch:
- (a) Switching on the three-phase motor which drives the hydraulic pump on the antenna assembly truck. After this, the mast sub-frame can be locked, the truck supports extended and the hinged mast elevated.
 - (b) Switching on the electronic equipment needed for aligning the radar system to north.
 - (c) Switching on the three-phase antenna rotating motor.



Fig 4-31 Pictorial View of Control Box 'North'

92. Certain of these processes can take place simultaneously, depending on switch positions.

Re (a) above: the three-phase motor for the hydraulic pump is started manually via a star-delta circuit. A non-return mechanism is incorporated between the star and delta positions of the switch. The three-phase current is fed from the equipment truck to the motor via plug MV 1 the star-delta circuit and the hydraulics and hydraulics 2 connection

Re (b) above: in order to avoid the necessity for complicated north seeking of the antenna whenever the equipment is re-sited, the control unit includes a circuit which permits a North bearing to be obtained straightforwardly with the aid of a north seeking gyroscope. A reference signal of varying voltage depending on the positions of the synchro drum in the antenna pedestal and

the synchro in the gyroscope is shown on the meter in the control box. By means of a servo motor which turns the synchro drum in the antenna pedestal and thus the synchro housings (1:1 and 1:11) of the angle transmission system and of the shaft encoder" The synchro drum can be moved to the same bearing as the gyroscope after this has found North. When this has been done the reference signal voltage is 0 and the system is on a North bearing. By means of a 'tumbler switch' the servo motor can be run either slow or fast in both directions. The servo motor control circuit and the meter circuit are housed within the servo motor control unit (5800689-821.). In switch position E (north seeking) the POWER ON lamp (523) comes on. The switch (510) supplies power to transformers 508 and 509. Fuse 505 protects them against overheating. The north seeking gyroscope is switched on at the illuminated button (521). The button remains illuminated until the gyroscope has arrived at its North bearing. To prevent the synchro drum from being turned too far past the 360° position, the antenna pedestal incorporates limit switches which cause the synchro drum to run back in the opposite direction during the north seeking procedure. During automatic runback RETURN LAMP lights up and the tumbler switch (518) becomes inoperative.

Re (c) above: when the program switch is set to ROTATION (R) the three-phase power supply at connector MV 1 passes to multi-pin plug QM 6 via switch contacts 22, 24 and 26, and thence to the antenna pedestal.

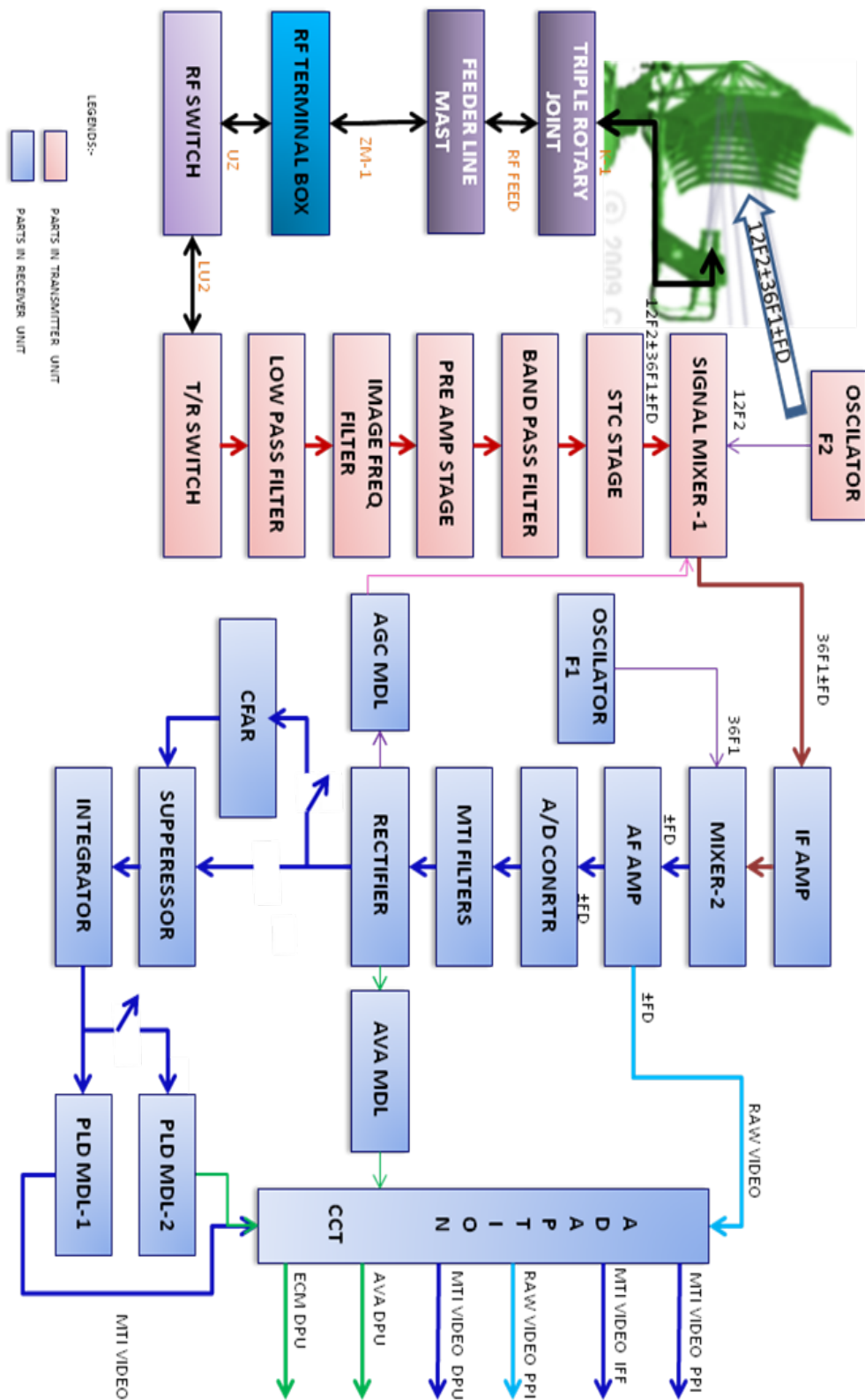


Fig 4-32 Block diagram of Receive Path

Receive Path

93. The echo signals from the antenna pass to the TR switch (5801232-1113) in the transmitter (Fig. 4-32) which switches over shortly after each transmitter pulse and thus feeds the echo signals via a harmonic filter (low pass filter. 5801232-115) into the receive path. A subsequent image frequency filter (5801232-116) in the preamplifier unit (5801232-117), which is tuned to the relevant transmitter frequency, rejects interference signal from external equipment which might otherwise be fed into the receiver due to cross modulation.

94. The receive signals are then fed to a preamplifier (5801232-118) incorporating a low noise amplifier. The following band-pass filter (5801232-113) suppresses the preamplifier noise power at the image frequency. Mechanically completed with these last three subassemblies is the STC stage (sensitivity time control stage, 5801232-119) which greatly reduces the gain at the beginning of each receive period and does not attain its full value until later. This prevents overdriving of the receiver by signals reflected from relatively close objects (targets and clutter). In addition, this measure reduces the possibility of receiving target echoes from side-lobes of the antenna radiation pattern.

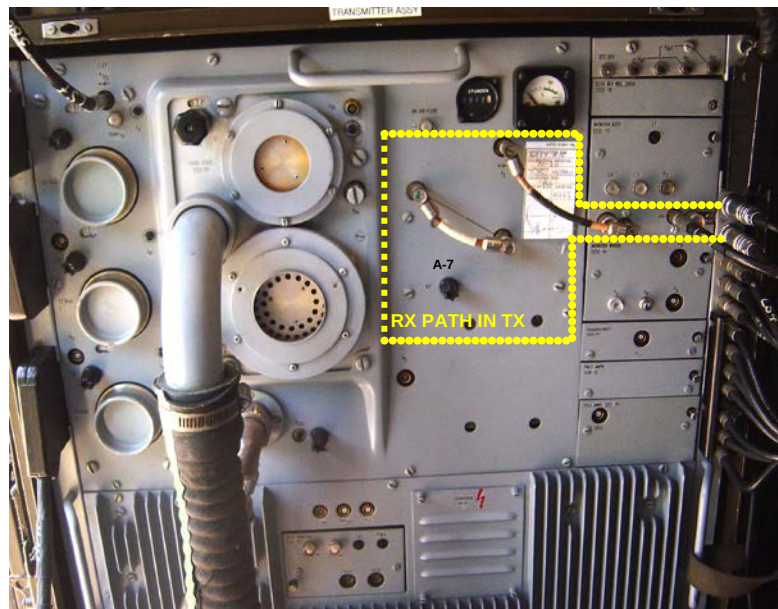


Fig 4-33 Pictorial View of Receive Path in Transmitter

95. The control signal for the STC is derived from the pulse amplifier (5801232-15). A pulse-shaping circuit in the STC stage converts the control signal into an approximately saw-tooth shaped pulse, which controls the gain periodically at the pulse repetition frequency. In the signal mixer 1 (5801232-18) the received frequency $12(f_2 \pm 3f_1)$ is mixed with the local oscillator frequency $12f_2$ to produce the first intermediate frequency $36f_1$. Sensitivity time control takes place also in the signal mixer 1. The local oscillator frequency $12f_2$ is derived via a X 12 multiplier (5801232-1212) from the same oscillator f_2 which generates the transmitter frequency.

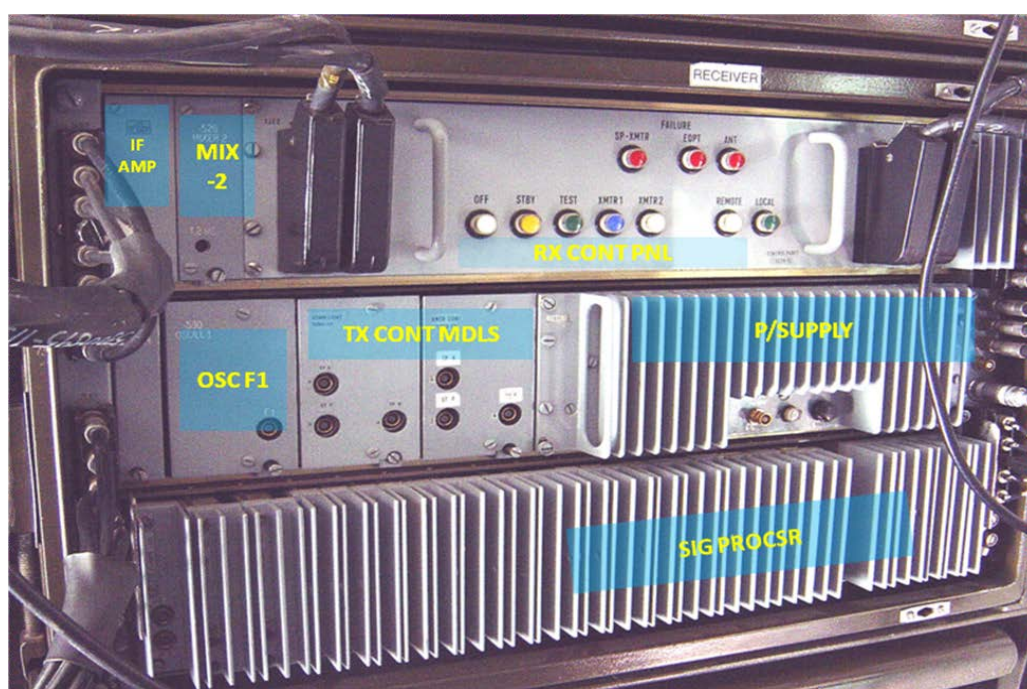


Fig 4-34 Pictorial View of Receiver Unit

Receiver Unit

Purpose

96. Apart from 2nd down-conversion, amplification and detection of echo signal, receiver unit performs functions such as radar control & monitoring, basic frequency generation, timing / clock generation for the whole system, signal processing (MTI evaluation) and ECCM function.

Radar Control & Monitoring

97. The control center for all switching and signaling processes in the radar set is the control panel on the receiver (Fig. 4-35). This control panel is used to put the radar set into operation, and to select the operating modes REMOTE or LOCAL CONTROL, OFF, STBY (RECEIVE), TEST and XMTR 1 (transmitter 1) or XMTR 2 (transmitter 2). The faults are also indicated on the receiver control panel: Lamp FAILURE SP-XMTR: Standby transmitter not operable, Lamp FAILURE EQPT, steady light: Operation not possible, Lamp FAILURE EQPT, flashing light: Limited operation possible (reduced range), Lamp FAILURE ANT: Antenna rotation fault (or switched off).



Fig 4-35 Pictorial View of Radar Control & Monitoring

Basic Frequency Generation

98. Process of basic frequency generation has been described above in Para 18 to 19 and refer Fig 4-2

Clock / Timing Generation

99. Process of Clock / Timing Generation has been described above in Para 20 to 24 and refers Fig 4-4 & 4-5.

2nd down-conversion, amplification and detection of echo signal

100. The path taken by the echo signal in the receiver (5801239-5) is as follows the IF signal with the frequency $36f_1 \pm f_d$ passes from the two signal mixers 1 (5801232-18) in the two transmitters to IF amplifier 1 (5801239-31). The latter combines the receiving paths of the two transmitters via one of the two separate inputs. After amplification the echo signal is passed to mixer 2 (5801239-526) for a second frequency conversion, Here the echo signals are mixed with the frequency

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36f1 (=43.2 MHz) from oscillator F1 (5801239-530). The IF signal at the input of mixer 2 is amplified, limited by diodes and brought down to the video position. The mixer output signal consists of echo signals from stationary objects and moving targets in the video range. After amplification, the received signal is passed to an analog filter in the AF amplifier (5801239-525).

101. The signal path is now split into two paths, one in which the signals from moving targets are evaluated (MTI video via AD converter) and a second for normal video in which the echo signals from all objects - stationary objects and moving targets - are amplified, limited and directly displayed on the indicator. The two paths are separated after analog matched filter in AF amplifier and recombined in the adaption circuit (5801239-538001).

102. Both paths operate continuously and their signals can be displayed simultaneously or separately on the PPI screen by choice of the operator.

103. Radial delays caused by processing times in the MTI path are compensated in the adaption circuit (5801239-538001).

MTI SIGNAL PROCESSING WITH SUPPRESSION OF MTA ECHOES

104. The video signal to be processed first passes to two identical analog-to-digital converters (AD converter 5801239- 527) which alternately scan the values of all echo signals of a receive period at equal time intervals (3.33 μ sec each) and convert them into equivalent 12 bit parallel binary-coded numerical values. The dummy-connector unit (5801239-533) only serves as a bridge between the AD converter units and the MTI filter. It can be removed for test purposes. After the video signal has been converted from analog to digital form, is passed to MTI filters. Five MTI filters are used which basically works on the well-known delay-line- canceller principle. MTI filters first converts the 12 bits data to 16 bits data for further processing then pass the signals which cross the minimal radial velocity limits. The I/P 16 bit data from MTI filters is rectified, limited and amplified in Rectifier module and passes to three paths: 1st to AGC module, 2nd to CFAR module through a changeover switch of the Rectifier module which can be used to bypass the CFAR module as output of CFAR after processing is again available as a 2nd input of the changeover switch and further fed to ISU (inter-pulse suppressor unit), 3rd to AVA module.

105. The AGC module (automatic gain control, 5801239-522) is used for automatic gain control of the receiver path, thus, ensuring that the noise level at the output of the MTI filter is held constant. It obtains its samples from the rectifier output in the CFAR module. After the signal has been suitably conditioned and reconverted into an analog signal it is used to control the gain in the IF section of the mixer 1. Correct operation is shown by a green LED on the front panel of the AGC-module. If the control voltage rises or falls considerably above/below the permissible value operation is switched to the second transmitter. At the same time the fault signaling circuit operates. The diode on the front panel of the AGC module extinguishes and the fault relay TS in the control panel (5801239-12) of the receiver causes the lamp SYSTEM FAULT to flash. As already mentioned, the AGC module stabilizes the noise level at the output of the MTI filter. However, the control function should not become effective if the level increases due to a noise jammer of an interference source which appears briefly, but is not powerful enough to be received over the side-lobes of the antenna. These interference signals, which are restricted to certain sectors, are evaluated by the CFAR. The AGC circuit, connected after a DA converter, is therefore fitted with an analog integrator whose time constant is automatically controlled by a differentiator at the integrator output. If the noise signal increases, the differentiator chooses a large time constant if it decreases, it chooses a low time constant.

106. A weak source of interference, which only reaches the receiver via the main lobe of the radar antenna and hence will only be of short duration, will not affect the control circuit: a powerful source of interference, on the other hand, which is also picked up via the side-lobes and is thus almost always present, will eventually cause the AGC to reduce the gain and hence the receiver sensitivity. When the powerful source of interference disappears, the differentiator selects the low time constant and the sensitivity quickly resets itself to the normal value. The AGC can be switched off when receiver measurements are carried out. The sensitivity can also be controlled manually via a connection provided for this purpose on Dummy-Connector (-533) by Fixed AGC module (5801239-555).

107. The **CFAR** control circuit keeps the error alarm rate of the radar set constant in the event of a fault, and the AGC circuit suppresses any interference signals which are received; target signal whose levels are still higher by an amount equal to the S/N ratio are indicated. The input interference signal is regulated to a constant output value by a multiplier. The output signal is limited in the control circuit and compared with the nominal value in the comparator. The output signal is fed to the **ISU** via the above-mentioned changeover switch.

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108. The moving target echo is further fed via the switch of ISU to post-detection Integrator designed as digital low pass filter, which provides feed back to ISU regarding suppression of MTA objects.

109. During one period of illumination of MTA object there are about 120 hits per scan by the MPDR 45/E. If 120 echoes are received the 120 echoes are locked up in the same range gates by the use of one PRF, by the use of three PRF's only 40 echoes a third part of the 120 echoes fall within the same range gates. The echoes of an object within the uniquely are however always received in the same range gates independent of the number of applied PRF's. After the integrator the amplitude of MTA echoes thus show only the third part of amplitude of an object within uniquely defined detection range under the condition of using three PRF's and receiving the same amplitude of the both types of echoes at input of receiving path.

110. Inter pulse suppression unit also suppresses short, non-synchronous single interference pulses.

111. The **AVA** module (AVA, amplitude versus azimuth; 5801239-529) follows the rectifier module; it displays the magnitude and direction of **CW** or noise interference in the shape of a lobe which cannot completely obscure the targets. In addition, it generates a constant interference signal indication when interference is present.

112. The integrator is followed by the pulse length discriminators (5801239-520 and 5801239-536) which detect and suppresses intermittent CW or noise sources. In addition, it is intended to maintain the sensitivity of the radar set in the undisturbed receiving range and to display a radar target moving within the disturbed receiving range in accordance with the signal-to-noise ratio. Genuine radar targets and interference signals are distinguished from each other by their duration. The maximum depth given for targets is the depth beyond which the signal in question is interpreted as an interference signal.

113. The amplitude values of the signals received in the range gates to be evaluated are fed to the pulse length discriminator 1. As the number of range channels to be evaluated are 179. The maximum target depth is 7 range channels. If a signal occurs, which exceeds duration of 7 channels, indication of an interference source is given. If several interference sources are picked up during a radar period, a continuous interference source indication is sufficient. If the duration is less than 8 channels, the signal is interpreted as a target.

114. The whole MTI signal processing is controlled by the central clock pulse module-1. The adaption circuit (5801239-538001) processes the incoming signals from the AF amplifier, pulse length discriminator and AVA module and conducts them via a 50 Ω line driver to the PPI, to radar data processing equipment set, to the IFF interrogator and to the control circuit. As long as the AF amplifier supplies at least one signal to the minimum level supervision per antenna rotation, a continuous signal occurs and is switched through to the output by enabling signal. *A green LED shows correct operation (upper LED).*

115. The digital Signal value available at the adaption circuit is accepted by a 12-position register in synchronism with clock pulse T4 and compared with a fixed digital reference value. If the latter is exceeded, a target indication signal appears. It is synchronized with clock pulse T4. It is fed to outputs: MTI-EXTR, MTI-IFF and MTI-PPI via a multiplexer.

116. The signals from the pulse length discriminator 1 are fed to the MTI outputs via the second input of the multiplexer. A decrease of the noise level at the inputs of the adaption circuit by approx. 3 dB below the desired value causes a fault message. *A second green LED shows correct operation (lower LED).*

Receiver Power Supply (5801239-531)

117. The power supply "Receiver" delivers all DC voltages required by the receiver. The power supply also delivers 6V (12V pp) supply voltage for the oscilloscope unit. The power supply "RECEIVER" contains the following PC boards.

- (a) The **CHOPPER CONTROL** is used to synchronize the DC to DC inverter of the power supply with the PRF of radar device.
- (b) The **EXCITER MODULE** is designed as a controlled inverter. It is fed with 150V DC from the 150V regulator. It is triggered by 3 KHz clock of the chopper control. It drives the current limiter and it also supplies the operating voltage for the current limiter.
- (c) The **CURRENT LIMITER CONTROLS** the conduction of the inverters in inverter module according to the output load. In case of over current drawn, the conduction of the inverter transistor ceases and this state is signaled by lamp.
- (d) The **CURRENT TRANSFORMER** in the output circuit of power supply unit supply a control voltage for the current limiter in order to protect the transistors of inverter in the event of over load or short circuit in the output section.

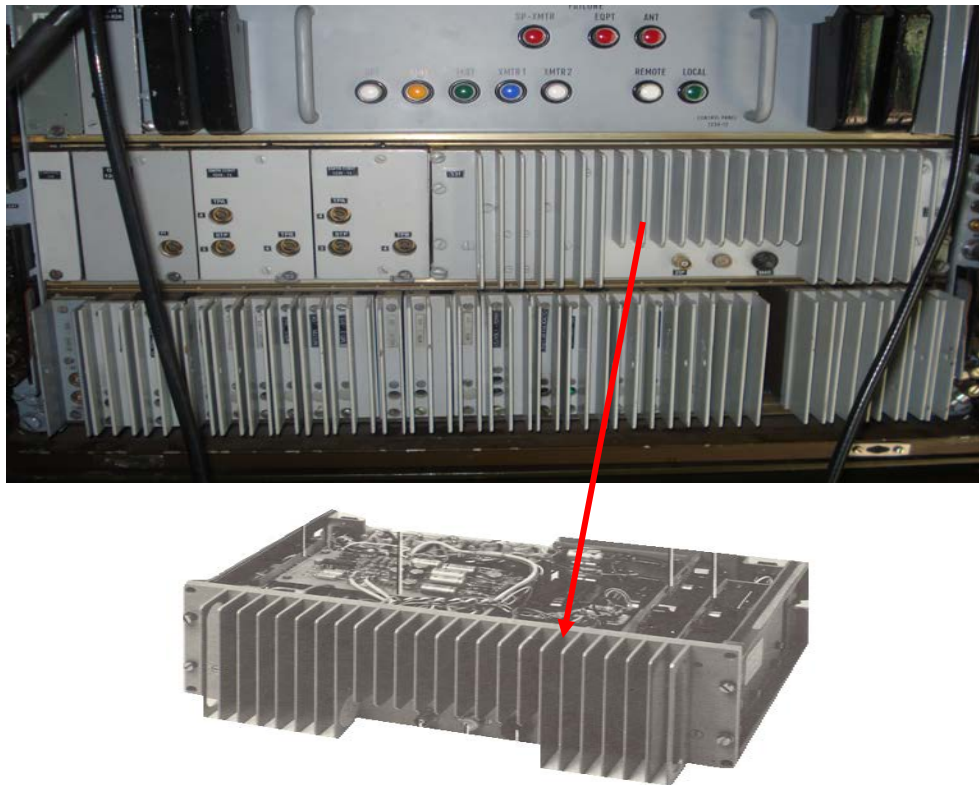


Fig 4-36 Pictorial View of Receiver Power Supply

PLAN POSITION INDICATOR UNIT

118. The targets echoes are displayed on cathode ray tube in polar coordinates with the radar antenna at the center. The display on the screen is produced by intensity modulating the electron beam with the target echoes, while simultaneously deflecting the electron beam from the center to the edge of the screen. Simultaneously with this radial deflection the trace rotates in synchronism with the antenna and thus produces the PPI display on the screen.

119. The principle items required for the display on the indicator screen are the MTI video, the start pulse, the mile markers and the antenna azimuth. It is also possible to feed in normal video civil or military emergency and identified MTI video (zebra pattern). In addition, the symbols from radar data processing equipment set are superimposed. The azimuth angle can be adjusted between target and north with a hand cursor and be read out throughout 360° divisions at the edge of screen



Fig 4-37 Pictorial View of Plan Position Indicator Unit

Time base generation process

120. The start pulse for the beam deflection with the mile markers for the range rings is generated by the clock pulse generator 1 in the receiver. The mile markers with opposite positive polarity follow at 5 mile intervals. The pulse train is passed via input PE1-1 to the multivibrator (5801240-114) in the indicator, which divides it between two paths. As a result of the opposite polarity of the start pulse compared with the mile markers, the multivibrator is triggered by the start pulse only. The start pulse causes the multivibrator to go into the operating state. The multivibrator then supplies an extended pulse of only half the frequency (1.5 kHz) of the start pulse. It triggers the sawtooth generator (5801240-115) for the beam deflection on the screen. In this way a new scanning cycle begins with every other start pulse. The sawtooth generator produces the sawtooth voltage with which the pre and final amplifiers for the beam deflection are driven. Frequency and deflection time (sweep speed) are determined by the trigger pulses. If the start pulse fails, a monitoring circuit incorporating a multivibrator produces an auxiliary control pulse to start the sawtooth generator (which then oscillates with reduced amplitude). The function of the clamping stage (5801240-128) is to reference the deflection voltages used to drive the deflection amplifiers to a particular potential. This results in starting the time base from the centre of CRT. In the subsequent pulse former module (5801240-116) the deflection voltages are pre-distorted in order to correct linearity errors caused by the CRT geometry. The preamplifier and the power amplifier (5801240-117 and -

129) change the saw-tooth voltages waveform into saw-tooth current waveform necessary to deflect the beam in the magnetically deflected CRT. The operating voltages (+ 24 V, -24 V) for the preamplifier and the power amplifier are monitored. If one of the operating voltages varies by more than 33% it is switched off.

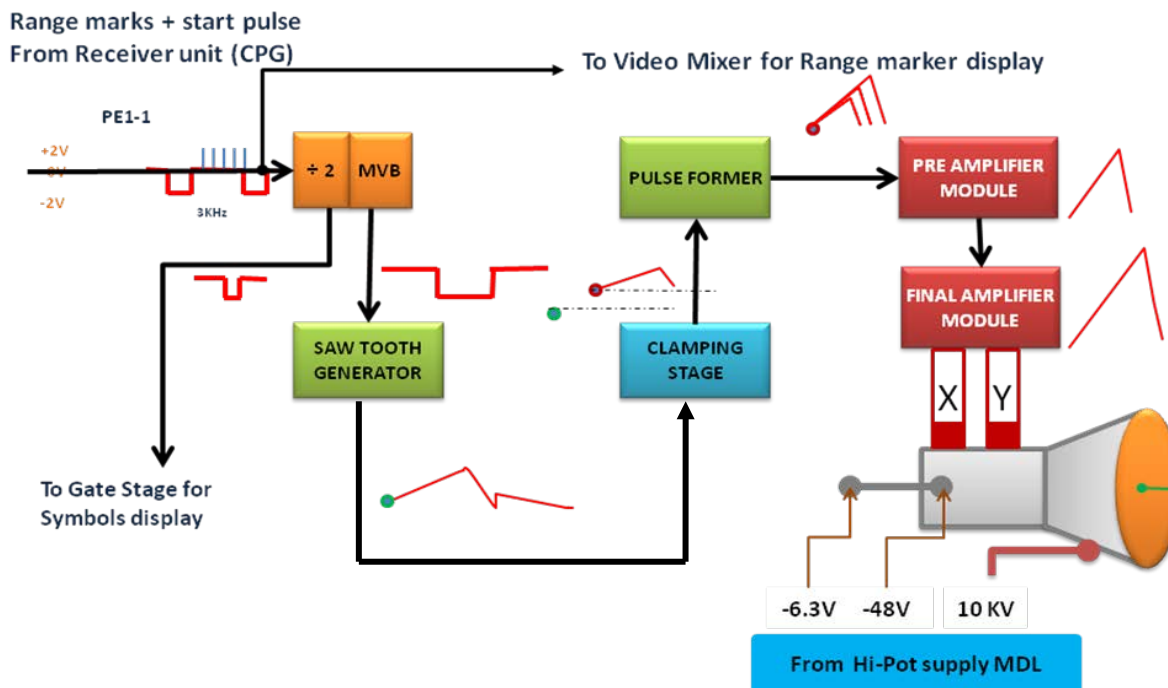


Fig 4-38 Block Diagram of Time base generation process

Time Base Rotation and Synchronization Process

121. Data relating to the antenna position is necessary in order to display the target azimuth on the indicator. This information is transmitted from the antenna to the resolver driver in the indicator by means of a synchr system consisting of two synchr transmitters and two synchr receivers. One pair of synchros (transmitter and receiver) forms a coarse synchr system and the other pair the fine synchr system. The deflection for the trace is produced by resolving the sawtooth voltage in an X and a Y component in a coordinate converter rotating synchronously with the antenna (resolver converts polar coordinates into Cartesian coordinates X, Y). After suitable amplification, these components are fed to the X and Y deflection coils of the CRT and deflect the trace in synchronism with the antenna rotation. The recurrence rate of the indicator time base is half to the system PRF. The voltages are fed from the resolver drive to the clamping stage.

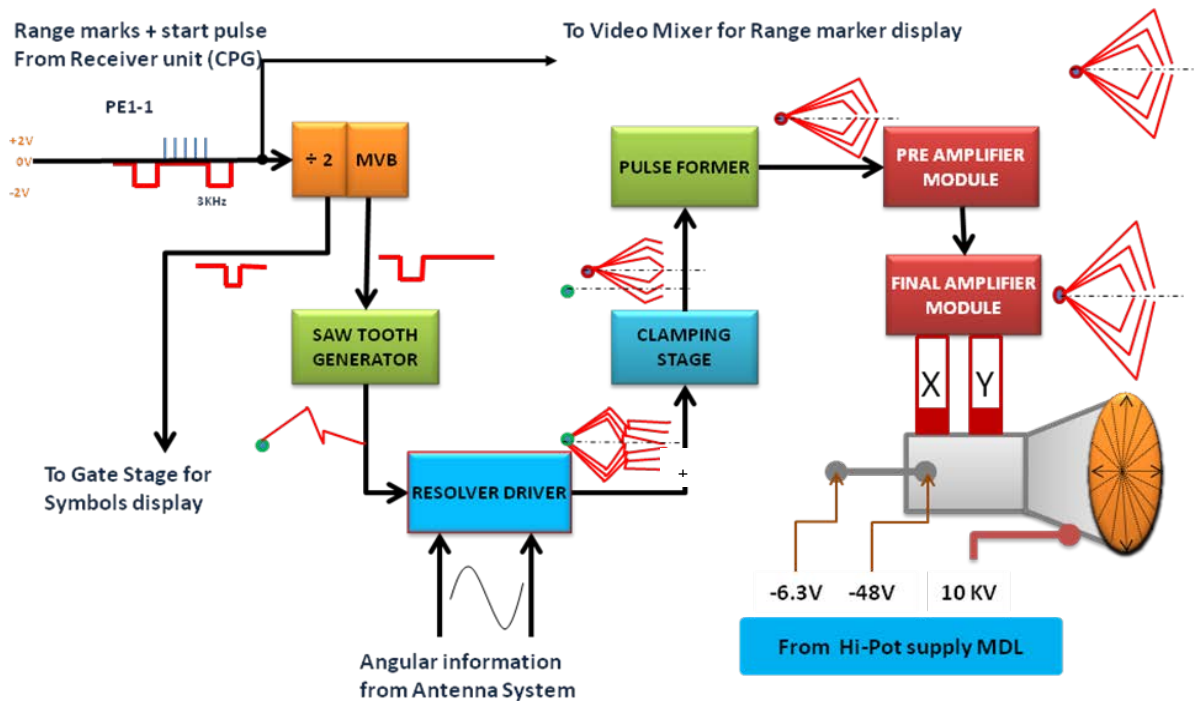


Fig 4-39 Block Diagram of Time Base Rotation and Synchronization Process

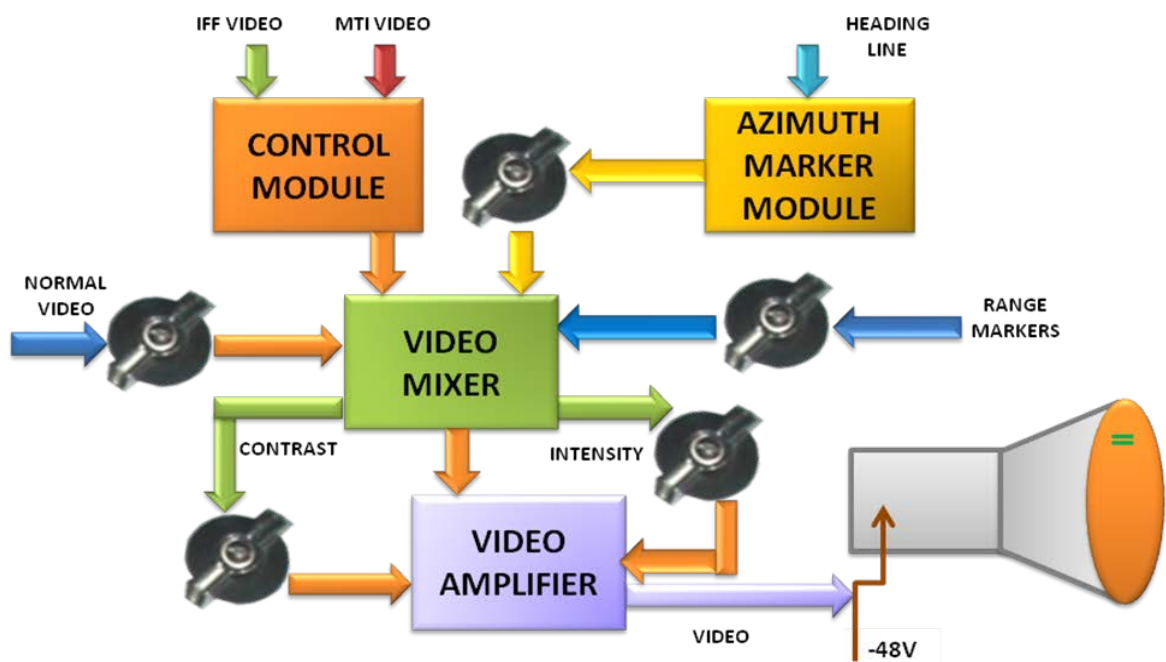


Fig 4-40 Block Diagram of Video Generation Process

Video Generation Process

122. The MTI and normal video signals supplied by the signal processing in the receiver are passed to the indicator via a video cable. The MTI video signal passed with positive TTL level via input PE1-3 is extended in the control module (5801240-512) so that perfect target display can be ensured. Since the MTI video signal is present in digital form, it is converted to an analog voltage in an analog OR gate of the DA converter.

123. The multivibrator module also controls the azimuth marker module (5801240-124), used to mark a particular azimuth bearing on the indicator tube by providing a bright-up pulse for the trace. The heading line information is supplied by a switch in the antenna Pedestal which is directly actuated by the rotating antenna. Via input PVa8 the switch in the azimuth marker module triggers a signal which brightens the trace for 2 ms at each antenna revolution and thus produces a clear marker. The azimuth marker module is then blocked in order to prevent the display of a false heading line, which could result from bounce of the heading line switch contacts. The output pulse from the last flip-flop in the azimuth marker module switches the pulses from the multivibrator module to output of the azimuth marker module for about 2 msec.

124. The subsequent video mixer (5801240-121) combines the signals (MTI and normal video, range marker and heading line) applied to its inputs and amplifies them. The composite signal is limited to a maximum value which is such that the signals appear with optimum brightness on the screen in a darkened room. The amplified composite signal is fed via an electronic switch to output. The function of this diode switch is to interrupt the composite signal while a symbol is being written.

125. Every input signal, with the exception of the MTI video with the IFF marker, can be switched off from the indicator and can be separately adjusted in amplitude, so that any desired ratio of intensity to the MTI video can be set. The composite signal at output of the video mixer is fed via the CONTRAST control on the front panel (5801240-134) and the video amplifier (5801240-122) to the cathode of the CRT for intensity modulation. The MTI video cannot be switched off and its brightness is adjusted by means of the "CONTRAST". The composite video signal is further amplified in the video amplifier and simultaneously the pulses necessary for bright-up and blanking out of the various signal paths are generated.

Symbol Generation Process

126. A timing pulse of 1.5 KHz (which is obtained through divider 1:2 in the MV) provided by the multivibrator module is amplified in the gate stage, inverted and fed

to the symbol generator, where it is used in the generation of the priming pulse.

127. If symbols are to be written, a priming pulse is fed from the symbol generator via input PY4 to the gate stage. The inverted switching pulse blocks all transistor switches in the other signal paths during symbol writing. The driving stage (5801240-118) is used to switch through the deflection voltages for antenna, and symbol presentation at the correct time. The symbol signals are fed via the unit inputs PY 2 and PY 3 to the driving stage and from there likewise to the outputs. During normal operation the transistor switches of the driving stage are so controlled that only one signal path (symbol) is connected to the outputs.

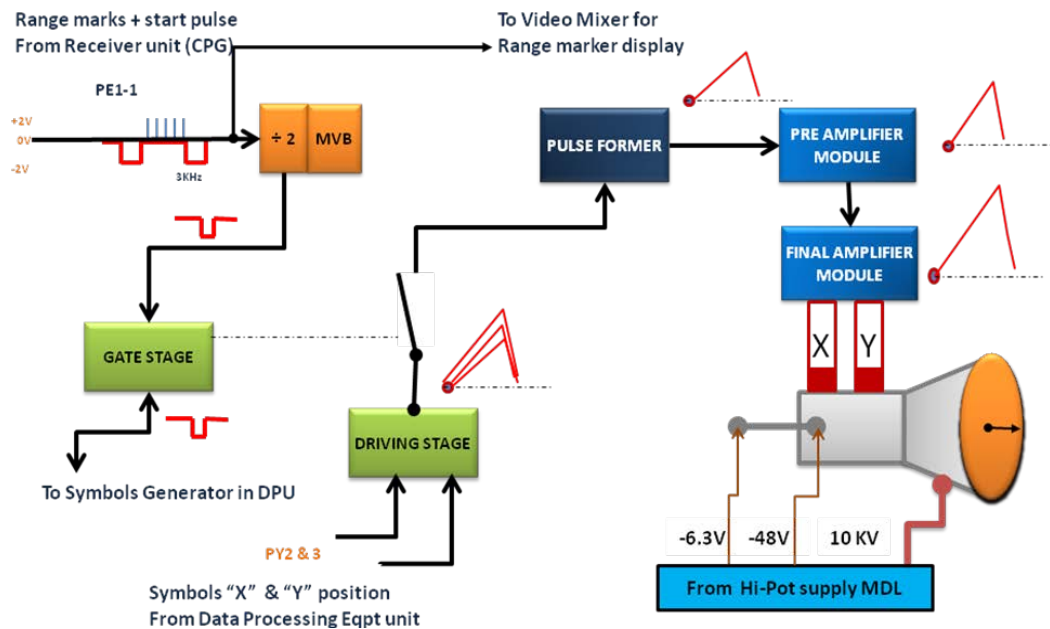


Fig 4-41 Block Diagram of Symbol Generation Process

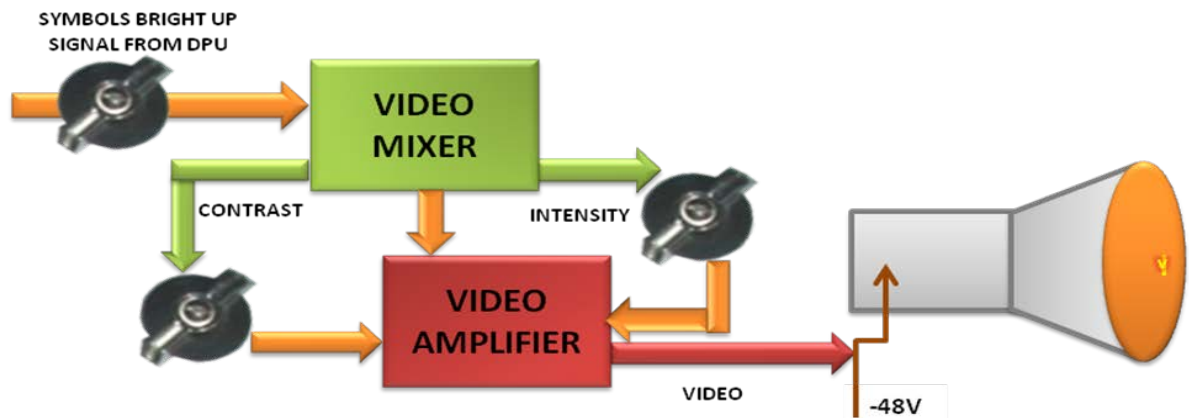


Fig 4-42 Block Diagram of Symbol Generation Process

CRT and HI-Pot Power Supply Module

128. The voltages necessary for operation of the display assembly (5801240-132) such as -48 V, 300V and 10KV are produced in the high voltage power supply (5801240-135). The cathode ray tube employed has a screen diameter of 300 mm (of which about 250 mm are used), the screen of which is aluminized to improve the intensity and afterglow. The screen has an orange afterglow. The deflection angle is about 70°, and the tube is therefore very short. Readjustment of the focus is not necessary. As already mentioned, the display unit (PPI) contains a protection circuit (5801240-126) which prevents damage when faults occur in the unit (failure of deflection voltages and +12V, -12 V supplies) or a special mode of operation is selected. If the antenna is stopped, for instance, the trace is deflected and intensity modulated on a single radius of the screen: In order to prevent damage to the screen phosphors a higher negative voltage is applied by the protection circuit to the control grid of the CRT and the brightness thus reduced when the antenna rotation falls below a certain speed. When the deflection falls, the background intensity of the CRT is also reduced by the protection circuit.

129. Due to the direct coupling, the video stages require several operating voltages (+ 12 V, -12 V, -24 V). When the + 12 and -12 V supplies fail, the CRT is blanked off by the protection circuit, otherwise the full bright-up voltage would appear on the cathode of the tube.

130. Since at warm-up period of a least 15 sec is prescribed for the CRT, a high voltage delay circuit ensures that adequate negative voltage is applied to the control grid of the CRT before the high voltage is switched on or the power supply is switched off.

131. A rotatable disc with parallel engraved lines and a 360° compass rose with adjustable illumination are arranged in front of the screen to facilitate measurement of the target azimuth angle.

PPI POWER SUPPLY

132. The power supply (5801241-5) for the indicator is connected directly to the output of the power distributor. In addition, the 150 V from the voltage regulator (5801234) is fed to the converter (5801241-14) of the indicator power supply. The 380V three-phase power is fed via socket NV and a fuse element in each phase to two transformers in the transformer module. The output voltages of one transformer is rectified in a three phase bridge circuit and fed to the indicator (illumination, signaling) and via connector NB 24 V to the IFF interrogator, AC outputs at 36 V, 18V and 3V with different ratings are taken directly from the winding of the other

POWER SUPPLY GENERATION AND DISTRIBUTION

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(5801236-5 and 5801237), power distributor (5801235), voltage regulator (5801234), PPI power supply (5801241) and the power supply modules (5801239-531, 5801231-14, 5800828-147) in the receiver, the exciter and the IFF interrogator. Distribution, routing and layout of power supply components vary according to the type of MPDR. The power requirement of MPDRs radar system is 380 V / 3 ϕ 47~63 Hz. As mobile equipment, the Radar System is normally powered up by the associated diesel generator. In longer stationary operation periods it may be powered by the "Mains" power supply, while the generator serves as emergency power supply. The safety requirements for the personnel can be secured only if the following safety conditions are compiled by the local power station.

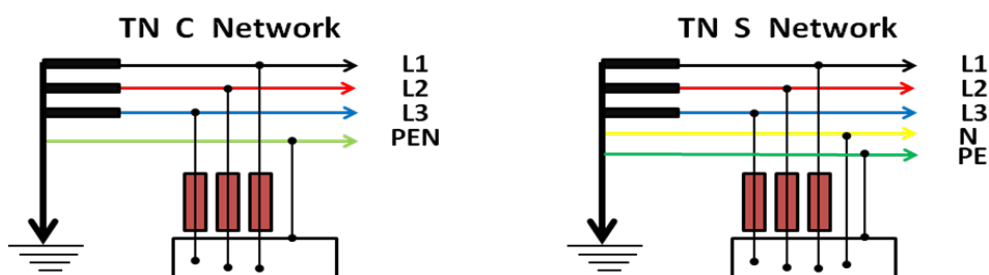


Fig 4-44 Grounding connection scheme for power supply

135. The neutral point of the transformer's low voltage winding must be grounded directly and grounding resistance must not exceed 2 ohms. The protection rod (PE) is connected to the neutral point and wired through to the mains junction or the neutral line may be wired separately. The user of Radar System is obliged to check the mains supply so that it is in accordance with the safety requirements mentioned above. If this cannot be guaranteed, operation of Radar System on the "Mains" is not allowed.

136. Following are the protective measures; those are to be kept in mind during feeding the power to radar system.

- (a) Check grounding of equipment and generators before use.
- (b) If the ground is dry, pour sufficient water on to the grounding connector.
- (c) Check and ground measuring equipment before use.
- (d) Turn the main switch to "off" before connecting the power supply.
- (e) Always first connect cable to the radar equipment (load) and then to the power source.
- (f) Open all ventilation flaps on the radar equipment and generator

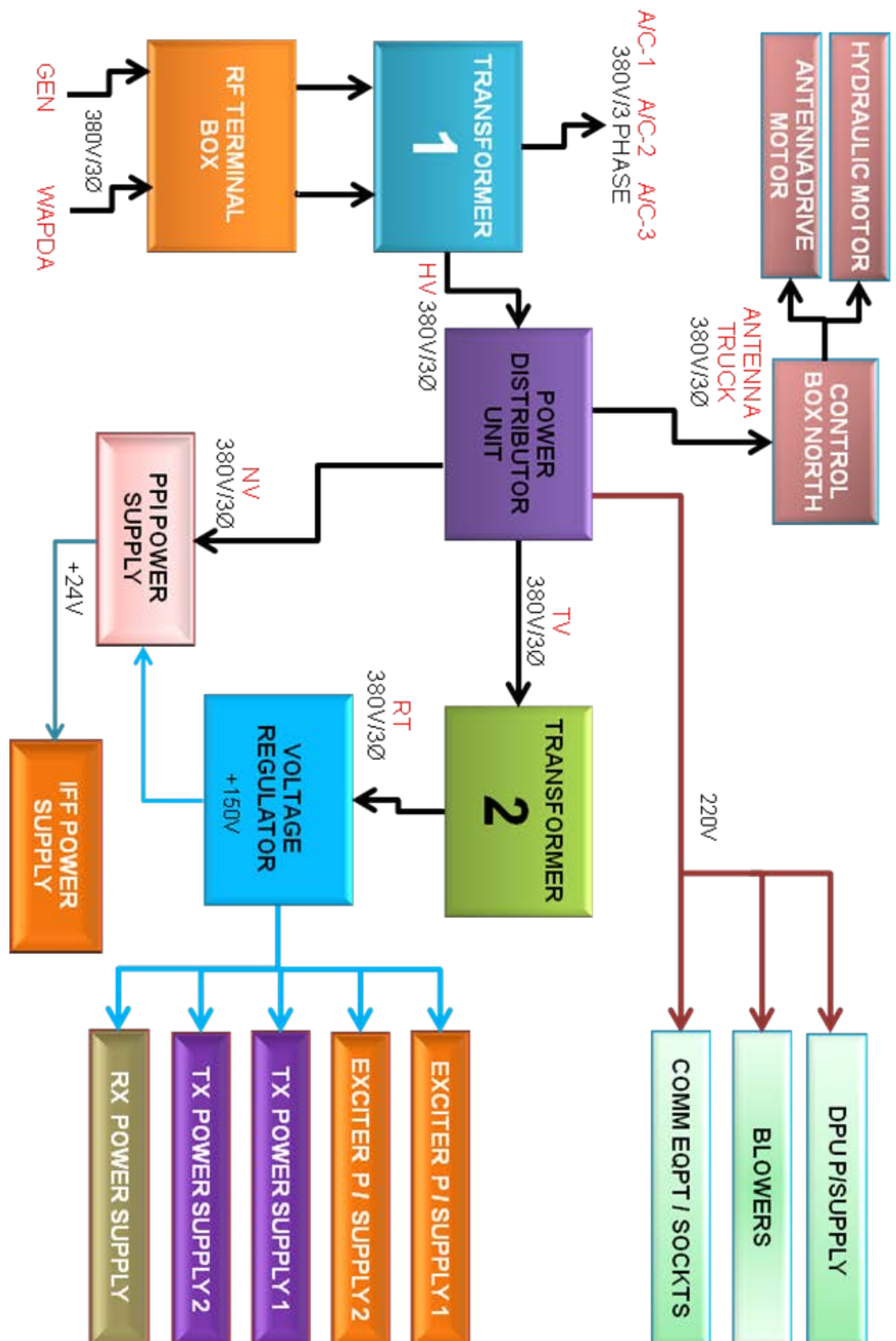


Fig 4-45 Block Diagram of Power Supply Generation and Distribution

137. The generator set supplies 380 V output voltages with 50 or 60 Hz $\pm 5\%$ at an output of 20 KV A; its static voltage stability is $\pm 5\%$. The public ac or generator voltage is applied via the RF terminal box to the switch GENERATOR / POWER MAIN of transformer 1. Transformer 1 functions as an isolating transformer with a turns ratio of 1: 1. Due to the neutral current rating, the output windings are arranged as a zigzag connection. The three phases (R, S, T) are fed via separate radio interference suppression filters to the power distributor, where meters indicate the three phase voltages and currents.

138. The ac line voltage is fed via the power distributor to the ac power loads (antenna, radar data processing equipment set, blower units and indicator power supply), as well as via transformer 2 to the voltage regulator, which feeds a rectified and regulated 150 V supply to the radar equipment loads . Transformer 2 isolates these radar equipment loads from the power loads. It is of similar construction to transformer 1. Both transformers incorporate filters to reduce ac power interference.

139. In most of the power supply units modules the output currents are measured via current transformers. Limiter circuits then protect the units in the event of over-current and short circuits.

140. The power supply units (transformers 1 and 2, power distributor, voltage regulator and PPI power supply) are housed in separate equipment cases. The supply modules are located in the exciter, in the receiver and in the IFF interrogator. They are accessible from the front. No tubes are employed in the power supply units and modules.

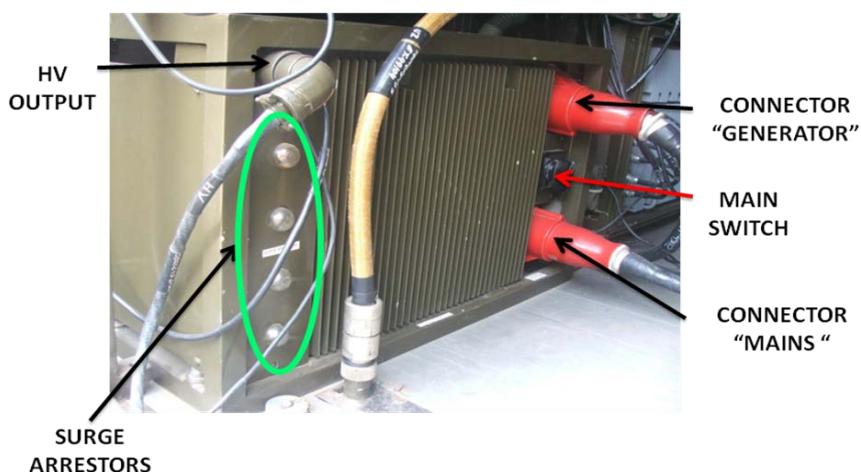


Fig 4-46 Pictorial View of Transformer-1

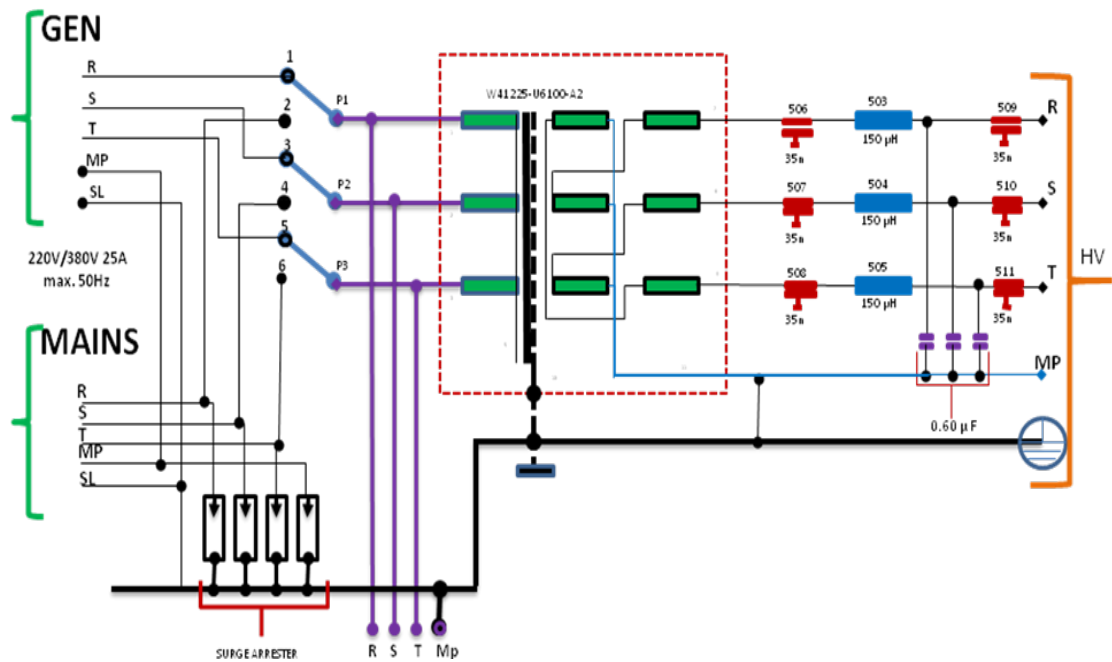


Fig 4-47 Connection Diagram of Transformers

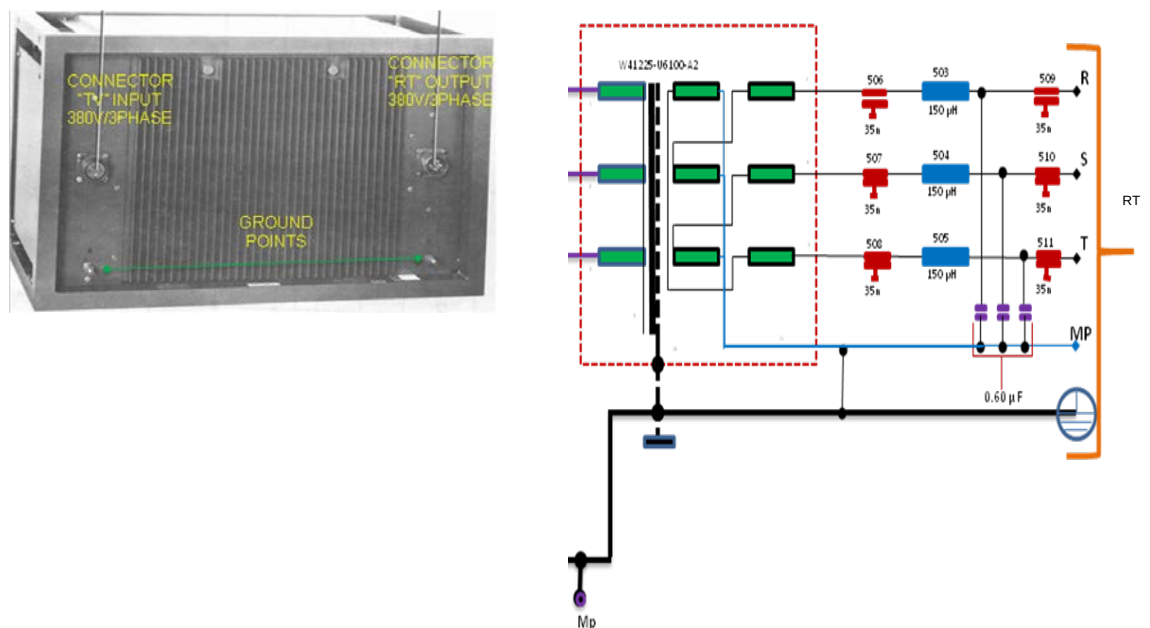


Fig 4-48 Pictorial View and Block Diagram of Transformers

Power Distributor Unit

141. The power distributor (5801235, Fig. 4-47) supplies the three phase power for the radar equipment truck and the antenna truck. It also monitors the phase symmetry of the ac supply and the currents drawn by the three phase loads. The equipment is automatically switched off in the event of a fault - short circuit or overload. An exception to this is the power supply for the system alarm signaling equipment. There is a automatic circuit breaker, by means of this automatic circuit breaker the input of the engage module (5801235- 62) is disconnected when the key OFF on one of the control panels is actuated, and this also occurs automatically in the event of an overload. The lamp OVER CURRENT then lights.

142. The phase-sequence monitor (5801235-214) in the turn-on unit supervises the phase sequence of the three phase net- work. If the phase sequence is incorrect (counterclockwise rotation), the turn-on unit is blocked. The lamp phase sequence WRONG lights. The system cannot be switched on until the phase sequence has been corrected by means of phase switch PHASE SEQUENCE.

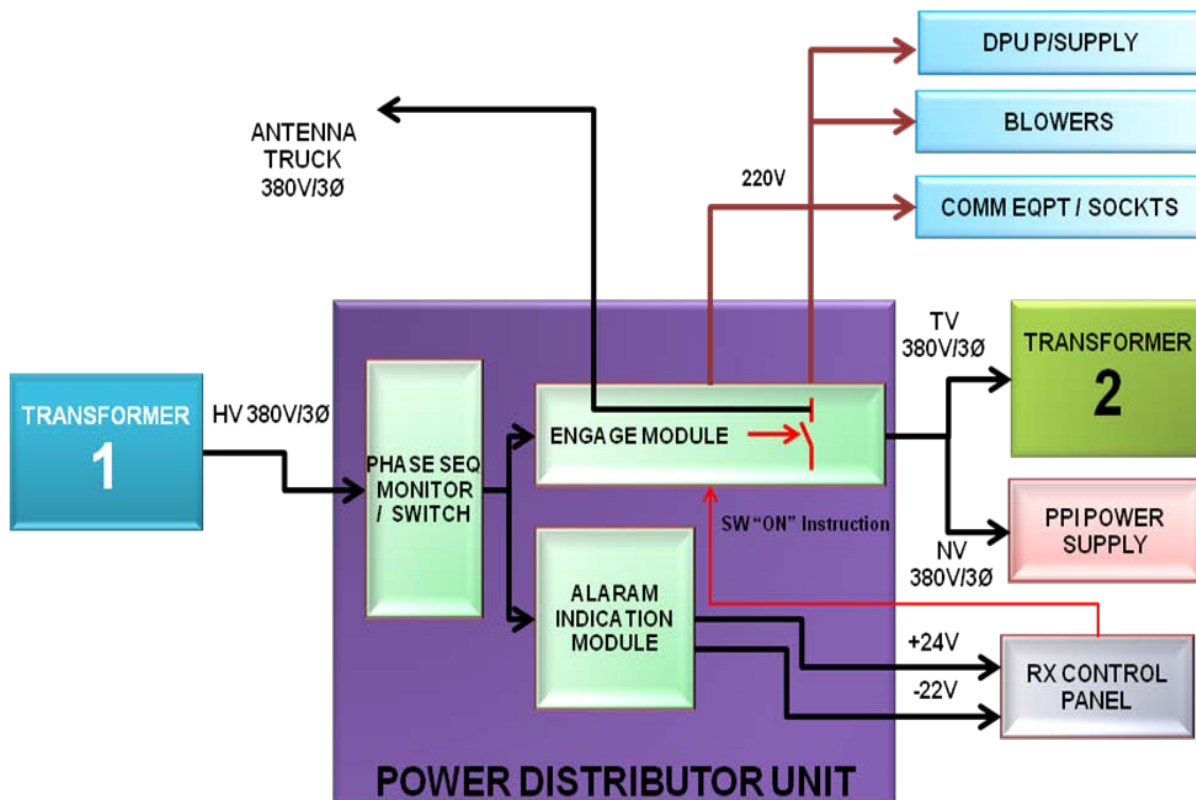


Fig 4-49 Block Diagram of Power Distributor Unit

143. The alarm indication module (5801235-3) of the power distributor also produces two dc supplies 22 V (unregulated) and 24 V (regulated) which feed a relay in the power distributor (E relay) and one in the control panel, as well as other control and monitoring devices in the system. The transformer producing the alarm indication voltage is fed directly from the input of the power distributor. An output of the alarm indication module is connected to a bell which operates in the event of a fault in the radar set. 220 V receptacles are provided on the front panel of the power distributor for the connection of lighting, the blower units, equipment in the control room and the communication equipment.

144. The outlets (4 x 220 V) 01 for, lighting and maintenance work sockets and 03 for the communication equipment are connected preceding the engage module (5801235-62). The power cable to the antenna truck is following the engage module (modified) and protected by a circuit breaker and can be switched on and off by means of the Pacco switch.

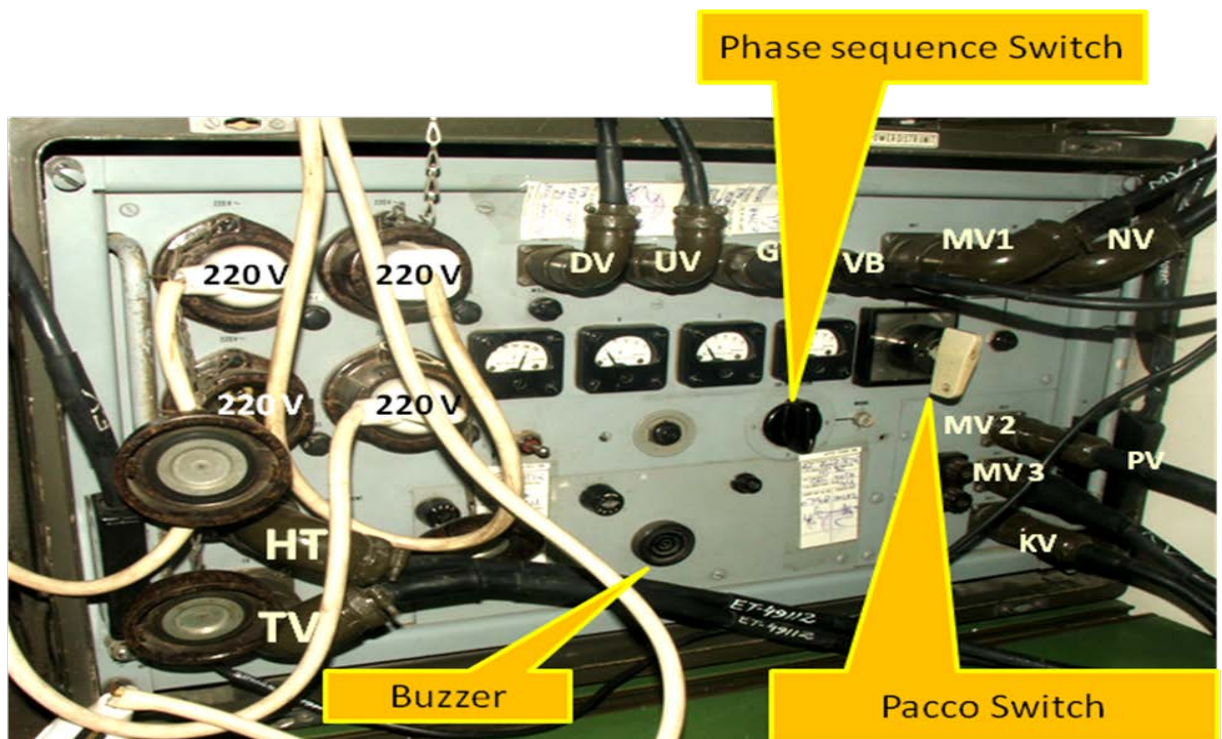


Fig 4-50 Pictorial View of Power Distributor Unit

Voltage Regulator +150V

145. The voltage regulator (5801234, Fig. 4-50) rectifies the voltage from transformer 2 and stabilizes the resulting 150 V. Silicon-controlled rectifiers (thyristors) are also used here for the rectification, regulation (phase control) and overcurrent disconnection. The lines for the three phases R, S, T are connected from transformer 2 (5801237) to the thyristor module (5801234-15), which is so driven by the regulator converter (5801234-12) that a pulsating dc voltage is produced at the output with a mean value of about 158 V. Following the subsequent filter module (5801234-21) the residual ripple on the dc voltage is reduced to a few parts per thousand in transistor module (5801234-23) by the fine regulator (5801234-17), so that the output voltage of the regulator is stabilized at 150 V. If the current in the output line exceeds the permissible value, the fine regulator blocks the transistor module.



Fig 4-51 Pictorial View of Voltage Regulator +150V

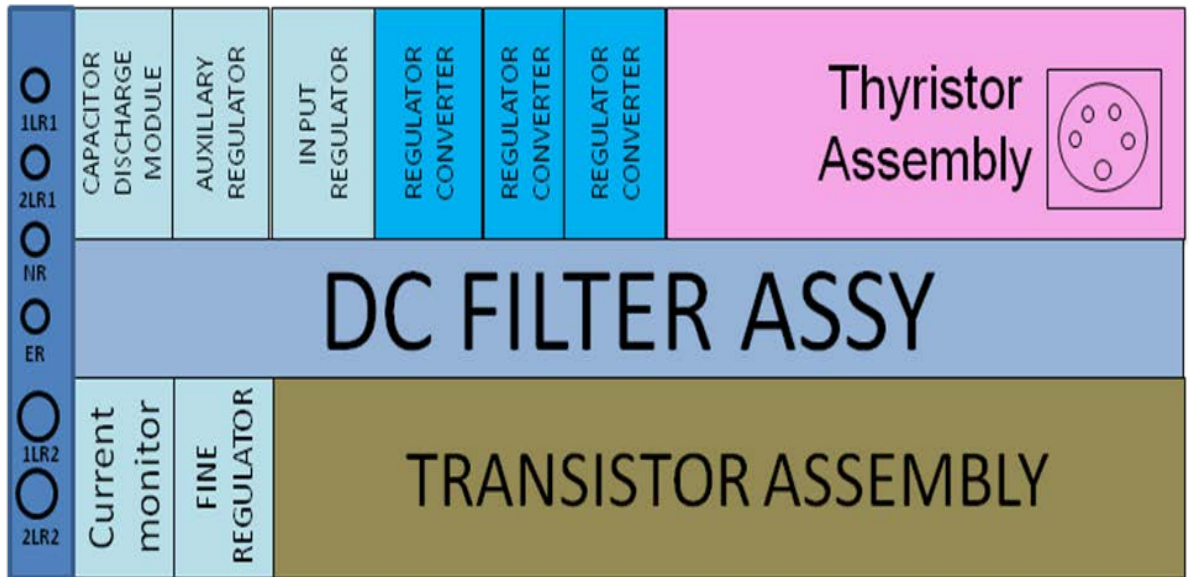


Fig 4-52 Block Diagram of Voltage Regulator +150V

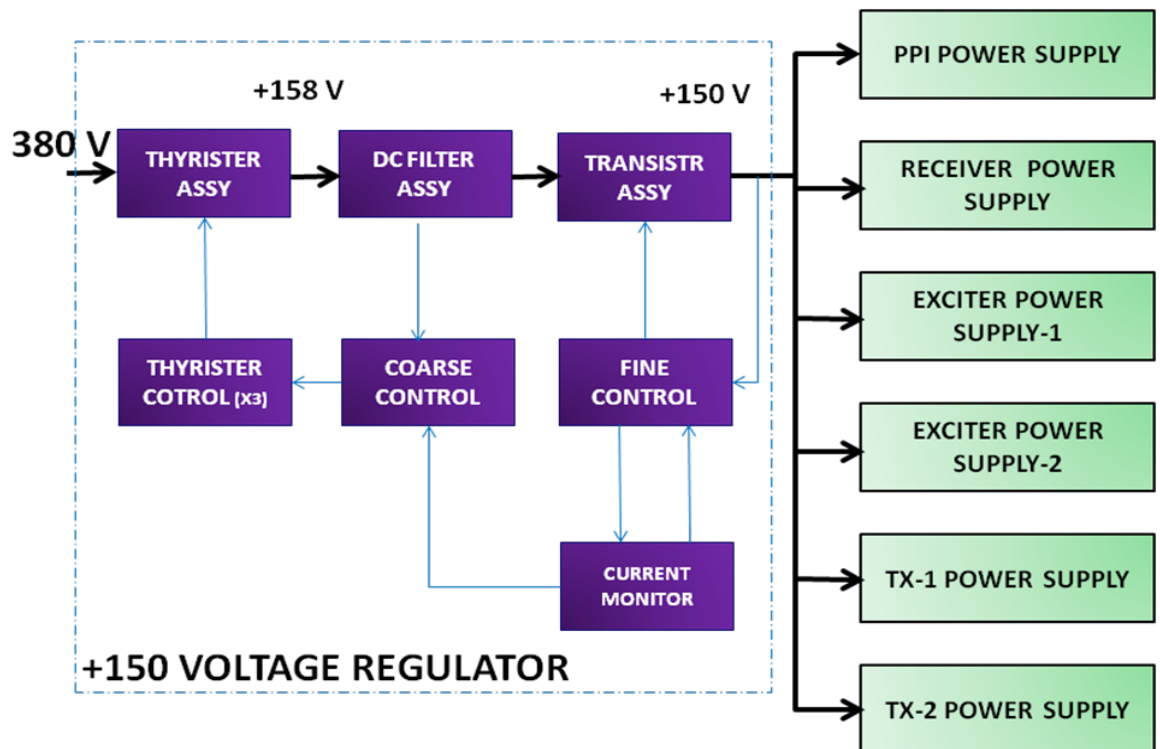


Fig 4-53 Circuit Diagram of Voltage Regulator +150V

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146. A signal is simultaneously fed to the input regulator (5801234-13) which then blocks the thyristor module. A test circuit in the current monitor module (5801234-16) attempts to switch the equipment on again at 5 sec intervals. This process is repeated as long as the current overload exists.

147. When the voltage regulator is disconnected the capacitors in the filter module are discharged via the capacitor discharge module (5801234-22).

148. The lines connected to outputs jacks 1LR1 and 2LR1 lead to the power supplies in the exciter units via transmitter unit. Those connected to 1LR2 and 2LR2 to the HV power supplies in the transmitters, the line connected to socket ER leads to the receiver power supply and the line connected to socket NR to the PPI power supply.

TEST OSCILLOSCOPE

149. The mode of operation of the oscilloscope (Fig 4-53) is the same as that of a normal cathode ray oscilloscope. The following signals and pulses occurring in the radar set can be checked by means of the oscilloscope at receptacles (test jacks) provided on the fronts of the units: In the transmitter: Thyatron keying pulses (U_{RES}), charging converter pulses (LU), keying voltages for final stage (U_s), tripler (U_v), quadrupler (TPB) and TR switch ($U_{S/E}$); the current pulses for tube 4 (driver stage, Jk4), (U_{Thy}) and the charging converter (U_{Lu}), as well as the forward (P_s) and reflected (P_R) transmitter pulses. In the receiver: Keying pulses A, B and the control pulse in the transmitter control module, output of oscillator 1, 2, as well as the output pulse (chopper pulse of the power supply. The CRT employed in the oscilloscope as a screen diameter of 7cm.

150. The various modules, forming the deflection section (sweep generator, X and Y amplifiers), the bright up section (trigger circuit, bright-up multivibrator time marker amplifier), and the indicator section (frame, power supplies 1 and 2) are accommodated on both sides of the CRT.



Fig 4-53 Pictorial View of Test Oscilloscope

CHAPTER No 5

IFF INTERROGATOR MSR-400/8

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CHAPTER No-5**IFF INTERROGATOR MSR-400/8****Basic Principles**

1. The function of IFF Interrogator MSR 400/8 is to identify the friendly aircraft detected by Radar Set MPDR 45/E. To this end the IFF interrogator is coupled with the radar set. Whereas the radar set transmits and also receives single pulses, the IFF interrogator transmits coded interrogations in the form of double pulses at an internationally agreed frequency (1030 MHz). When an aircraft is located in the acquisition range of the radar set and if it is a friendly aircraft with an operable transponder, the airborne transponder decodes the interrogation and transmits a coded reply (active re-radiation method). The interrogator receives this reply (1090 MHz), processes it and marks the moving target on the indicator screen with a zebra pattern if the reply agrees with the reply code set in the interrogator. If an interrogated target does not reply or replies with the wrong code, it is not identified and the corresponding blip is not marked by the zebra pattern.



Fig 5-1 Pictorial view of MSR 400/8 IFF Interrogator

2. In order to get time correlation between the radar echo and the reply pulse, the IFF interrogator must be triggered in synchronism with the transmitter pulse of the radar set. The IFF interrogation is therefore triggered by a synchronizing pulse supplied from the radar set. To avoid mutual interference, the IFF interrogator and

the radar set transmit at different times and on different frequencies. In addition, the signal radiated and received by the IFF interrogator is vertically polarized, whereas the radar set operates with horizontal polarization (crossed polarization).

3. In order to prevent the transponders in the aircraft responding to interrogations radiated via the side-lobes of the IFF directional antenna instead of via the main beam, interrogation path side-lobe suppression is employed with the aid of an additional omnidirectional antenna (SLS antenna). By comparing the amplitudes of the signals received in the aircraft from the directional and omnidirectional antennas, the transponder in the aircraft is able to determine whether the interrogation took place via the main beam or via a side-lobe of the IFF directional antenna.

4. The reply path side-lobe suppression method (RSLS method) is used to suppress replies received via the side-lobes of the IFF directional antenna. In this method the IFF directional antenna is so combined with the SLS antenna that a separate receive path results for each antenna (two channel receiver). The reply signal is only evaluated when the amplitude of the reply signal from the directional antenna is greater than that from the omnidirectional antenna. IFF Interrogator MSR 400/8 employs both the ISLS method to prevent replies being triggered via the side-lobes of the IFF of the IFF directional antenna, as well as the RSLS method to prevent reception of replies via the side-lobes of the IFF of the IFF directional antenna.

5. The IFF directional antenna is combined with the radar antenna and the reflector of the surveillance antenna is used jointly by the radar set and by the IFF interrogator. The SLS omnidirectional antenna is arranged symmetrically above the reflector. Variation of the gain as a function of time (gain time control GTC) improves the side-lobe suppression and prevents overloading of the receiver by replies from near targets. In addition, interrogation only takes place at bearings on which the radar set has detected and indicated a target.

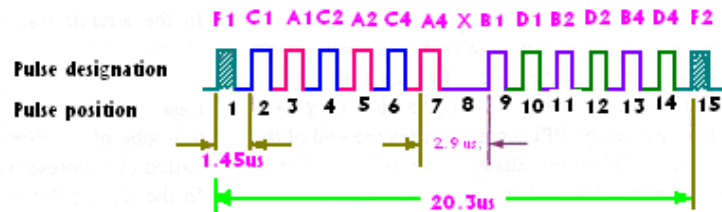
Interrogation pulse pair PI, P3

and ISLS control pulse P2

between (possible pulse combinations $2^{12} = 4096$)

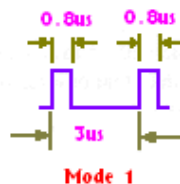
The reply consists of the two hatched framing pulses

and an arbitrary combination of the 12 pulses in



a. Please designations in IFF system Mark X-A

Interrogation



Reply

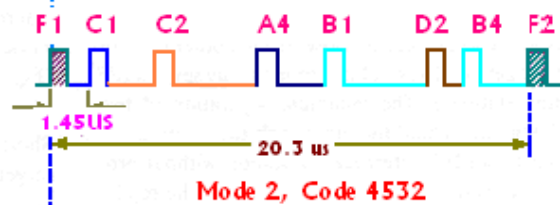
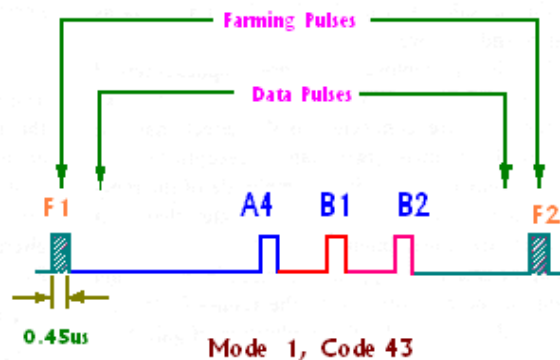


Fig 5-2 Pulses of IFF System Mark X-A

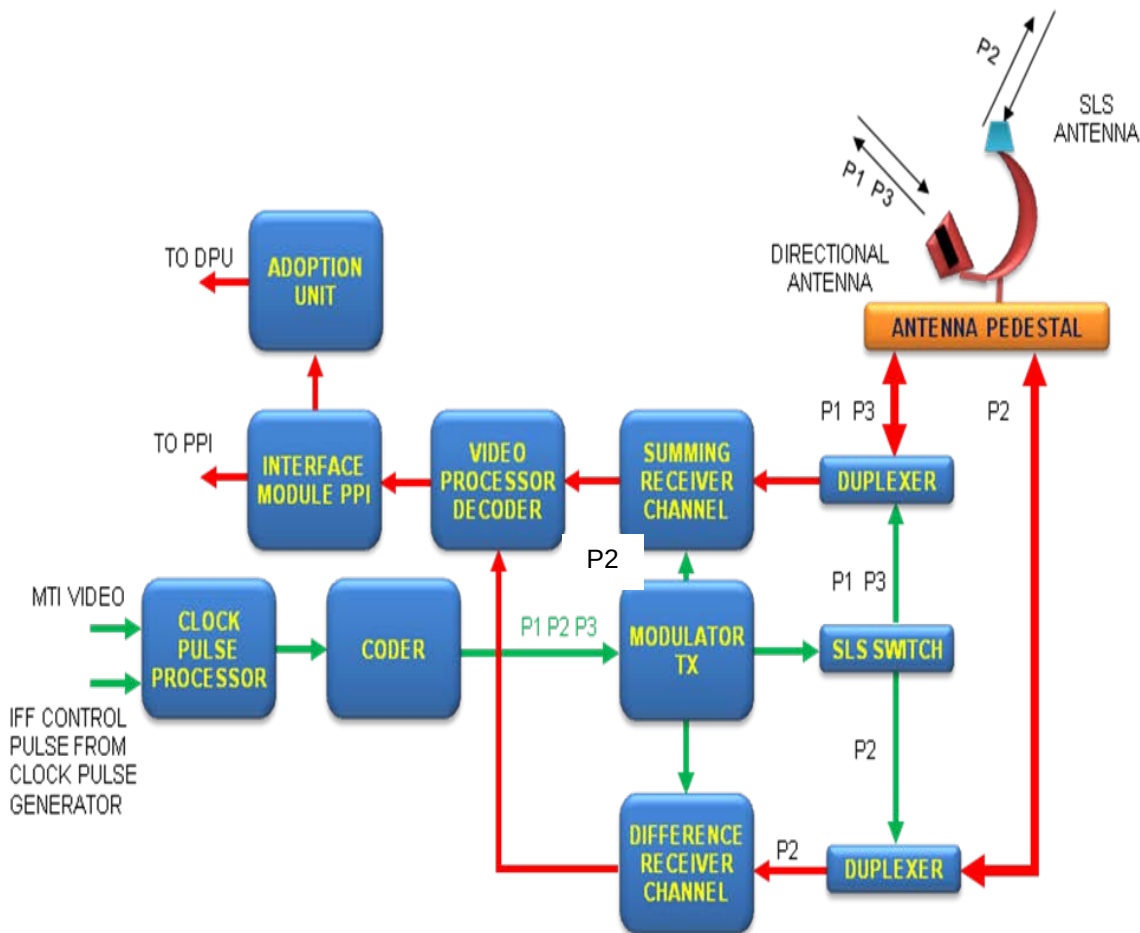


Fig 5-3 Block Diagram of IFF Set MSR400/8

Principle of Operation

6. Fig. 5-3 shows the block diagram of IFF Interrogator MSR 400/8. The signals necessary for initiating the IFF interrogations (IFF pre-trigger, MTI video) are fed from the radar set to the clock pulse processor of the IFF interrogator, where they produce trigger and timing signals. The trigger pulses for controlling the sequence of operations in the IFF interrogator are derived by frequency division from the synchronizing pulses of the radar set. This reduction of the interrogation rate compared with the radar PRF reduces the possibility of excessive interrogation of the transponders in the aircraft.

7. Due to the fact that the IFF-PRF is derived from a crystal frequency which is the same for all radar systems MPDR 45/E, mutual interference of two adjacent IFF stations could take place:

Case 1: If interrogation is performed almost at the same instant, the transponder accepts only the first interrogation and is then inhibited for other interrogations. This could result in failure of any reply to the second interrogating station.

Case 2: With major time delay between the interrogations of two stations, one station can simulate the presence of an IFF target to the second (synchronous fruit). To avoid this, the IFF-PRF is jittered 'pseudo-statistically', i. e. the division ratio used for deriving the IFF- PRF from the radar PRF changes at random between 3000/5, 6, 7, 8, 9, from one interrogation to the next.

8. Depending on the setting of the mode switch on the IFF control unit, the code trigger initiates in the coder interrogation pulses in mode 1,2 or 3/A. The interrogation pulses P1, P3 and control pulse P2 are fed to the modulator which keys the driver and the transmitter. The driver unit (frequency generator) provides the fundamental frequency from which the transmitter frequency and the local oscillator frequency for the signal mixers in the receivers are produced by a 9 X multiplier. The transmitter signal is keyed in the driver unit by the interrogation pulses from the modulator. The transmitter raises the power of the pulse-modulated signal to that required for the output, and feeds the pulses P1 and P3 via the SLS switch to the sigma channel duplexer and from there to the IFF directional antenna.

9. In addition to the actual interrogation by means of pulses P1 and P3 via the IFF directional antenna, the control pulse P2 lying between the two interrogation pulses is fed via the SLS switch and the delta channel duplexer to the SLS antenna, which has an omnidirectional radiation pattern. The function of the SLS switch is to separate the control pulse P2 from interrogation pulses P1 and P3. The SLS switch is controlled by the SLS trigger from the coder and switches the transmitter output to the omnidirectional antenna for the duration of control pulse P2 and to the directional antenna during interrogation pulses P1 and P3.

10. The transponder in the aircraft determines by comparison of the amplitudes of P1 and P2 whether the interrogation was received via the main beam or via a side-lobe of the directional antenna, and transmits or suppresses a reply accordingly.

11. After the formation of the pulse reply in the transponder and a delay proportional to the range, the directional and SLS omnidirectional antennas receive the reply at a frequency of 1090 MHz. The signal transmitted by the aircraft transponder passes from the IFF directional antenna to the sigma channel receiver. The reply simultaneously received by the SLS omnidirectional antenna (RSLS signal) is fed to the receiver in the delta channel (delta channel receiver), which is

of similar design to the receiver in the sigma channel. The duplexer decouples the receive path from the transmit path. The reply is formed by two framing pulses 20.3 μ sec apart providing 12 pulse positions in between for the reply code. The received frequency is 1090 MHz. Through heterodyning with the local-oscillator frequency of 1030 MHz the IF of 60 MHz is derived. After amplification, filtering and demodulation the video signals are present at the outputs of the receiver which are coupled to the video processor. There the amplitudes of the video signals from both channels are compared. If the signal received via the directional antenna is smaller than the one received via the omnidirectional antenna, decoding is not performed (Side-lobe suppression on the receive path RSLS). If the video signal from the directional antenna is larger than the signal from the omnidirectional antenna, the reply is passed to the decoder. Apart from comparison of signal amplitudes gain time control (GTC) function is also performed over here.

12. The reply code is compared with the code set on the IFF control unit. If the two agree, the target is considered to be known (friendly). If the target does not reply or replies with the wrong code, it is considered to be unknown (hostile). In addition, the IFF interrogator contains a civil and a military emergency decoder, the videos of which are linked to the MTI-video of the primary radar.

13. The civil emergency decoder is preceded by a degarbling circuit that suppresses the emergency decoding when accidentally two replies are received at the same time. The military emergency decoder does not utilize the degarbling circuit in order to prevent destruction of the emergency code by a second target reply lying shortly behind it in range.

14. The pulse repeater PPI interface module increases the recurrence frequency of the decoded replies to make it the same as the PRF of the radar pulses, in order to clearly mark the "friendly" targets on the indicator. The IFF information at the output of the pulse repeater is range quantized and the "friendly" signal is thus correctly related in time and consequently in range to the MTI video. The "friendly" signal initiates the marking of the corresponding MTI video on the PPI, and the radar blip of the responding target appears on the indicator screen intensity modulated in a radial direction (zebra pattern) so that the associated range channel is divided by two or three dark bars, parallel to the range rings. The videos coming from the emergency-decoder at IFF pulse frequency are doubled and then marked positive on the PPI screen behind the target video (out in range). The videos from the military emergency decoder appear on the PPI with larger radial proportions than the civil emergency marks. Both emergency marks appear darker on the PPI than the radar target.

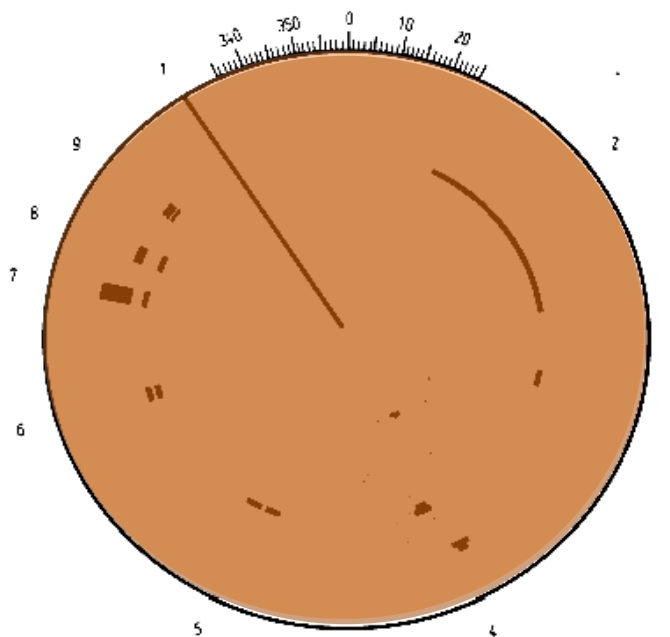


Fig. 5-4 Presentation of moving targets on the PPI

- 1 Azimuth Marker (corresponding to heading of antenna pedestal)
- 2 Range ring arc
- 3 Target, in one of the range elements
- 4 Two targets, each falling in two range elements
- 5 Two targets in one range element with minimum angular separation
- 6 Two targets, each in one range element, separated by one range element
- 7 Target with military IFF emergency
- 8 Target with civil IFF emergency
- 9 Target with IFF marker (zebra pattern)

Transmit Path

15. The coder, modulator, driver unit, transmitter, SLS switch and duplexer lie in the transmit path. The coder is driven by the clock pulse processor and produces the interrogation pulse pair P1/P3 with the intervening control pulse P2, and passes these pulses to the modulator. The SLS trigger, which causes the SLS switch to transfer from the IFF directional antenna to the SLS antenna for the duration of the control pulse, is synchronized to these pulses. The GTC trigger is also generated in the coder. All these pulses are produced by a PROM in the coder. The modulator

amplifies the pulses from the coder and uses them to key the penultimate and final amplifiers of the transmitter. A further output is used to modulate the driver unit, in order to achieve adequate suppression of the carrier voltage at the transmitter output when the transmitter is turned off.

16. The driver unit (freq gen 5800828-133) supplies the keyed transmit signal of 1030 MHz to drive the transmitter input stage and a continuous signal at 1030 MHz, which is used as the local oscillator signal in the receiver signal mixers of the sigma and delta channels.

17. The transmitter (5800828-134) incorporates two identical cascaded stages employing planar triodes to amplify the RF signal from the driver unit to the required output pulse power. The RF power is coupled out of the input stage capacitively and fed via a coaxial cable to the final stage.

Receive Path and Decoding

18. The receiver section is divided into two channels. In IFF Interrogator MSR 400/8 the first receiver channel is used for evaluation of the replies, and the second channel for enabling or inhibiting the evaluation (RSLS).

19. The replies are received simultaneously by the IFF directional antenna and the SLS omnidirectional antenna and converted to the intermediate frequency in the associated receivers (sigma channel receiver 5800828-136 and delta channel receiver 5800828-135) of the sigma and delta channels. In addition to being amplified, the amplitudes of the two signals are compared, and if the amplitude of the reply signal supplied by the sigma channel is larger than that of the delta channel, the relevant reply was received via the main lobe of the directional antenna. Only such replies are processed further. Each receiver channel includes in the "receiver" module a filter, a mixer, an IF preamplifier and a logarithmic IF amplifier. A test signal generator for testing the sigma channel is also included in the sigma channel receiver.

20. For the receive signal at 1090 MHz the duplexer represents a band-pass filter, which provides more than 50 dB of attenuation for the transmitter signal (1030 MHz) in the direction of the receiver. The signals from the output of the duplexer are fed to the mixer diode, at which the IF difference frequency (60 MHz) is produced from the received and local oscillator signals (1030 MHz). The local oscillator signal is fed from the driver unit to the mixer in the receiver via a coaxial filter, which attenuates spurious signals from the driver unit and restricts the noise bandwidth. In order to test the receive path (receiver, decoder) of the sigma

channel during the dead time, the test signal generator (5800828-136111) feeds a pulse train at a frequency of 970 MHz corresponding to the expected reply to the receiver mixer, where it is converted to pulses at a frequency of 60 MHz. The choice of the image frequency, 970 MHz, for the test signal ensures that it is not radiated from the antenna via the 1090 MHz receive filter.

21. The decoding of the A, B, C and D pulses in the mark X-A system takes place in a two-channel passive decoder consisting of plug-in modules decoder unit 1 (5800828-338), decoder unit 2 (5800828-339) and a degarbling unit (5800828-342).

Interfacing

22. The function of the interface module PPI (5801117-353) is to condition the decoded signals of the standard IFF interrogator and those of the civilian and military emergency calls for display on the screen.

23. The decoded reply and emergency call signals are outputted via line drivers on a 75 ohms line, the decoded reply being positive (+ 4 V) and the emergency signals negative (-4 V). A test circuit automatically checks the interface module PPI for operability (also see 1-13.6). Since a radar target does not normally fall in only one range channel, but in three or even four, the quantized reply signal is not only assigned to the channel representing the true range but also to the channels immediately preceding and following this channel. This ensures that the entire radar target on the indicator screen appears modulated with the zebra pattern. A target at zero range, for instance, is read out in channels -1, 0 and +1 instead of just in channel + 1. (The primary radar-blip normally has four channels in range duration).

24. The adaption circuit (5801117-344) generates the signals required in the manual test to test the emergency call functions (civilian and military) and test gate 2. The quantized input signals of the interface module PPI (civilian and military emergency call, decoded reply) are also fed via buffers circuit.

Automatic Test and Supervisory Equipment

25. The automatic test and supervisory equipment continuously monitor all the basic functions of the IFF interrogator.

26. The transmit path is monitored at each interrogation by the test module, which checks the power level of the interrogation signal, the reflection on the antenna feeder line (VSWR resulting of forward and reflected waves) as well as the number of interrogation pulses. If the SLS switch is operating correctly, the interrogation must consist of two pulses (P1, P3). A further check is made to determine whether the power fed to the antenna at full or reduced transmitter power is correct (about 3 dB down on the nominal value). If the transmitter is not operating correctly, the test module causes the fault lamp on the IFF control unit to light.

27. The receive path is tested during the dead-time at the end of each receive period. For this purpose a simulated reply in the currently valid code (test code) is fed from the receiver test signal generator (5800828-136111) into the receive path.

28. If both codes preset on the IFF control unit are valid during the code change transition period, code 2 is used for testing.

29. For the test, the test signal generator supplies a signal in the form of a reply which is capacitively coupled to the receiver mixer diode in the receive path. This signal has a level which is only slightly above a threshold in the video processor. If the receiver is functioning correctly, the decoder supplies decoded test signal 1. This is fed during the test period (test period gate) from the decoder to the test module. Evaluation of the stored test results takes place during the test enabling gate, which is produced by the test module at about four-second intervals. A counter in the test module records the number of successful tests. The length of the test enabling gate corresponds to 13 interrogation periods, so that 13 tests take place. A minimum of four successful tests indicate satisfactory operation of the IFF interrogator. Less than four successful tests indicate an equipment malfunction or the effect of random answers, and the lamp "FAULT" lights. This indication is retained until the next test enabling gate. The test module also checks the pulse repeater in the interface module PPI (5801117-353). The interface module PPI generates a test target at the start of each IFF interrogation; the target is stored in the memory (RAM) for IFF replies. A test is carried out in the following radar periods to determine whether the test target is read out (pulse repetition). At the same time the address counter of the memory undergoes a function test.

30. Signals recognized as faults are stored and called up by the test module. In the event of a fault, this is indicated on the IFF control panel.

Manual Test and Supervisory Equipment

31. Depending on the switch position on the IFF control panel, a fixed code is generated on the adaption circuit the code follows the same path as the test code during the automatic test. In switch position TEST MIL. EMERG, on the IFF control panel, the code is so formed that a military emergency call is decoded, when the military emergency decoder (5800828-343) is intact and several civilian emergency calls are decoded when the degarbling stage (5800828-342) is defective. Only one wide "ring" appears on the screen if no fault is recognized but, one wide ring and several narrow rings appear when there is a fault. In switch position TEST CIV EMERG, on the IFF control panel, the code is so designed that SIX civilian emergency calls are produced, five of which are shown as narrow rings on the screen, when the decoder for 'civilian emergency calls in the degarbling stage (5800828-342) is in order. At the start of these tests all memories (RAM) of the interface module PPI are automatically checked for full operability. When faults are detected, alarm signaling is initiated on the IFF control panel.

Power Supply

32. The power supply module (5800828-147) of the IFF interrogator produces the various operating voltages for the modules (Fig. 1-41) as well as the heater voltage (6 V dc) and the anode voltage (1100 V) for the transmitter tubes.

33. The outputs of the power supply are protected against short circuit and overload. The power supply is designed for connection to the + 24 V from the power supply module PPI. The input power is about 60 W. The power supply eliminates the voltage variations at the unit and decouples the outputs and supplies load voltages of -52V, -12 V, +5 V, +6 V, + 12V and +25 V) in order to restrict the no-load voltages to the nominal values, five of the circuits (-52 V, -12 V, +6 V, +25 V and 6 V heater) include limiters. The 6 V rectifier circuit supplies the dc voltage for the transmitter tube heaters). In the event of an overload, a current monitor at the output of the inverter derives a control voltage which is fed to the inverter to suitably reduce the mean current at the inverter output. This monitor only operates when a reference voltage is exceeded. The power supply for the IFF interrogator is protected by a fuse in the power supply unit PPI (5801241). The power supply is turned on with the ac supply when the radar set is switched on.



Fig 5-6 MSR 400/8 IFF (Power Supply)

IFF Control Panel

34. The control and indicating elements necessary for the operation of the IFF interrogator are located on the IFF control panel.

35. There is set the type of interrogation, the interrogation mode, the reply code and the desired test mode. In addition, the IFF interrogator can also be regulated from of the MCC (priority). Two lamps (STAND BY, FAULT) on the IFF control panel indicate the operating conditions "operable" (interrogation inhibited) and "fault". The lenses of signal lamps are in the form of push buttons (PRESS TO TEST) in order to facilitate testing. When the lens is pressed, 24 V dc is applied to the lamp concerned, which lights if it is in order. In addition, the digit windows in the code-setting switches corresponding to the mode selected with mode selector switch are also illuminated.



Fig 5-7 MSR-400/8 (IFF) Control Panel Interrogator

36. One of the three modes 1, 2 or 3/ A can be selected by means of mode selector switch S1. The position of code selector switch S2 determines which of the two preset codes 1 or 2 is valid as "friendly" reply or whether both (codes 1 and 2) are admissible as friendly replies. All the Mark X-A replies are received in the switch position AC; the code is unimportant in this case.

37. Two anticipated reply codes can be set by means of the two code-setting switches.

NOTE

When setting the code number it should be noted that the thumb wheel switches are arranged from left to right in the same sequence as the pulse designations A, B, C and D (Fig. 1-4). Since only data pulses A and B are used for mode 1, code setting is affected by means of the two left-hand thumbwheel switches. Accordingly, the code for mode 2 and 3/A is set by means of all

38. Interrogation switch S3 has three positions: STAND BY, AUTOMATIC and MANUAL. In position STAND BY the IFF interrogator is turned on but does not transmit interrogations. Target initiated interrogation occurs in position AUTOMATIC. Interrogation takes place independently of moving targets as long as the switch is held in the spring-loaded position MANUAL. Switch S4 have four positions: In position AUTOMATIC a self test of the IFF interrogator is performed every 4 seconds. In position EMERGENCY MIL the military emergency decoder and the degarbling stage, in position EMERGENCY CIV the civil emergency decoder is tested (Code 7700) with the aid of the test rings on the PPI (see Technical Handbook part 1, section 6-4). In this case the switch "interrogation" must be held in position MANUAL. In the spring-loaded position ALARM OFF the buzzer in the IFF control panel released during FAULT can be switched off. The buzzer remains cut off held by a relay until the fault is cleared, even when the TEST switch is returned to its initial position. The red lamp FAULT cannot be switched off. It always lights until the fault is cleared.

39. The IFF control panel is provided with three connectors via which multi-core cables are coupled to the IFF interrogator and the radar equipment. The series numbers 5801117-32 or -62 of control panel both panels are compatible.

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CHAPTER No 6

RADAR DATA PROCESSING EQUIPMENT

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CHAPTER No-6

RADAR DATA PROCESSING EQUIPMENT

General Introduction

1. The radar data processing equipment set is a subsystem of radar system MPDR which forms a connecting link between the radar set and the radar data transmission equipment which permits narrowband transmission of the radar data to the MCC. The radar data processing equipment set evaluates the data coming from radar set MPDR in order to define the target position (range & azimuth) and correlates the signals from IFF interrogator set for the respective identification information. The operation of complete system is controlled by processing unit control panel provided with push buttons, switches and indicators. In manual mode of operation, the positioning of symbol on PPI is accomplished with the help of rolling ball assembly. Power supply generates the operating voltages required for the radar data processing equipment set.

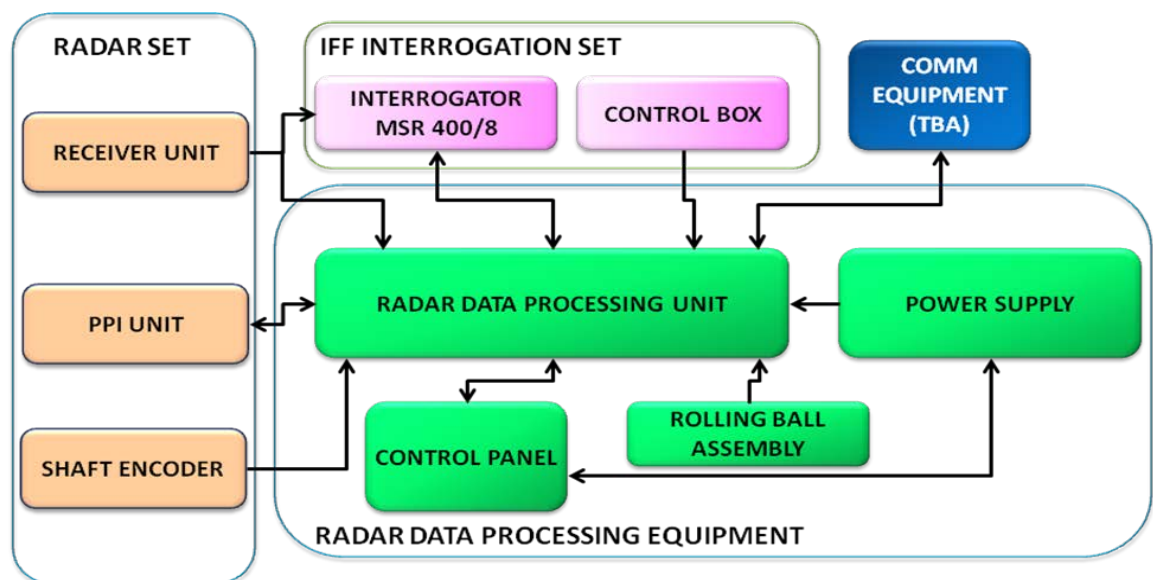


Fig 8-1 Block diagram of Data Processing Equipment Set

Main parts of Radar Data Processing Equipment

2. The Radar Data Processing equipment set consists of the following items:-

- (a) *Radar Data Processing Unit* contains the entire electronic circuits

necessary for the data processing and transmission to MCC.

(b) *DPU Control Panel* contains all push switches and indicators necessary for the operation of radar data processing equipment set.

(c) *Rolling Ball Assembly* is used to position symbols during manual tracking mode.

(d) *Power Supply* which supplies the necessary voltages for the operation of DPE from 220 V mains.

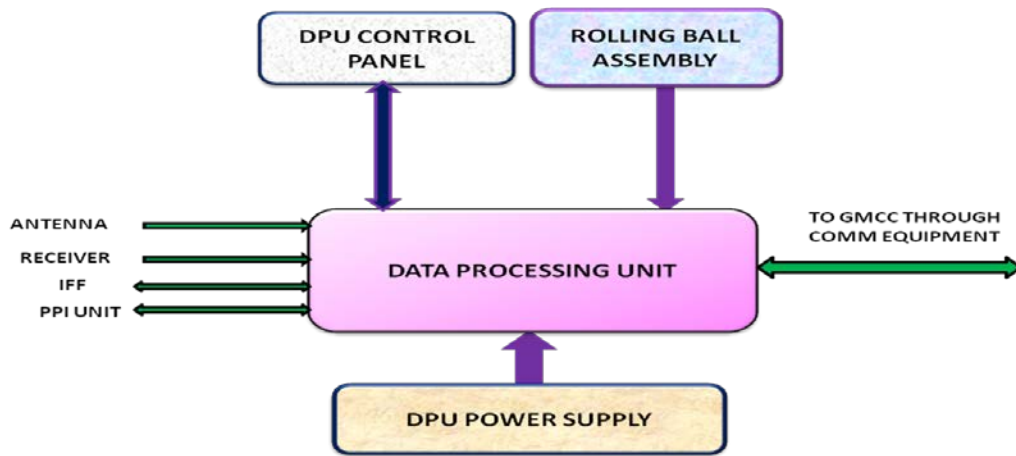


Fig 8-2 Main parts of Data Processing Equipment Set

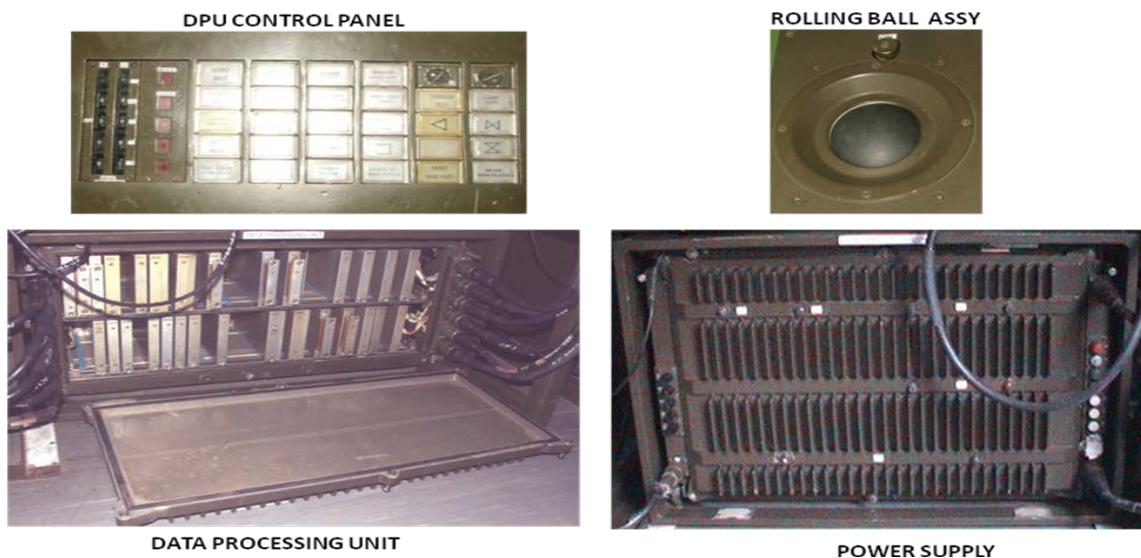


Fig 8-3 Pictorial view of the Main parts

Radar Data Processing Unit

3. To fulfill the above mentioned tasks, the following functions are performed by the different functional groups of the data processing equipment set.
- Target data extraction for target center determination from radar data and for correlation of identifying information coming from the IFF set.
 - Target data preparation and control of the different modes of operation by means of the built-in microprocessor.
 - Target data transmission for data exchange between radar site and MCC.
 - Processing and display of the synthetic symbols on the PPI.
 - Processing of the IFF Remote control data (interrogation modes and codes) for control of the IFF interrogator from MCC.
 - Self test of individual functional groups and data transmission (BITE)

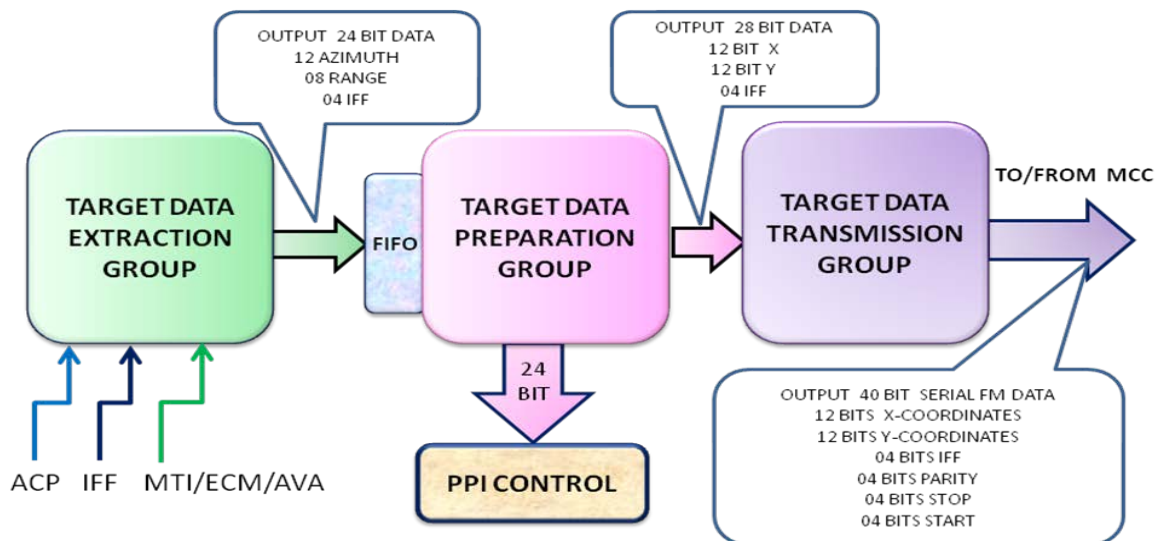


Fig 8-3 Main functional groups of Data Processing Unit

Target Data Extraction Group

4. The functional group within the radar data processing unit which is operating in real time and determining the target coordinates of the radar information is called

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extractor. The extractor determines the position of the target in form of polar co-ordinates (range + azimuth) from the information received from the primary radar MPDR and assigns the IFF signals to the corresponding primary radar signals. It also determines the direction of jammers detected by the primary radar and their intensity by evaluating the ECM and AVA signals from the primary radar MPDR. By means of built-in test equipment (BITE), which simulates the input signals of the radar set and the IFF set, correct operation of the extractor, can be checked. Evaluation of the test results is performed by microcomputer assembly (target data preparation group) of the data processing unit. The interface between target data extraction and preparation is formed by a buffer storage (FIFO-memory; FIFO = first in first out) which is necessary to adapt the different data processing speeds of the two groups. In detail the extraction group performs the following functions:-

- (a) Angle processing
- (b) Target range determination
- (c) Target azimuth determination
- (d) IFF correlation
- (e) Determination of the direction of jammer
- (f) Generation of the test signals

5. In order to perform abovementioned function the extraction group contains the following subassemblies

- | | | |
|-----|--------------------------|--------------|
| (a) | Azimuth counter assembly | 5801180-8112 |
| (b) | Range counter assembly | 5801180-8113 |
| (c) | Extractor memory module | 5801180-8114 |
| (d) | Adder module | 5801180-8115 |
| (e) | Status logic module | 5801180-8116 |
| (f) | IFF correlator module | 5801180-8117 |
| (g) | Control stage | 5801180-8118 |
| (h) | Test assembly | 5801180-8119 |

Interfacing with MPDR and IFF set

6. The Extractor is series connected to MPDR. In case of target acquisition, the receiver unit of radar set supplies a train of echo pulses with constant amplitudes. The position in time within the radar period indicates the target range. The radar periods are divided into range gates each having a width of 250 meters, by quantizing the period into time intervals of $1.66 \mu\text{sec}$ in duration. A radar period is defined by radar zero pulse and its repetition frequency is the pulse repetition frequency (PRF) of radar set and its width is identical with the width of one range gate. Its time position is that of a 'zero-target echo'. Due to the 500 meters range resolution of the radar set and the 3 kHz pulse repetition frequency (PRF), one radar period consists of 200 range gates. Only 180 of them are used due to the radar range of 45 km. The outer 181 ~ 200 fall within the outer dead time and inner dead time comprises the gates 1~6.

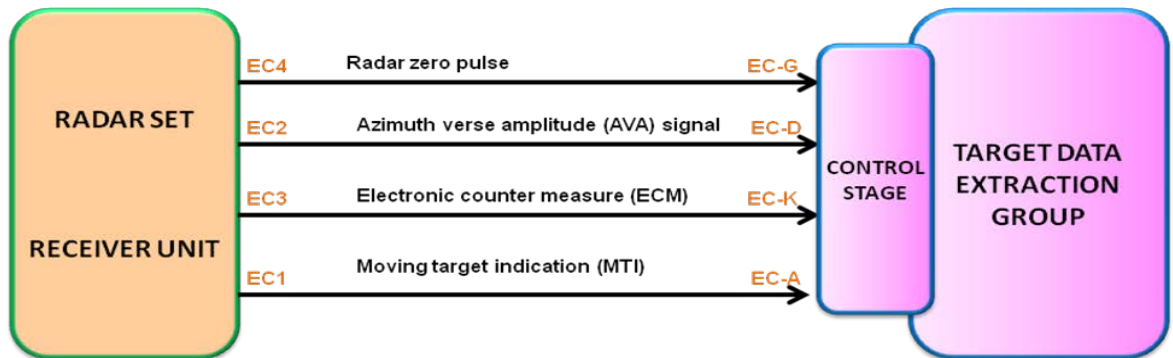


Fig 8-4 Interfacing with Radar set

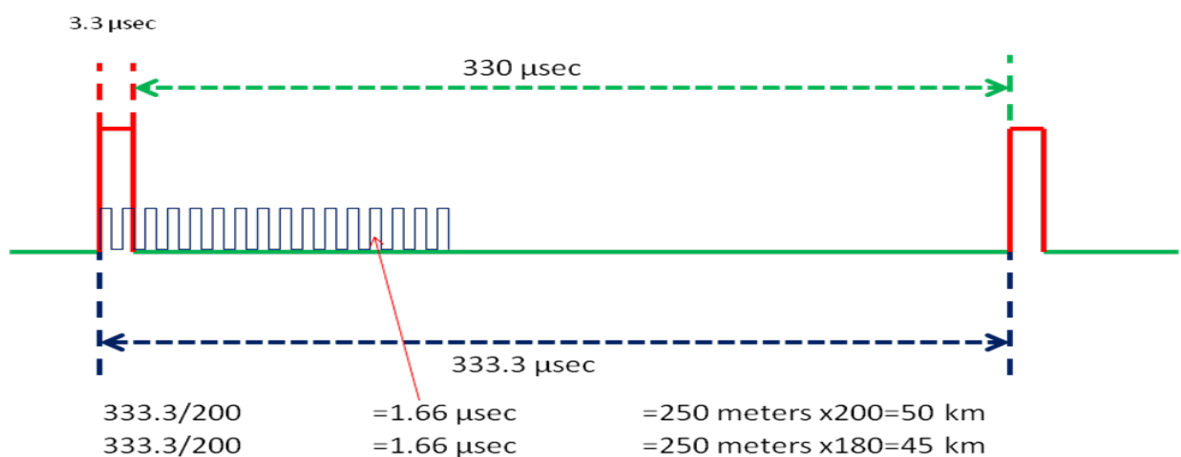


Fig 8-5 Range Gates of one Radar Period

7. The radar data processing equipment and IFF set are series connected. The IFF interrogator set evaluates the target response following an interrogation (target controlled) and if the response is correct, indicates it as 'friend'. In this case it produces pulses with an average PRF of 400 Hz. Time position of these pulses with regards to the radar zero pulse determines the target range.

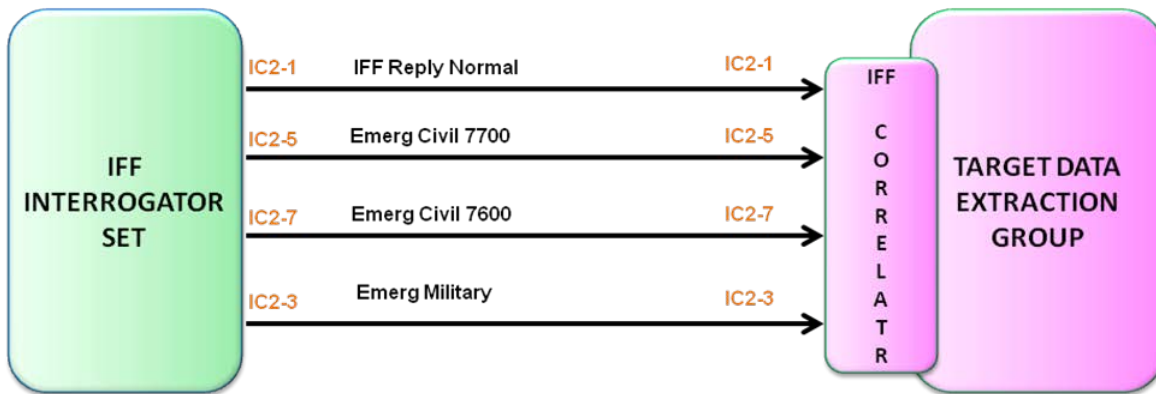


Fig 8-6 Interfacing with IFF set

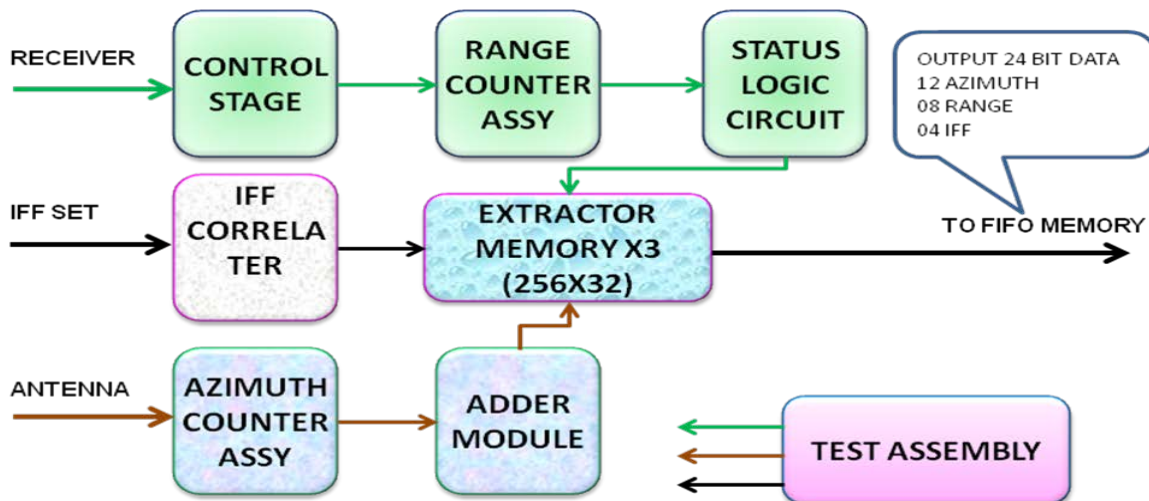


Fig 8-7 Block diagram of Target Data Extraction Group

8. The *control stage* of the extractor generates several clock and control signals which are used for clock controlled processing of the target signals in time multiplexing operation.

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9. The *range counter assembly* is incremented by a clock signal from the control stage and provides the respective binary coded range gates, from which a range determination is performed. Initially the PRF signal from radar set resets counter regardless of the momentary state and then permits a new counting operation to start. As the main pulse of radar set has the width of two range gates ($1.66 \times 2 = 3.3 \mu \text{ sec}$) and as the target can have any value within the range system, the echo signals reflected from a target results in energy values of various amplitudes in several neighboring range gates. This fact means that MTI signals of these targets start and stop at different times during time on target. In the extractor, the range gate which provides best the target range is determined and the signal in the neighboring gates are suppressed. In this manner, only one target message is generated each a target is detected by radar set. Neighboring gates suppression necessary for determination of target range is carried in the status logic circuit.

10. The *Azimuth counter assembly* converts the output pulses from shaft encoder on the antenna rotary axis into a binary coded, parallel arranged angle position. For this purpose, the shaft encoder produces three signals.

- (a) UAO = Antenna zero pulse, generated by shaft encoder at every transit of antenna through north and it resets the counter of azimuth counter assembly to zero
- (b) UA1 = Square wave pulses (1:1) with 1024 pulses per shaft encoder revolution.
- (c) UA2 = Square wave pulses (1:1) with 1024 pulses per shaft encoder revolution but are phase shifted by 90° with UA1.

11. These abovementioned signals are translated to TTL level in the azimuth counter assembly and from these, coarse direction of antenna rotation is determined. Next, the signals are converted to count pulses which are used to determine the even fine direction of antenna position.

12. The MTI signals supplied by the radar set MPDR during time on target provide the direction of the detected target only inaccurate due to the antenna beam width. For this reason, the extractor not only performs the neighboring gates suppression for an improvement of the range information but also determines exactly the target azimuth angle. For this purpose, the start and end of the target signal are determined for each range gate in status logic circuit and the arithmetical average of the corresponding angles is generated in *Adder module*.

14. For the tracking processes in the computer of the MCC the extractor provides a reference message once per antenna rotation, always at the same point during one time operation. The extractor therefore processes a so-called north message in the auto mode. It is produced in the control stage and synchronized with antenna zero pulse (UA0). It is processed in the 185th range gate, it is given to the MTI input and it is marked with its own identification symbol (Δ) at display unit at outer range marker and in the MCC as ($\sim$).



15. Target Data Preparation Group processes the data coming from target data extractor and passes them on to, after suitable processing, to the functional group 'transmission group and PPI control'. The functional group 'target data preparation' consists of following modules.

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- (a) FIFO memory
- (b) Processor module
- (c) Memory module
- (d) Input stage control panel
- (e) Output stage control panel
- (f) Output stage symbols
- (g) Lamp driver module
- (h) Test I/O module (optional)
- (i) Rolling ball register

16. The controlling element for the data processing and functional control is the microcomputer that is connected to all modules of preparation group via the data and address buses. The data coming from the extractor is buffered in the *FIFO* memory, processed with aid of the processor module and the memory module, passed to the PPI control via output stage symbol and via the data bus to the 'target data transmission'. The plug-in module rolling ball register is the connecting link between the rolling ball assembly and site coordinate switch on the one hand and the microprocessor bus on the other. The input stage control panel and output stage control panel are the connecting elements between the control panel and various control signals and the microcomputer bus.

17. The micro-computer consists of Processor module and Memory module. The micro-computer bus consists of a 16 bit address & control bus and 8 bit bidirectional data bus; this permits transfer of 8 bit data word either from the processor to some output interface or from an input stage to the processor. The plug-in memory module contains a 4096 bit ROM and 1024 bit RAM. The ROM contains all programs of the micro-computer and can neither be erased nor overwritten. The micro-computer programs are consist of a main program 'PU', various subroutines for the performance of the individuals jobs of the micro-computer and an interrupt program 'INTER' for the display of symbols on the PPI. When power supply is switched "ON" a reset signal is generated by the processor module, this set up the initial status in the micro-computer and in the inputs and outputs stages. Subsequently all subprograms are continuously run in accordance with the mode and appropriate operating state.

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18. Control for the operating modes of the radar data processing unit is carried out from control panel. Contacts of the push switches generates logic signals which represents the state of the push switches and pushbuttons “actuated” or “not actuated”. These logic signals are connected to control panel input stage module which has a connection to microprocessor bus.

19. The micro-computer first reads in the push button signals of the control panel from control panel input stage. It then evaluates which operating mode is requested by the operator and whether this mode has changed. In addition, the priority of the operating modes is determined by the program. After this evaluation, a bit pattern is generated to represent- illumination of the lamps in the control panel and- the control signals for the extractor section. This bit pattern is passed to a register in the control panel output stage. From further the lamps signals are passed on to the control panel. The remaining control signals are connected directly to the IFF correlator, Control stage, and test assembly of the extractor hardware.

20. As already mentioned, buffering of extracted data is necessary as the extractor operates in real time whereas the subsequent micro-computer carries out further processing in the rhythm of the data transmission to the MCC. The **FIFO** (first in-first out) memory (storage capacity 64 words of 32 bits each) is an asynchronous buffer into which 64 words can be entered sequentially and then read out again in the same manner. It is used to buffer the target messages from extractor, which can arrive in a very rapid sequence (with a minimum spacing of 3 μ sec) as the speed of micro-computer is not adequate to register and process targets at this minimal time interval. The micro-computer only obtains a target message from FIFO memory, if the target out-put memory is empty. It is thus possible to transfer a new target data to the functional group ‘target data transmission. The transmission duration for a target message to the MCC is 33 milli-secs at a rate of 1200 bit/sec. each extracted target is entered in the FIFO memory as a 24 bit word. The output of this memory is connected to the bus of microcomputer. Each time the main program “PU” is executed by the microcomputer, the system asks whether the conditions are fulfilled. As soon as they are fulfilled, the 24 bit target word is read out by the microcomputer in the form of three bytes. These three bytes, which represents the polar coordinates and the identification of the target, are now processed further via program “ZIEL”.

WORD IDENTIFICATION					SIGNIFICATION	SYMBOLS	STAGE	AUTO MODE	MAN MODE
K5	K4	K3	K2	K1					
0	0	0	0	0	UNKNOWN 1	V		x	
0	0	0	0	1	UNKNOWN 2	V		x	
0	0	0	1	0	UNKNOWN EMERGENCY	X		x	
0	0	0	1	1	ECM, Fictitious Jammer Position	X		x	x
0	0	1	0	0	FRIENDLY 1	M		x	
0	0	1	0	1	FRIENDLY 2	J		x	
0	0	1	1	0	FRIENDLY EMERGENCY	C		x	
0	0	1	1	1	CIVIL EMERGENCY	#		x	
0	1	0	0	0	UNKNOWN 1	X		x	x
0	1	0	0	1	UNKNOWN 2	V		x	x
0	1	0	1	0	MASS RAID	A		x	x
0	1	0	1	1	transm. test, spec. symbol	A		x	x
0	1	1	0	0	FRIENDLY 1	X		x	x
0	1	1	0	1	FRIENDLY 2	U		x	x
0	1	1	1	0	END OF SCAN	N		x	x
0	1	1	1	1	LAND MARK	T		x	x
1	0	0	0	0	ROLLING BALL	O		x	x

Fig 8-9 Target Symbols and Codes

21. At first the target information delivered in the form of polar coordinates is converted into Cartesian coordinates, conversion from meters into miles being carried out at the same time. The converted coordinates and unchanged identification of the target are now changed to a further 'three bytes format' and stored in the target display memory, part of the RAM of the microcomputer. From here, a symbol corresponding to the target can be displayed on the display unit with aid of the interrupt program "INTER". Prior to transmission of the target data to the MCC, the radar site coordinates must be added to the Cartesian coordinates of the target. This operation is also carried out by the microcomputer. The resulting five byte word is stored in the target output memory, which is also a part of the RAM in the microcomputer. The subprogram "ZAUS" accesses this area and transfers target data to the transmission module as soon as the appropriate conditions are fulfilled.

PPI Control

22. The display unit is controlled by an interrupt program "INTER" via the output stage symbols. In every second display period of the display unit, the symbol output stage generates an interrupt signal which results in immediate interruption of the currently executed microcomputer program. As a result of this, the interrupt program INTER is started. This at first determines whether a target or one of 17 symbols are to be displayed, resets the interrupt signal and passes on the appropriate record from the target display memory or symbols display memory to the symbol output stage. If a symbol is to be displayed, then the bit 'SYMBOL bright' is set in the 'three byte data word'. As a result of this, the output stage symbols, first generates a signal for switching the display unit to external control, followed by the control pulses for trapezoidal and character generator. If no symbol

is to be displayed as a result of interrupt signal, then a data record in which the bit 'SYMBOL bright' is not set is not issued by the interrupt program. In this case, the display unit is not switched to external control and generation of control pulses for trapezoidal and character generator is suppressed. Only the interrupt signal is reset. At the end of the interrupt program, a time counter is decremented. This is used to inform the main program 'PU' that one of the set symbols is to be transmitted to the MCC for that purpose the main program 'PU' calls the subroutine ZAUS, which provides for output of the symbol coordinates and of the symbol code to the functional group 'target data transmission'.

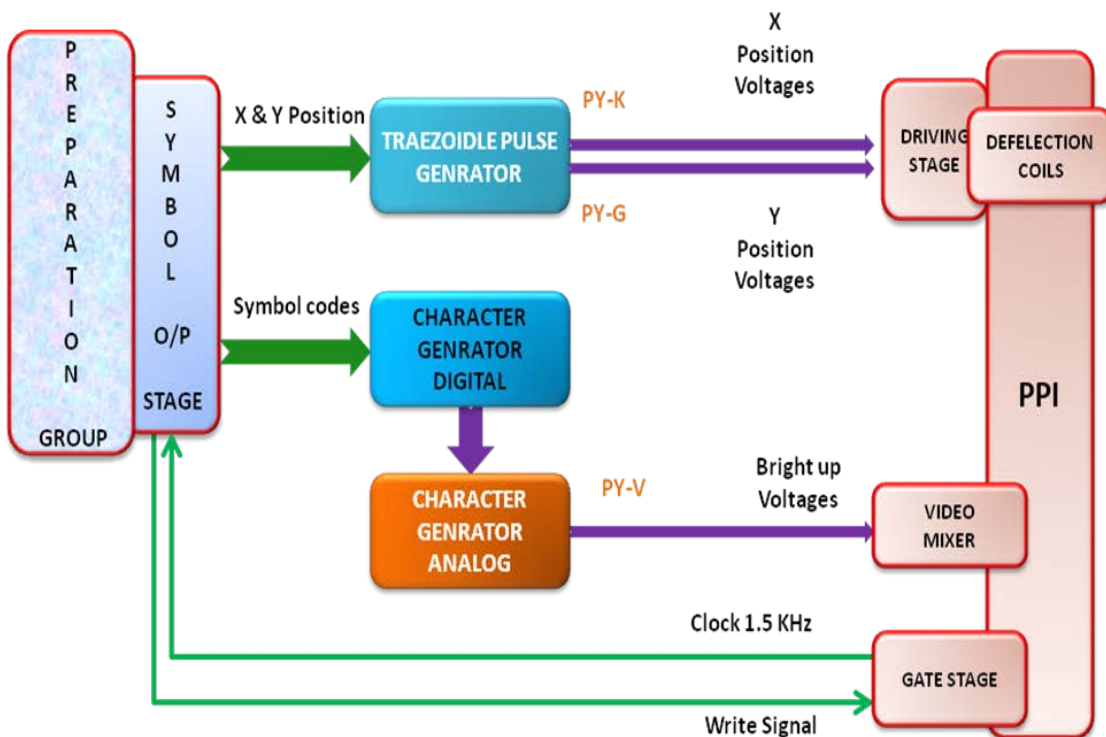


Fig 8-10 Block diagram of PPI control Group

Target Data Transmission Group

23. For the control of the PPI, the processing unit provides deflection voltages for writing of the different symbols. They are produced in the character generator which has a digital/ analog converter connected to its output side. The character generator generates the deflection and unblinking voltages necessary for display of the different symbols and rolling ball symbol on the PPI screen. The symbols are displayed by the electron beam in form of short straight lines arranged one after other, the visible portions being un-blanked whereas the junction lines are blanked. The character generator analog also converts the digital signals generated by the digital character generator to corresponding analog voltage step for x and y

deflection. After completion of the character, the character generator supplies a stop pulse to the trapezoidal wave generator to reset the positioning voltages for the displayed character to zero. Thereby the electron beam is traced back to its origin.

24. The assemblies forming the 'target data transmission' functional group are used to process the target data which has been prepared by the 'target data preparation' group for transmission to the MCC.

25. The 'Data Transmitter Module first adds 12 bits (04 bit start, 04 bit stop and 04 bit parity) then converts the parallel TTL signals into serial data signals of 40 bits of 04 packets and then outside the DAP the data to be transmitted are frequency modulated, whereas within processing unit the signal are processed in a digital form. Therefore they are applied to the 'Data Channel Modulator' where the conversion from TTL to FSK format (frequency shift keying) takes place. The center frequency is 1700Hz; 1700-400 Hz represents logic 1 and 1700+400 Hz represents logic 0. The following level 'Level Adaption Circuit is required for compensation of the losses introduced by the SILLACS system interconnection cables. The level adaption circuit is connected between 'transmission group' and the radar link equipment. It converts the unbalanced signal into balanced signal and it also contains an attenuator stage to control the level (-14.8 dB or -6.1 dB) of outgoing data.

26. During execution of the processing unit test, the output of the data channel modulator is connected to the input of the line equalizer assembly via switching sub-module in level adaption circuit. Thus permits testing of the data transmission and reception path.

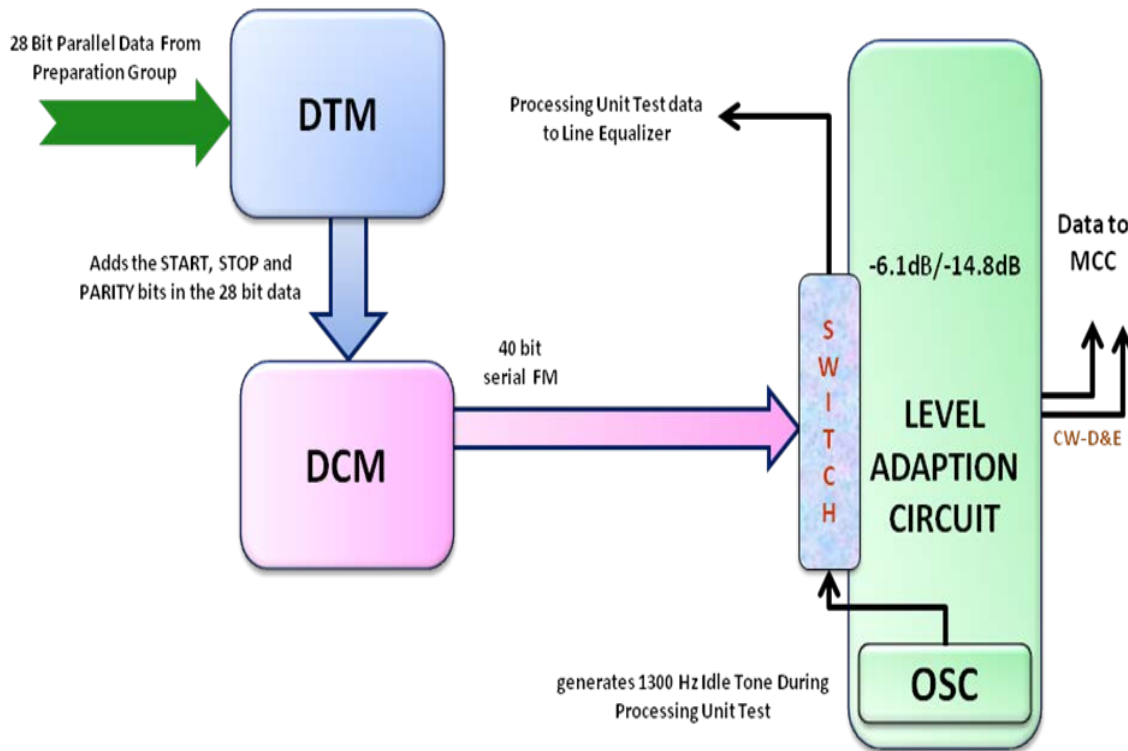


Fig 8-11 Signal Flow diagram of Target Data Transmission Group (To MCC)

IFF Remote Control Processing

27. The interrogation mode and the expected code of the IFF interrogator set can be controlled from IFF control box in the operator consol and IRCP (IFF remote control panel) in the MCC. The 'IFF remote control' modules are housed in the radar data processing unit. The main parts are the receiver part, the IFF remote control and the IFF display control in the processing unit and the display module in the control panel of processing unit.

28. The data supplies from the MCC in binary frequency modulated form are first applied to a 'line equalizer assembly'. The line equalizer assembly is used to compensate for long delay attenuation distortion generated during transmission. The output of 'line equalizer assembly' is connected to 'data channel demodulator' which converts the received binary signals into digital TTL signals. These are applied to the 'data receiver module'. This module detects the data to be processed from the applied bit pattern, eliminates the faulty ones and provides the data words at its output in a parallel 28-bit form. Further multiple word errors and amplitude variations of incoming data are processed by IFF Remote control module and Display Control module.

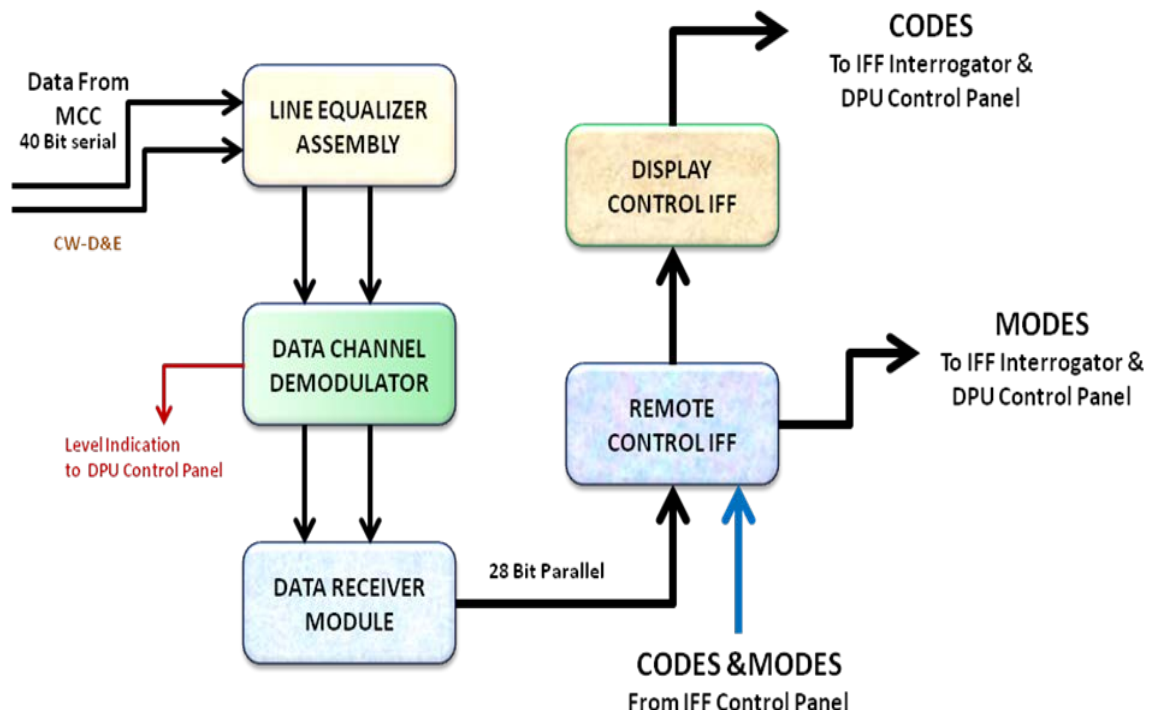


Fig 8-12 Block diagram of IFF Remote Control Processing (From MCC)

29. As a part of the control panel the display module is used for the display of codes and modes under remote control status and test operation condition.

Modes of Operation

30. Different modes of operation of operations are possible for extraction, preparation transmission of radar data:-

- (a) **Automatic Mode.** In automatic mode of operation, the target appearing at the PPI are automatically extracted, prepared and transmitted to the MCC.
- (b) **Manual Mode.** In case of overload due to radar jamming, the operator, can switch from automatic mode to manual mode of operation. It is possible to mark targets on the PPI unit with symbols and to track them manually. There are 7 or 8 different switches and 16 symbols can be generated on the PPI with the help of rolling ball assembly.

(c) **Mixed Mode.** The automatic and manual mode of operation can also be combined. Therefore, it is possible to mark targets whose data are extracted automatically by additional symbols. This possibility of identification of course can also be extended to symbolize land marks, disturbance and jammer sources.

(d) **Test Mode.** For checking the functions of the processing unit the appropriate input signals are simulated by means of built in test equipment. The results of the individual test cycles are compared with a reference and leads to a failure indication on the control panel. Furthermore, a test figure is generated on the PPI, which must be controlled by the operator.

DPU Control Panel

31. The control panel of the radar data processing equipment set serves as an input, control and display unit. The processing unit control panel incorporates all pushbuttons, push-switches and indicators for the monitoring and the control of the processing unit. The following functions are performed by the DPU control panel:-

- (a) Switching ON/OFF power supply
- (b) Selection of individual operating modes
- (c) Start of transmission and processing unit test
- (d) Selection of symbols to be set manually
- (e) Selection of position co-ordinates
- (f) Fault reporting
- (g) Indication of IFF Mode and code set in the MCC/GMCC
- (h) Indication of memory status

32. When pressing the push buttons or push switches a 0 V control signal is supplied via the appropriate switching contact of the control panel to the control panel input stage, which is connected to the micro processor bus. After processing of control signals, the corresponding driver in the lamp driver is controlled by the micro processor via the control panel output stage. The lamp driver causes illumination of the lamp of the engaged push button.

33. On the rear of control panel, a buzzer is provided, which is used for acoustic representation of all symbols set on the PPI and transmitted to the MCC. Switch, "ON" and switch, "OFF" off the buzzer is possible via the bush switch Audio Indication.



Fig 8-13 Front View of DPU Control Panel

Self test Bite

34. The radar data processing unit contains two built-in 'Off-line' test facilities.
- (a) The 'processing unit test' performs a check of 'data processing equipment' during this test data output to MCC is inhibited during this test.
 - (b) With the 'transmission test' the whole system is checked functionally and in addition, 'the transmission line to MCC'. This test can also be called up from MCC. During both test processing if the normal target data is not possible.
35. For signaling of the faults during normal operation, three indicators are provided on the control panel.
- (a) The '*antenna rotation*' indicator illuminates when either the radar antenna does not rotate or angle processing in the respective subassembly is incorrect.
 - (b) The '*clock*' warning lamp illuminates when the main program 'PU' of the microprocessor failed cycling through its loop.
 - (c) The '*level*' warning light indicates that the signal voltages input from the MCC is too low for proper operation the IFF remote control.
36. The 'processing unit test' is called up by depressing the appropriate push-switch on the control panel. This program (PUtest) possesses second highest priority and only can be interrupted by the 'transmission test' program. Immediately after starting the 'processing unit test', a red and green lamp light up within the depressed switch, indicating that the test is running. After 10 sec, if no fault is detected the red lamp extinguishes and green lamp stays "ON" and it indicates a successful conclusion of the test. Also a test pattern is shown on the PPI as follows.

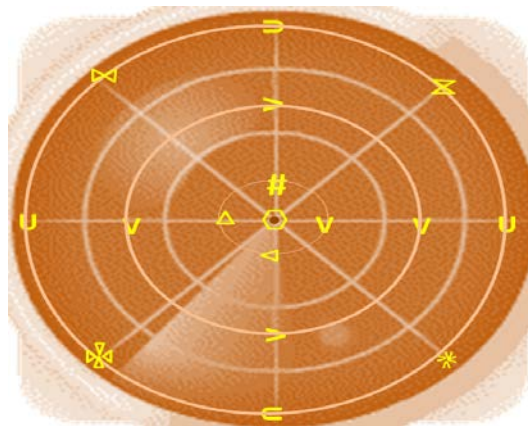


Fig 8-14 Pictorial view of test pattern (PUTest) on PPI

37. If a fault arises during test sequence, only the red lamp will continue to be lit and test pattern is displayed too. For further examination of fault 'Test I/O Module' must be plugged into a connector provided for this purpose in the processing unit. If the test program is now restarted, the test program sequences are indicated in the 'Test I/O Module' indicator. When fault occurs, the indication stops at the last step performed and an error code is displayed. From test step and fault number the defective subassembly can be determined.

	✓ .	X	X
Test step	2	0	1
Error code	3	4	2
	0	0	0
	0	3	4

Fig 8-15 Pictorial view of Test I/O Module indicator

38. During the '*processing unit test*' program following checks are performed.

- (a) RAM test
- (b) ROM test
- (c) Interrupt test

- (d) FIFO test
- (e) Extractor test
- (f) Symbol O/P test
- (g) Data transmission test
- (h) Remote control test
- (i) Test display on PPI test

39. The '*transmission test*' is used for testing the entire system, including extractor and angle encoder within the radar site as well as the plot display at MCC. During this test, no processing of MTI, ECM or AVA signals from the MPDR radar set is performed because the TRANSM TEST mode of control excludes the AUTO MODE of control. During TRANSM TEST simulation, signals are processed and that in the same manner as normally the target signals. For this purpose the simulation signals are applied to the general MTI input in the control stage. The processed signals initiate messages which are provided with a separate identification.

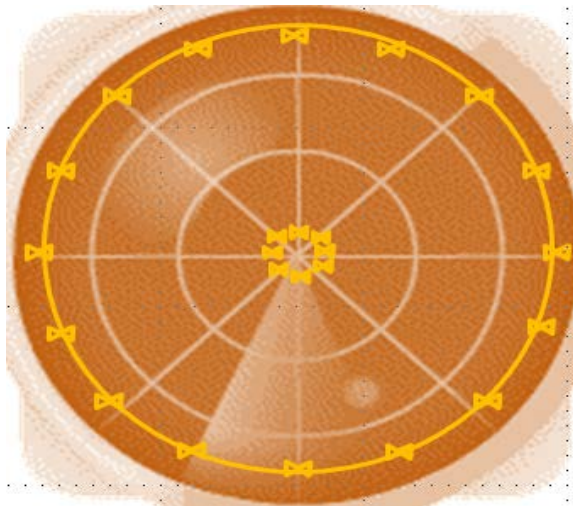


Fig 8-16 Pictorial view of test pattern (TRANSMTEST) on PPI

Rolling Ball Assembly

40. The rolling ball assembly, which is inserted in the desk, provides pulses in proportion to path changes. These pulses are generated by the interruption of light

barrier by a slotted disk. The light barrier falls on to a photo transistor. The signal for the sign in the case of directional change is determined by a comparison of the output voltages of two photo-transistors arranged next to each other.



Fig 8-17 Pictorial view Rolling Ball Assembly

Power Supply Unit

41. It requires an input voltage of 220 V AC $\pm$ 20% 47-63 Hz. It operated on the principle of electronic switch over of the power transformer by means of series opposed and controlled semiconductors. This switch over at the power transformer provides for a stabilization of the supply voltage of $\pm$ 4%. For fine control down to less than 2% the voltage regulator for 5V/11A and 12V/2A are subsequently added. The unregulated 24 V is also being provided.

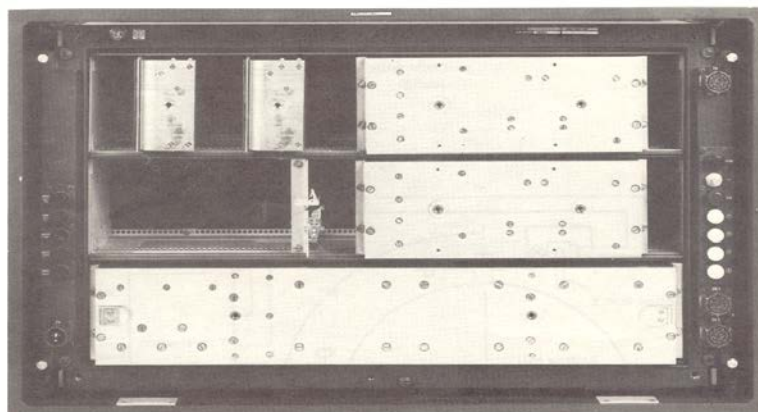


Fig 8-18 DPU Power Supply Unit

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CHAPTER No 7

ASSEMBLY AND DISMANTLING

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CHAPTER No 7

ASSEMBLY AND DISMANTLING

NOTES ON PREVENTION OF ACCIDENTS

WARNING

- *This equipment develops lethal high voltages. Extreme caution should therefore be exercised during all operations.*
- *The radar set should first be switched off if units or individual modules have to be withdrawn from the equipment cases*

1. Suspend step ladder securely from the rear of radar equipment truck. Step onto antenna assembly truck only via the steps on the dropped front supporting legs and hold on to the driver's cab.
2. Clean dirt and snow from steps and roof of the trucks as well as the structure of the antenna assembly truck.
3. Don't leave any parts or tools upon the platform of the antenna assembly truck to avoid causing injury to personnel.
4. Do not leave tools and parts under the auxiliary supports or under the outrigger where they might be left forgotten.
5. Prior to connecting the power supply, ground the radar system correctly. Check that all ground cables are securely connected to the equipment.
6. Damaged connecting cables must be replaced immediately.
7. Units may only be withdrawn from the equipment cases by trained personnel. Caution high voltages.

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8. If the indicator is withdrawn from the equipment case, implosion of the CRT may occur unless carefully handled.
9. Tools and the like must on no account be allowed to come into contact with the glass envelope of the CRT.
10. Before any work on the antenna, ensure that the safety switch on the antenna pedestal is set to 0 (horizontal position) (Resultantly transmitter power and antenna motor will be switched off).
11. During work on the antenna, ensure that the illuminated key STBY (RECEIVE) or OFF in the receiver is pressed (lamps light) the transmitter is then inoperable.



NOTE

The transmitter is automatically switched off antenna rotation will be stopped when the antenna reflector is lowered to a height of approximately 1 m above the drivers cab

12. If not absolutely required, switch off the antenna power supply (set Program switch at control box on the antenna assembly truck to "0").
13. If the antenna is stationary with the transmitter switched on, avoid the immediate radiation area of the antenna between the horn radiator and the reflector.
14. Do not erect the antenna mast as long as side sections are still in transport position to avoid damage to the reflector.
15. Do not erect the antenna mast with the rear left outrigger support still retracted and the support spindle withdrawn, otherwise damage to the RF feeder in the cable duct of the antenna mast will occur.
16. Always remove the RF cables ZM1 from transport position before erecting the antenna mast.

17. Erection of the antenna mast and disassembly of the antenna reflector should be performed only at wind speed not exceeding 60 km/h. Lowering of the antenna mast is possible at higher wind velocities, but on no account must the antenna be dismantled.

18. If wind speeds are expected with gusts exceeding 160 km/h, retract antenna mast.

NOTE

Special measures in the event of icing of the antenna are not necessary if the thickness of the ice on all sides does not exceed 1 cm and the wind speed is not greater than 95 km/h.

19. Check that the ground is suitable when selecting the site for the radar system.

SITING THE RADAR SYSTEM

20. For reliable acquisition of all low flying objects within the surveillance area of the radar system the selection of the antenna site is of primary importance. The antenna must be erected to such height that the radar beam can reach the ground almost everywhere within the radar range and will not be obstructed by trees, hills or other obstacles. The radar antenna must be sited on raised ground. The area for setting up the two trucks and the trailers should not be less than 100 m² and the gradient of ground below the antenna assembly truck should not exceed 15 %./The radar equipment truck should, whenever possible, be placed in a level position for easy operation of the equipment. The nature of the soil must be solid and must not soften even in rainy weather. If necessary prepare the ground where the outrigger supports are positioned, with stones or blocks. In winter time remove ice and snow in order to ensure the outrigger supports do not move. If necessary, the tilt of the antenna reflector maybe adjusted within the limits of -1.5° to + 8.5° by means of the large turnbuckle. The tilt is indicated on the scale on the right bearing of the reflector center section.



Fig 8-1 Radar site

POSITIONING THE TRUCK

21. The maximum distance between the trucks and between the generator and the radar equipment truck are determined by the length of interconnections cables. Therefore a distance of 20 m between radar equipment truck and antenna truck and 25 m between radar equipment truck and genera must not be exceeded when positioning the vehicles. Two three-phase supply net-works can be connected to the RF terminal box. On transformer-1 it is possible to change over to the power used. A dual rotary joint, rotatable in two planes perpendicular to each other, is provided on the radar equipment truck for connection of the thick RF coaxial cable ZM 1. Depending upon the distance between radar equipment truck and antenna assembly truck one or four coaxial cables ZM 1 is used which can be coupled directly without adapters. The connection to the base of the antenna mast is facilitated by a swiveling cable support and a cable slip. The hook wrench (C-spanner) is utilized for tightening or loosening the coaxial connector. The wrench is contained in the case "accessories and spares".

- (a) Check nature of ground. Minimum compressive strength of the ground must be $25 \cdot 10^4 \text{ N/m}^2$ (2.5 kp/cm^2).

NOTE

- *Normally this value is reached with certainty, if the antenna truck can drive over the ground without sinking in deeply. Note that the stated compressive strength is required only at those points where the outrigger supports are positioned.*
- *The supplied support blocks must always be used.*
- *At the support points the ground must be such that the outrigger support discs are equally supported at all points.*

CAUTION

Outrigger supports and auxiliary supports must not stand on sewer covers! Tilting of the antenna truck can occur, if the ground is not checked carefully prior to set-up.

- (b) The slope of the ground must not exceed 15%, i.e. an angle of 8.5° or a gradient of 15 cm over a distance of 1 m.
- (c) Position antenna assembly truck
 - (i) Wheels on one side of the truck should be standing on a horizontal line (truck parallel to slope).
 - (ii) Clean steps. Step on to antenna truck only via the dropped front outrigger supports.
- (d) Position radar equipment truck
 - (i) Position truck in such manner that tight rear corner is within 20 m from the middle of the left side of the antenna truck.
 - (ii) Pullout step beneath the right side-door of radar equipment truck.



Fig 8-2 Setting of the step and step ladder

- (iii) Open rear door.
- (iv) Remove step ladder from rear of equipment truck and engage into hook beneath the rear door: Remove snow or unevenness from below step ladder in order to be able to open and close the door.
- (e) Open terminal boxes
 - (i) Take allen wrench from drawer in the equipment room.
 - (ii) Use it to open RF terminal box, communication antenna connection box, AF terminal box on right side of truck.
 - (iii) Flap the lids of the terminal boxes up and secure them with the corresponding straps and press buttons.



Fig 8-3 Opening of terminal boxes

- (f) Shut off vehicle engines

22. Leave battery master switches of both trucks switched on. The lever switch under the passenger's seat must be placed in the horizontal position. The lever is removable.

NOTE

The battery switch of the radar equipment truck also connects the field communication battery (24 VDC) to the control panel 24 V for the illumination.

Connecting and Switching ON the Ac Power

23. In order to quickly set up the radar system, the crew should be divided into two groups; one operating the hinged antenna mast, the other laying the cables between the trucks and connecting them. Inside the trucks, the cables interconnecting the different units must be connected, where necessary. Disassembly can be performed in the reverse order, also by 2 teams. If assembly of radar system is performed by two groups, one group erects the antenna, the other switches the radar unit.

Connect Power Supply

24. Put the cover on the threshold of the rear door so that it is possible to enter and leave freely and to prevent the RF cable (UZ) from being damaged.

NOTE

The connectors of the cables must not be dirty or damaged. Do not place cable ends on ground without protection caps plugged on. If necessary, put something underneath

- (a) Unroll cable MV 1 from cable reel and, having removed the protection cap, connect it between the RF terminal box and the control unit "North".
- (b) Connect power supply line to the generator and equipment truck.

Ground System Correctly

- (a) Remove ground auger from mounting in left aisle of radar equipment truck.
- (b) Screw ground auger into soil as far as possible, within 10 m of right side of radar truck.
- (c) Release ground cable from rear right wheel arch.
- (d) Secure cable terminal at one end of cable beneath screw on ground auger head using 17 mm open end wrench. Position terminal at other end of cable under wing nut in RF terminal box and tighten wing nut.



Fig 8-4 Connecting the grounding cable

- (e) Ground the generator as described in step (d) with a second ground auger, which is located in a mounting on the floor of the generator trailer. Switch "ON" generator and wait for 5 minutes to be stabilized then check the parameters such as voltages and frequency of output after it switch on the circuit breaker.

Switch ON Transformer 1

25. Turn main switch to input GENERATOR or POWER-MAINS to which the power supply line is connected in the RF terminal box.



Fig 8-5 Switching ON of Transformer 1 and Checking of phase sequence

Switch ON Illumination

26. For switching on the indoor illumination, set the rotary switch on the battery control panel 24 V in the operator's console from position DRIVE to position FM OPERATION BATT. If necessary, switch on the rectifiers 1 and 2 in the side aisle. Actuate the light switches in both rooms.

Check Phase Sequence on Power Distributor

- (a) Check lamp phase sequence WRONG. If lamp is illuminated the phase sequence is incorrect.
- (b) Then correct the phase sequence by turning the switch PHASE SEQUENCE through 180° into position I or II. The lamp must extinguish.

Close Antenna Switch

27. Set the antenna operation PACCO switch on the power distributor to "I". The arrow on the removable knob must points upwards.



Fig 8-6 Setting the antenna operation PACCO switch

SUPPORTING THE ANTENNA MAST SUBFRAME

Switch ON the Antenna Assembly Truck Hydraulic Pump

28. Move the pump operating program switch at the right under the protective flap on the control box "North" initially from the OFF position clockwise to "Y" (star) and after a few seconds further round to 'Δ' (delta). This will start the three-phase hydraulic pump motor.

29. Check if the green telltale lamp, on the connector panel at the right of the control box has come on (24 V dc electrical system and hydraulic system starting circuit on the antenna assembly truck operational).



Fig 8-7 Switching ON the antenna assembly truck hydraulic pump

30. If the lamp fails to come on then perform following checks:

- (a) Check if the lamp is not dimmed by turning the lamp holder. If this does not cause the lamp to bum brightly, check that the battery master switch under the passenger's seat is at on.

- (b) If this check is also positive. Stop the pump again immediately and trace the fault by reference to the Technical Handbook SILLACS 5801245-6-2.
- (c) If the motor does not run up (no starting noise audible) then perform following checks.
- (d) Check the control levers in front of the control box "North" are in neutral (central position) and,
- (e) The antenna drive Pacco switch on the power distributor unit in the equipment truck is at I (top).
- (f) The automatic circuit breaker (over current circuit breaker for antenna and antenna assembly truck) next to the Pacco switch on the power distributor unit is pressed in (if not, press the button), and if the left meter on the power distributor unit reads at least 190 V for all three phases (R, S and T). To check, move toggle switch up for R and down for T. Failing this, trace the fault by reference to the Technical Handbook SILLACS 5801245-6-2.

Operate the Antenna Assembly Truck Frame Interlock and Suspension Locks

31. To do so, press the "Movement 1" control lever rearwards.



If the hydraulic motor comes almost to a standstill under load, switch off again immediately and investigate the cause.

32. Check that the interlocking pin at the front left of the sub-frame has engaged with the chassis. Insert your head between the driver's doorway and step to check if the pin has engaged.



Fig 8-8 Operating the Frame Interlock and Suspension Locks

NOTE

- It is possible to check if the piston rod (Fig. 8-8) of the suspension locks between the two rear wheels have engaged in their forked housing either by inserting a hand to feel the relative positions of the components or visually by observing the way in which the rear wheels lift clear of the ground after auxiliary supports have been extended ("Movement 2").
- Operation of the suspension locks also provides verification that the frame interlocking pins have engaged correctly.
- If the equipment is being set up on very uneven or steep sloping ground, it is possible for the interlocking pins at the frame and the suspension locks to fail to engage. In this event, the interlocks are to be returned to the travel position, the program switch on control box "North" set to 0, cable MV 1 detached again, the truck started and driven a short distance to a flatter area of ground. Here the engine is to remain running, cable MV 1 to be reconnected, the pump motor restarted and the interlocking lever again operated to engage the frame interlocks and suspension locks. After stopping the pump motor (switch on control box "North" to 0) and detaching cable MV 1, the truck is to be driven very carefully back to the originally chosen site, the engine stopped and cable MV1 reconnected
- When the interlocks are engaged, the truck engine cannot be started.

CAUTION

The "Movement 1" control lever for the frame interlocking must not be moved while the vehicle is being leveled or supported.

Extend the Auxiliary Supports.

CAUTION

Make sure that no cables are trapped under the supports.

33. Partly extend the auxiliary supports by pulling the four left control levers ("Movement 2") towards the operator. The supports controlled by each lever are marked on the identification plate.



Fig 8-9 Placing of the support blocks

34. Remove the support blocks from the accessory box and place the two thicker ones at the front and the two thinner ones at the rear exactly under the auxiliary supports.

35. Extend the auxiliary supports until the truck's wheels are slightly clear of the ground.



Fig 8-10 Extending the auxiliary supports

CAUTION

Always check if the rear wheels lift clear immediately the auxiliary supports are extended. If this does not happen, the suspension locks have not engaged. In this case, retract the auxiliary supports slightly and repeat as mentioned above.

36. Continue to pull the control levers singly or in pairs towards the operator until the spirit level under the cover flap between the levers is in equilibrium. The air bubbles must be in the center of the circles.



Fig 8-11 Pictorial view of spirit level

Layout the Support Blocks for the Outrigger Supports.

37. Release the splint pin retaining the large support blocks under the right side of the truck.

38. Unscrew and remove the retaining bar and stow it between the mast and the accessory box.
39. Position the support blocks at the points where outrigger support discs will probably be located. The side of the blocks with the recessed metal strips should be uppermost.

Swing Down the Outrigger Supports

CAUTION

- *Do not operate the "Movement 1" control lever during leveling and outrigger extension.*
- *Do not pull out the outrigger support locking pin unless the load has been fully lifted from the outrigger support (or the actuating rod may fracture).*

40. Unlock all four outrigger supports. This is done by releasing and lifting the handle of each locking pin and withdrawing the locking pin.



Fig 8-12 Unlocking and lowering of the outrigger supports

41. Lower the outrigger supports by pressing the "Movement 3" control lever away from the operator.

NOTE

- *The outrigger supports extend under their own weight, pulling out the hydraulic rams at the same time.*
- *The outrigger supports with the least friction in the pivots will move first.*
- *If friction is high, it may be necessary to pull the outrigger support by hand to start it moving.*



Fig 8-13 Lowering of the outrigger supports by hand



Fig 8-14 Location of the supports blocks

42. Before the load is taken by the outrigger supports, move the support blocks under the center of the support discs so that the latter are fully supported.

43. After extending the outrigger supports reinsert the locking pins until they engage positively.

NOTE

If necessary, raise the outrigger support slightly to enable the pin to be inserted, or screw out the support disc spindle sufficiently for the pin to be pushed through.

Screw out the Support Discs

44. Screw out the support disc spindles until the auxiliary which have so far taken the vehicle weight are relieved of load. The discs are screwed out by inserting one of the Tommy bars through the hole in the spindle and screwing the spindle out manually until stiff resistance is encountered.



Fig 8-15 Screwing out the support disc spindles

45. Finally, lengthen the Tommy bar (by attaching to it the hand pump lever kept at the center of the truck next to the wing (Fender) and apply a few more turns.

Retract the Auxiliary Supports

46. Push the four control levers away from the operator until the auxiliary supports are fully retracted.

Check Leveling

47. If necessary, alter the height of the spindles on the outrigger supports to correct the mast sub-frame leveling, watching the spirit level all the time.

ALIGNMENT TO NORTH USING OPTICAL AIDS

48. Alignment of the angle transmission system of the surveillance antenna to north must be performed every time the position / direction of the radar antenna truck is changed. Generally, the antenna truck will be moved to a position already surveyed and the angle transmission system be adjusted by means of the north seeking gyroscope on the antenna assembly truck. Alignment to north can also be performed with the aid of the telescope mounted on the antenna turntable or with an aiming circle (or theodolite) in conjunction with the telescope. The synchro drum is set to north via a servo motor or manually.



WARNING

Place safety switch on antenna pedestal to "0" before manually operating the synchro drum to avoid injury to personnel due to lethal high voltages and /or antenna rotation.

49. Determine from the map the angle between north, own position, and a bearing point (Landmark) marked on the map, starting from north in a clockwise direction. Then proceed as follows:

- (a) Align antenna manually to the bearing point, using the telescope. To turn antenna utilize hand-wheel below drive motor or turn reflector by hand. The telescope is located under the console of the equipment truck.
 - (b) Place safety switch at antenna pedestal to 0.
 - (c) Remove small cover on front of antenna pedestal after removing the screws
 - (d) Turn synchro drum by means of the knurled wheel on synchro drum accessible through opening' in large pedestal cover until the angle previously determined is in line with the zero mark on the vernier on the scale. The synchro system is thus adjusted to north (antenna points to north, if scale points to 0).
50. Aim the antenna telescope at the theodolite. Use hand wheel below drive motor for convenient adjustment. The inner scale (vernier) inside the window on the antenna pedestal follows the motion. Put on handbrake on antenna pedestal.

51. Rotate the synchro drum to adjust the stator of the synchros and the housing of the shaft encoder to the angle previously determined. Then the synchro system is aligned to north. For this purpose, turn synchro drum by means of the servo motor electrically through tumbler switch or the knurled wheel until the angle measured with the theodolite is lined up with the zero mark on the vernier scale of rotary portion.

ELEVATING THE ANTENNA

Attach the Reflector Side Sections

52. Release the retaining strap between the two reflector side sections mounted on the antenna assembly truck, and stow the strap in the cabin.



Fig 8-16 Release the retaining strap and two levers on one reflector side section

53. Release the two levers on one reflector side section.



Fig 8-17 Attaching the reflector side sections

54. The first man should position himself on the edge of the antenna assembly truck between the reflector center and the stowed side section. He should lift the front edge of the reflector side section (nearest him). He should keep hold of this

edge of the side section as reflector assembly progresses. The second man should lift the side section at the other (rear) edge and move that edge towards the cab. The reflector mesh will now be pointing towards the rear. The side section should then be lowered at the rear so that the reflector mesh points down. A third man should hold the outer edge of the side section as this is done. The second man then climbs on to the truck via the cab, guides the side section into its mounting on the reflector center section (working together with the first man) and moves both locking levers over.

55. The same procedure is repeated on the second reflector side section.

Release the Horn Radiator Lock

56. Turn the locking bolt under the horn radiator and pull the pin back, working from the right side of the truck.

Release the Reflector Lock

57. Release the lock for the reflector center section at the top of the mast next to the SLS antenna. Swing the clamp out and secure it (first man).

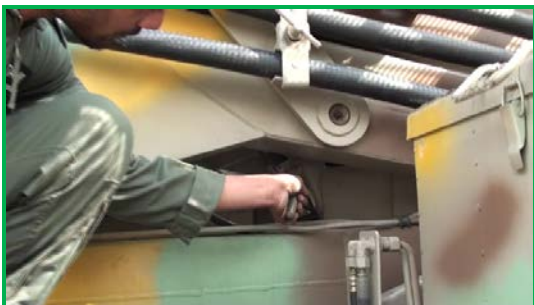


Fig 8-18 Releasing the Horn Radiator Lock and Reflector Lock

Swing up the Reflector

58. Release the large turnbuckle (second man) and swing over. The smaller turnbuckles remain for the moment in the transport position.

59. The first man should stand on the folded mast and the second should assist him from the platform of the antenna assembly truck to raise the reflector.

NOTE

- *The reflector cannot tip over backwards as a cross bar on the center section restricts the tilt angle.*
- *Make sure that no undue pressure is exerted on unsupported areas of the reflector mesh.*
- *Do not use SLS antenna as a handgrip.*



Fig 8-19 Swinging up the Reflector

60. Lock the large turnbuckle to the turntable section of the antenna pedestal (third man). Set the antenna tilt angle, if necessary. The adjustable tie bar for setting the tilt angle is located on the rear side of the reflector. When the elevation operation has been completed the turnbuckle, which serves as a support between the reflector and the upper part of the rotating device, should be attached the corresponding cross-tail stud on the scale and be locked in position. The desired angle of elevation (tilt angle) is obtained by extending or shortening the turnbuckle. The scale on the side of the antenna base indicates the maximum permissible angle of beam elevation from the horizontal (direction of propagation).

61. In order to achieve optimum coverage at ground level on the radar site, the antenna has to be tilted so that the scale shows an angle of $+3.50^\circ$.

62. The angle relevant to the direction of propagation of the antenna is indicated on this scale. If the antenna truck is located on a plain or plateau, the scale should be set to read 3.5° in order to ensure that the upper height limit of the radar's area of coverage is the same for an equal distance in all azimuths.

63. If the radar site is located on a hill or mountain top or in a depression or valley, the adjustment for optimum radar performance may vary by -1.5° to $+8.5^\circ$.

64. In order to set the antenna reflector to the required angle, loosen the lock nuts and turn the tie bar until the required angle appears on the scale, then retighten the lock nuts.

CAUTION

If the antenna tilt angle has been adjusted after operation, turn the antenna carefully by hand to check if it can rotate freely.

65. Raise the antenna boom slightly, release the two small turnbuckles at the outsides of the two boom arms (third man), lower and lock.

CAUTION

Do not alter the settings of the two small turnbuckles or else the radiation pattern of the antenna will be changed.

Make the Antenna Ready to Rotate

CAUTION

Operate the antenna brake only when the antenna is at a standstill.

66. Release the brake on the antenna turntable. Move the brake lever to a

tangential position and press it down until it presses the switch.

67. Move the safety switch on the antenna pedestal to I (vertical).

Start the Hydraulic Pump

68. Restart the pump motor as described above.

Elevate the Mast



CAUTION

- *If the antenna mast is elevated although the reflector side sections are still in the transport position, there is the risk of damage to the reflector side sections.*
- *Similarly, if the left rear outrigger support is in the retracted position with the spindle screwed in, the RF cable duct on the mast may be damaged.*

69. Press the safety button to the right of the control box "North" and, at the same time, operate the right control lever (2) ("Movement 5"). Only the upper mast arm should move at first.

70. When the antenna has been elevated far enough to permit retracting it without damaging the truck cab (blocking by the lower limit switch) on the control cam is removed), release the safety button.



NOTE

- *When the upper arm of the mast has been elevated sufficiently for the upper limit switch to be released by the control cam, the lower arm should also begin to rise.*
- *When the lower arm reaches its fully raised position, the upper arm*

71. After the desired mast height has been reached, reset the control lever to

the central (neutral) position.

72. Turn the program switch on the control box "North" to R (Rotation).

CONNECTING THE REMAINING CABLES

Lay Down the Thick RF Cables ZM1

73. Release the six retaining clips for the RF cables on the right side of the mast and place next to the truck.



Fig 8-20 Releasing and connecting thick RF-Cable ZM 1.

Connect Thick RF-Cable ZM 1.

CAUTION

- When connecting the two thick RF-cables together and to the rotary joints on the antenna and radar equipment truck, check for clean contacts to avoid arcing.
- Place cable connector connecting the four RF- cables on a support, if ground is wet or dirty.

(a) Connect the four 5 m sections of RF cable together, if required. Remove protective caps from one cable end with a coupling nut and from another cable end without a coupling nut and join the two ends. Take care not to lose the gasket.

- (b) Release the swiveling cable support in front of gyroscope assembly on radar assembly truck and fold forward.
- (c) Unscrew protective cap from the cable end with coupling nut.
- (d) Place cable end into swiveling cable support under cable clip.
- (e) Connect cable end to rotary joint of cable duct at mast base and secure. Check that rubber ring is seated correctly.
- (f) Release dual rotary joint from transport bracket in RF terminal box of radar equipment truck.
- (g) Unscrew protective cap from the cable end with- out coupling nut and connect cable to dual rotary joint in RF terminal box.

CAUTION

Do not kink cables during installation; observe minimum radius of curvature of 1 m. Otherwise transmitter switches off by reflections.

- (h) Tighten all coupling nuts of RF cable with hook wrench (C-Spanner).
- (i) Place joint of RF cables on the ground in such a way that it does not lie in water; support, if necessary, with canister stone or wooden block.

DISMANTLING THE RADAR SYSTEM

- 74. Switch off the radar set. Set the radar system to LOCAL at the receiver control panel and then to OFF.
- 75. Switch off the radar data processing equipment set.
- 76. Switch off and stow the air conditioners
 - (a) Set switch on the 3 air conditioners to OFF.
 - (b) Set switch on power distribution box to OFF.
 - (c) Pull the air conditioners slowly inside the vehicle, flaps close

automatically; secure the air conditioners with the screws provided.

77. Switch off illumination.
78. Start the hydraulic pump.
 - (a) Turn on the battery master switch below the passenger's seat of the antenna assembly truck (knob horizontal).
 - (b) Lift the protective cover on the control box "North".
 - (c) Start the hydraulic pump drive motor by moving the program switch on the control unit "North" to "Y" and after a few seconds further round to "Δ".
79. If the pump motor fails to start, check if
 - (a) The control levers in front of the control box are in their central (neutral) positions.
 - (b) The ANTENNA overcurrent circuit breaker in the power distribution unit has not tripped, if so, reset by pressing the button.
 - (c) The ANTENNA switch next to the overcurrent circuit breaker is turned off.
 - (d) The meter on the power distribution unit shows a reading of at least 190 V for all three phases. Check by moving the toggle switch to the positions R, Sand T. The green oil pressure telltale lamp on the control panel to the right of the control box "North" must light up. If it does not come on, the pump must be stopped again immediately and the fault must be traced by reference to the Technical handbook, section 4, SILLACS 5801245-6-2.

NOTE

If the pump has fault, the mast can be retracted with the hand pump.

- (e) Retracting and folding the antenna mast

80. Pull the control lever for mast movement ("Movement 5") towards the operator until the lower edge of the antenna pedestal is 1 m above the cab; at this point the mast movement will cut out automatically.

CAUTION

- *Observe the mast movement to ensure that mast lowering and antenna rotation in this height zone automatically cut out in good time.*
- *Do not use the brake to halt antenna rotation.*
- *Never lock the control lever on during lowering as it may be necessary to open it in a hurry if the automatic cutout fails to work.*

81. Return the control lever to the center position.

NOTE

If the mast was only partly elevated and then moved up and down fairly often, it is possible for the upper arm to retract to its rest position before the lower arm. In this event the antenna pedestal could damage the cab roof if the red safety button is pressed to continue mast retraction. Instead, the mast should be elevated slightly again and retraction of the mast should be resumed or the movement of the upper arm should be blocked by pressing the hand blocking lever on the mast base at a right angle to the pipe until the lower arm is folded in sufficiently.

82. Set the safety switch on the antenna pedestal to "0".
83. Turn the antenna reflector to face to the rear and that the red marks on the left side of the base and on the turntable match. One man must remain on the platform as the mast is lowered further to check that the horn does not strike the mast fork. Apply the brake lever in a strong wind.
84. When the antenna pedestal can no longer strike the cabin and the horn radiator is correctly positioned, press in the red button to the right of the control box North (Movement 5) until the antenna pedestal is lowered firmly on to the platform.
85. Disconnect the thick RF cable ZM 1
- (a) Unscrew RF cable ZM 1 from the radar equipment and antenna assembly trucks, part it, if necessary, and screw the protective caps on the cable connectors and the flanges on the antenna assembly and radar equipment trucks.



CAUTION

Keep the cable connectors clean.

- (b) Turn the dual rotary joint in the RF terminal box of the equipment truck to the transport position and screw to the mounting in the RF terminal box.
- (c) Secure the thick RF cables ZM 1 to the right of the upper arm of the hinged mast so that the ends do not project beyond the vehicle. Secure the clips.
86. Stow the antenna reflector
- (a) Raise the horn radiator slightly (first man) and flex it a little at the same time. Pull out the pins in the small turnbuckles by pressing the button on the corresponding handle third man.

NOTE

- *Release the large pin only when the personnel have secured the reflector.*
- *The pins have a ball catch. The ball is released when the button in the center of the handle is pressed in.*
- *If the pins cannot be pulled out, flex the reflector slightly for withdrawing the large pin and flex the boom for withdrawing the small pins.*
- *Make sure that the pins are always well greased.*
- *Never strike the end of the pin with a hammer.*

- (b) Raise and secure the turnbuckles.
- (c) Slowly lower the horn radiator.
- (d) Position the horn precisely in the center of the recess.
- (e) Position at least one man (first man) on the lowered mast to hold the reflector.
- (f) Pull out the pin of the large turnbuckle by pressing the button on the handle.
- (g) Lower the reflector. During the last third of its movement, the second man should also hold the reflector.

NOTE

Make sure that no undue pressure is applied to unsupported areas of the reflector mesh and that the reflector does not rest on the clamp.

- (h) Lock the horn radiator into position from the right side of the truck.

NOTE

If the horn radiator was lowered crookedly and is resting on the locker raise the reflector slightly and alter its position until the horn drops into the lock.

- (i) Apply the brake lever on the antenna pedestal turntable.
- (j) Lock the reflector center section to the upper mast arm.
- (k) Release the reflector side sections, detach them from the center section, turn them round and secure them to the transport mountings.

NOTE

- *This is done by positioning the first man in front of the reflector and the second man behind it (positions on vehicle referred to forward travel direction). The third man should support the end of the reflector side section (Fig. 5-22). The engaged locking levers should then be raised and the reflector side section pulled out slightly. The man at the rear (first man) keeps hold of the reflector and does not shift his position.*
- *The front man (second man) gets down from the vehicle and the side section securing face IS then swung down. After this, the second man takes hold of the reflector side section again at the same point as before, moves to the rear turning the reflector side section through 180°. It is then placed on the transport mounting with the convex face inwards and locked into position by turning the levers.*

- (l) Pass the strap tightly over the mast so that it does not trap any branches. Secure the strap to the reflector side sections and lash firmly.

87. Extend auxiliary supports

- (a) Place small support blocks beneath the auxiliary supports make sure that no cables are trapped beneath the supports or the blocks.
- (b) Operate the four left control levers ("Movement 2") to extend the auxiliary supports until the load is taken off the outrigger supports. To extend the auxiliary supports, the operator must pull the levers towards his body.

88. Retract outrigger supports

- (a) Lift and pull the outrigger support locking pins; it may be necessary that a man must raise the end of the outrigger slightly.
- (b) Insert a Tommy bar and screw the support disc spindles fully in.

NOTE

The truck will not be within the permitted transport dimensions unless the outrigger support discs are screwed in fully.

- (c) Secure the support spindle Tommy bars at the rear left of the truck with the rubber loops provided.
- (d) The hand pump lever should be in its mounting ahead of the left rear wing (fender).
- (e) Fold in the outrigger supports. This is done by pulling the "Movement 3" control lever of hydraulic control system towards the operator. The outrigger supports fold in one after the other.

WARNING

The outriggers must not be folded in by hand, or else the hydraulic rams will not retract and the ropes will remain slack. If the locking pins are removed, the outriggers will fall outwards suddenly and may lead to injury.

- (f) Secure the outrigger supports by pushing in the locking pins.
- (g) Engage the safety hook in the locking pin handles.

89. Retract the auxiliary supports

- (a) Retract the auxiliary supports completely using the four left control levers until the catches engaged. The control levers must be pushed away from the operator.
- (b) Check if all four auxiliary supports have engaged (the discs must be raised fully against the rams).

90. Release the truck suspension locks and frame interlock.

- (a) Pull the "Movement 1" control lever towards the operator to release the suspension locks and frame interlocks.
- (b) Check if the suspension locks have been fully released. When the ignition key is turned, the SUSPENSION LOCK telltale lamp on the driver's instrument panel must not light.

NOTE

The truck engine cannot be started until this lock has been fully released (starter relay inhibited). If the lock has not been fully released, the SUS- PENSION LOCK lamp will be illuminated when the ignition key is turned.

91. Stow the support blocks on the antenna assembly truck

- (a) Place the four small support blocks in the large accessory box.
- (b) Stack the large support blocks in front of the accessory box.
- (c) Insert the retaining bar through the support blocks and screw tight.
- (d) Secure with a split pin at the bottom.

92. Switch off the control box "North"
- (a) Return the program switch under the protective cover on the control box "North" to 0.
 - (b) Close the cover.
93. Turn off the power supply.
- (a) Reset the main switch at transformer 1 on the equipment truck to 0.
 - (b) Switch off the motor of the generator set. The two switch-on contactors will cut out automatically.
94. Loosen cable and coil on cable reel.
- (a) Disconnect the cables ZM 2 and ZM 3, MV 2 and MV 3, MV 1 from their connectors on the control unit "North", control panel for radar antenna.
 - (b) Cover sockets and cable connectors with protective caps.
 - (c) Coil the cables in the reverse order, on their respective reels, if possible, inside the equipment truck.
 - (d) Close the Camlocs on the cable racks.
95. Disconnect ground cables
- (a) Unscrew cables from the ground auger (17 mm open end wrench).
 - (b) Pull out the earth drills and ground augers and stow the earth drills on the floor of the generator set trailer and secure the ground augers in mounting in the equipment truck.
 - (c) Coil up the cables and secure them in the equipment truck.
96. Close the terminal boxes on the equipment truck
- (a) Close the RF and AF terminal boxes outside on the right of the equipment truck.
 - (b) Place the key in the mounting at the rear door in the equipment truck.

97. Stow truck equipment kit in both trucks
- (a) If used, stow the tools, camouflage net, erecting poles, spade etc.
 - (b) Detach the sight telescope from the antenna and stow under the console of the equipment truck.
98. Prepare the system for transportation.
- (a) Check if all the equipment in the equipment truck is firmly located and lash down as necessary.
 - (b) Close the "Accessories and spares" case in the left aisle of the equipment room by shutting the lid.
 - (c) Strap the pivoting lights above the console to the ceiling.
 - (d) Screw down the swivel chairs.
99. Prepare the generator set trailer for transportation
- (a) Close the generator set cover.
 - (b) Secure the canisters.
100. Prepare the trucks to move.
- (a) Set the rotary switch on the battery control panel of the console to the position DRIVE
 - (b) Stow the ladder and step for the equipment truck correctly.
 - (c) Check if all vehicles are ready to move off.

CAUTION

When ready to travel, the antenna truck exceeds the rail loading gauge for Europe by approx. 30 cm. accordingly, for rail travel on European routes, the side reflector sections must be removed from the truck and stowed separately on the goods wagon.

DISMANTLING IN THE EVENT OF AN OPERATION FAULT IN THE ELECTRICAL OR HYDRAULIC SYSTEM

Dismantling in the Event of 24 V Supply Failure

101. Identifying the fault: the green oil pressure telltale lamp does not come on when the pump is running, although it is possible to move the auxiliary supports (left-hand levers).

(a) First of all, attempt to restore the 24 V supply. The battery master switch under the passenger's seat must be horizontal. Check the fuse in the junction box under the gyroscope platform, after unscrewing the lid of the box.

(b) Switch on the truck lighting briefly to check if the battery is not flat. If the lights do not come on, unroll the external charging cable from the reel on the far left in the equipment truck and connect the battery outlets of both trucks. The truck battery outlets on the antenna assembly and equipment trucks are under the passenger's seat. The telecommunication battery outlet is under the equipment truck passenger seat. Both battery master switches must be on. If the light still does not come on, remove the cable again immediately. If the oil pressure telltale lamp now comes on, continue to dismantle normally.

(c) If the lamp does not come on, pull the "Movement 5" control lever towards the operator. First of all press the two push buttons on the solenoid control valves below the right side of the control panel. Remove the rubber caps which are mounted on the middle of the valves. The valve can be actuated by pressing the pushbuttons or with a tool which has a diameter of less than 4mm.



NOTE

The upper solenoid control valve need only be kept pressed in until the upper limit switch contacts the control cam. The red knob for mast movement on the control panel next to the control box "North" need not be pressed.

(d) Make sure that the horn radiator does not strike the mast fork.

(e) Stop retracting the mast before it reaches cab height and turn the reflector so that the horn radiator can enter the mast fork and the red marks on the left side of the base and the turntable match. Check also if, as a result of the upper solenoid control valve not having been pressed sufficient the upper arm has not already reached its limit position, so that the antenna pedestal can strike and damage the cab. In this event, perform the movement of the upper arm by pressing only the lower solenoid control valve button while moving the mast with the "Movement 5" control lever.

(f) If the horn radiator is in the correct position and the upper arm not folded too far, retraction of the mast can continue.

DISMANTLING IN THE EVENT OF A HYDRAULIC FAILURE



NOTE

- *Hand pump operation in an emergency is intended only for retracting the mast.*
- *The hand pump is located on the left side of the vehicle in the vicinity of the control panel, and is operated by attaching a pump lever. Operation of the controls, when the emergency hand pump is in use, does not differ from the procedure described for normal working; with the motor-driven pump. When the upper arm is two-thirds retracted, the hand blocking lever must be set at a right angle to the pipe to shut off the oil flow until the lower mast arm has retracted fully. The lever can then be reset to open valve and allow oil to flow again so that the upper arm can be lowered.*

DISMANTLING IN THE EVENT OF FAILURE OF BOTH THE 24 V SUPPLY AND THE HYDRAULIC PUMP

102. While lowering the mast with the hand pump, keep the two pushbuttons for the solenoid control valves pressed.

DISMANTLING WITH A DAMAGED HYD- RAULIC RAM

103. Identification: the upper or lower mast arm will not move. One half of the mast arm is pulled over at an angle.

(a) In this event, the jammed hydraulic ram can be retracted by slightly opening the bleed screw (marked white) on the return side of the ram (the side with the piston protected by the gaiter). Open the screw only until a little oil emerges. Never remove the screw completely. Oil then escapes to the atmosphere from the piston. Catch the escaping oil.

(b) Perform the mast retraction slowly to prevent the mast from Jamming at an angle.

INSTRUCTIONS FOR TRANSPORTATION AND STORAGE

104. When the vehicles are in motion, no persons must remain in the equipment and control room of the radar equipment truck.

TRAVEL

105. Controls.

(a) Check if the auxiliary supports on the antenna assembly truck are locked in position. If necessary, raise them with the aid of the hand pump. The pump lever is secured with straps ahead of the left rear wing (fender).

(b) Observe the maximum vehicle width of 2.50 m.

(c) Observe the maximum height of 3.20 m.

(d) The maximum admissible gradient for towing a trailer is 38% (21°), the admissible lateral tilt 20% (11°).

(e) Ground clearance beneath the auxiliary supports on the antenna assembly truck is 51 cm at the front and 42 cm at the rear (spare wheel 36 cm).

LOADING ON RAIL WAGONS

106. In the road going condition, the antenna assembly truck exceeds the European loading gauge (3.07 m) by approx. 30 cm. For this reason, if the truck is to be conveyed by rail on European routes, the side reflector sections stowed on the truck must be removed and carried separately on the rail wagon.

107. To guarantee military security, take the following precautions.

- (a) Remove the crystal cassette from the drawer in the radar equipment truck.
- (b) Remove all Technical Handbooks.
- (c) Remove all operating documentation including over-lay films.
- (d) Lock and seal (with lead seals) the doors of the equipment room.

LOADING INTO AIRCRAFT

108. Antenna assembly truck: detach the reflector center section and remove the reflector side sections. Drive forwards into the aircraft.

109. Equipment truck: fit the special air transport wheels (rims).

STORAGE

110. If the radar system is taken out of service for a lengthy period and stored in the open air, the operating personnel must take the following precautions.

Equipment truck

- (a) Detach all visible cables from the front of the units.
- (b) Check if the covers for the RF and AF terminal boxes on the right side of the equipment truck are locked.
- (c) Remove the case covers from their mountings in the left aisle of the equipment room.
- (d) Fit covers of all cases to make an airtight seal.
- (e) Close the partition door inside the vehicle.
- (f) Close and lock the doors of the vehicle.
- (g) Remove both batteries (truck and telecommunication batteries) and return them to the store. If batteries are not removed, they must be recharged every four weeks.

Antenna assembly truck

- (a) Check if all covers on the antenna pedestal, on the function boxes under the gyroscope platform and on the connector flanges and cables are securely closed.
- (b) Check if the cover on the control box "North" and the unit's threaded connection to the case are tight.
- (c) Grease all pins and bearings.

TRUCK REPAIRS

111. If the radar equipment truck is repaired in a civilian workshop, the measures laid down for guaranty military security must be taken.

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CHAPTER No 8

OPERATION

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CHAPTER No 08

OPERATION

CHECKS FOR INITIAL SETTINGS ON RADAR SET

1. The radar set in the radar equipment truck is switched on and checked during installation of the antenna mast.
 - (a) The procedure described in the two following sections must be carried out only if the equipment is put into operation for the first time or after a change of position or after a longer rest period or if a transmitter case or the frequency has been changed.
 - (b) In general, for normal use, the equipment must be reset to check maximum transmitter power only after it has been in operation for 1 hour.
 - (c) The radar set is normally completely cabled. If this is not the case (e.g. after long storage), connect cables.
 - (d) Ensure that the cables from the transmitter 1 (1LE1/1LE2) and from transmitter 2 (2LE1/2LE2) are not interchanged on the receiver.
 - (e) Ensure that all ground connections are correctly made inside the radar equipment truck.
 - (f) Check if the coupling nuts of all cables are secured.

Maintaining Minimum Channel Spacing of Radar Set

2. If the distance to the adjacent radar system operating in the same frequency band is less than 2.3 km, the channel spacing must be checked. If the given distance for the channel spacing between the radar systems is smaller than the transmitters must be tuned to a larger channel spacing. E.g. for a distance of 1km between two radar systems, the transmitter of one radar system must be tuned to a Channel number which differs from that of the other by at least 10.

To avoid damage to the radar transmitter (receiver input), parallel operation at a distance 50 m is not permissible.

-
- | Distance (km) | 0.05 | 0.10 | 0.16 | 0.30 | 0.50 | 0.68 | 1.0 | 1.6 | 2.5 | 2.9 | 3.2 | 3.5 | >3.6 |
|--------------------|-------------|------|------|------|------|------|-----|-----|-----|-----|-----|-----|------|
| Channel difference | not allowed | 16 | 14 | 12 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 0 |

[illegible]

Précis Type Rating Course MPDR

INITIAL SETTINGS ON EXCITER

5. Initial setting on the exciter and power transmitters are also possible during the warm-up phase.
6. If the desired values cannot be reached for the following settings on the exciter, there is a failure which must be eliminated according to the Technical Handbook SILLACS 5801245-6-2. Make sure that the correct crystal with the prescribed channel number (transmit frequency) is in the oscillator of the exciter. Open the flap of the oscillator and check the channel number indicated on the crystal. Maintain distance between adjacent radar systems.
7. If necessary, take crystal with prescribed channel number from the chest and insert it in the oscillator.
 - a. Press key U_{osc} at exciter and read meter indication on the power transmitter. The pointer should be in the green zone of the scale.
 - b. Set rotary knob A1 (U_{su}) in the exciter to the tuning number in the graduated dial. The tuning number corresponds to the channel. (Initial settings of image frequency filter). Press key U_{su} at transmitter converter and turn rotary knob A1 until maximum meter deflection is reached (fine tuning). The pointer must be in the green zone of the transmitter meter.

INITIAL SETTINGS ON POWER TRANSMITTER**Tuning Numbers for Transmitter Settings**

8. Set all dials A2 to A7 on the transmitter to the tuning number.

NOTE

Two tuning positions are possible for almost every crystal. However, only tune to the tuning number specified.

Switch on Receive Path

- (a) After switching on transformer 1 only illuminated keys OFF and LOCAL CONTROL should light on the receiver control panel. If this is not the case, press the keys referred to.
- (b) Press illuminated key STBY on the receiver control panel. The illuminated keys STBY and XMTR 1 or 2 flash. If the equipment is operating correctly, all lamps with the exception of the lamps in the illuminated keys and the green LED's in the receiver must extinguish after a few seconds. The illuminated key OFF extinguishes.

NOTE

- *During the first seconds after switching on, there is a high pitched whistling from the buzzers in the power distribution case and this indication continues until the fault lamps have extinguished or one of the FAILURE keys in the receiver has been pressed.*
- *The main fault lamps in the receiver indicator light until all switch-on operations have been completed.*
- *The lamp indicating the actual cause of the fault always lights on the defective unit.*
- *In the clock pulse module 1 and AGC module in the receiver, failure is signaled when the green LED's go out.*
- *During switch-on, the lamp SYSTEM FAULT (FAILURE EQPT) and the lamp ANT always light first.*
- *The lamp on the 150 Voltage Regulator and the lamp on the transmitter power supply only light briefly and extinguish after a few seconds.*
- *If the system is only equipped with a single transmitter, the lamp STANDBY XMTR FAULT (SP-XMTR FAILURE) also lights.*

- (c) Check if all lights in the transmitters, in the receiver and in the power

supply and PPI unit are out. Only the lamps in the illuminated keys on the receiver and indicator control panels and the lamp in the illuminated key REMOTE CONTROL on the indicator should light. If other transmitter lamps light, press key "LT" at the top right of the transmitter.

(d) Check if the lamp NO AIR FLOW is out in both transmitters. If necessary, turn on the rotary switch on the blower unit (below transmitter).

NOTE

The lamp NO AIR FLOW goes out when the required air pressure is maintained by blower

9. If the lamp NO AIR FLOW does not go out, then perform following checks.
- (a) Check for any leakage of air in air hose and dust trap filter at the outlet of blower and repair or replace as required.
 - (b) Check and clean / replace the filter at the inlet of blower

CAUTION

Do not put hand or a piece of paper at the outlet (exhaust) of final stage as this keeps hot air inside, resultantly required cooling for electron tubes decreases and causes the pre-mature failure / less operation hours of electron tubes

INITIAL SETTINGS AT INDICATOR DISPLAY

10. When the antenna assembly truck has been set up and the antenna rotates, the operation of the deflection circuits, the range rings and the azimuth marker can be checked in the indicator.

CAUTION

- *The cathode ray tube is coated with the long persistence orange phosphorus P19. This kind of phosphorus must be protected against overload (danger of ion spots.)*
- *The individual intensity controls should only be turned clockwise far enough to achieve an image just visible in a completely darkened room.*

- (a) Set the control CONTRAST fully clockwise.
- (b) Adjust background intensity with control INTENSITY so that the trace is just invisible.
- (c) Set switches for RANGE MKR and AZIMUTH MKR on front panel of the indicator to ON.
- (d) Adjust the associated controls to give the correct intensity.
- (e) On indicator read angular position of azimuth marker from 360⁰ scale and record in the log for later on checking of north alignment.

INITIAL SETTINGS ON IFF CONTROL PANEL

11. Turning on the radar unit also feeds a 24 V supply voltage to the IFF interrogation set. To put the equipment into operation, proceed as follows.

- (a) Check if all cables have been correctly connected and plugs are tight. Reconnect as necessary.
- (b) Check if the IFF interrogator is switched on (yellow) lamp "STAND BY" out) and no faults signaled; switch INTERROGATION must be set to AUT.
- (c) Press the lamp lenses on the IFF control panel in order to check the indicator lamps (PRESS TO TEST Lamps). The lamps must then light.
- (d) Set the required mode on the IFF control panel. Set mode switch to

mode 1, 1 + 2 or 3/A.

- (e) Set code selector switch to 1, 1+2 or 2 or AC (all codes).
- (f) Set thumbwheels of code-setting switch 1 (left) to ensure code, thumbwheels of code-setting switch 2 (right) to future code. For mode 1, set the code on the relevant code-setting switch with the two left-hand thumbwheels; for mode 2 and 3/A with all thumbwheels.

SWITCHING ON THE TRANSMITTER AND DETERMINING THE OPTIMAL ANTENNA HEIGHT

Switch on Transmitter

12. The transmitter can be switched on provided that the thick RF cables ZM 1 are connected to the equipment and antenna assembly truck, the antenna mast has been raised to a position where the circuit breaker on the pedestal is released and the 3 minute time delay after initial switching on has expired.

- (a) Press illuminated key XMTR 1 (or XMTR 2) on the receiver control panel.

NOTE

- *A time switch passes the start command to the relevant transmitter after a warm-up period of three minutes.*
- *The warm-up period begins immediately after the illuminated keys TEST, RECEIVE or XMTR are pressed.*
- *Certain adjustments and test (points b to d) can be carried out during the warm-up period in order to reduce the time necessary to get the set into service.*
- *At the end of the warm-up period the equipment carries out all set functions automatically.*
- *The transmitter is disabled (XMTR lamp flashes) as long as the antenna mast is not raised high enough to release the circuit breaker on the pedestal.*

- (b) After the remaining operating voltages have been switched on, check transmitter tuning with the aid of the meter built into the transmitter. The pointer deflection must lie in the green zone. The transmitter power is then adequate.
- (c) If necessary, retune transmitter.

Switch on Fixed Target Echoes Display

- 13. If the transmitter output power is adequate (no fault signal on transmitter).
 - (a) Set NORM VIDEO switch on indicator to ON in order to mix in permanent echo display.
 - (b) Turn control NORM VIDEO clockwise until the intensity of the permanent echoes on the screen is correct.
 - (c) If no permanent echoes are obtained although the radar set is in a favorable site and the antenna is able to radiate, retune image-frequency filter.

NOTE

- ◆ *When the permanent echo display is switched on, all areas scanned by the radar beam including the earth's surface are displayed bright, unless they are shielded by other objects.*
- ◆ *Thus, radar sites and mast height settings are preferred which produce the permanent echo display consisting of the maximum possible number of bright areas over the entire screen, since only in such locations can targets flying just above the earth's surface be indicate.*

Adjustment of Threshold of Normal Video Channel Monitor in AF Amplifier in Receiver

- 14. Pull out AF amplifier (5801239-525 in third line of receiver, left hand); remove cover; reinsert AF amplifier with AF adapter (in case "accessories and spare").

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15. Set switch AUTO FREQ on PPI control panel to OFF.
16. Switch system to transmit, either XMTR 1 or XMTR 2.
17. Adjust potentiometer in AF amplifier so that the lamp FAILURE EQPT in the receiver control panel just stops blinking.
18. Switch over the other XMTR.
19. The lamp FAILURE EQPT must remain dark. If the lamp blinks, repeat step 17.
20. Switch system to off. Pull out AF adapter with AF amplifier. Remove AF adapter. Attach cover and reinsert AF amplifier.

Determination of the Optimal Antenna Height

21. When the radar system operates, retract antenna mast while observing the permanent echo display, until some of the permanent echoes just start to black-out. Gradually retract the antenna mast as follows.

- (a) Place program switch on control box "North" to "ΔR" via "Y".
- (b) Pull right control lever towards operator and fold down hinged mast to desired height (leave radar system at XMTR 1 or XMTR 2).
- (c) Observe PPI until the antenna reaches the optimal height for acquisition of targets.
- (d) Stop mast at this height.
- (e) (5) Switch off fixed target echo display by placing NORM VIDEO switch to OFF.
- (f) Place program switch on control box "North" to R.

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- (g) Close cover on control box "North".
- (h) Switch off battery switch below driver's seat in antenna truck (lever in vertical position).

Switch to Remote Control

- 22. For changeover of control from the receiver to the PPI perform the following.
 - (a) Press illuminated key REMOTE on receiver control panel. The lamp in the key LOCAL extinguishes the lamp in the key REMOTE lights.
 - (b) Check if the red lamp REMOTE CONTROL on indicator control panel extinguishes after the key REMOTE on receiver has been pressed.
 - (c) Check if system is controllable from PPI: press keys XMTR 1, XMTR 2, STAND BY and TEST and observe switching action when lamp in the pressed key lights.

SWITCHING ON THE RADAR SYSTEM

- 23. The information given in this section is based on the assumption that the radar system has already been set up i.e. the antenna is erected; the equipment is wired up and aligned to north.

Switching on After Short Period Out of Service

- 24. In order to switch the radar system on again after the equipment has been temporarily out of service (illuminated key OFF on control panel pressed, i.e. radar and IFF equipment switched off), press illuminated key TRANSMITTER 1 or TRANSMITTER 2 on the indicator control panel (key REMOTE on the receiver control panel must be pressed and must light). The radar set is ready for service again after the expiration of the warm-up period of three minutes and starts to transmit.
- 25. Retuning of the transmitter in accordance with section 6-2 shortly after switching on is only necessary if lamp Ps signals a fault. In this case the tuning must be rechecked after about an hour.

Switching on After Long Period Out of Service or After Maintenance

NOTE

The following information applies to operation of the set-up equipment after a long interruption or after severe weather conditions.

Checks before Switching on

26. Check if all cables are correctly connected and plugs are tight; reconnect in accordance with Fig. 5-38, if necessary. In particular, check if the cables from transmitter 1 (1LE1/1LE2) and transmitter 2 (2LE1/2LE2) are not interchanged at the receiver.
27. Check if all ground connections are made.
28. Check oil level by means of the oil level inspection glass in antenna pedestal above the drive motor.
29. Check whether the oil in the antenna pedestal must be replaced (summer and winter oil, capacity about 2.5 liter). For operation in ambient temperatures of 0 to 55 °C lubricating oil NATO reference X-15 (0-184). For operation in ambient temperatures from + 10 to -40 °C lubricating oil NATO reference 0-186.
30. Check oil and fuel level on generator (see Technical Handbook SILLACS 5801245-63-1).
31. Check if all switches on the antenna drive are on.
32. Check if all cables in the transmit path between transmitter and antenna are properly connected and the connectors are firmly screwed together.

SWITCHING ON THE RADAR EQUIPMENT SET

- (a) Switch on generator.
- (b) Switch on transformer 1.
- (c) Check phase sequence.
- (d) Switch on antenna drive.
- (e) Switch on receive path and check with noise generator

33. Only the illuminated key OFF and LOCAL CON.: TROL should light on the receiver control panel, if this is not the case, press the keys referred to.

34. Consecutively press illuminated keys TRANSMITTER 1 and TEST on the receiver control panel. This turns on the noise generator in blower unit 1 to check the receive path via transmitter 1. If the equipment is operating correctly, all lamps with the exception of the lamps in the pressed keys TEST, TRANSMITTER 1 and LOCAL must go out after a few minutes.

NOTE

- ♦ *Illuminated key TEST flashes until the ionization time (2 minutes) of the noise generator has expired. (See section 5-13.2.b).*
- ♦ *In addition, the green LED's in the clock pulse module 1, in the mixer 2 and in the AGC module must be lit.*

35. Check if the lamp NO AIR FLOW is out on both transmitters. If necessary, turn on the rotary switch on the blower unit.

36. Check if all lamps on the transmitters, on the receiver, on the power supply and on the indicator are out. Only the lamps in the control buttons of the receiver and indicator and the lamp in the illuminated key REMOTE TROL on the indicator should light. The 4 green LED's of the receiver described above are also illuminated. If other lamps on the transmitter light, press key LT at the top right of the transmitter.

37. Wait until the lamp in the key TEST lights. The generator has then fired. The ECCM switch of the PPI control panel is in position OFF. If possible, a large distance range towards the centre of the PPI should be painted bright.

Check Settings of the PPI

38. If the antenna is rotating (check) the deflection, the range rings and the azimuth marker can be checked on indicator.

39. If there is no trace to be seen, set the CONTRAST .01 (on the left near the screen) fully clockwise and the INTENSITY control clockwise until the trace is invisible (range rings should be visible).

40. Set the RANGE MARKERS and the AZIMUTH MARKERS on the front panel of the indicator to ON. Set relevant controls to produce the correct brightness, but turn as far as necessary as otherwise the beam can the screen. The CONTRAST control must be turned

Switching on and checking transmitter

41. If the thick RF cable is connected to the equipment truck and antenna truck and if at least three minutes elapsed since initially switching on, press the illuminated key XMTR 1 or XMTR 2 on the control panel to select transmitter.

42. After the remaining operation voltages have been switched through this occurs automatically when key XMTR 1 or XMTR 2 is pressed - check the transmitter tuning. If the pointer lies in the green zone, it should be retuned only if work is to be done on the transmitter or if it has warmed up. The pointer of the meter in the transmitter must lie in the green zone. If necessary, retune the transmitter. The pointer must always be below green zone when key P_R (reflected transmitter power) is pressed.

43. If the transmitter produces enough output power (no fault signal on transmitter), set the NORM VIDEO switch on the indicator to ON in order to mix in the permanent echo display. Ensure that control VIDEO GAIN is fully on.

44. Turn the control NORM VIDEO clockwise until the brightness of the fixed target echoes on the screen is as required. If the usual fixed target echoes do not appear on the screen and the antenna has not been resited, retune the image frequency filter.

Switching to Remote Control

45. In order to switch control from the receiver to the indicator, actuate illuminated key REMOTE CONTROL on the receiver control panel. The lamp in the key LOCAL CONTROL goes out and the lamp in the key REMOTE CONTROL lights.
46. After pressing key REMOTE CONTROL on the receiver, check if lamp REMOTE CONTROL on the indicator control panel goes out and the lamp XMTR 1 or XMTR 2 lights.
47. Check the control of the system from the indicator. Set ECCM switch to OFF.
48. Read the position of the azimuth marker and compare with notation in log. If there is a difference, realign the RADAR SET TO NORTH.

TUNING THE TRANSMITTER



- The procedure described in this section should be carried out only when the set is first put into service or switched on after change of position or switched on after being out of service for a long period of time or when a transmitter equipment case has been replaced or the frequency has been changed.*
- In normal use the units should not be re-tuned to check the maximum transmitter output until they have been operating for at least 1 hour*

49. Tune the exciter.
50. Set the tuning number for the transmitter path. Set all dials on the transmitter to the tuning number.

Tuning the Image Filter

51. Consecutively press key XMTR 1 or XMTR2 of the desired transmitter and key TEST on the receiver control Panel.

CAUTION

The transmitter is switched on. The noise generator is switched to the desired receive path by pressing the key TEST. Set switch STC to ON, switch ECCM and switch AUTO-FREQ to OFF.

52. Set knob A 7 on the power transmitter to the tuning number which corresponds to the relevant channel (presetting of the image frequency filter).
53. After the noise generator has fired (about 2 minutes key TEST lights) turn the knob A7 until the intensity on the PPI reaches its maximum value (fine tuning).

Tuning the Power Transmitter

NOTE

If the required transmitter power is not obtained after 25 seconds fault lamps P_s on the monitor lights and changeover from position AUTO-FREQ ON to the other transmitter takes place a few seconds later.

NOTE

- *On the receiver control panel the lamps in the push-button SP-XMTR FAILURE and in the illuminated key EQPT FAILURE light. This last indication can be canceled together with the buzzer in the power distributor by pressing one of the illuminated keys FAULT.*
- *Switching off is delayed for about 25 seconds by pressing one of the test keys J_{K1} to J_{K5} on the transmitter.*

- *When the transmitter has switched off the fault signal stored in the monitor must be canceled by pressing the key LT on the monitor. The transmitter can now be switched on again by pressing the relevant transmitter key, but a new warm-up time of 3 minutes must first expire.*
- *If only one transmitter is in service (lamp SP-XMTR FAILURE lights), the transmitter is not turned off but the lamp EQPT FAILURE flashes*

54. Press key XMTR 1 or XMTR2. If the warm-up time has expired, lamps Ps in the transmitter monitor and SYSTEM FAULT (EQPT FAILURE) on the control panel flash if the transmitter power is too low (e. g. transmitter not tuned).

55. As the automatic turn-off might start, start tuning as soon as the flashing begins. Press key JK2 to JK5 on the transmitter one after the other and turn the knobs A2 to A5 which have identical indices until the maximum meter deflection is reached in each case. c. Turn knob A6 (without key) to obtain maximum meter deflection (tuning the power stage for maximum output power). The transmitter power is continuously indicated on the meter and it is unnecessary to press a key for this purpose.



The delay of 25 seconds before the transmitter is switched off due to insufficient output power is not automatically available during this tuning operation. One of the keys J_{K1} to J_{K5} can be pressed briefly during this delay period in order to postpone switching off by

56. Then turn knob ANT MATCHING on the power transmitter to produce maximum meter indication (matching transmitter to antenna).

57. Repeat the tuning with knobs A2 to A6 and ANT MATCHING without pressing the associated keys, tuning for maximum meter indication in each case.

58. After the equipment has been operating for about 1 hour, readjust all tuning elements for maximum meter indication.

TRANSMITTER CHANGEOVER AND FREQUENCY CHANGE AT THE RADAR SET

Transmitter Changeover

59. Two transmitters are employed for normal operation. If the transmitter power falls below a limiting value (80% of nominal power) set in the transmitter monitor, changeover to the standby transmitter takes place automatically.

60. Changeover from one transmitter to the other can also be affected, by pressing the relevant illuminated keys on the receiver control panel (with illuminated key LOCAL pressed) or on the indicator (illuminated key REMOTE on receiver control panel pressed).

61. In general, it is not necessary to tune the transmitter which has just been put into operation, as the standby transmitter too has been tuned to maximum power during the last transmission operation.

62. During operation with only one transmitter the lamp SYSTEM FAULT (EQPTFAILURE) and STANDBY XMTR FAULT (SP-XMTR) light on the control panel. Cancel lamp SYSTEM FAULT (EQPT FAILURE) by pressing key. When the power falls below a certain value (80%) signal lamp Ps on the transmitter monitor flashes. Changeover to the standby transmitter is blocked so that in spite of reduced transmitter power operation with reduced range is possible. The fault indication at the indicator or receiver remains unchanged (lamp STANDBY XMTR FAULT or SP-XMTR FAILURE lights).

Frequency Change

63. If the frequency of a transmitter is to be changed, replace the crystal in the driver by another.

64. Then retune the transmitter.

MONITORING DURING OPERATION

Automatic Monitoring

75. Faults which endanger or interrupt operation of the radar set are indicated by signaling equipment in the equipment room and on the indicator console. Certain

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malfunctions simultaneously cause buzzers to sound in the power distribution case and indicator and produce lamp signals on the receiver and indicator control panels. The buzzer can be switched off briefly by pressing the illuminated fault key. Any further faults that occur do not cause the buzzer to operate again. The lamp in the pushbutton lights until the cause of the fault has been eliminated. If the fault lamp flashes, it could indicate a fault which permits only limited radar operation.

76. Excessive temperatures in the equipment room may cause Individual monitoring circuits in the cases to signal faults or to interrupt the operation.

(a) Excessive temperature in the receiver only causes the lamp SYSTEM FAULT to flash. In this case check if the blower in front of the receiver case is working.

(b) Excessive temperature in the transmitter produces automatic changeover to the second transmitter. The lamps TEMP in the module HIPOT monitor (5801232-177) light during the phase initiating the changeover.

(c) Inadmissibly high temperatures in the regulator 150 V cause the regulator to be cut off before the transistors could be damaged. In this case, the main failure lamp on the regulator lights.

(d) In addition to the main fault lamps, a lamp in the equipment case containing the component causing the fault usually lights. Exceptions are the green LED s in the receiver which go out in the event of failure in the relevant module.

(e) If the entire power supply fails, all lamps are out, including the lamps in the illuminated keys indicating whether the equipment is on or off.

(f) The IFF interrogator has a separate test sequence. In the event of a fault, only the red lamp FAULT lights on the IFF control unit and a buzzer sounds in the IFF control unit.

Monitoring by the Operator at the Indicator

77. Additional monitoring is carried out by the operator at the indicator. Variations from the normal presentation on the CRT screen indicate a fault. This can be due to an external interference source. The fault can also be a malfunction in the radar equipment.

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78. The following faults can occur on the indicator screen without causing the monitoring facilities to respond.

- (a) The screen IS completely dark although the controls for contrast, intensity and sensitivity are turned fully up; the mile markers and the fixed target echoes (NORM VIDEO) are switched on and the associated controls are turned up.
- (b) Video signals are missing. The rotating trace is visible,' but no mile markers, azimuth marker, fixed or moving targets can be seen although all controls are turned up.
- (c) No mile markers although the switch RANGE MKR is set to ON and the associated control is turned up. Fixed and moving targets are displayed.
- (d) No moving targets appear m spite of flying activity.
- (e) The trace moves in steps or moves backward and forward.
- (f) The trace does not rotate although the antenna is rotating.
- (g) The display is displaced to one side.
- (h) The trace only paints half the screen diameter.
- (i) The azimuth marker rotates.

79. The receive path of the radar set should be checked several times, per day from the operator's console.

Testing with Fixed Target Echoes (NORM VIDEO)

80. Set switch NORM VIDEO to ON. Turn control VIDEO GAIN fully clockwise.

81. Leave controls or contrast, intensity, sensivity and normal video in the normal operating position. The fixed targets appropriate to the location should then appear.

82. Repeat the above test with the second transmitter.

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Testing with Noise Generator

83. Press illuminated key TEST.
84. Controls CONTRAST, INTENSITY, NORM VIDEO and VIDEO GAIN in their normal operating position.
85. Set switch NORM VIDEO, ECCM OFF and AUTO-FREQ to OFF. About two minutes after switching on the noise generator, the screen must be painted very bright in sections (the innermost display range remains dark due to STC is on) when the control ECCM and AUTO FREQ on the indicator are set to OFF. If key STC on the transmitter is pressed ("off"), the innermost display range too must be painted bright.
86. Set switch NORM VIDEO to ON, VIDEO GAIN fully clock wise. Remove one A/D converter in the receiver. The screen must be painted very bright. Reducing the gain with control VIDEO GAIN the test signal appears on the screen.
87. Repeat the above test with the second transmitter. This is done by pressing key TEST and the key for the second transmitter (TRANSMITTER 1 or TRANSMITTER 2) on the receiver control panel.
88. Reinsert the A/D converter in the receiver.

TESTING THE ECCM CIRCUIT

89. Operation of the automatic ECCM circuit can be checked with the key TEST (noise generator on) pressed on the receiver control panel.
 - (a) Set switches NORM VIDEO, ECCM and AUTO- FREQ to OFF. Key TEST is pressed. When the noise generator has fired a cross bar pattern must be visible on the PPI. The innermost display range remains dark due to the STC.
 - (b) Set the STC switch to OFF; the innermost display range must also brighten up.
 - (c) Set ECCM switch to CF AR. The cross bar pattern disappears. The interference alarm is displayed outside the 25 miles ring.

- (d) Set ECCM switch to PLD. Same display as in point C
- (e) Set switch ECCM to CFAR or PLD and switch AUTO-FREQ to ON. An automatic changeover from one to the other XMTR in a rhythm of about one minute must take place. This changeover can be interrupted by pressing the FOOT SWITCH.
- (f) Set switch ECCM and switch AUTO -FREQ to ON. Press FOOT SWITCH. No automatic changeover must take place. Gain must be controllable with the control VIDEO GAIN.

OPERATION UNDER SPECIAL CONDITIONS

"System Fault" Signal

90. If EQPT FAILURE or SYSTEM FAULT is signaled on the receiver or indicator control panel and a buzzer sounds, operation can be maintained under the following conditions.

- (a) There is no excess temperature in the receiver.
- (b) The system is equipped with two transmitters.

91. Changeover to the other transmitter then takes place automatically, the illuminated key SP-XMTR FAILURE or STANDBY-XMTR FAULT in the receiver or indicator lights.

- (c) The system is equipped with one transmitter or the blower of the second transmitter is switched off. The system is then not switched off and lamp P_s in the transmitter monitor flashes; the transmitter continues to operate but at reduced output power. The lamp STANDBY-XMTR FAULT or SP-XMTR FAILURE lights continuously. The system is switched off for all other transmitter faults.

ELECTRONIC COUNTER-COUNTERMEASURES

92. If the receive path of the radar set MPDR45/E is disturbed by signals coming from jamming transmitters (ECM, electronic countermeasures) in such a way that part of the screen would be painted very bright, the echoes received from targets might be masked. This is counteracted by an ECCM control unit so that the received jamming signals are not evaluated. To this end, it is necessary to select the appropriate operation mode on the control panel of the PPI to counteract jamming signals.

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- (a) CW interference (e.g. sector completely bright) switch on CFAR.
- (b) Pulse interference (e.g. multiple channel interference) switch on PLD.

OPERATION UNDER DIFFICULT CLIMATIC CONDITIONS

93. If gusts with wind speeds exceeding 140 km/h are anticipated, the antenna drive must be switched off (antenna switch on power distribution case in horizontal position to 0).

94. If gusts in excess of 160 km/h are anticipated, the antenna mast must be retracted to just about the driver's cab. The antenna brake must not be pulled on, so that the antenna can align itself in the direction of minimum air distance.

95. If the antenna assembly truck is positioned on soft ground, it may be necessary to retract the mast even at low wind speeds.

96. If thick ice forms on the antenna reflector, the mast must be retracted and the ice carefully removed, Take care not to bend the mesh forming the reflector.

97. The antenna truck equipment may be operated at ambient temperatures from 328 K to 233 K.

98. The radar set is designed for operation at ambient temperatures from 248 K to 328 K. The interior of the radar equipment truck should be warmed up by air condition before switching on when the ambient temperature is below 248 K.

99. The air conditioner must be on when the doors are closed. If the doors are open for a longer period of time when the air conditioner is in operation, the air conditioner becomes ineffective and may be damaged as a result of continuous operation.

100. Keep the antenna truck and the roofs of the drivers' cabs free of snow at all times, especially when it starts to thaw.

101. Ensure that the oil in the antenna pedestal is suitable.

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SWITCHING OFF THE EQUIPMENT

102. Press illuminated key OFF on the indicator/ RX control panel. This switches off all units of the radar and IFF equipment with the exception of transformer 1.

103. Set main switch at transformer 1 to position 0.

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LIST OF ABBREVIATION USED IN THIS TEXT

ac	Alternating current	n.m.	Nautical Mile
AGC	Automatic gain control	NSK	North seeking gyroscope
AVA	Amplitude versus azimuth	P	Positive
BITE	Built-in-test equipment	PCB	Printed circuit board
CFAR	Constant false alarm rate	PLD	Pulse length discriminator
CRT	Cathode ray tube	PPI	Plan Position Indicator
dc	direct current	PRF	Pulse Repetition Frequency
ECCM	Electronic Counter-Counter Measure	PROM	Programmable ROM
ECM	Electronic Counter Measure	RAM	Random Access Memory
EQPT	Equipment	ROM	Read Only Memory
EXT	External	RSLS	Reply Path Side-Lobe Suppression
FR	Enable signal	SCV	Sub-Clutter Visibility
GTC	Gain Time Control	SIF	Selective Identification Feature
IC	Integrated Circuit	SILLACS	Siemens Low Level Air Control System
IFF	Identification Friend or Foe	SLS	Side-Lobe Suppression
ISLS	Interrogation path Side-Lobe Suppression	SSR	Secondary Surveillance Radar
ISU	Inter-pulse Suppression amplitude	STAMPLE	Interference
LED	Light-emitting Diode	STBY	Stand By
M	Minus	STC	Sensitivity Time Control
ML	Maintenance Level	STM	Fault indication
MOS	Metal Oxide Silicon	STP	Start Pulse
MPDR	Mobile Pulse Doppler Radar	TTL	Transistor-Transistor-Logic
MPUE	Minimum level supervisions	TR	Transmitter/ Receiver

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MSR	Mobile Secondary Radar
MTA	Multiple Time Around
MTI	Moving Target Indication
NKE	North seeking gyro system

ÜS	Carry
XMTR	Transmitter
ZERH	chopper pulse

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aircraft employ ground-mapping Radars to image a scene. The radio altimeter is also Radar used to indicate the height of an aircraft above the terrain and as a part of self contained guidance system over land.

Ship Safety

75. There is one of the least expensive, most reliable and largest applications of Radar. Radar is found on ships and boats for collision avoidance and to observe navigation buoys, especially when the visibility is poor. The automatic detection and tracking equipments are available with these Radars for collision avoidance. Similarly shore based Radars of moderate resolution are used from harbor surveillance.

Space

76. Radars are used for rendezvous and docking and was used for landing on the moon. The large ground based Radars are used for detection and tracking of satellites. The satellite-borne Radars used for remote sensing (SAR, Synthetic Aperture Radar).

Remote Sensing

77. All Radars are remote sensors, and used for sensing geophysical objects (the environment). The Radar astronomy is used to probe the moon and planets. The earth resources monitoring Radars measure and map sea conditions, water resources, ice cover, agricultural land use, forest conditions, geological formations, environmental pollution (Synthetic Aperture Radar, SAR and Side Looking Airborne Radar, SLAR).

Law Enforcement

78. The Radar speed meter, familiar to many, is used by police for enforcing speed limits. Radar has been considered for making vehicles safer by warning of pounding collision, actuating the air bag, or warning of obstructions or people behind a vehicle or in the side blind zone. It is also employed for detection of intruders.

Military

79. Radar is an important part of Air Defence Systems as well as the operation of offensive missiles and other weapons. In Air Defence it performs the functions of surveillance and weapon control. Surveillance includes target detection, target recognition, target tracking, and designation to a weapon system. Weapon-control Radars track targets, direct the weapon to an intercept, and assess the effectiveness of the engagement (called battle damage assessment). A missile system might employ Radar methods for guidance and fusing of the weapon. High resolution imaging Radars, such as synthetic aperture Radar, have been used for reconnaissance purpose and for detecting fixed and moving targets on the battle field.